TRIPATHI STUDY ZONE

JEE MAIN | NEET | AIIMS | KVPY | BITSAT

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CHAPTER 1 RELATIONS & FUNCTIONS

1. INTRODUCTION

Relation is a word used in common language. In our daily life we come across many relations.
For example : (i) Delhi is the capital of India. (ii) 5 is a divisor of 1. (iii) Shyam is a son of Sohan (iv) Triangle *ABC* is congruent to triangle *DEF* (v) The set B is a subset of the set *A*.
In this chapter, we will study different types of relations and functions, composition of functions, invertible functions and binary operations.

2. TYPES OF RELATIONS

Empty relation : A relation *R* in a set *A* is called empty relation, if no element of *A* is related to any element of *A*,
i.e., $R = f \subset A \times A$.
Universal relation : A relation *R* in a set *A* is called universal relation, if each element of *A* is related to every element of A,
i.e., $R = A \times A$.
Both the empty relation and the universal relation are some times called trivial relations.
Reflexive relation : A relation *R* defined on the set *A*, is said to be reflexive if each element of *A* is related to itself. In other words, *R* will be reflexive if $(a, a) \in R \ \forall \ a \in A$. If there is any element in A which is not related to itself, the given relation will not be reflexive.
In other words, a is essentially related to itself but it may also be related to other elements of *A*, whereas in case of identity relation, a is related to a and only to *a*. Hence every identity relation is reflective but every reflexive relation is not an identity relation.
Symmertic relation : Let *R* be a relation, defined on the set *A*. Then *R* is said to be symmetric if under *R*, *a* related to *b* implies, *b* related to *a*, i.e. *R* will be symmetric, if

$$(a,b) \in R \Rightarrow (b,a) \in R \ \forall \ a,b \in R$$

Clearly a relation defined in *A* will be symmetric, if there exist at least two elements *b* in *A* such that $(a, b) \in R$ but not $(b, a) \in R$
If *R* is a symmetric relation defined on a set *a*, then $R = R^{-1}$.

Since $(x,y) \in R \Rightarrow (y,x) \in R \Rightarrow (x,y) \in R^{-1}$ $\quad \therefore \quad R \subseteq R^{-1}$ ..... (1)

Similarly, $(x,y) \in R^{-1} \Rightarrow (x,y) \in R \Rightarrow (x,y) \in R$ $\quad \therefore \quad R^{-1} \subseteq R$ ..... (2)

From (1) and (2) $R = R^{-1}$.
Anti symmetric relation : A relation *R*, defined in a set *A* is said to be antisymmetric if *a* related to *b* and *b* related to *a* implies
$a = b$ where $a, b \in A$.
i.e., $(a, b) \in R$ and $(b, a) \in R \Rightarrow a = b \ \forall \ a, b \in A$
Clearly a relation *R*, defined on a set *A* will not be antisymmetric, if there exist at least elements *a* and *b* in *A* such that
$(a, b) \in R$ and $(b, a) \in R$ but $a \neq b$.
Transitive relation : Let *R* be a relation defined on the set *A*. Then *R* is said to be transitive if under *R*, a related to *b* and *b* related to *c* implies a related to *c*. i.e.

$(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R, \ \forall \ a, b, c \in A$

Equivalence relation

A relation *R* defined on a set *A* is said to be an equivalenceelation, if
(i) *R* is reflexive, i.e. $(a, a) \in R \ \forall \ a \in A$

(ii) R is symmetric, i.e. $(a, b) \in R \Rightarrow (b, a) \in R \; \forall \; a, b \in A$

(iii) R is transitive, i.e., $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R, \; \forall \; a, b, c \in A$

Partial order relation

A relation R defined on a set A is said to be partial order relation, if :

(a) R is reflexive, (b) R is antisymmetric, (c) R is transitive,

If a partial order relation is defined on a set A, then A is called a partially ordered set.

Total order relation : A relation R defined on a set A is said to be a total order relation, if :

(a) R is a partial order relation.

(b) Any two elements of A are comparable, i.e. for any $a, b \in A$, either $(a, b) \in R$ or $(b, a) \in R$ or $a = b$.

A set A on which a total order relation is defined is called a totally ordered set.

Illustration 1: Show that the relation R in the set {1, 2, 3} given by R = {(1, 1), (2, 2), (3, 3), (1, 2), (2, 3)} is reflexive but neither symmetric nor transitive.

Solution : R is reflexive, since (1, 1), (2, 2) and (3, 3) lie in R. Also, R is not symmetric, as $(1, 2) \in R$ but $(2, 1) \in R$. Similarly, R is not transitive, as $(1, 2) \in R$ and $(2, 3) \in R$ but $(1, 3) \notin R$.

Illustration 2 : Examine whether the following relations R and P are reflexive, symmetric and transitive.

(i) $a \; R \; b \Leftrightarrow b$ is divisible by a, in N.

(ii) $a \; R \; b \Leftrightarrow a \perp b$ where a and b are lines in the same plane.

Solution : (i) By definition of R, $a \; R \; b \Leftrightarrow b$ is divisible by a; $a, a, b \in N$ Reflexive : R is reflexive since for every $a \in N, a|a$

Symmetric : R is not symmetric, for $a, b \in N$, if a divides b then b cannot divide a (unless $a = b$).

Transitive : R is transitive since for $a, b, c \in N$, $a \; | b$ and $b \; | c \Rightarrow a | c$.

(ii) By definition of R, $a \; R \; b \Leftrightarrow a \perp b$

Reflexive : R is not reflexive because for any line a, $a \perp a$, is not true

Symmetric : R is symmetric because for two lines a and b , $a \perp b \Rightarrow b \perp a$

Transitive : R is not transitive since for three coplanar lines, a, b and c , and $a \perp b$ and $b \perp c \Rightarrow a \; || \; c$

Illustration 3 : If R is an equivalence relation defined on a set A, prove that its inverse relation R^{-1} is also an equivalence relation on A.

Solution : Since R is a relation on A.

$\therefore \; R \subseteq A \times A \Rightarrow R^{-1} \subseteq A \times A$

Hence R^{-1} is also a relation in A.

We shall prove that R^{-1} is an equivalence relation.

(i) Reflexive : Let $a \in A$ then $a \in A \Rightarrow (a, a) \in R \Rightarrow (a, a) \in R^{-1}$

Thus $(a, a) \in R^{-1}, \; \forall \; a \in A$

Hence, R^{-1} is reflective.

(ii) Symmetric : Let $(a, b) \in R^{-1}$

then $(a, b) \in R^{-1} \Rightarrow (b, a) \in R \Rightarrow (a, b) \in R \Rightarrow (b, a) \in R^{-1}$ [$\because$ R is symmetric]

Thus, $(a, b) \in R^{-1} \Rightarrow (b, a) \in R^{-1}$

Hence, R^{-1} is symmetric.

(iii) Transitive : $(a, b) \in R^{-1}$ and $(b, c) \in R^{-1}$

Now, $(a, b) \in R^{-1}$ and $(b, c) \in R^{-1} \Rightarrow (b, a) \in R$ and $(c, b) \in R$

$\Rightarrow$ $(c, b) \in R$ and $(b, C) \in R \Rightarrow (c, b) \in R$ and $(b, a) \in R$

$\Rightarrow$ $(c, b) \in R$ [$\therefore$ R is transitive]

$\Rightarrow$ $(a, c) \in R^{-1}$

Thus, $(a, b) \in R^{-1}$ and $(b, c) \in R^{-1} \Rightarrow (a, c) \in R^{-1}$

Hence, R^{-1} is transitive. Hence R^{-1} is also an equivalence relation.

Student Task

Q.1 Show that the relation R in the set R of real numbers, defined as $R = \{(a, b) : a \leq b^2\}$ is neither reflexive nor symmetric nor transitive.

Q.2 Show that the relation R in R defined as $R = \{(a, b) : a \leq b\}$, is reflexive and transitive but not symmetric.

Q.3 Show that the relation R in the set {1, 2, 3} given by $R = \{(1, 2), (2, 1)\}$ is symmetric but neither reflexive nor transitive.

Q.4 Show that the relation R in the set $A = \{1, 2, 3, 4, 5\}$ given by $R = \{(a, b) : |a - b| \text{ is even}\}$, is an equivalence relation. Show that all the elements of {1, 3, 5} are related to each other and all the elements of {2, 4} are related to each other. But no element of {1, 3, 5} is related to any element of {2, 4}.

Q.5 Show that the relation R in the set A of points in a plane given by $R = \{(P, Q) :$ distance of the point P from the origin is same as the distance of the point Q from the origin$\}$, is an equivalence relation. Further, show that the set of all points related to a point $P \neq (0, 0)$ is the circle passing through P with origin as centre.

Q.6 Show that the relation R defined in the set A of all polygons as $R = \{(P_1, P_2) : P_1$ and P_2 have same number of sides$\}$, is an equivalence relation. What is the set of all elements in A related to the right angle triangle T with sides 3, 4 and 5?

Q.7 Let R be the relation in the set {1, 2, 3, 4} given by $R = \{(1, 2), (2, 2), (1, 1), (4,4), (1, 3), (3, 3), (3, 2)\}$. Choose the correct answer.

(a) R is reflexive and symmetric but not transitive
(b) R is reflexive and transitive but not symmetric
(c) R is symmetric and transitive but not reflexive
(d) R is an equivalence relation.

Q.8 Let R be the relation in the set N given by $R = \{(a, b) : a = b - 2, b > 6\}$. Choose the correct answer.

(a) $(2, 4) \in R$ (b) $(3, 8) \in R$ (c) $(6, 8) \in R$ (d) $(8, 7) \in R$

ANSWERS

7. (b) 8. (c)

3. PERIODIC FUNCTION

A function $f(x)$ is said to be periodic function if, there exists a fixed positive real number T independent of x, such that, $f(x + T) = f(x)\ \forall\ x \in$ Domain. T is called one of the period of the function.

In other words, a function is said to be periodic function if its each value is repeated after a definite interval.

Here the least positive value of T is called the fundamental period of the function.

Clearly $f(x) = f(x + T) = f(x + 2T) = f(x + 3T) = \ldots$

For example :

(a) sin x, cos x, sec x and cosec x are periodic functions with period 2π.
(b) tan x and cot x are periodic functions with period π.
(c) $|\sin x|, |\cos x|, |\tan x|, |\cot x|, |\sec x|, |\text{ cosec } x|$ are periodic functions with period π.
(d) $\sin^n x, \cos^n x, \sec^n x, \text{cosec}^n x$ are periodic functions with period 2π or π according as n is odd or even.
(e) $\tan^n x$ and $\cot^n x$ are periodic function with period π whether n is odd or even.

PROPERTIES OF PERIODIC FUNCTION

(i) If $f(x)$ is periodic with period T, then **(a)** $f(x + c)$ is periodic with period T. **(b)** $f(x) \pm c$ is periodic with period T.

(c) $f(ax + b)$ has period $T/|a|$, i.e., period is affected only by coefficient of x, where; a, b, c, are constants with $a,\ b \neq 0$.

(ii) If $f(x)$, $g(x)$ are periodic functions with periods T_1, T_2 respectively then; we have $h(x) = f(x) + g(x)$ has period as,

(a) LCM of $\{T_1, T_2\}$; if $f(x)$ and $g(x)$ can not be interchanged by adding a least positive number less than the LCM of $\{T_1, T_2\}$.

(b) k; if $f(x)$ and $g(x)$ can be interchanged by adding a least positive number k ($k <$ LCM of $\{T_1, T_2\}$).

(iii) A constant function is periodic but does not have a fundamental period.

(iv) If g is periodic then fog will always be a periodic function. Period of fog may or may not be the period of g.

While taking LCM we should always remember,

(a) LCM of p and q always exist if p/q is a rational quantity.

(b) LCM of rational with rational is possible

Ex. LCM of $\left(\frac{a}{b}, \frac{c}{d}, \frac{e}{f}\right) = \frac{LCM\ of\ (a,c,e)}{HCF\ of\ (b,d,f)}$; Where $a, b, c, d, e, f \in I$ with $b, d, f \neq 0$

(c) LCM of irrational with irrational is possible (but not always)

(d) But LCM of rational and irrational is not possible.

e.g., LCM of $(2\pi, 1, 6\pi)$ is not possible as $2\pi, 6\pi$ are irrational and 1 is rational.

All periodic functions can be analyzed over an interval of one period within the domain as the same pattern shall be repetitive over the entire domain.

Illustration 4 : Find the period of $f(x) = |\sin x| + |\cos x|$.

Solution : $|\sin x|$ has period π, $|\cos x|$ has period π

Hence, according to the rule of LCM, period of $f(x)$ must be π.

But $\left|\sin\left(\frac{\pi}{2}+x\right)\right| = |\cos x|$ and $\left|\cos\left(\frac{\pi}{2}+x\right)\right| = |\sin x|$

Since, $\frac{\pi}{2} < \pi$, period of f (x) is $\frac{\pi}{2}$

Illustration 5 : If $f(x) = \sin x + \cos ax$ is a periodic function, show that a is a rational number.

Solution : Period of $\sin x = 2\pi = \frac{2\pi}{1}$ and period of $\cos ax = \frac{2\pi}{a}$

$\therefore$ Period of $\sin x + \cos a x$ = L.C.M of $\frac{2\pi}{1}$ and $\frac{2\pi}{a}$ = $\frac{\text{LCM of } 2\pi \text{ and } 2\pi}{\text{H.C.F. of 1 and } a} = \frac{2\pi}{\lambda}$ where λ is the H.C.F. of 1 and a.

Since 1 is the H.C.F of 1 and a, $\frac{1}{\lambda}$ and $\frac{a}{\lambda}$ should be both integers.

Suppose $\frac{1}{\lambda} = m$ and $\frac{a}{\lambda} = n$, then $\frac{a/\lambda}{1/\lambda} = \frac{n}{m}$, where $n, m \in I$ i.e., $a = n/m$.

Hence, a is rational number

Student Task

Q.1 The period of the function, $f(x) = [\sin 3x] + \frac{1}{2}\cos 6x$, (where [.] denotes the greatest integer less than or equal to x), is

(a) π (b) $2\pi/3$ (c) 2π (d) $\pi/2$

Q.2 If $f(x) = \sin \sqrt{[a]}\, x$, (where $[x]$ denotes the greatest integer function, has π as it's fundamental period, then

(a) $a = 1$ (b) $a \in [1, 2)$ (c) $a = 9$ (d) $a \in [4, 5)$

Q.3 Fundamental period of the function, $f(x) = \cos(\tan x + \cot x) \cdot \cos(\tan x - \cot x)$, is

(a) $\pi/4$ (b) $\pi/2$ (c) π (d) 2π

Q.4 $\sin 2\sqrt{x+1}$ is period

(a) 2π (b) π (c) $\pi/2$ (d) None of these

Q.5 If the period of the function $f(x) = \tan(\sqrt{[k]}x)$, where [.] denotes the greatest integer function, is π, then

(a) $1 < k < 2$ (b) $1 \le k < 2$ (c) $k = 1, 2$ (d) none of these

Q.6 Period of function $f(x) = \frac{1}{3}(\sin 3x + |\sin 3x| + [\sin 3x])$ where [.] denotes the greatest integer function is

(a) $\pi/3$ (b) $2\pi/3$ (c) $4\pi/3$ (d) π

ANSWERS

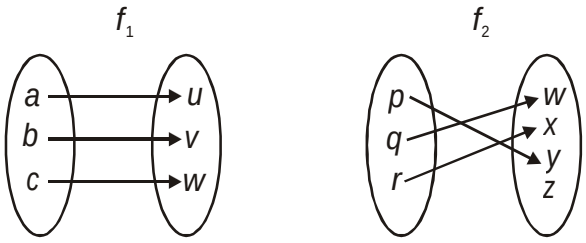
1. (b) 2. (d) 3. (b) 4. (d) 5. (b) 6. (b)

4. TYPE OF FUNCTION

One -One function or Injection :

A function $f : A \to B$ is said to be one-one if different elements of A have different images in B. Therefore for any two elements x_1, x_2 of a set A

$x_1 \neq x_2 \Rightarrow f(x_1) \neq f(x_2)$ or $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ then function is one - one

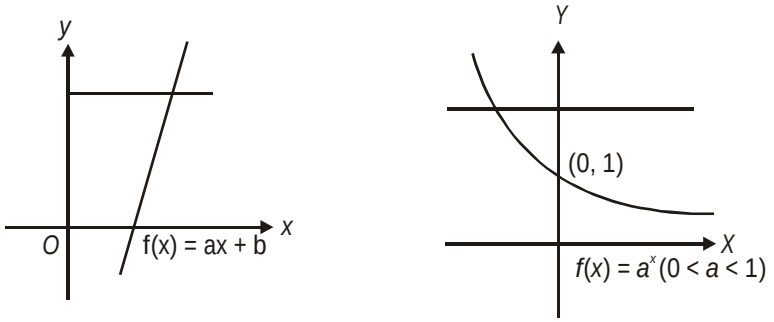


The above given diagrams show f_1 & f_2 one-one function.

Note :- (a) If function is given in the form of ordered pairs and if no two ordered pairs have same second element then function is one-one.

(b) If the graph of the function $y = f(x)$ is given, and each line parallel to x- axis cuts the given curve at maximum one point then function is one - one.

(c) If function is always increasing, (f'(x) > 0) or decreasing (f'(y) < 0) in their whole interval then function is one-one.



Examples of one-one functions

(i) $f : R \to R$, $f(x) = x$ (ii) $f : R \to R$, $f(x) = ax + b$,

(iii) $f : R \to R$, $f(x) = ax^n + b$, n is odd positive integer (iv) $f : R \to R$, $f(x) = x\,|x|$

(v) $f : R \to R$, $f(x) = e^x$, (vi) $f : R \to R$, $f(x) = a^x$ ($a > 0$)

(vii) $f : R_0 \to R$, $f(x) = 1/x$, (vii) $f : R^+ \to R$, $f(x) = \log x$,

(ix) $f : R^+ \to R$, $f(x) = \log_a x$ ($a > 0$)

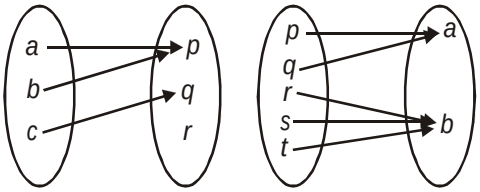
Many - One Function

A function $f : A \to B$ is called many- one, if two or more different elements of A have the same f - image in B.

Therefore for any two elements x_1, x_2 of a set A

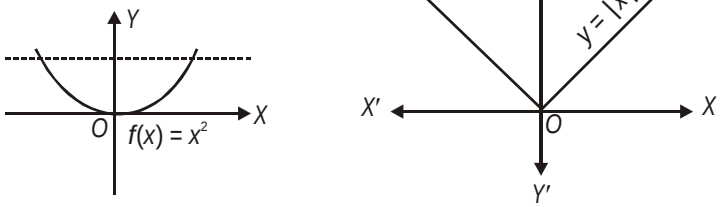
$x_1 \neq x_2 \Rightarrow f(x_1) = f(x_2)$ then function is many one.

The above given arrow-diagrams show many-one function.



(a) If function is given in the form of set of ordered pairs and the second element of at least two ordered pairs are same then function is many-one

(b) If the graph of $y = f(x)$ is given and the line parallel to x- axis cuts the curve at more than one point then function is many-one



Example of many - one function :

(i) $f : R \to R$, $f(x) = C$, where C is a constant (ii) $f : R \to R$, $f(x) = x^2$

(iii) $f : R \to R$, $f(x) = ax^2 + b$ (iv) $f : R \to R$, $f(x) = |x|$

(v) $f : R \to R$, $f(x) = x + |x|$ (vi) $f : R \to R$, $f(x) = x - |x|$

(vii) $f : R \to R$, $f(x) = [x]$

(viii) $f : R \to R$, $f(x) = x - [x]$ Where $[x]$ is greatest integer function.

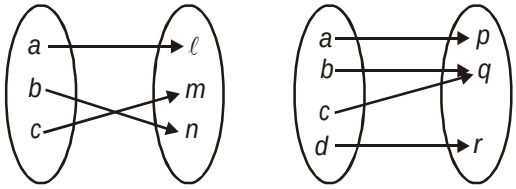
Onto function or Surjection

A function $f : A \to B$ is onto if each element of B has its pre image in A.

Therefore if $f^{-1}(y) \in A \ \forall\ y \in B$ then function is onto.

In other words, Range of f = co-domain of f.

The following arrow-diagrams show onto function.



Example of onto function

(i) $f : R \to R$, $f(x) = x$ (ii) $f : R \to R$, $f(x) = ax + b$, $a \neq 0$, $b \in R$

(iii) $f : R \to R$, $f(x) = x^3$ (iv) $f : R \to R$, $f(x) = x\,|x|$

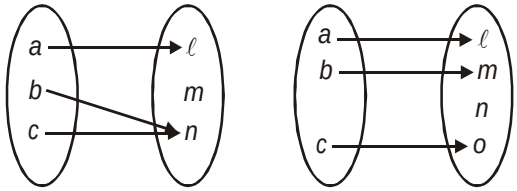
(v) $f : R \to R$, $f(x) = e^x$ (vi) $f : R^+ \to R$, $f(x) = \log x$

Into function

A function $f : A \to B$ is into if there exit at least one element in B which is not the f - image of any element in A.

Therefore, at least one element of B such that $f^{-1}(y) = f$ then function is into. In other words Range of f^+ co-domain of f

The following arrow-diagrams show into function.



Example of into function

(i) $f : R \to R$, $f(x) = x^2$ (ii) $f : R \to R$, $f(x) = |x|$

(iii) $f : R \to R$, $f(x) = c$ (c is constant) (iv) $f : R \to R$, $f(x) = \sin x$

(v) $f : R \to R$, $f(x) = \cos x$ (vi) $f : R \to R$, $f(x) = e^x$

(vii) $f : R \to R$, $f(x) = a^x$, $a > 0$

Note : For a function to be onto or into depends mainly on their co-domain.

Ex. $f : R \to R$ then $f(x) = |x|$ is a into function

$f : R \to R^+ \cup \{0\}$ then $f(x) = |x|$ is a onto function

Ex. $f : [0, \pi] \to [-1, 1]$ then $f(x) = \sin x$ is into function

f : $[0, \pi] \to [-1, 1]$ then $f(x) = \cos x$ is onto function

Now, we can classify the function further into four categories :

One -One Onto function or bijection :

A function f is said to be one-one onto if f is one-one and onto both .

Examples :

(i) If $f : R^+ \to R^+$ then $f(x) = |x|$ is a one- one onto function.

(ii) If $f : [-\pi/2, \pi/2] \to [-1, 1]$ then $f(x) = \sin x$ is a one- one onto function.

One-One Into function :

A function is said to be one-one into if f is one-one but not onto

Examples :

(i) If $f : [-\pi/2, \pi/2] \to R$, $f(x) = \sin x$

(ii) $f : R^+ \to R$, $f(x) = |x|$

($\therefore$ co-domain and range are not equal so function is not onto)

Many One -Onto function :

A function $f : A \to R$ is said to be many one- onto if f is onto but not one-one.

Examples :

(i) $f : R \to R^+ \cup \{0\}$, $f(x) = x^2$

(ii) $f : R \to [0, \infty)$, $f(x) = |x|$

(iii) $f : R \to [-1, 1]$, $f(x) = \sin x$

Many One -Into function :

A function is said to be many one-into if it is neither one-one nor onto.

Examples :

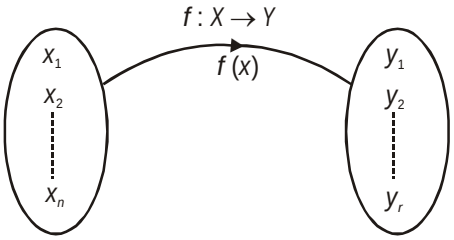
(i) $f : R \to R$ $f(x) = \sin x$

(ii) $f : R \to R$ $f(x) = |x|$

NUMBER OF FUNCTION

Let $X = \{x_1, x_2, x_3, \ldots\ldots\ldots x_n\}$ (i.e. n elements)

and $Y = \{y_1, y_2, y_3 \ldots\ldots\ldots y_r\}$ (i.e. r elements)



(a) Total number of functions = r^n = (Number of elements in co-domain)$^{\text{number of elements in domain}}$

(b) Total number of one to one function = $\begin{cases} {}^{r}C_{n}.n! & r \geq n \\ 0 & r < n \end{cases}$

(c) Total number of onto function = $\sum_{k=1}^{r} {}^{r}C_{k}(-1)^{k}(k)^{n}$

(d) Total Number of one-one-onto function = $\underline{\lfloor n}$, n = r

Illustration 6 : Show that the function $f : N \in N$, given by $f(1) = f(2) = 1$ and $f(x) = x - 1$, for every $x > 2$, is onto but not one-one.

Solution : f is not one-one, as $f(1) = f(2) = 1$. But f is onto, as given any $y \in N$, $y \neq 1$, we can choose x as $y + 1$ such that
$f(y + 1) = y + 1 - 1 = y$. Also for $1 \in N$, we have $f(1) = 1$.

Illustration 7 : et A be the set of all 50 students of Class X in a school. Let $f : A \to N$ be function defined by $f(x)$ = roll number of the student x. Show that f is one-one but not onto.

Solution : No two different students of the class can have same roll number. Therefore, f must be one-one. We can assume without any loss of generality that roll numbers of students are from 1 to 50. This implies that 51 in N is not roll number of any student of the class, so that 51 can not be image of any element of X under f. Hence, f is not onto.

Illustration 8 : how that an onto function $f : \{1, 2, 3\} \to \{1, 2, 3\}$ is always one-one.

Solution : Suppose f is not one-one. Then there exists two elements, say 1 and 2 in the domain whose image in the co-domain is same. Also, the image of 3 under f can be only one element. Therefore, the range set can have at the most two elements of the co-domain {1, 2, 3}, showing that f is not onto, a contradiction. Hence, f must be one-one.

Illustration 9 : Show that the function $f(x) = \dfrac{x^2 - 8x + 18}{x^2 + 4x + 30}$ is not one-one.

Solution : A function is one-one if $f(x_1) = f(x_2) \Rightarrow x_1 = x_2$ (only)

Now $f(x_1) = f(x_2) \Rightarrow \dfrac{x_1^2 - 8x_1 + 18}{x_1^2 + 4x_1 + 30} = \dfrac{x_2^2 - 8x_2 + 18}{x_2^2 + 4x_2 + 30}$

$\Rightarrow 12x_1^2x_2 - 12x_1x_2^2 + x_1^2 - 12x_2^2 - 8x_2 + 18 = 0 \Rightarrow (x_1 - x_2)\{12x_1x_2 + 12(x_1 + x_2) - 312\} = 0$

$x_1 = x_2$ or $x_1 = \dfrac{26 - x_2}{1 + x_2}$ $\therefore$ $f(x)$ is not one-one.

Student Task

Q.1 Let $A = \{1, 2, 3\}$, $B = \{4, 5, 6, 7\}$ and let $f = \{(1, 4), (2, 5), (3, 6)\}$ be a function from A to B. Show that f is one one.

Q.2 Let $f : R \to R$ be defined as $f(x) = x^4$. Choose the correct answer.

(a) f is one-one onto (b) f is many-one onto

(c) f is one-one but not onto (d) f is neither one-one nor onto.

Q.3 Let $f : R \to R$ be defined as $f(x) = 3x$. Choose the correct answer.

(a) f is one-one onto (b) f is many-one onto

(c) f is one-one but not onto (d) f is neither one-one nor onto.

Q.4 Let $A = R - \{3\}$, $B = R - \{1\}$ and let $f : A \to B$ defined by $f(x) = \dfrac{x-2}{x-3}$ is f bijective ?

Q.5 A mapping $f : A \to [-1, 1]$ defined by $f(x) = \sin x$, $\forall x \in R$, where A is a subset of R (the set of all real numbers) is one-one and onto if A is

(a) $[0, 2\pi]$ (b) $\left[-\dfrac{\pi}{2}, \dfrac{\pi}{2}\right]$ (c) $[-\pi, \pi]$ (d) $[0, \pi]$

Q.6 Let $f : R \to R$ be a function defined by $f(x) = \dfrac{e^{|x|} - e^{-x}}{e^{x} + e^{-x}}$, then

(a) f is one-one and onto both (b) f is one-one but not onto

(c) f is not one-one but onto (d) f is neither one to one nor onto

Q.7 The function $f : R \to R$; where R is the set of all real numbers defined by $f(x) = 2x + 3$ is

(a) both one-one and onto (b) one-one but not onto

(c) f is onto but not one-one (d) f is neither one-one nor onto

ANSWERS

2. (d) 3. (a) 4. yes 5. (b) 6. (d) 7. (a)

5. EVEN AND ODD FUNCTIONS

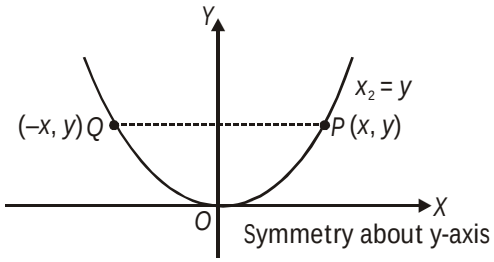
(i) Even Function : A function $y = f(x)$ is said to be an even function if $f(-x) = f(x)$ $\forall$ $x \in D_f$.

e.g. $f(x) = \cos 2x$ is even because

$f(-x) = \cos(-2x) = \cos 2x = f(x)$

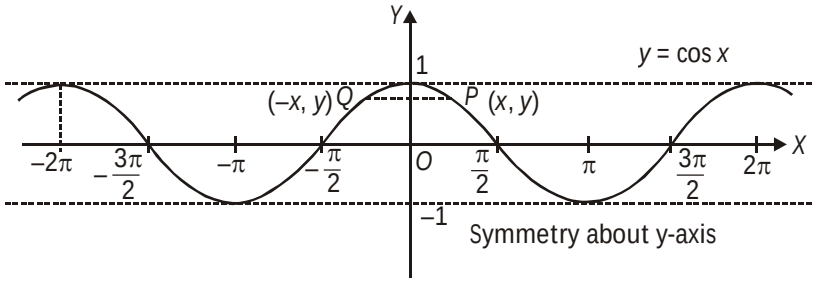
The graph of an even function $y = f(x)$ is **symmetrical about the y-axis**. i.e., if point (x, y) lies on the graph then $(-x, y)$ will also lie on the graph or we say that if any point lies on the graph then its image about y-axis also lies on the graph. e.g.

(1)



Symmetry about y-axis

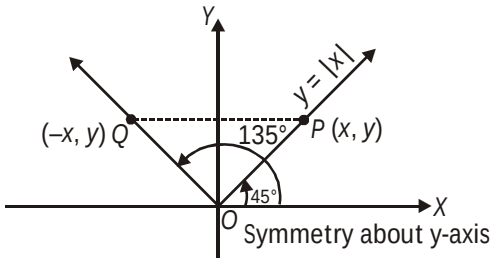
(2)



Symmetry about y-axis

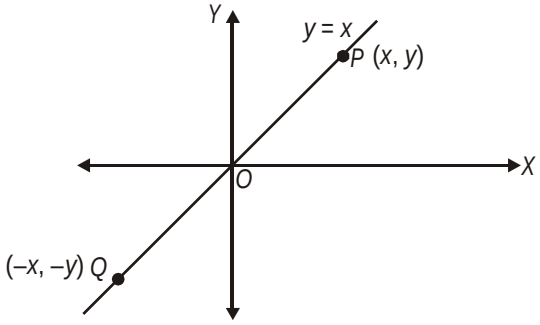
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(3)

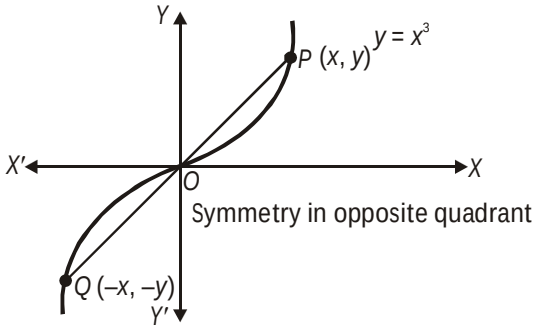


(ii) Odd Function : A function $y = f(x)$ is said to be an odd function if $f(-x) = -f(x)\ \forall\ x \in D_f$
e.g. $f(x) = x^3 + x$ is odd because $f(-x) = -x^3 - x = -f(x)$
The graph of an odd function $y = f(x)$ is **symmetrical in opposite quadrants**. i.e., if point (x, y) lies on the graph then $(-x, -y)$ will also lies on the graph or we say that if any point lies on the graph then its image about origin also lies on the graph. e.g.

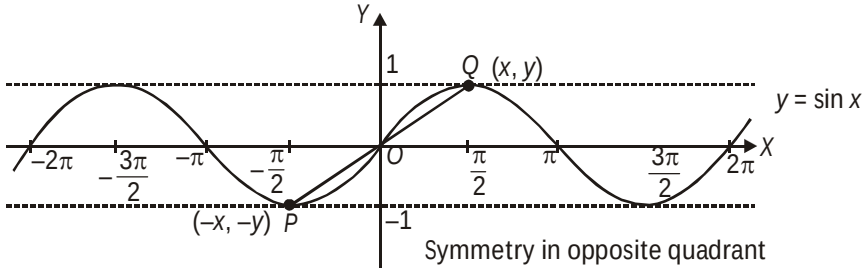
(1)



(2)



(3)



Notes :

1. Inverse of an even function is not defined.
2. Every function can be expressed as the sum of an even and an odd function.

   i.e., $f(x) = \frac{1}{2}\{f(x) + f(-x)\} + \frac{1}{2}\{f(x) - f(-x)\}$

   = {Even function} + {Odd function}
3. If $f(x) - f(-x) = 0$ then $f(x)$ is an even function and if $f(x) + f(-x) = 0$ then $f(x)$ is an odd function
4. A function may neither be even nor odd.
5. $f(x) = 0$ is the only function which is defined on the entire number line is even and odd at the same time.

6. Every odd continuous function passes through origin.
7. Every even function $y = f(x)$ are not one-one $\forall\ x \in D_f$.
8. The derivative of an odd function is an even function and derivative of an even function is an odd function.
9. If *f* and *g* both are even or both are odd then the function *fog* will be even but if any one of them is odd then *fog* will be odd.
10. The square of an even or an odd function is always an even function.

Table of two functions which are attached

$f(x)$	$g(x)$	$f(x) + g(x)$	$f(x) - g(x)$	$f(x)\,g(x)$	$\frac{f(x)}{g(x)}$	$(fog)\,x$
Even	Even	Even	Even	Even	Even	Even
Even	Odd	Neither even nor odd	Neither even nor odd	Odd	Odd	Even
Odd	Even	Neither even nor odd	Neither even nor odd	Odd	Odd	Even
Odd	Odd	Odd	Odd	Even	Even	Odd

Illustration 10 : Prove that $f(x) = \ell n(x + \sqrt{(1+x^2)})$ is an odd function.

Solution : $\because \quad f(x) = \ell n(x + \sqrt{(1+x^2)})$

$\therefore \quad f(-x) = \ell n(-x + \sqrt{1+(-x)^2})$

Now $f(x) + f(-x) = \ell n(\sqrt{(1+x^2)} + x) + \ell n(\sqrt{1+x^2} - x)$

$= \ell n\ (1 + x^2 - x^2)$

$= \ell n\ 1 = 0$

$\Rightarrow \quad f(-x) = -f(x)$.

Hence $f(x)$ is an odd function.

Alternative Method :

$\because \quad f(x) = \ell n(x + \sqrt{1+x^2})$

$\therefore \quad f'(x) = \dfrac{1}{\sqrt{(1+x^2)}} = f'(-x) \qquad \left\{\because \int \dfrac{dx}{\sqrt{(1+x^2)}} = \ell n\{x + \sqrt{1+x^2}\}\right\}$

$\Rightarrow \quad f'(x)$ is an even function.

Hence $f(x)$ is an odd function.

Illustration 11 : If $f(t) = \dfrac{t}{e^t - 1} + \dfrac{t}{2} + 1$, show that $f(t)$ is an even function.

Solution : **Since** $f(t) = \dfrac{t}{e^t - 1} + \dfrac{t}{2} + 1$ .... (1)

$\therefore \quad f(-t) = \dfrac{-t}{e^{-t} - 1} - \dfrac{t}{2} + 1$

$= \dfrac{te^t}{e^t - 1} - \dfrac{t}{2} + 1$ .... (2)

Subtracting (2) from (1), we get

$$f(t) - f(-t) = \frac{t}{(e^t - 1)}(1 - e^t) + t$$

$= -t + t = 0$

$\therefore$ $f(-t) = f(t)$. Hence $f(t)$ is an even function.

Illustration 12 : Express ae^x as the sum of an even and an odd function.

Solution : Let $f(x) = ae^x$, then $f(-x) = ae^{-x}$

Now $$f(x) = \frac{1}{2}\{f(x) + f(-x)\} + \frac{1}{2}\{f(x) - f(-x)\}$$

$$= \frac{1}{2}(ae^x + ae^{-x}) + \frac{1}{2}(ae^x - ae^{-x})$$

= (Even function) + (odd function)

Illustration 13 : Prove that $x^2 - \text{arc cos } y = \pi$ is an explicit function of x, Also find the domain of definition of the explicit function.

Solution : Since $x^2 - \text{arc cos } y = \pi$

$\Rightarrow$ $x^2 - \cos^{-1} y = \pi$ or $\cos^{-1} y = x^2 - \pi$

$\Rightarrow$ $y = \cos(x^2 - \pi) = \cos(\pi - x^2) \Rightarrow$ $y = -\cos x^2$

Hence $x \in [-\sqrt{2\pi}, -\sqrt{\pi}] \cup [\sqrt{\pi}, \sqrt{2\pi}]$

Illustration 14 : If 'f' is an even function defined on (–5, 5) then find the four real values of x satisfying the equation $f(x) = f\left(\frac{x+1}{x+2}\right)$.

Soluiton : Since, $f(x) = f\left(\frac{x+1}{x+2}\right)$

$\Rightarrow$ $x = \frac{x+1}{x+2}$ $\Rightarrow$ $x^2 + x - 1 = 0$

$\therefore$ $x = \frac{-1 \pm \sqrt{1+4}}{2}$ $\Rightarrow$ $x = \frac{-1 \pm \sqrt{5}}{2}$

Now $-x = \frac{x+1}{x+2}$ $\Rightarrow$ $-x^2 - 2x = x + 1$

$\Rightarrow$ $x^2 + 3x + 1 = 0$ $\therefore$ $x = \frac{-3 \pm \sqrt{(9-4)}}{2}$

Hence four real values of x are $\frac{-1 \pm \sqrt{5}}{2}, \frac{-3 \pm \sqrt{5}}{2}$

EXTENSION OF A FUNCTION

If a function $f(x)$ is defined on the interval $[0, a]$, it can be extend on $[-a, a]$ so that $f(x)$ is either even or odd function on the interval $[-a, a]$.

(i) Even Extension : If a function $f(x)$ is defined on the interval $[0, a]$.

$0 \le x \le a \Rightarrow -a \le -x \le 0$

$\therefore$ $-x \in [-a, 0]$

We define $f(x)$ in the interval $[-a, 0]$ such that $f(x) = f(-x)$. Let g be the even extension, then

$$g(x) = \begin{cases} f(x) & ; \ x \in [0, a] \\ f(-x) & ; \ x \in [-a, 0] \end{cases}$$

(ii) **Odd Extension :** If a function $f(x)$ is defined on the interval $[0, a]$

$0 \le x \le a \Rightarrow -a \le -x \le 0$

$\therefore \quad -x \in [-a, 0]$

We define $f(x)$ in the interval $[-a, 0]$ such that $f(x) = -f(-x)$. Let g be the odd extension, then

$$g(x) = \begin{cases} f(x) & ; \ x \in [0,a] \\ -f(-x) & ; \ x \in [-a,0] \end{cases}$$

Illustration 15 : Let $f(x) = e^x + \sin x$ be defined on the interval $[-4, 0]$. Find the odd and even extension of $f(x)$ in the interval $[-4, 4]$

Solution : **Odd Extension :** Let g_0 be the odd extension of $f(x)$, then

$$g_0(x) = \begin{cases} f(x) & ; \ x \in [-4,0] \\ -f(-x) & ; \ x \in [0,4] \end{cases}$$

$$= \begin{cases} e^x + \sin x & ; \ x \in [-4,0] \\ -e^{-x} + \sin x & ; \ x \in [0,4] \end{cases}$$

Even Extension : Let g_e be the even extension of $f(x)$, then

$$g_0(x) = \begin{cases} f(x) & ; \ x \in [-4,0] \\ f(-x) & ; \ x \in [0,4] \end{cases}$$

$$= \begin{cases} e^x + \sin x & ; \ x \in [-4,0] \\ e^{-x} - \sin x & ; \ x \in [0,4] \end{cases}$$

Illustration 16 : Let the function $f(x) = x^2 + x + \sin x - \cos x + \ell n\,(1 + |x|)$ be defined over the interval $[0, 1]$. Find the odd and even extensions of $f(x)$ in the interval $[-1, 1]$.

Solution : **Odd Extension :** Let g_0 be the odd extension of $f(x)$, then

$$g_0(x) = \begin{cases} f(x) & ; \ x \in [0,1] \\ -f(-x) & ; \ x \in [-1,0] \end{cases}$$

$$= \begin{cases} x^2 + x + \sin x - \cos x + \ell n(1+|x|), & x \in [0,1] \\ -x^2 + x + \sin x + \cos x - \ell n(1+|x|), & x \in [-1,0] \end{cases}$$

Even Extension : Leg g_e be the even extension of $f(x)$, then

$$g_e(x) = \begin{cases} f(x) & ; \ x \in [0,1] \\ f(-x) & ; \ x \in [-1,0] \end{cases}$$

$$= \begin{cases} x^2 + x + \sin x - \cos x + \ell n(1+|x|), & x \in [0,1] \\ x^2 - x \sin x \cos x + \ell n(1+|x|), & x \in [-1,0] \end{cases}$$

6. EQUAL OR IDENTICAL FUNCTION

Two functions f and g are said to be equal if :

(i) The domain of f = the domain of g.

(ii) The range of f = the range of g

(iii) $f(x) = g(x)$, for every x belonging to their common domain.

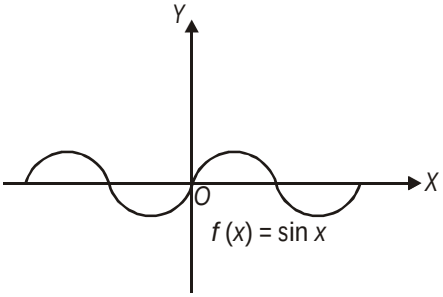
e.g. $f(x) = \dfrac{1}{x}$, $g(x) = \dfrac{x}{x^2}$ are identical functions.

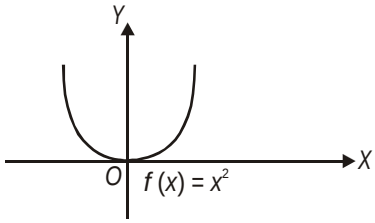
Continuous and Discontinuous Function :

Continuous Function : A function is said to be continuous function in an interval I if we are not required to lift the pen or pencil off the paper i.e. there is no gap or break or jump in the graph.

Example :

$f(x) = x^2$, $f(x) = \sin x$, $f(x) = |x|$, $f(x) = \cos x$ all are continuous functions.

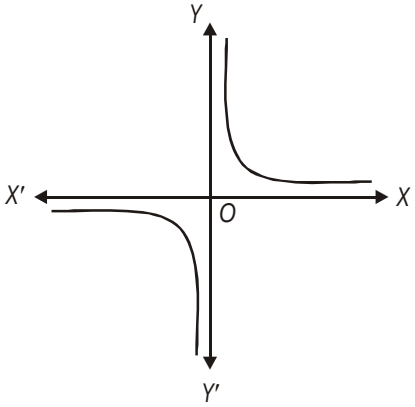




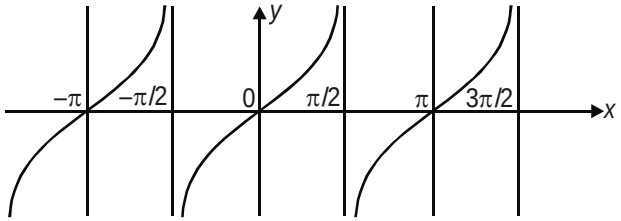
Discontinuous Function : A function is said to be discontinuous if there is a break or gap or jump in the graph of the function at any point.

Example :

$f(x) = 1/x$, $f(x) = \tan x$, $f(x) = [x]$ are discontinuous functions.



$f(x) = 1/x$



$f(x) = \tan x$

EXERCISE

Q.1 Draw the graphs of the following function, where [ ] denotes the greatest integer function.

(i) $f(x) = x + [x]$ (ii) $y = x^{[x]}$

(iii) $y = \text{sgn}[x]$ (iv) $y = \text{sgn}(x - |x|)$

Q.2 Identify the pair(s) of functions which are identical? (where [ ] = G.I.F. and { } = F.P.F.)

(i) $f(x) = \text{sgn}(x^2 - 3x + 4)$ and $g(x) = e^{[\{x\}]}$ (ii) $f(x) = \sqrt{\dfrac{1-\cos 2x}{1+\cos 2x}}$ and $g(x) = \tan x$

(iii) $f(x) = \ell n(1 + x) + \ell n(1 - x)$ and $g(x) = \ell n(1 - x^2)$

(iv) $f(x) = \dfrac{\cos x}{1-\sin x}$ and $g(x) = \dfrac{1+\sin x}{\cos x}$

Q.3 Find whether the following functions are even or odd or none

(a) $f(x) = \log\left(x + \sqrt{1+x^2}\right)$ (b) $f(x) = \dfrac{x(a^x+1)}{a^x-1}$

(c) $f(x) = \sin x + \cos x$ (d) $f(x) = x \sin^2 x - x^3$

(e) $f(x) = \sin x - \cos x$ (f) $f(x) = \dfrac{(1+2^x)^2}{2^x}$

(g) $f(x) = \dfrac{x}{e^x-1} + \dfrac{x}{2} + 1$ (h) $f(x) = [(x + 1)^2]^{1/3} + [(x - 1)^2]^{1/3}$

Q.4 Write explicitly, functions of y defined by the following equations and also find the domains of definition of the given implicit function :

(a) $10^x + 10^y = 10$ (b) $x + |y| = 2y$

Q.5 The function $f(x)$ is defined on the interval [0, 1]. Find the domain of definition of the functions

(a) $f(\sin x)$ (b) $f(2x + 3)$

Q.6 Given that $y = f(x)$ is a function whose domain is [4, 7] and range is [–1, 9]. Find the range and domain of

(a) $g(x) = \frac{1}{3} f(x)$ (b) $h(x) = f(x - 7)$

Q.7 Prove that the function define as $f(x) = \begin{cases} e^{-\sqrt{|\log\{x\}|}} - \{x\}^{\sqrt{\frac{1}{|\log\{x\}|}}} & ; \text{ whenever it exists} \\ \{x\} & ; \text{ otherwise} \end{cases}$, then $f(x)$ is odd as well as even. (where { } = F.P.F.)

Q.8 Let $f(x) = \begin{cases} 0 & \text{if } x \text{ is rational} \\ x & \text{if } x \text{ is irrational} \end{cases}$ and $g(x) = \begin{cases} 0 ; & \text{if } x \text{ is irrational} \\ x ; & \text{if } x \text{ is rational} \end{cases}$, then the function $(f - g)x$ is

(a) Odd (b) Even (c) Neither odd nor even (d) Odd as well as even

Q.9 If the solution set for $f(x) < 3$ is $(0, \infty)$ and the solution set for $f(x) > -2$ is $(-\infty, 5)$, then the true solution set for $(f(x))^2 \geq f(x) + 6$, is

(a) $(-\infty, \infty)$ (b) $(-\infty, 0]$ (c) $[0, 5]$ (d) $(-\infty, 0] \cup [5, \infty)$

Q.10 Which of the following represent an odd function?

(a) $f(x) = \frac{(1+e^x)^2}{e^x}$ (b) $g(x) = \sec^{-1}(\sec x)$

(c) $h(x) = \cos(\cos^{-1} x)$ (d) $k(x) = \cot^{-1}(\cot x)$

Q.11 $f(x) = \frac{x}{\ell n x}$ and $f(x) = \frac{\ell n x}{x}$. Then identify the correct answer

(a) $\frac{1}{g(x)}$ and $f(x)$ are identical functions (b) $\frac{1}{f(x)}$ and $g(x)$ are identical functions

(c) $f(x) \times g(x) = 1 \ \forall \ x > 0$ (d) $\frac{1}{f(x) \cdot g(x)} = 1 \ \forall \ x > 0$

Q.12 The domain of definition of the function $f(x) = \left[2\tan \pi x\right]^{\log_{[2\tan \pi x]}\left(\frac{x^2+2x-3}{4x^2-4x-3}\right)}$ where [ ] = G.I.F. is given by the interval $\left[n + \frac{1}{4}, n + \frac{1}{2}\right)$ where $n \in I$ then n can be equal to

(a) –5 (b) –1 (c) 0 (d) 1

Q.13 Which of the following function(s) is/are transcendental?

(a) $f(x) = 5 \sin \sqrt{x}$ (b) $f(x) = \frac{2\sin 3x}{x^2 + 2x - 1}$ (c) $f(x) = \sqrt{x^2 + 2x + 1}$ (d) $f(x) = (x^2 + 3) \times 2^x$

Q.14 Match the following where [ ] = G.I.F. and { } = F.P.F.

Column-I	**Column-II**
(A) $f(x) = e^{\ell n[1 + \{x\}]}$	(p) Range consists of only one natural
(B) $g(x) = \sin^{-1}(\sin x)$	(q) Periodic
(C) $h(x) = e^{-\|x\|}$	(r) Domain is $x \in (-\infty, \infty)$
(D) $k(x) = \sqrt{[x] + [-x]} + \sqrt{2 - \|x\|} + \frac{1}{x^2}$	(s) Symmetric about y-axis

Q.15 Which of the following functions are odd functions?

(a) $f(x) = \sqrt{1+x+x^2} - \sqrt{1-x+x^2}$ (b) $f(x) = x\left(\frac{a^x+1}{a^x-1}\right)$

(c) $f(x) = \log_{10}\left(\frac{1-x^2}{1+x^2}\right)$ (d) $f(x) = k$ (constant)

(e) $f(x) = \sin\left\{\log_{10}\left(x+\sqrt{x^2+1}\right)\right\}$ (f) $f(x) = 1 + x + 2x^3$

(g) $f(x) = \begin{cases} x^4 \tan\frac{\pi x}{2} & , \ |x|<1 \\ x|x| & , \ |x|\geq 1 \end{cases}$

Q.16 If f is an even function defined on the interval (–5, 5), then find the total number of real values of x satisfying the equation $f(x) = f\left(\frac{x+1}{x+2}\right)$, is

(a) 1 (b) 2 (c) 4 (d) None of these

Q.17 If a real valued function $f(x)$ satisfying the equation $f(x + y) = f(x) + f(y)$ $\forall$ $x, y \in R$ then $f(x)$ is

(a) A periodic function (b) An even function

(c) An odd function (d) None of these

Q.18 A function whose graph is symmetrical in opposite quadrants is

(a) $f(x) = e^x + e^{-x}$ (b) $f(x) = \log_e x$ (c) $f(x + y) = f(x) + f(y)$ (d) $f(x) = \cos x + \sin x$

Q.19 Which of the following functions are odd

(a) $f(x) = \begin{cases} x^2 \sin\frac{\pi x}{2} & ; \ |x|<1 \\ x|x| & ; \ |x|\geq 1 \end{cases}$ (b) $g(x) = \frac{f(x)-f(-x)}{2}$

(c) $g(x) = \frac{f(x)+f(-x)}{2}$ (d) $g(x) = \left[\left|f(x)\right|+2\right]$

Q.20 If $f : \left[-\pi/2, \pi/2\right] \to R$ given by $f(x) = \cos x + \sin\left[\frac{x+1}{\lambda}\right]$ is an even function. Then, the set of values of λ ($\lambda > 0$) is (where [.] = G.I.F.)

(a) $(-\pi/2, \pi/2) - \{0\}$ (b) $\left(\frac{\pi+2}{2}, \infty\right)$

(c) $\left(0, \frac{\pi+2}{2}\right) \cup \left(\frac{\pi+2}{2}, \infty\right)$ (d) $\left(-\frac{\pi}{2}, \frac{\pi+2}{2}\right) - \{0\}$

Q.21 (a) If $f(t) = \frac{t}{e^t-1} + \frac{t}{2} + 1$, show that $f(t)$ is an even function

(b) Express ae^x as the sum of an even and an odd function

(c) Prove that $x^2 - \text{arc}\cos y = \pi$ is an explicit function of x. Also find the domain of definition of the explicit function

(d) Let the function $f(x) = x^2 + x + \sin x - \cos x + \ell n\,(1 + |x|)$ be defined over the interval [0, 1]. Find the odd and even extensions of $f(x)$ in the interval [-1, 1]

Q.22 Prove that : $f(x) = \dfrac{2x(\sin x + \tan x)}{2\left[\dfrac{x+2\pi}{\pi}\right]-3}$ is an odd function where [.] denotes the greatest integer function.

Q.23 Which of the following function(s) is/are identical

(a) $f(x) = \dfrac{x}{x^2}$ and $g(x) = \dfrac{1}{x}$

(b) $f(x) = \dfrac{x^2}{x}$ and $g(x) = x$

(c) $f(x) = \ell n\, x^4$ and $g(x) = 4\, \ell n\, x$

(d) $f(x) = \ell n\, \{(x-1)(x-2)\}$ and $g(x) = \ell n\,(x-1) + \ell n\,(x-2)$

Q.24 (a) Let $F(x) = \left[\dfrac{f(x)+f(-x)}{g(x)-g(-x)}\right]^n$, show that $F(x)$ is even when n is even and is odd when n is odd

(b) $f : [-20, 20] \to R$, defined by $f(x) = \left[\dfrac{x^2}{a}\right]\sin x + \cos x$ ([.] = G.I.F.) is an even function, then set of values of 'a' is given by

(a) R (b) [0, 400] (c) (400, ∞) (d) Null set

ANSWERS

2. (i), (iii) are identical.
3. (a) Odd (b) Even (c) Neither odd nor even (d) Odd
   (e) Neither odd nor even (f) Even (g) Even (h) Even
4. (a) $y = \log(10 - 10^x)$, $-\infty < x < 1$ (b) $y = x/3$ when $-\infty < x < 0$ and $y = x$ when $0 \le x < \infty$
5. (a) $2k\pi \le x \le 2k\pi + \pi$ where $k \in I$ (b) $[-3/2, -1]$
6. (a) Range : $[-1/3, 3]$, Domain = [4, 7] (b) Range $[-1, 9]$ and domain [11, 14]
8. (a) Odd 9. (d) 10. (c) 11. (a)
12. (a,c) 13. (a, b, d) 14. A $\to$ p, q, r; B $\to$ p, q, r; C $\to$ p, r, s; D $\to$ p,s
15. (a) Odd (b) Even (c) Even (d) Even
    (e) Odd (f) Neither odd nor even (g) Odd
16. (c) 17. (c) 18. (c)
19. (a) Odd;
    (b) Odd; (c) Even; (d) Even
20. (b)
21. (b) $ae^x = \dfrac{1}{2}\left(ae^x + ae^{-x}\right) + \dfrac{1}{2}\left(ae^x - ae^{-x}\right)$ (c) $[-\sqrt{2\pi}, -\sqrt{\pi}] \cup [\sqrt{\pi}, \sqrt{2\pi}]$;

(d) $g_0(x) = \begin{cases} x^2 + x + \sin x - \cos x + \ell n\left(1+|x|\right), x \in [0, 1] \\ -x^2 + x + \sin x + \cos x - \ell n\left(1+|x|\right), x \in [-1, 0] \end{cases}$

$g_e(x) = \begin{cases} x^2 + x + \sin x - \cos x + \ell n\left(1+|x|\right), \\ x^2 - x - \sin x - \cos x + \ell n\left(1+|x|\right), x \in [-1, 0] \end{cases}$

23. (a) 24. (c)

7. COMPOSITION OF FUNCTIONS

The basic operations of addition, subtraction, multiplication, and division can be defined for functions. However, for our purposes in this text, there is an additional operation, called composition, that we will use a bit later.

The composition of functions f and g is defined by

$$(f \circ g)(x) = f(g(x)),$$

for all x in the domain of g such that $g(x)$ is in the domain of f.

The left side, $(f \circ g)(x)$, of the equation can be read as the composition of f and g and the right side,

$f(g(x))$, can be read as f of g of x. It may also be helpful for you to mentally picture Definition as two function machines hooked together to produce another function (often called a composite function) as illustrated in Figure. Notice that what comes out of the function g is substituted into the function f. Thus, composition is sometimes called the substitution of functions.

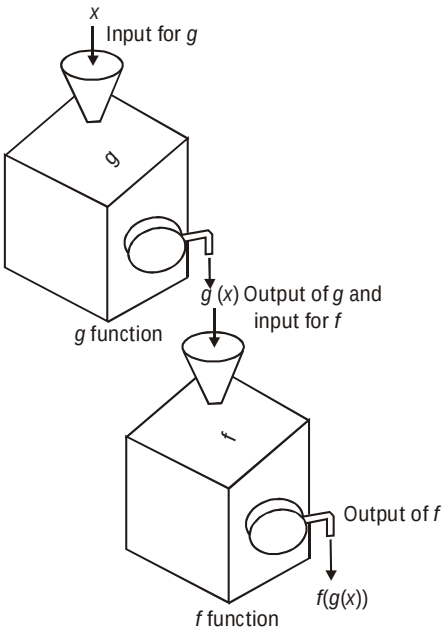


Figure also vividly illustrates the fact that fog is defined for all x in the domain of g such that $g(x)$ is in the domain of f.

In other words, what comes out of g must be capable of being fed into f.

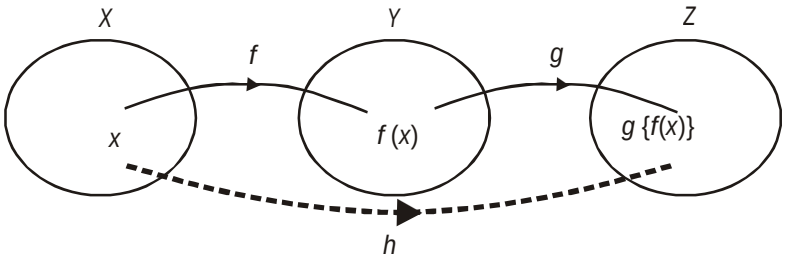
consider two functions

$f : X \to Y$, $g : Y \to Z$

one can define $h : X \to Z$ such that $h(x) = g\{f(x)\}$

Domain of gof (x) i. e. $g\{f(x)\} = \{x : x \in \text{Dom } f, f(x) \in \text{Dom } g\}$

Domain of fog $(x) = \{x : x \in \text{Dom } g, g(x) \in \text{Dom } f\}$



Note :- (a) Function gof will exist only when range of f is the subset of domain of g.
(b) gof (x) is simply the g- image of f (x), where f (x) is f - image of elements $x \in A$
(c) fog does not exist here because range of g is not a subset of domain of f.

Properties of composite functions :

(i) The composite of function is not necessarily commutative :

Let $f : A \to B$ and $g : B \to C$ be two functions. Then the composite of functions (gof) : $A \to C$ exists and is well defined because the range of f is sub-set of the domain of g. But under the condition, (fog) does not exist because the range of the function g is not a sub-set of the domain A of f. Hence if $C \not\subset A$, (fog) will not exist.

If $C = A$, then $f : A \to B$ and $g : B \to A$

In this condition (gof) : $A \to A$ and (fog) : $B \to B$ will bot exist, but the two are not necessarily equal.

For example : If $f : R \to R$, $f(x) = 2x + 3$ and $g : R \to R$, $g(x) = x^3$

then (gof) : $R \to R$ and (fog) : $R \to R$

But gof $(x) = g[f(x)] = g(2x + 3) + (2x + 3)^3$

and (fog) $(x) = f[g(x)] = f(x^3) = 2x^3 + 3$

Hence (gof) $\neq$ (fog)

Note : (gof) and (fog) may be equal only under special circumstances. For example if

$f : R \to R$, $f(x) = x^2$

$g : R \to R$, $g(x) = x^3$ then (gof) : $R \to R$, (fog) : $R \to R$

and (gof) $(x) = g[f(x)] = g(x^2) = (x^2)^3 = x^6$

(fog) $(x) = f[g(x)] = f(x^3) = (x^3)^2 = x^6$

Hence (fog) = (gof)

(ii) Composite of functions is associative :

Theorem : If three functions f, g, h be such that fo (goh) and (fog) oh are defined, then fo (goh) = (fog) oh

Proof : Let the three functions, f, g, h be defined as follows :

$h : A \to B$, $g : B \to C$, $f : C \to D$

Then both fo (goh) and (fog) oh, are defined from A to D.

i.e., fo (goh) : $A \to D$ and (fog) oh : $A \to D$

Clearly the domain of both is A and the co-domain of both is D. Hence to prove them to be equal, we must prove that

$[fo\ (goh)]\ (x) = [\ (fog)\ oh]\ (x)\ \forall\ x \in A$

Let $x \in A$, $y \in B$, $z \in C$ such that

$h(x) = y$ and $g(y) = z$

[fo (goh)] $(x) = f[(goh)\ (x)] = f[g\{h\ (x)\}] = f[g\ (y)] = f\ (z)$

$\therefore$ [fo (goh)] $(x) = f\ (z)$ ..... (1)

Again, $[\ (fog)\ oh]\ (x) = (fog)\ [h\ (x) = (fog)\ (y)] = f[g\ (y)] = f\ (z)$ ..... (2)

Hence from (1) and (2)

$[fo\ (goh)]\ (x) = [(fog\ oh]\ (x),\ \forall\ x \in A \quad \therefore\ [fo\ (goh)] = (fog)\ oh$

(iii) The composite of two bijections is a bijection

Theorem : If f and g be two bijections such that (gof) is defined, then (gof) is also a bijection.

Proof : Let f : $A \to B$ and $g : B \to C$ be two bijections. Then the composite of function (gof) is defined from A to C,

i.e., (gof) : $A \to C$

We have to prove that (gof) is also a bijection.

One-one : To prove that (gof) is one-one. Let $a_1, a_2 \in A$ such that

(gof) (a_1) = (gof) (a_2)

$\Rightarrow$ $g[f(a_1)] = g[f(a_2)]$

$\Rightarrow$ $f(a_1) = f(a_2)$ [$\because$ g is one-one]

$\Rightarrow$ $a_1 = a_2$ [$\because$ f is one-one]

$\therefore$ (gof) is one-one

Onto : To prove that (gof) is onto. Let $c \in C$ then

$c \in C \Rightarrow \exists\ b \in B$ such that $g(b) = c$ [$\because$ g is onto]

again $b \in B \Rightarrow \exists\ a \in A$ such that $f(a) = b$ [$\because$ f is onto]

Thus, $c \in C \Rightarrow \exists\ a \in A$ such that

(gof) $(a) = g[f(a)] = g(b) = c$

Hence every element of C is the image of some element of A. Hence (gof) is onto. Hence (gof) is a bijection.

(iv) If $A \to B$ then $fo\ I_A = I_B$ of = f, where I_A and I_B are the identity functions defined on A and B, i.e. we composite any function, with the identity function, we get the same function.

Proof : Let $I_A : A \to A$ and $f : A \to B$ $\therefore$ $(foI_A) : A \to B$

Let $x \in A$, then

$(fo\ I_A)(x) = f[I_a(x)] = f(x)$ [$\because$ $I_A(x) = x\ \forall\ x \in A$]

$\therefore$ $fo\ I_A = f$

Again, $f : A \to B$ amd $I_B : B \to B$ $\therefore$ $(I_B\ \text{of}) : A \to B$

Let $x \in A$, and $f(x) = y$ where $y \in B$

$\therefore$ $(I_B\ of)(x) = I_B[f(x)] = I_B(y) = y$[$\because$ $I_B(y) = y\ \forall\ y \in B$]

$\therefore$ $(I_B\ of) = f$

HOMOGENEOUS FUNCTIONS :

A function is said to be homogeneous with respect to any set of variables when each of its terms is of the same degree with respect to those variables.

For example $5x^2 + 3y^2 - xy$ is homogeneous in x and y. Symbolically if,

$f(tx, ty) = t^n . f(x, y)$ then $f(x, y)$ is homogeneous function of degree n.

BOUNDED FUNCTION

A function is said to be bounded if $|f(x)| \le M$, where M is a finite quantity.

IMPLICIT AND EXPLICIT FUNCTION

A function defined by an equation not solved for the dependent variable is called an implicit function. For eg. the equation

$x^3 + y^3 = 1$ defines y as an implicit function. If y has been expressed in terms of x alone then it is called an explicit function.

8. INVERSE FUNCTION

Let $f : X \to Y$ be a function defined by $y = f(x)$ such that f is both one-one and onto, then there exists a unique function $g : Y \to X$ such that for each $y \in Y$, $g(y) = x \Leftrightarrow y = f(x)$. The function g so defined is called the inverse of f.

Further, if g is the inverse of f, then f is the inverse of g and the two functions f and g are said to be the inverse of each other. For the inverse of a function to exist, the function must be one-one and onto.

Some standard functions given below along with their inverse functions

	Function	**Inverse Function**
(i)	$f : [0, \infty) \to [0, \infty)$ defined by $f(x) = x^2$	$f^{-1} : [0, \infty) \to [0, \infty)$ defined by $f^{-1}(x) = \sqrt{x}$
(ii)	$f : \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \to [-1, 1]$ defined by $f(x) = \sin x$	$f^{-1}[-1, 1] \to \left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ defined by $f^{-1}(x) = \sin^{-1} x$

(iii) $f : [0, \pi] \to [-1, 1]$ defined by $f(x) = \cos x$ $\quad f^{-1} [-1, 1] \to [0, \pi]$ defined by $f^{-1}(x) = \cos^{-1}x$

Note :- For the existence of inverse function, it should be one -one onto.

Properties of Inverse function :

Let $f : X \to Y$ and $g : Y \to Z$

(a) Inverse of a bijection is also a bijection function.

(b) Inverse of a bijection is unique.

(c) $(f^{-1})^{-1} = f$

(d) If f and g are two bijections such that (gof) exists then $(gof)^{-1} = f^{-1} og^{-1}$

Method of finding inverse of a function :

If you are asked to check whether the given function $y = f(x)$ is invertible, you need to check that $y = f(x)$ is one-one and onto.

If you are asked to find the inverse of a bijective function $f(x)$, you do the following : if f^{-1} be the inverse of 'f' , then $f(f^{-1}(x)) = x$. Apply the formula of f on $f^{-1}(x)$ and use of the above identity to solve for $f^{-1}(x)$.

Illustration 17 : If $f(x) = x^2$ and $g(x) = x - 3$, find $(f \circ g)(x)$ and determine its domain.

Solution : $(f \circ g)(x) = f(g(x)) = f(x - 3) = (x - 3)^2$

Because g and f are both defined for all real numbers, so is $f \circ g$.

Illustration 18 : If $f(x) = \sqrt{x}$ and $g(x) = x - 4$, find $(f \circ g)(x)$ and determine its domain.

Solution : $(f \circ g)(x) = f(g(x)) = f(x - 4) = \sqrt{x-4}$

The domain of g is all real numbers but the domain of f is only the nonnegative real numbers. Thus $g(x)$, which is $x - 4$, has to be nonnegative. So $x - 4 \geq 0$; $x \geq 4$ and the domain of $f \circ g$ is $D = \{x | x \geq 4\}$.

Definition with f and g interchanged defines the composition of g and f as $(g \circ f)(x) = g(f(x))$.

Illustration 19 : If $f : R \to R$, $f(x) = 2x + 3$ and $g : R \to R$, $g(x) = x^2 + 1$, then find the value of $gof(x)$.

Solution : $gof(x) = g\{f(x)\}$

$f(x) = 2x + 3$

$g\{f(x)\} = g(2x + 3) = (2x + 3)^2 + 1 = 4x^2 + 9 + 12x + 1$

$gof(x) = 4x^2 + 12x + 10$

Illustration 20 : If $f(x) = x^2$ and $g(x) = x - 3$, find $(g \circ f)(x)$ and determine its domain.

Solution : $(g \circ f)(x) = g(f(x)) = g(x^2) = x^2 - 3$

Since f and g are both defined for all real numbers, the domain of $g \circ f$ is the set of all real numbers.

The result of above examples demonstrate an important idea, namely, that the composition of functions is not a commutative operation. In other words, it is not true that $f \circ g = g \circ f$ for all functions f and g. However, as we will see later, there is a special class of functions where $f \circ g = g \circ f$.

Illustration 21 : If $f(x) = 2x + 3$ and $g(x) = \sqrt{x-1}$, determine each of the following

(a) $(f \circ g)(x)$ **(b)** $(g \circ f)(x)$ **(c)** $(f \circ g)(5)$ **(d)** $(g \circ f)(7)$

Solution : **(a)** $(f \circ g)(x) = f(g)) = f(\sqrt{x-1}) = 2\sqrt{x-1} + 3$

(b) $(g \circ f)(x) = g(f(x)) = g(2x+3) = \sqrt{2x+3-1} = \sqrt{2x+2}$

(c) By using the composite functions formed in part (a) we obtain

$(f \circ g)(5) = 2\sqrt{5-1}+3 = 2\sqrt{4}+3 = 2(2)+3 = 7$

(d) By using the composite function formed in part (b) we obtain

$(g \circ f)(7) = \sqrt{2(7)+2} = \sqrt{16} = 4$

Illustration 22 : If $f(x) = x^2 + 1$, $g(x) = \frac{1}{x-1}$, then find $(fog)(x)$ and $(gof)(x)$.

Solution : Given, $f(x) = x^2 + 1$ .....(1) $g(x) = \frac{1}{x-1}$ .....(2)

Now $(fog)(x) = f(g(x)) = f\left(\frac{1}{x-1}\right) = f(z)$, where $z = \frac{1}{x-1} = z^2 + 1$

$[\because f(x) = x^2 + 1]$

$= \left(\frac{1}{x-1}\right)^2 + 1 = \frac{1}{(x-1)^2} + 1$

Domain of $fog(x)$ is $x \in R - \{1\}$

$(gof)(x) = g(f(x)) = g(x^2 + 1) = g(u)$, where $u = x^2 + 1$

$= \frac{1}{u-1} = \frac{1}{x^2+1-1} = \frac{1}{x^2}$

Domain of $gof(x)$ is $x \in R - \{0\}$

Illustration 23 : Consider functions f and g such that composite gof is defined and is one-one. Are f and g both necessarily one-one.

Solution : Consider $f : \{1, 2, 3, 4\} \to \{1, 2, 3, 4, 5, 6\}$ defined as $f(x) = x$, $\forall\ x$ and $g : \{1, 2, 3, 4, 5, 6\} \to \{1, 2, 3, 4, 5, 6\}$ as $g(x) = x$, for $x = 1, 2, 3, 4$ and $g(5) = g(6) = 5$.

Then, $gof(x) = x\ \forall\ x$, which shows that gof is one-one. But g is clearly not one-one.

Illustration 24 : Find the inverse of the function $f(x) = \log_a\left(x + \sqrt{(x^2+1)}\right)$; $a > 1$ and assuming it to be an onto function.

Solution : Given $f(x) = \log_a\left(x + \sqrt{x^2+1}\right)$

$\therefore \quad f'(x) = \frac{\log_a e}{\sqrt{(1+x^2)}} > 0$

Which is a strictly increasing functions.

Thus, $f(x)$ is injective, given that $f(x)$ is onto. Hence the given function $f(x)$ is invertible.

Now $f(f^{-1}(x)) = x \Rightarrow \log_a\left(f^{-1}(x) + \sqrt{\left(f^{-1}(x)\right)^2 + 1}\right) = x$

$\Rightarrow \quad f^{-1}(x) + \sqrt{(f^{-1}(x))^2 + 1} = a^x$ .....(i)

and $\sqrt{(f^{-1}(x))^2 + 1} - f^{-1}(x) = a^{-x}$ .....(ii)

From (i) and (ii), we get $f^{-1}(x) = \frac{1}{2}(a^x - a^{-x})$

Illustration 25 : If $f : Q \to Q$, $f(x) = 2x$; $g : Q \to Q$, $g(x) = x + 2$, then $(fog)^{-1}(20)$ equals

Solution : $\therefore \quad f^{-1}(x) = x/2$, $g^{-1}(x) = x - 2$

$\therefore \quad (fog)^{-1}(20) = (g^{-1} of^{-1})(20) = g^{-1}[f^{-1}(20)] = g^{-1}(10) = 10 - 2 = 8$

Student Task

Q.1 If R is a set of real numbers and $f : R \to R$ is given by the relation $f(x) = \sin x$, $x \in R$ and mapping $g : R \to R$ by the relation $g(x) = x^2$, $x \in R$, then prove that $f \circ g \neq g \circ f$.

Q.2 Given $f(x) = \log\left(\frac{1+x}{1-x}\right)$ and $g(x) = \frac{3x + x^3}{1 + 3x^2}$ then $fog(x)$ equals

(a) $-f(x)$ (b) $3f(x)$ (c) $[f(x)]^3$ (d) none of these

Q.3 Let $f : [-1, 1] \to [-1, 1]$ defined by $f(x) = x|x|$, find $f^{-1}(x)$.

Q.4 Which of the following functions has inverse

(a) $f : Z \to Z$ defined by $f(x) = x + 2$ (b) $f : Z \to Z$ defined by $f(x) = 2x$

(c) $f : Z \to Z$ defined by $f(x) = x$ (d) $f : Z \to Z$ defined by $f(x) = |x|$

Q.5 Show that $f : [-1, 1] \to R$, given by $f(x) = \frac{x}{(x+2)}$ is one-one. Find the inverse of the function $f : [-1, 1] \to$ Range f.

Q.6 Consider $f : R_+ \to [-5, \infty)$ given by $f(x) = 9x^2 + 6x - 5$. Show that f is invertible with $f^{-1}(y) = \left(\frac{(\sqrt{y+6}) - 1}{3}\right)$

Q.7 Let $f : X \to Y$ be an invertible function. Show that the inverse of f^{-1} is f, i.e., $(f^{-1})^{-1} = f$.

Q.8 If $f : R \to R$ be given by $f(x) = (3 - x^3)^{1/3}$, then $fof(x)$ is

(a) $x^{1/3}$ (b) x^3 (c) x (d) $(3 - x^3)$.

Q.9 Let $f : R - \{-4/3\} \to R$ be a function defined as $f(x) = \frac{4x}{3x+4}$. The inverse of f is the map g : Range $f \to R - \{-4/3\}$ given by

(a) $g(y) = \frac{3y}{3-4y}$ (b) $g(y) = \frac{4y}{4-3y}$ (c) $g(y) = \frac{4y}{3-4y}$ (d) $g(y) = \frac{3y}{4-3y}$

ANSWERS

2. (b) 3. $f^{-1}(x) = \begin{cases} -\sqrt{-x}, & -1 \le x \le 0 \\ \sqrt{x}, & 0 \le x \le 1 \end{cases}$ 4. (AB) 8. (C) 9. (B)

9. BINARY OPERATION

Let S be a non-void set. A function defined from $S \times S$ to S is known as a binary operation on S. In other words a binary operation on the set S is a rule or a law which gives for any ordered pair (a, b) of elements of S, a unique element of S. Generally, a binary operation is devoted by * or o or by $\oplus$ symbols. Under the operation *, the element associated with the element $(a, b) \in S \times S$ is denoted as $a * b$.

Types of binary operations :

(i) Commutative operations :

Let S be a non-void set and let * be a binary operation defined on S. If $a, b \in S$, then we know that $(a, b) \neq (b, a)$ unless $a = b$. Therefore it is not necessary that the image of (a, b) and (b, a) under the operation * are equal. In other words, it is not necessary that $a * b = b * a, \forall a, b \in S$ then the binary operation * is said to be commutative on S

A binary operation * defined on a set S is said to be commutative if $a * b = b * a, \forall a, b \in S$.

(ii) Associative operation :

Let * be a binary operation, defined on a set S. Let $a, b, c \in S$. Let us consider the expression $a * b * c$. Since the binary operation * can operate only on two elements at a time and we have three elements. Therefore, we should consider either $a * (b * c)$ or $(a * b) * c$.

It is not necessary that $a * (b * c) = (a * b) * c$ is always $\forall a, b, c \in S$.

If $a * (b * c) = (a * b) * c \; \forall a, b, c \in S$ then the binary operation * is said to be associative.

A binary operation *, defined on a set S is said to be associative if

$$a * (b * c) = (a * b) * c \quad \forall a, b, c \in S.$$

Identity element for an operation :

Let S be a non void set on which a binary operation * is defined. If there exists an element e_1 in S such that $e_1 * a = a \; \forall a \in S$,

then e_1 is called a left identity in S with respect the operation *. Similarly, if there exists an element e_2 in S, such that $a * e_2 = a, \forall a \in S$, then e_2 is called a right identity in S with respect to the operation *.

This element is generally denoted as e. Hence, if there exist an element e in the set S such that $e * a = a * e = a, \forall a \in S$, then e is called the identity element in S for the operation.

Theorem : The identity element for a binary operation, if it exists, is unique.

Proof : Let * be a binary operation on a set S. If possible, let e and e' be two identities in S.

then $e*e' = e'$ [$\because$ e is identity in S and $e' \in S$] ..... (1)

Again, $e*e' = e$ [$\because$ e' is identity in S and $e \in S$] ..... (2)

From (1) and (2), $e = e'$

Inverse on an element :

Let * be a binary operation on S and let e be the identity in S. let $a \in S$. If there exists an element b in S, such that $a * b = b * a = e$, then b is called the inverse of a and it is denoted by a^{-1}.

If an element a of S has its inverse in S, then a is said to be an invertible element. Hence a is invertible $\Leftrightarrow a^{-1} \in S$

Note : Let e be the identity element in S for a binary operation *. Then $e*e = e*e = e$. Hence, the identity element in a set, if it exists, is invertible and is own inverse.

Illustration 26 : Show that subtraction and division are not binary operations on N.

Solution : $N \times N \to N$, given by $(a, b) \to a - b$, is not binary operation, as the image of (3, 5) under '–' is $3 - 5 = -2 \notin N$.

Similarly, $\div : N \times N \to N$, given by $(a, b) \to a \div b$ is not a binary operation, as the image of (3, 5) under ÷ is $3 \div 5 = 3/5 \notin N$

Illustration 27 : Show that the $\vee : R \times R \to R$ given by $(a, b) \to \max \{a, b\}$ and the $\wedge : R \times R \to R$ given by $(a, b) \to \min \{a, b\}$ are binary operations.

Solution : Since $\vee$ carries each pair (a, b) in $R \times R$ to a unique element namely maximum of a and b lying in R, $\vee$ is a binary operation. Using the similar argument, one can say that $\wedge$ is also a binary operation.

Illustration 28 : Show that addition and multiplication are associative binary operation on R. But subtraction is not associative on R. Division is not associative on R.

Solution : Addition and multiplication are associative, since $(a + b) + c = a + (b + c)$ and $(a \times b) \times c = a \times (b \times c)$ $\forall$ $a, b, c \in R$.
However, subtraction and division are not associative, as $(8 - 5) - 3 \neq 8 - (5 - 3)$ and $(8 \div 5) \div 3 \neq 8 \div (5 \div 3)$.

Illustration 29 : Show that zero is the identity for addition on R and 1 is the identity for multiplication on R. But there is no identity element for the operations $- : R \times R \to R$ and $\div : R_* \times R_* \; ? \; R_*$.

Solution : $a + 0 = 0 + a = a$ and $a \times 1 = a = 1 \times a$, $\forall$ $a \in R$ implies that 0 and 1 are identity elements for the operations '+' and '×' respectively. Further, there is no element e in R with $a - e = e - a$, $\forall$ a. Similarly, we can not find any element e in R_* such that $a \div e = e \div a$, $\forall$ a in R*. Hence, '–' and '÷' do not have identity element.

Student Task

Q.1 Which of the following definition of * is a binary operation on the set given against them. Give reason in support of your answer.

(i) $a * n = \sqrt{2a^2 - b^2}$ $in R$ (b) $a * b = a - b$ in R

Q.2 In each of the following cases, determine whether * is commutative and associative.
(i) * in N, where $a * b = ab$ (ii) * in $Q \times Q$ where $(a, b) * (c, d) = (ac, ad + b)$

Q.3 Find the following sets have identity element for given operations. In case the identity exists, find its value.
(i) For multiplication in Q_0
(ii) For union of sets in $P(S)$, the power set of S

Q.4 On the set R of real numbers, a binary operation * is defined $a * b = a + b - ab$ $\forall a, b \in R$
(i) Examine if * is commutative.
(ii) Find the identity element for *, if it exists.
(iii) Find the invertible element of R with respect to *.

Q.5 On the set Q^+ of positive rationals, two binary operations are defined as follows :

(i) $a * b = \frac{ab}{3}, \forall a, b \in Q^+$ (ii) $a * b = \frac{ab}{4}, \forall a, b \in Q^+$

Prove that both the operations are commutative and associative. Find their identities. Find the inverse of an element with respect to these operations.

ANSWERS

1. (i) No, (ii) Yes
2. (i) Neither commutative nor associative (ii) Associative but not commutative
3. (i) 1 (ii) f
4. (i) Commutative and associative (ii) 0 (iii) $a \in R$ $(a \neq 1)$ is commutative
5. (i) Identity = 3, $a^{-1} = 9/a$ (ii) Identity = 4, $a^{-1} = 16/a$

USEFUL TIPS

1. If domain of $f(x)$ is continuous and $\frac{dy}{dx} > 0$, $\forall$ x in domain then f is One - One

2. If domain of $f(x)$ is continuous and $\frac{dy}{dx} < 0$, $\forall$ x in domain then f is One - One

3. If a continuous function $f(x)$ which has either local minima or local maxima or both then $f(x)$ will be Many - One

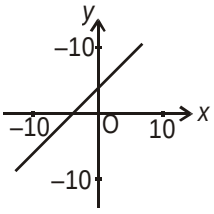
4. Every even function is many-one
5. Every periodic function is many-one.
6. If a function is not onto, it is automatically into. Further a function $f : A \to B$ is said to the a one-one onto function or a bijection of it is one-one as well as onto.
7. If $A \to B$ is a bijection, then $f^{-1} : B \to A$ is defined by $f(x) = y \Leftrightarrow f^{-1}(y) = x$ and is itself a bijection.
8. The composite of functions is associative i.e., $(fog)\, oh = fo\,(goh)$
9. Let * be a binary operation on S. Then the inverse of an element a of S is an element of b of S such that $a * b = b * a = e$.
10. The inverse of an element S for the operation * will exist only if the identity element for * exists in S.
11. If f is both injective and surjective, then its called a bijective mapping.
12. If a set A contains n distinct elements then the number of different functions defined from $A \to A$ is n^n and of it $n!$ are one-one.
13. The composite of function is not commutative i.e. $gof \neq fog$.
14. The composite of two bijections is a bijection i.e. if f and g are two bijections such that gof is defined, then gof is also a bijection.
15. The inverse of a bijection is unique.

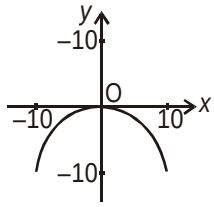
EXERCISE

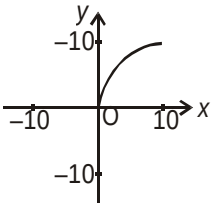
Q.1 Let f be a real valued function such that $f(x) + 2f\left(\frac{2002}{x}\right) = 3x$ for all $x > 0$. Find $f(2)$

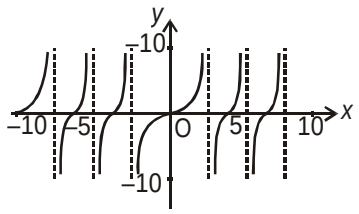
(a) 1000 (b) 2000 (c) 3000 (d) 4000

Q.2 Which one of the following depicts the graph of an odd function?

(a) 

(b) 

(c) 

(d) 

Q.3 Let f be a function satisfying $f(xy) = \frac{f(x)}{y}$ for all positive real numbers x and y. If $f(30) = 20$, then the value of $f(40)$ is

(a) 15 (b) 20 (c) 40 (d) 60

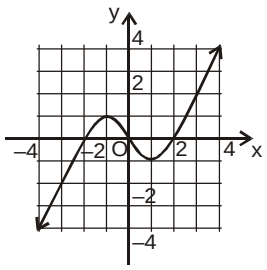
Q.4 If $f(x, y) = (\max(x, y))^{\min(x, y)}$ and $g(x, y) = \max(x, y) - \min(x, y)$, then $f\left(g\left(-1, -\frac{3}{2}\right), g(-4, -1.75)\right)$ equals

(a) –0.5 (b) 0.5 (c) 1 (d) 1.5

Q.5 If the solution set for $f(x) < 3$ is $(0, \infty)$ and the solution set for $f(x) > -2$ is $(-\infty, 5)$, then the true solution set for $(f(x))^2 \,^3\, f(x) + 6$, is

(a) $(-\infty, +\infty)$ (b) $(-\infty, 0]$ (c) $[0, 5]$ (d) $(-\infty, 0] \cup [5, \infty)$

Q.6 The graph of the function $y = g(x)$ is shown. The number of solutions of the equation $||g(x)| - 1| = \frac{1}{2}$, is



(a) 4 (b) 5 (c) 6 (d) 8

Q.7 Consider the function $f : X \to Y$ and $g : Y \to Z$ then which of the following is/are incorrect?

(a) If f and g both are injective then gof : $X \to Z$ is injective

(b) If f and g both are surjective then gof : $X \to Z$ is surjective

(c) If gof : $X \to Z$ is bijective then f is injective and g is surjective

(d) None of these

Q.8 Which of the following statements are incorrect?

I. If $f(x)$ and $g(x)$ are one to one then $f(x) + g(x)$ is also one to one.

II. If $f(x)$ and $g(x)$ are one-one then $f(x) \times g(x)$ is also one-one.

III. If $f(x)$ is odd then it is necessarily one to one.

(a) I and II only (b) II and III only

(c) III and I only (d) I, II and III

Q.9 Let $f(x) = \begin{cases} 0 & \text{If } x \text{ is rational} \\ x & \text{If } x \text{ is irrational} \end{cases}$ and $g(x) = \begin{cases} 0 & \text{If } x \text{ is irrrational} \\ x & \text{If } x \text{ is rational} \end{cases}$. Then the function $(f - g)x$ is

(a) Odd (b) Even

(c) Neither odd nor even (d) Odd as well as even

Q.10 The function f is one to one and the sum of all the intercepts of the graph is 5. The sum of all of the intercept of the graph of $y = f^{-1}(x)$ is

(a) 5 (b) $\frac{1}{5}$ (c) $\frac{2}{5}$ (d) –5

Q.11 The range of the function, $f(x) = (1 + \sec^{-1} x)(1 + \cos^{-1}x)$ is

(a) $(-\infty, \infty)$ (b) $(-\infty, 0] \cup [4, \infty)$ (c) $\{1, (1 + \pi)^2\}$ (d) $[0, (1 + \pi)^2]$

Q.12 If $f(x) = x^2 + bx + c$ and $f(2 + t) = f(2 - t)$ for all real numbers t, then which of the following is true?

(a) $f(1) < f(2) < f(4)$ (b) $f(2) < f(1) < f(4)$ (c) $f(2) < f(4) < f(1)$ (d) $f(4) < f(2) < f(1)$

Q.13 The solution set for $[x]\{x\} = 1$ where $\{x\}$ and $[x]$ are fractional part & integral part of x, is

(a) $R^+ - (0, 1)$ (b) $R^+ - \{1\}$

(c) $\left\{m + \frac{1}{m} / m \in I - \{0\}\right\}$ (d) $\left\{m + \frac{1}{m} / m \in N - \{1\}\right\}$

Q.14 If $f(x) = px + q$ and $f(f(f(x))) = 8x + 21$, where p and q are real numbers, then $p + q$ equals

(a) 3 (b) 5 (c) 7 (d) 11

Q.15 The range of the function $y = \frac{8}{9 - x^2}$ is

(a) $(-\infty, \infty) - \{\pm 3\}$ (b) $\left[\frac{8}{9}, \infty\right)$ (c) $\left(0, \frac{8}{9}\right)$ (d) $(-\infty, 0) \cup \left[\frac{8}{9}, \infty\right)$

Q.16 Range of the function $f(x)=\frac{\{x\}}{1+\{x\}}$ where $\{x\}$ denotes the fractional part function is

(a) [0, 1) (b) $\left[0,\frac{1}{2}\right]$ (c) $\left[0,\frac{1}{2}\right)$ (d) $\left(0,\frac{1}{2}\right)$

Q.17 Let $[x]$ denote the greatest integer in x, Then in the interval [0, 3] the number of solution of the equation, $x^2 - 3x + [x] = 0$ is

(a) 6 (b) 4 (c) 3 (d) 0

Q.18 The domain of function, satisfying $f(x)+f(x^{-1})=\frac{x^3+1}{x}$, is

(a) An empty set (b) a singleton set (c) a finite set (d) an infinite set

Q.19 A function $f(x)=\sqrt{1-2x}+x$ is defined from $D_1 \to D_2$ and is onto. If the set D_1 is its complete domain then the set D_2 is

(a) $\left[-\infty,\frac{1}{2}\right)$ (b) $(-\infty, 2)$ (c) $(-\infty, 1)$ (d) $(-\infty, 1]$

Q.20 Domain of the function $f(x)=\frac{1}{\sqrt{\ln\cot^{-1}x}}$ is

(a) (cot 1, ∞) (b) $R - \{\cot 1\}$

(c) $(-\infty, 0) \cup (0, \cot 1)$ (d) $(-\infty, \cot 1)$

Q.21 Let $f(x) = a + b\cos^{-1}x$ ($b > 0$). If domain and range of $f(x)$ are the same set then $(b - a)$ is equal to

(a) $1-\frac{1}{\pi}$ (b) $-\left(\frac{2}{\pi}+1\right)$ (c) $1-\frac{2}{\pi}$ (d) 2

ANSWERS

1.	(b)	2.	(d)	3.	(a)	4.	(d)	5.	(d)	6.	(d)	7.	(d)
8.	(d)	9.	(a)	10.	(a)	11.	(c)	12.	(b)	13.	(d)	14.	(b)
15.	(d)	16.	(c)	17.	(c)	18.	(c)	19.	(d)	20.	(d)	21.	(b)

10. SOME IMPORTANT RESULTS

1. If x, y are independent variables, then

(i) $f(xy) = f(x) + f(y) \Rightarrow f(x) = k\,\ell n\, x$ or $f(x) = 0$

(ii) $f(xy) = f(x) \cdot f(y) \Rightarrow f(x) = x^n$, $n \in R$

(iii) $f(x + y) = f(x) \cdot f(y) \Rightarrow f(x) = a^{kx}$

(iv) $f(x + y) = f(x) + f(y) \Rightarrow f(x) = kx$, where k is a constant.

2. If $f(x) = a\sin x + b\cos x + c\ \forall\ x \in R$ and a, b, c are constants then $c-\sqrt{a^2+b^2} \le f(x) \le c+\sqrt{a^2+b^2}$.

Illustration 30 : Let $f(x) = \frac{9^x}{9^x+3}$. Evaluate the sum $f\left(\frac{1}{1996}\right)+f\left(\frac{2}{1996}\right)+....+f\left(\frac{1995}{1996}\right)$.

Solution : Given $f(x) = \frac{9^x}{9^x+3}$ ..... (i)

now $f(1-x) = \frac{9^{1-x}}{9^{1-x}+3}=\frac{3}{9^x+3}$ ..... (ii)

Adding (i) and (ii) $f(x) + f(1 - x) = 1$ ..... (iii)

Putting $x=\frac{1}{1996},\frac{2}{1996},.....,\frac{998}{1996}$ in (iii)

We get $f\left(\frac{1}{1996}\right)+f\left(\frac{1995}{1996}\right)=1$

$$f\left(\frac{2}{1996}\right)+f\left(\frac{1994}{1996}\right)=1$$

.....

.....

$$f\left(\frac{997}{1996}\right)+f\left(\frac{999}{1996}\right)=1$$

$$f\left(\frac{998}{1996}\right)+f\left(\frac{998}{1996}\right)=1 \Rightarrow f\left(\frac{998}{1996}\right)=\frac{1}{2}$$

Adding all, we get

$$f\left(\frac{1}{1996}\right)+f\left(\frac{2}{1996}\right)+.....+f\left(\frac{1995}{1996}\right) = (1 + 1 + ... + 997 \text{ times}) + \frac{1}{2} =$$

997.5

Illustration 31 : If $f\left(x+\frac{y}{8}, x-\frac{y}{8}\right)=xy$, then $f(m, n) + f(n, m) = 0$

(a) only when $m = n$ (b) only when $m \neq n$ (c) only when $m = -n$ (d) for all m and n

Solution : (d) $f\left(x+\frac{y}{8}, x-\frac{y}{8}\right)=xy$

put $\frac{y}{8}=k$, $f(x + k, x – k) = 8xk$

put $x + k = A$, $x – k = B$ or $x=\frac{A+B}{2}$ and $k=\frac{A-B}{2}$

Now, $f(A, B) = 8\left(\frac{A+B}{2}\right)\left(\frac{A-B}{2}\right)$

$f(A, B) = 2(A^2 – B^2)$; $f(B, A) = 2(B^2 – A^2)$

then $f(m, n) + f(n, m) = 0$ for all m, n

EXERCISE

Q.1 If $3f(x) + 5f\left(\frac{1}{x}\right) = \frac{1}{x} – 3, \forall x (\neq 0) \in R$, then $f(x) =$

(a) $\frac{1}{14}\left(\frac{3}{x}+5x-6\right)$ (b) $\frac{1}{14}\left(-\frac{3}{x}+5x-6\right)$ (c) $\frac{1}{14}\left(-\frac{3}{x}+5x+6\right)$ (d) none of these

Q.2 If $f(x) . f(y) = f(x) + f(y) + f(xy) – 2 \ \forall x, y \in R$ and if $f(x)$ is not a constant function, then the value of $f(A)$ is –

(a) 1 (b) 2 (c) 0 (d) – 1

Q.3 The function $y(x)$ given implicitly by the equation $3 \sin y + 5 \sin x = 1$ is real in-

(a) $\left[-\frac{\pi}{2}, \tan^{-1}\frac{2}{\sqrt{23}}\right]$ (b) $\left[-\tan^{-1}\frac{2}{\sqrt{21}}, \tan^{-1}\frac{4}{3}\right]$

(c) $\left[\tan^{-1}\frac{5}{3}, \frac{\pi}{2}\right]$ (d) None of these

Q.4 If f is a function such that $f(0) = 2$, $f(A) = 3$ and $f(x + 2) = 2f(x) – f(x + 1)$ for every real x then $f(5)$ is

(a) 7 (b) 13 (c) 1 (d) 5

Q.5 If $f(x+y) = f(x) + 2y^2 + kxy$ and $f(A) = 2$, $f(B) = 8$, then $f(x)$ is of the form
(a) $2x^2$ (b) $2x^2 + 1$ (c) $2x^2 - 1$ (d) x^2

Q.6 Given that, f(x) is a polynomial in x satisfying the condition

$f(x) + f\left(\frac{1}{x}\right) = f(x) \cdot f\left(\frac{1}{x}\right)$ and $f(5) = 126$, $f(6) =$

(a) 216 (b) 218 (c) 217 (d) –215

Passage (Q.7 - Q.9)

Mr. X is a teacher of mathematics. His students want to know the ages of his son's S_1 and S_2. He told that their ages are 'a' and 'b' respectively such that $f(x+y) - axy = f(x) + by^2 \ \forall x, y \in R$ after some time students said that information is insufficient, please give more information . Teacher says that $f(1) = 8$ and $f(2) = 32$.

Q.7 The age of S_1 will be
(a) 4 (b) 8 (c) 16 (d) 32

Q.8 The function f(x) is
(a) even (b) odd
(c) neither even nor odd (d) periodic as well as odd

Q.9 The age of S_2 will be
(a) 4 (b) 8 (c) 16 (d) 32

Q.10 Determine all functions f satisfying the functional relation.

$f(x) + f\left(\frac{1}{1-x}\right) = \frac{2(1-2x)}{x(1-x)}$ where $x \in R - \{0, 1\}$.

ANSWERS

1. (b) 2. (b) 3. (b) 4. (b) 5. (a) 6. (a) 7. (c)
8. (a) 9. (b) 10. $f(x) = \frac{x+1}{x-1}$

MISCELLANEOUS EXAMPLES

Example 1 :

Consider $f : N \to N$, $g : N \to N$ and $h : N \to R$ defined as $f(x) = 2x$, $g(y) = 3y + 4$ and $h(z) = \sin z$, $\forall$ x, y and z in N. Show that
$ho(gof) = (hog)$ of.

Solution :

We have, ho $(gof)(x) = h(gof(x)) = h(g(f(x))) = h(g(2x))$
$= h(3(2x) + 4) = h(6x + 4) = \sin(6x + 4) \ \forall \ x \in N$.
Also, $((hog)o\ f)(x) = (hog)(f(x)) = (hog)(2x) = h(g(2x))$
$= h(3(2x) + 4) = h(6x + 4) = \sin(6x + 4), \ \forall \ x \in N$.
This shows that $ho(gof) = (hog)$ of.
This result is true in general situation as well.

Example 2 :

Let $f : \left[-\frac{\pi}{3}, \frac{2\pi}{3}\right] \to [0, 4]$ be a function defined as $f(x) = \sqrt{3}\sin x - \cos x + 2$. Then $f^{-1}(x)$ is given by

(A) $\sin^{-1}\left(\frac{x-2}{2}\right) - \frac{\pi}{6}$ (B) $\sin^{-1}\left(\frac{x-2}{2}\right) + \frac{\pi}{6}$ (C) $\frac{2\pi}{3} + \cos^{-1}\left(\frac{x-2}{2}\right)$ (D) None of these

Solution : (B)

$f : \left[-\frac{\pi}{3}, \frac{2\pi}{3}\right] \Rightarrow [0, 4], \quad f(x) = \sqrt{3}\sin x - \cos x + 2$

$f(x) = 2\sin\left(x - \frac{\pi}{6}\right) + 2 \Rightarrow (f^{-1}(x)) = \sin^{-1}\left(\frac{x-2}{2}\right) + \frac{\pi}{6}$

Example 3 :

Let R be a relation on the set A of ordered pairs of positive integers defined by $(x, y)\ R\ (u, v)$ if and only if $xv = yu$. Show that R is an equivalence relation.

Solution :

Clearly, $(x, y)\ R\ (x, y)$, $\forall\ (x, y) \in A$, since $xy = yx$. This shows that R is reflexive.

Further, $(x, y)\ R\ (u, v) \Rightarrow xv = yu \Rightarrow uy = vx$ and hence $(u, v)\ R\ (x, y)$. This shows that R is symmetric.

Similarly, $(x, y)\ R\ (u, v)$ and $(u, v)\ R\ (a, b) \Rightarrow xv = yu$ and $ub = va \Rightarrow xy\frac{a}{u} \Rightarrow xv\frac{b}{v} = yu\frac{a}{u} \Rightarrow xb = ya$

and hence $(x, y)\ R\ (a, b)$.

Thus, R is transitive. Thus, R is an equivalence relation.

Example 4 :

Show that if $f : R - \left\{\frac{7}{5}\right\} \to R - \left\{\frac{3}{5}\right\}$ is defined by $f(x) = \frac{3x+4}{5x-7}$ and $g : R - \left\{\frac{3}{5}\right\} \to R - \left\{\frac{7}{5}\right\}$ is defined by $g(x) = \frac{7x+4}{5x-3}$, then $fog = \mathrm{I}_A$ and $gof = \mathrm{I}_B$, where, $A = R - \left\{\frac{3}{5}\right\}, B = R - \left\{\frac{7}{5}\right\}$; $\mathrm{I}_A = x,\ \forall\ x \in A$, $\mathrm{I}_B(x) = x,\ \forall\ x \in B$ are called identity functions on sets A and B, respectively.

Solution :

We have, $gof(x) = g\left(\frac{3x+4}{5x-7}\right) = \frac{7\left(\frac{(3x+4)}{5x-7}\right)+4}{5\left(\frac{(3x+4)}{5x-7}\right)-3} = \frac{21x+28+20x-28}{15x+20-15x+21} = \frac{41x}{41} = x$

Similarly, $fog(x) = f\left(\frac{7x+4}{5x-3}\right) = \frac{3\left(\frac{(7x+4)}{(5x-3)}\right)+4}{5\left(\frac{(7x+4)}{(5x-3)}\right)-4} = \frac{21x+12+20x-12}{35x+20-35x+21} = \frac{41x}{41} = x$

Thus, $(gof)(x) = x,\ \forall\ x \in B$ and $fog(x) = x,\ \forall\ x \in A$, which implies that $gof = \mathrm{I}_B$ and $fog = \mathrm{I}_A$.

Example 5 :

Consider the identity function $I_N : N \to N$ defined as $I_N(x) = x\ \forall\ x \in N$.

Show that although I_N is onto but $I_N + I_N : N \to N$ defined as $(I_N + I_N)(x) = I_N(x) + I_N(x) = x + x = 2x$ is not onto.

Solution :

Clearly I_N is onto. But $I_N + I_N$ is not onto, as we can find an element 3 in the co-domain N such that there does not exist any x in the domain N with $(I_N + I_N)(x) = 2x = 3$.

Example 6 :

Show that if $f : A \to B$ and $g : B \to C$ are onto, then gof : $A \to C$ is also onto.

Solution :

Given an arbitrary element z ? C, there exists a pre-image y of z under g such that $g(y) = z$, since g is onto. Further, for $y \in B$, there exists an element x in A with $f(X) = y$, since f is onto.

Therefore, $gof(x) = g(f(x)) = g(y) = z$, showing that gof is onto.

Example 7 :

If $f : R \to R$, $f(x) = 2x + 3$ then $f^{-1}(x)$

Solution :

Since f is a bijection therefore its inverse mapping exists and

$y = 2x + 3 \Rightarrow x = 2y + 3$

$\Rightarrow \quad y = \dfrac{x-3}{2} \quad \therefore \; f^{-1}(x) = \dfrac{x-3}{2}$

Example 8 : If $f(x) = \begin{cases} 2+x, \text{ if } x \geq 0 \\ 2-x, \text{ if } x < 0 \end{cases}$, then find $(fof)(x)$.

Solution : $(fof)(x) = f\{f(x)\} = \begin{cases} 2+f(x), & f(x) \geq 0 \\ 2-f(x), & f(x) < 0 \end{cases}$

$$= \begin{cases} 2+2+x, & 2+x \geq 0 \text{ and } x \geq 0 \\ 2-(2+x), & 2+x < 0 \text{ and } x \geq 0 \\ 2+2-x, & 2-x \geq 0 \text{ and } x < 0 \\ 2-(2-x), & 2-x < 0 \text{ and } x < 0 \end{cases} = \begin{cases} 4+x, & x \geq 0 \\ -x, & x \in \phi \\ 4-x, & x < 0 \\ x, & x \in \phi \end{cases}$$

Hence $(fof)(x) = \begin{cases} 4+x, & x \geq 0 \\ 4-x, & x < 0 \end{cases}$

Example 9 :

Let $f : \{1, 2, 3\} \to \{a, b, c\}$ be one-one and onto function given by $f(1) = a$, $f(2) = b$ and $f(3) = c$. Show that there exists a function $g : \{a, b, c\} \to \{1, 2, 3\}$ such that $gof = IX$ and $fog = IY$, where, $X = \{1, 2, 3\}$ and $Y = \{a, b, c\}$.

Solution :

Consider $g : \{a, b, c\} \to \{1, 2, 3\}$ as $g(a) = 1$, $g(b) = 2$ and $g(c) = 3$.

It is easy to verify that the composite $gof = I_X$ is the identity function on X and the composite $fog = I_Y$ is the identity function on Y.

Example 10 :

Find out the type of the function $f : R \to R - \{1\}$, $f(x) = \dfrac{x-2}{x-3}$

Solution :

One-one/many-one : Let $x_1 x_2 \in R - \{3\}$ are the elements such that

$f(x_1) = f(x_2)$: then $f(x_1) = f(x_2) \Rightarrow \dfrac{x_1-2}{x_1-3} = \dfrac{x_2-2}{x_2-3}$

$\Rightarrow \quad (x_1 - 2)(x_2 - 3) = (x_2 - 2)(x_1 - 3)$

$\Rightarrow \quad x_1x_2 - 2x_2 - 3x_1 + 6 = x_2x_1 - 2x_1 - 3x_2 + 6$

$\Rightarrow \quad -2x_2 - 3x_1 = -2x_1 - 3x_2$

$\Rightarrow \quad x_2 = x_1 \quad \therefore \quad f(x_1) = f(x_2) \Rightarrow x_1 = x_2$

$\Rightarrow \quad f$ is one - one function

Onto/into : Let $y \in R - \{1\}$ (co-domain)

Then one element $x \in R - \{3\}$ in domain is such that

$f(x) = y \Rightarrow \dfrac{x-2}{x-3} = y \Rightarrow x - 2 = xy - 3y \Rightarrow x = \left(\dfrac{3y-2}{y-1}\right) = x \in R - \{3\}$

$\therefore$ The pre-image of each element of co-domain R – {1} exists in domain R – {3}.

$\Rightarrow$ f is onto

Example 11 :

Which of the function defined below is one-one

(A) $f:(0,\infty)\to R$, $f(x)=x^2-4x+3$ (B) $f:[0,\infty)\to R$, $f(x)=x^2+4x-5$

(C) $f:R\to R$, $f(x)=e^x+\frac{1}{e^x}$ (D) $f:R\to R$, $f(x)=\ell n(x^2+x+1)$

Solution : (B)

By definition only $f(x)=x^2+4x-5$ with domain $[0,\infty)$ is one to one.

Example 12 :

The function $f:[2,\infty)\to(0,\infty)$ defined by $f(x)=x^2-4x+a$, then the set of values of 'a' for which $f(x)$ becomes onto is

(A) $(4,\infty)$ (B) $[4,\infty)$ (C) $\{4\}$ (D) f

Solution : (D)

$f(x)=x^2-4x+a$

always attains its minimum value. So its range must be closed.

So, $a=\{f\}$

Example 13 :

If $f:R\to R$, $f(x)=\begin{cases}x|x|-4, & x\in Q\\ x|x|-\sqrt{3} & x\notin Q\end{cases}$, then $f(x)$ is

(A) one to one and onto (B) many to one and onto

(C) one to one and into (D) many to one and into

Solution : (D)

$f(2)=f\left(3^{1/4}\right)\Rightarrow$ many to one function

and $f(x)\neq-\sqrt{3}\ \forall\ x\in R\Rightarrow$ into function

Example 14 :

Let $f(x)=\begin{cases}x+1, x\le 1\\ 2x+1, 1<x\le 2\end{cases}$ and $g(x)=\begin{cases}x^2, & -1\le x<2\\ x+2, & 2\le x\le 3\end{cases}$ Find fog.

Solution :

$$(fog)(x)=f(g(x))=\begin{cases}g(x)+1, & g(x)\le 1\\ 2g(x)+1, & 1<g(x)\le 2\end{cases}$$

Let us consider, $g(x)\le 1$:

(i) $x^2\le 1, -1\le x<2\Rightarrow -1\le x\le 1, -1\le x<2$

(ii) $x+2\le 1, 2\le x\le 3\Rightarrow x\in[-\sqrt{2},-1)\cup(1,\sqrt{2}], -1\le x<2\Rightarrow 1<x\le\sqrt{2}$

(iv) $1<x+2\le 2, 2\le x\le 3\Rightarrow -1<x\le x\le 3, x=f$

Thus $f(g(x))=\begin{cases}x^2+1, & -1\le x\le 1\\ 2x^2+1, & 1<x\le\sqrt{2}\end{cases}$

Example 15 :

Show that $-a$ is not the inverse of $a\in N$ for the addition operation + on N and $1/a$ is not the inverse of $a\in N$ for multiplication operation × on N, for $a\neq 1$.

Solution :

Since $-a\notin N$, $-a$ can not be inverse of a for addition operation on N, although $-a$ satisfies $a+(-a)=0=(-a)+a$.

Similarly, for $a \neq 1$ in N, $1/a \notin N$, which implies that other than 1 no element of N has inverse for multiplication operation on N.

Example 16 :

If $f : R \to R$ and $f(x) = ax + \sin x + a$, then

(A) $f(x)$ is one–one onto function if $a \in R$

(B) $f(x)$ is one–one onto function if $a \in R - [-1, 1]$

(C) $f(x)$ is one–one onto function if $a \in R - \{0\}$

(D) $f(x)$ is one–one onto function if $a \in R - \{-1\}$

Solution : (B)

If $a \neq 0$, Range of $f(x) = R$. Also, $f'(x) = a + \cos x$

If $a > 1$, $f'(x) > 0$, i.e. function is increasing

$a > -1$, $f'(x) < 0$ i.e. function is decreasing

$f(x)$ to be one to one it must be monotonic and is passible its $a < -1$ or $a > 1$

Example 17 :

Find the domain of the function $f(x) = \dfrac{x^2 + x - 6}{x^2 + 2x - 8}$. Find the inverse function, if it exists, and its domain.

Solution :

$f(x) \dfrac{(x-2)(x+3)}{(x-2)(x+4)} = \dfrac{x+3}{x+4}$ provided $x \neq 2$ so that the domain of the function is readily

$(-\infty, -4) \cup (-4, 2) \cup (2, \infty)$.

If $y = \dfrac{x+3}{x+4}$, $x = \dfrac{x-4y}{y-1}$ giving $f^{-1}(x) = \dfrac{3-4x}{x-1}$.

It can be seen that if $y = \dfrac{5}{6}$, $x = 2$ is not allowed for $f(x)$ and so $y \neq \dfrac{5}{6}$ in order that $x \neq 2$.

The domain of the inverse function $f^{-1}(x)$ is therefore $\left(-\infty, \dfrac{5}{6}\right) \cup \left(\dfrac{5}{6}, 1\right) \cup (1, \infty)$.

Example 18 :

Consider the following functions :

(i) $f(x) = \sin(\sin^{-1} 2x) + \operatorname{cosec}(\operatorname{cosec}(\operatorname{cosec}^{-1} 2x) + \tan(\tan^{-1} 2x)$

(ii) $g(x) = \sin^{-1}\{x\}$ where {x} = F.P.F.

(iii) $h(x) = \dfrac{2}{\pi}\operatorname{cosec}^{-1}(\sin x)$

(iv) $k(x) = \cos^{-1}(|\sin x + \cos x|)$

Comment for all these four functions w.r.t. Range, odd/even, periodicity and whether one-one or many-one. Marks will be given in each part if all the four comments are correct.

Solution :

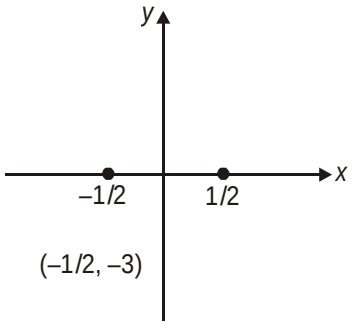
(i) $f(x) = 6x$ for $x = \pm\dfrac{1}{2}$ ($\sin \sin^{-1} 2x = 2x$, $-1/2 \leq x < 1/2$

range of $f(x)$ is {3, –3} (two integer) $\operatorname{cosec} \operatorname{cosec}^{-1} 2x = 2x$, $x \leq -1/2$ or $x \geq -1/2$

$f(x)$ is also odd and non periodic $\tan \tan^{-1} 2x = 2x \quad \forall \; x \in R$

$f(x)$ is also one-one function common x is $x = \pm 1/2$

graph of $f(x)$ is



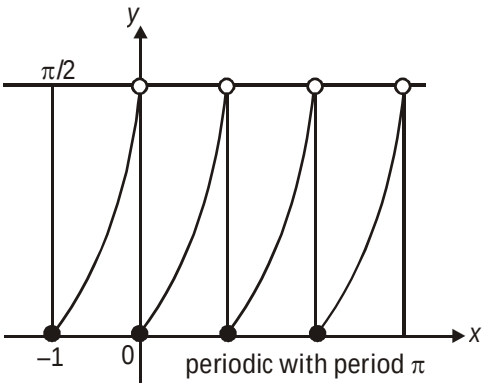
(ii) $g(x) = \sin^{-1}\{x\}$

$0 \le \{x\} < 1$ for all $x \in R$

$0 \le g(x) < \pi/2 \Rightarrow$ contain only two integers 0 and 1

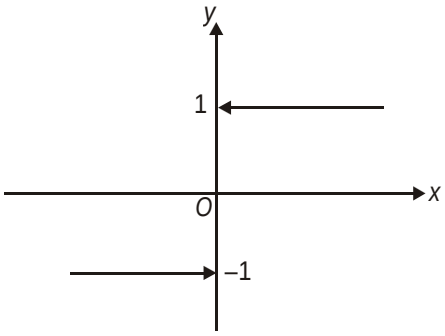
$g(x)$ is periodic with period $1 \Rightarrow$ many-one

range contain only positive value $[0, \pi/2) \Rightarrow$ function graph of $\sin^{-1}\{x\}$



(iii) $h(x) = \frac{2}{\pi}\text{cosec}^{-1}1 \qquad x > 0$ (sgn $x = 1$ *for* $x > 0$)

$= \frac{2}{\pi}\text{cosec}^{-1}0 \qquad x = 0 \Rightarrow$ not defined



$= \frac{2}{\pi}\text{cosec}^{-1}(-1) \qquad x < 0$ (*sgn* $x = -1$ for $x < 0$)

$h(x) = 1 \qquad x > 0$

$= -1 \qquad x < 0$

graph shows $h(x)$ is odd, range contain only two integer which is non one-one and non periodic

(iv) $k(x) = \cos^{-1}\left|\sqrt{2}\cos\left(x - \frac{\pi}{4}\right)\right|$

$k(x) \ne -k(-x)$

k (x) is periodic with period $\pi \Rightarrow$ many-one

$$\left|\sqrt{2}\cos\left(x-\frac{\pi}{4}\right)\right| \in [0, \sqrt{2}]$$

$\therefore \quad \cos^{-1}\left|\sqrt{2}\cos\left(x-\frac{\pi}{4}\right)\right|$ must lie between 0 to 1

$\therefore$ range of $\cos^{-1}\left|\sqrt{2}\cos\left(x-\frac{\pi}{4}\right)\right|$ will $[0, \pi/2]$, which contain only two integer 0 and 1.

Example 19 :

Let $f : X \to Y$ be a function. Define a relation R in X given by $R = \{(a, b): f(a) = f(b)\}$. Examine if R is an equivalence relation.

Solution :

For every $a \in X$, $(a, a) \in R$, since $f(a) = f(a)$, showing that R is reflexive.

Similarly, $(a, b) \in R \Rightarrow f(a) = f(b) \Rightarrow f(b) = f(a) \Rightarrow (b, a) \in R$.

Therefore, R is symmetric. Further, $(a, b) \in R$ and $(b, c) \in R \Rightarrow f(a) = f(b)$ and $f(b) = f(c) \Rightarrow$ f $(a) = f(c) \Rightarrow (a, c) \in R$, which implies that R is transitive. Hence, R is an equivalence relation.

Example 20 :

If $X = \{1, 2, 3, 4, 5\}$ and $Y = \{a, b, c, d, e, f\}$and $f : X \to Y$, then find the total number of

(a) Functions
(b) One to one function
(c) Many-one function
(d) Constant function
(e) Onto function
(f) Into function

Solution :

(a) Total number of functions = $6^5 = 7776$

(b) Total number of one to one function = ${}^6C_5.5! = 6! = 720$

(c) Total number of many-one function = $6^5 - 6! = 7056$

(d) Total number of constant function = 6

(e) Total number of onto function = 0 (as $r > n$)

(f) Total number of into function = $6^5 = 7776$

Example 21 :

If f : $R \to R$, $f(x) = x^3 + 2$ then find $f^{-1}(x)$

Solution :

$f(x) = x^3 + 2, x \in R$

Since this is a one-one onto function therefore inverse of this function (f^{-1}) exists.

Let $f^{-1}(x) = y \quad \therefore \quad x = f(y) \Rightarrow x = y^3 + 2 \quad \Rightarrow \quad y = (x - 2)^{1/3}$

$\therefore \quad f^{-1}(x) = (x - 2)^{1/3}$

Example 22 :

Find the inverse of the function $f(x) = \ell n(x^2 + 3x + 1); x \in [1, 3]$ and assuming it to be an onto function.

Solution :

Given $f(x) = \ell n(x^2 + 3x + 1) \quad \therefore \quad f'(x) = \frac{2x+3}{(x^2+3x+1)} > 0 \ \forall \ x \in [1, 3]$

Which is a strictly increasing function. Thus $f(x)$ is injective, given that $f(x)$ is onto. Hence the given function $f(x)$ is invertible.

Now $f(f^{-1}(x)) = x \Rightarrow \ell n((f^{-1}(x))^2 + 3(f^{-1}(x)) + 1) = x$

$\Rightarrow (f^{-1}(x))^2 + 3(f^{-1}(x)) + 1 - e^x = 0 \quad \therefore \quad f^{-1}(x) = \frac{-3 \pm \sqrt{9 - 4.1(1 - e^x)}}{2} = \frac{-3 \pm \sqrt{(5 + 4e^x)}}{2}$

$\Rightarrow f^{-1}(x) = \frac{-3 + \sqrt{(5 + 4e^x)}}{2}$ (as $f^{-1}(x) \in [1, 3]$)

Hence $f^{-1}(x) = \frac{-3 + \sqrt{(5 + 4e^x)}}{2}$

UNSOLVED EXERCISE

Section-A

(Straight objective type with single option correct)

Topic : 1 Type of Functions Based Questions

Q.1 The $f : R \to R$ be any function. Define $g : R \to R$ by $g(x) = |f(x)|$ for all x. Then g is
(a) onto if f is onto
(b) one–one if f is one-one
(c) continuous if f is continuous
(d) differentiable if f is differentiable

Q.2 Let function $f : R \to R$ be defined by $f(x) = 2x + \sin x$ for $x \in R$, then f is
(a) one-to-one and onto
(b) one-to-one but NOT onto
(c) onto but NOT one-to-one
(d) neither one-to-one nor onto

Q.3 Let S be the set of all triangles and R^+ be the set of positive real numbers. Then the function, $f : S \to R^+$, $f(\Delta)$ = area of the Δ, where $\Delta \in S$ is –
(a) injective but not surjective
(b) surjective but not injective
(c) injective as well as surjective
(d) neither injective nor surjective

Q.4 Let $f : R \to R$ be a function defined by $f(x) = \frac{e^{|x|} - e^{-x}}{e^x + e^{-x}}$. Then –
(a) f is both one-one and onto
(b) f is one-one but not onto
(c) f is onto but not one-one
(d) f is neither one-one nor onto.

Q.5 Let $E = \{1, 2, 3, 4\}$ & $F = \{1, 2\}$. Then the number of onto functions from E to F is
(a) 14 (b) 16
(c) 12 (d) 8

Q.6 If $f : [0, \infty) \to [0, \infty)$ and $f(x) = \frac{x}{1 + x}$ then f is
(a) one - one and into
(b) onto but not one - one
(c) one - one and onto
(d) neither one - one nor onto

Q.7 $f(x) = \begin{cases} x, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational} \end{cases}$, and $g(x) = \begin{cases} 0, & \text{if } x \text{ is rational} \\ x, & \text{if } x \text{ is irrational} \end{cases}$, then $f - g$ is
(a) one–one and into
(b) neither one–one nor onto
(c) many one and on to
(d) one–one and onto

Q.8 The function $f : R \to R$ given by $f(x) = 3 - 2\sin x$ is.
(a) one -one (b) onto
(c) bijective (d) none of these

Q.9 Let $f : (1, \infty) \to (1, \infty)$ be defined by $f(x) = \frac{x + 2}{x - 1}$. Then
(a) f is 1 - 1 and onto
(b)f is 1- 1 but not onto
(c) f is not 1-1 but onto
(d) f is neither 1 - 1 nor onto

Q.10 Let $f : R \to R$ be a function defined by $f(x) = x^{2009} + 2009x + 2009$ Then f(x)is
(a) one - one but not onto
(b) not one- one but onto
(c) neither one-one nor onto
(d) one - one and onto

Q.11 Let $f : R \to R$ be defined by $f(x) = \frac{x-a}{x-b}$, where $a \neq b$. Then f is
(a) Injective but not surjective
(b) Surjective but not injective
(c) Bijective
(d) none of these

Q.12 Let $f : R \to R$ be a function such that $f(x) = \begin{cases} x+5, & x \in Q \\ x-\sqrt{5}, & x \in Q^c \end{cases}$, then
(a) f is one–one and onto
(b) f is one–one and into
(c) f is many one and into
(d) f is many one and onto

Q.13 If $f : [0, \infty) \to [0, \infty)$ and $f(x) = \frac{x}{1+x}$ then f is
(a) one - one and into
(b) onto but not one - one
(c) one - one and onto
(d) neither one - one nor onto

Q.14 The function $f : R \to R$ defined by $f(x) = (x-1)(x-2)(x-3)$ is
(a) one-one but not onto
(b) onto but not one-one
(c) both one-one and onto
(d) neither one-one nor onto

Q.15 If $f : R \to R$ be a function such that $f(x) = x^3 + x^2 + 3x + \sin x$. Then
(a) f is one-one into
(b) f is one-one and onto
(c) f is many one and into
(d) f is many one and onto

Q.16 The function $f : N \to N$ (N is the set of natural numbers) defined by $f(n) = 2n + 3$ is
(a) surjective (b) not surjective
(c) bijective (d) none of these

Q.17 Function $f : R \to R$ $f(x) = x + |x|$ is
(a) one-one (b) onto
(c) one-one onto (d) none of these

Q.18 Let $f : \left[\frac{\pi}{3}, \frac{2\pi}{3}\right] \to [0, 4]$ be a function defined as $f(x) = \sqrt{3}\sin x - \cos x + 2$. Then $f^{-1}(x)$ is given by

(a) $\sin^{-1}\left(\frac{x-2}{2}\right) - \frac{\pi}{6}$

(b) $\sin^{-1}\left(\frac{x-2}{2}\right) + \frac{\pi}{6}$

(c) $\frac{2\pi}{3} + \cos^{-1}\left(\frac{x-2}{2}\right)$

(d) None of these

Q.19 If $f : [1, \infty) \to [2, \infty)$ is given by, $f(x) = x + \frac{1}{x}$, then $f^{-1}(x)$ equals

(a) $\frac{x + \sqrt{x^2-4}}{2}$ (b) $\frac{x}{1+x^2}$

(c) $\frac{x - \sqrt{x^2-4}}{2}$ (d) $1 - \sqrt{x^2-4}$

Topic : 2 Composition of Functions Questions

Q.1 Let $f(x) = \sin x + \cos x$ and $g(x) = x^2 - 1$, then domain for which gof is invertible, is

(a) $\left(0, \frac{\pi}{2}\right)$ (b) $\left(\frac{\pi}{2}, \frac{2\pi}{3}\right)$

(c) $\left(-\frac{\pi}{2}, \frac{\pi}{3}\right)$ (d) $\left(-\frac{\pi}{4}, \frac{\pi}{4}\right)$

Q.2 If $f(x) = \begin{cases} x; \text{ when x is rational} \\ 1-x; \textit{ when } x \text{ is irrational} \end{cases}$, then fof (x) is given as
(a) 1 (b) x
(c) $1 + x$ (d) None of these

Q.3 If the function $f : R \to A$ given by $f(x) = \frac{x^2}{x^2+1}$ is a surjection, then A is
(a) R (b) [0, 1]
(c) (0, 1] (d) [0, 1)

Q.4 Let $f(x) = \frac{\alpha x}{x+1}$, $x \neq -1$. Then for what value of α is $f(f(x)) = x$
(a) $\sqrt{2}$ (b) $-\sqrt{2}$
(c) 1 (d) -1

Q.5 $f(x) = |x-1|$, $f : R^+ \to R$, $g(x) = e^x$, $g : [1, \infty) \to R$, if the function fog (x) is defined, then its domain and range respectively are –
(a) $(0, \infty)$ & $[0, \infty) \to R$
(b) $[-1, \infty)$ & $[0, \infty)$

(c) $[-1, \infty)$ & $\left[1 - \frac{1}{e}, \infty\right]$

(d) $[-1, \infty)$ & $\left[\frac{1}{e} - 1, \infty\right]$

Q.6 Let f be a function with domain $[-3, 5]$ and let $g(x) = |3x + 4|$, Then the domain of $(fog)(x)$ is

(a) $\left(-3, \frac{1}{3}\right)$ (b) $\left[-3, \frac{1}{3}\right]$

(c) $\left[-3, \frac{1}{3}\right)$ (d) none of these

Q.7 Let $f(x) = \frac{2}{x+1}, g(x) = \cos x$ and $h(x) = \sqrt{x+3}$ then the range of the composite function fogoh, is

(a) R^+ (b) $R - \{0\}$
(c) $[1, \infty)$ (d) $R^+ - \{1\}$

Q.8 The domain of the function $f(x) = \sqrt{x^{14} - x^{11} + x^6 - x^3 + x^2 + 1}$ is –

(a) $(-\infty, \infty)$ (b) $[0, \infty)$
(c) $(-\infty, 0]$ (d) R

Q.9 *gof* exists when

(a) codomain of g = domain of f
(b) codomain of f = domain of g
(c) co domain of g = domain of g
(d) codomain of g = co domain of f

Topic : 3 Inverse Functions Based Questions

Q.1 If the function $f : [1, \infty) \to [1, \infty)$ is defined by $f(x) = 2^{x(x-1)}$, then $f^{-1}(x)$ is

(a) $\left(\frac{1}{2}\right)^{x(x-1)}$

(b) $\frac{1}{2}\left(1 + \sqrt{1 + 4\log_2 x}\right)$

(c) $\frac{1}{2}\left(1 - \sqrt{1 + 4\log_2 x}\right)$

(d) not defined

Q.2 If $f(x) = 3x - 5$, then $f^{-1}(x)$

(a) is given by $\frac{1}{3x-5}$

(b) is given by $\frac{x+5}{3}$

(c) does not exist because f is not one-one

(d) does not exist because f is not onto

Q.3 Let $f : X \to Y$ be a function such that $f(C) = \{f(x) : x \in C\}, C \subseteq X$ and $f^{-1}(D) = \{f^{-1}(x) : x \in D\}, D \subseteq Y$, then

(a) $f^{-1}(f(A)) = A$ only if $A \subseteq D$

(b) $f(f^{-1}(B)) = B$ only if $B \subseteq D$

(c) $f(f^{-1}(A)) = A$

(d) $f(f^{-1}(B)) = B$

Q.4 Function $f\ [6\pi, 7\pi] \to [-1, 1]$ which function is one one onto

(a) sin x (b) cos x
(c) tan x (d) cot x

Q.5 Which of the following functions is inverse of itself

(a) $f(x) = \frac{1-x}{1+x}$ (b) $g(x) = 5^{\log x}$
(c) $h(x) = 2^{x(x-1)}$ (d) none of these

Q.6 If $f(x) = \frac{1-x}{1+x}$, then domain of $f^{-1}(x)$ is

(a) R (b) $R - \{-1\}$
(c) $(-\infty, -1)$ (d) $(-1, \infty)$

Q.7 If $f(x) = \log_{e^2 x}\left(\frac{2\ln x + 2}{-x}\right)$ and $g(x) = \{x\}$ then range of $g(x)$ for the existence of $f(g(x))$ is

(a) $\left(0, \frac{1}{e}\right) \sim \left\{\frac{1}{e^2}\right\}$ (b) $\left(0, \frac{2}{e}\right) \sim \left\{\frac{1}{e^2}\right\}$
(c) $\left(0, \frac{3}{e}\right) \sim \left\{\frac{1}{e^2}\right\}$ (d) none of these

Q.8 $f : [2, \infty) \to (-\infty, 4]$, where $f(x) = x(4 - x)$ then $f^{-1}(x)$ is

(a) $2 - \sqrt{4-x}$ (b) $2 + \sqrt{4-x}$
(c) $-2 + \sqrt{4-x}$ (d) $-2 - \sqrt{4-x}$

Q.9 If $f : I \to I$ be defined by $f(x) = [x + 1]$, where [.] denotes the greatest integer function, then $f^{-1}(x)$ is equal to

(a) $x - 1$ (b) $[x + 1]$
(c) $\frac{1}{[x-1]}$ (d) $\frac{1}{x+1}$

Q.10 Let $f : R \to [0, \pi/2)$ (where R is the set of real numbers) be a function defined by $f(x) = \tan^{-1}(x^2 + x + a)$. If f is onto then a equals

(a) 0 (b) 1
(c) 1/2 (d) 1/4

Q.11 In the following, onto function is

(a) $f : Z \to Z, f(x) = |x|$ (b) $f : R_0 \to R^+, f(x) = |x|$
(c) $f : N \to Z, f(x) = |x|$ (d) none of these

Q.12 If $f(x) = \begin{cases} \lambda \text{ when } x \text{ is rational number} \\ 0 \text{ when } x \text{ is irrational number} \end{cases}$ then fof $(\sqrt{2})$ is equal to

(a) 1 (b) 0
(c) 1 (d) none of these

Q.13 If f and g are two bijection such that (gof) exist then

(a) $(gof)^{-1} = g^{-1} o f^{-1}$ (b) $(fog)^{-1} = g^{-1} o f^{-1}$

(c) $(gof)^{-1} = f^{-1} o g^{-1}$ (d) none of these

Q.14 If $f : R \to R\ f(x) = 2x + 1$ and $g : R \to R, g(x) = x^3$, then $(gof)^{-1}$ (27) equals

(a) -1 (b) 1

(c) 0 (d) 2

Q.15 If $f : [1, \infty) \to [2, \infty)$ is given by, $f(x) = x + \frac{1}{x}$, then $f^{-1}(x)$ equals

(a) $\frac{x + \sqrt{x^2 - 4}}{2}$ (b) $\frac{x}{1 + x^2}$

(c) $\frac{x - \sqrt{x^2 - 4}}{2}$ (d) $1 - \sqrt{x^2 - 4}$

Q.16 Let $f(x) = (x+1)^2 - 1, (x \geq -1)$. Then the set $S = \{x : f(x) = f^{-1}(x)\}$ is, if f is onto

(a) $\left\{0, -1, \frac{-3 + i\sqrt{3}}{2}, \frac{-3 - i\sqrt{3}}{2}\right\}$

(b) {0, 1, –1}

(c) {0, –1}

(d) empty

Q.17 Let $f : X \to Y$ be a function such that $f(C) = \{f(x) : x \in C\}, C \subseteq X$ and $f^{-1}(D) = \{f^{-1}(x) : x \in D\}, D \subseteq Y$, then

(a) $f^{-1}(f(A)) = A$ only if $A \subseteq D$

(b) $f(f^{-1}(B)) = B$ only if $B \subseteq D$

(c) $f(f^{-1}(A)) = A$

(d) $f(f^{-1}(B)) = B$

Q.18 If the function $f : R \to R$ be such that $f(x) = x - [x]$, where $[y]$ denotes greatest integer less than or equal to y, then f–1 (x) is –

(a) $\frac{1}{x - [x]}$ (b) $[x] - x$

(c) not defined (d) none of these

Q.19 If $f : R^- \to [1, \infty)$ is defined by $f(x) = x^2 + 1$, the value of $f^{-1}(17)$ and f^{-1} (C) are respectively.

(a) $\pm 4, \pm\sqrt{2}$ (b) $4, \sqrt{2}$

(c) $-4, \sqrt{2}$ (d) $-4, -\sqrt{2}$

Topic : 4 Perodic Even & Odd Functions Based Questions

Q.1 If (x)= sin x - cos x is written as $f_1(x) + f_2(x)$ where $f_1(x)$ is even and $f_2(x)$ is odd then

(a) $f_1(x) = \cos x$ (b) $f_1(x) = -\cos x$

(c) $f_2(x) = -\sin x + \cos x x$ (d) $f_2(x) = \sin(2\pi - x)$

Q.2 The function $f(x) = x^2 \log\frac{1-x}{1+x}$ is

(a) a periodic function

(b) a bounded function

(c) an odd fucntion

(d) an even function

Q.3 The value of $n \in Z$ for which the function $f(x) = \frac{\sin nx}{\sin(x/n)}$ has 4π as its peiod is

(a) 2 (b) 3

(c) 4 (d) 5

Q.4 The function $f(x) = \frac{x}{e^x - 1} + \frac{x}{2} + 1$ is

(a) even (b) periodic

(c) odd (d)neither even nor odd

Q.5 Let $f(x) = \sin\sqrt{x}$, then

(a) f(x) is periodic with period $\sqrt{2\pi}$

(b) f(x) is periodic with period $\sqrt{\pi}$

(c) f(x) is periodic with period $4\pi^2$

(d) none of these

Q.6 Which of the following functions are odd fucntions

(a) $f(x) = x\left(\frac{a^x + a^{-x}}{a^x - a^{-x}}\right)$ (b) $f(x) = \frac{a^x + x}{a^x + x}$

(c) $f(x) = \frac{a^x - 1}{a^x + 1}$ (d) $f(x) = x\log_2\left(x + \sqrt{x^2 + 1}\right)$

SECTION-B
(PREVIOUS YEARS QUESTIONS)

Q.1 Let f be a real valued function defined on the interval $(-1, 1)$ such that

$$e^{-x} f(x) = 2 + \int_0^x \sqrt{t^4 + 1}\cdot dt, \forall x \in (-1, 1)$$

and let f^{-1} be the inverse function of f. Then $[f^{-1}(2)]'$ is equal to

(a) 1 (b) $\frac{1}{3}$

(c) $\frac{1}{2}$ (d) $\frac{1}{c}$

Q.2 Domain of definition of the function $f(x) = \sqrt{\sin^{-1}(2x) + \frac{\pi}{6}}$ for real valued x, is

(a) $\left[-\frac{1}{4}, \frac{1}{2}\right]$ (b) $\left[-\frac{1}{2}, \frac{1}{2}\right]$

(c) $\left(-\frac{1}{2}, \frac{1}{9}\right)$ (d) $\left[-\frac{1}{4}, \frac{1}{4}\right]$

Q.3 Range of the function $f(x) = \frac{x^2 + x + 2}{x^2 + x + 1}$; $x \in R$ is

(a) $(1, \infty)$ (b) $\left(1, \frac{11}{7}\right)$

(c) $\left(1, \frac{7}{3}\right]$ (d) $\left(1, \frac{7}{5}\right)$

Q.4 Let $f(x) = (1 + b^2)x^2 + 2bx + 1$ and let $m(b)$ be the minimum value of $f(x)$. As b varies, the range of $m(b)$ is

(a) [0, 1] (b) $\left[0, \frac{1}{2}\right]$

(c) $\left[\frac{1}{2}, 1\right]$ (d) (0, 1]

Q.5 The domain of definition of $f(x) = \frac{\log_2(x+3)}{x^2 + 3x + 2}$ is

(a) $R/\{-1, -2\}$ (b) $(-2, \infty)$

(c) $R/\{-1, -2, -3\}$ (d) $(-3, \infty)/\{-1, -2\}$

Q.6 The domain of definition of the function $y(x)$ is given by the equation $2^x + 2^y = 2$, is

(a) $0 < x \le 1$ (b) $0 \le x \le 1$

(c) $-\infty < x \le 0$ (d) $-\infty < x < 1$

Q.7 Let $f(\theta) = \sin\theta(\sin\theta + \sin 3\theta)$. Then, $f(\theta)$

(a) ≥ 0, only when $\theta \ge 0$

(b) ≤ 0, for all real θ

(c) ≥ 0, for all real θ

(d) ≤ 0, only when $\theta \le 0$

Q.8 The domain of definition of the function $y = \frac{1}{\log_{10}(1-x)} + \sqrt{x+2}$ is

(a) $(-3, -2)$ excluding -2.5

(b) [0, 1] excluding 0.5

(c) $(-2, 1)$ excluding 0

(d) None of these

Q.9 The function $f : R \to \left[-\frac{1}{2}, \frac{1}{2}\right]$ defined as $f(x) = \frac{x}{1+x^2}$ is

(a) invertible

(b) injective but not surjective

(c) surjective but not injective

(d) neither injective nor surjective

Q.10 Let $a, b, c \in R$. If $f(x) = ax^2 + bx + c$ be such that $a + b + c = 3$ and $f(x + y) = f(x) + f(y) + xy$, $\forall x, y \in R$, then $\sum_{n=1}^{10} f(n)$ is equal to

(a) 330 (b) 165

(c) 190 (d) 255

Q.11 If $f_k(x) = 1/k(\sin^k x + \cos^k x)$, where $x \in R$ and $k \ge 1$, then $f_4(x) - f_6(x)$ is equal to

(a) 1/6 (b) 1/3

(c) 1/3 (d) 1/12

Q.12 If $f(x) = x^2$ and $g(x) = \sin x$ for all $x \in R$. Then, the set of all x satisfying $(fogogof)(x) = (gogof)(x)$, where $(fog)(x) = f(g(x))$, is

(a) $\pm\sqrt{n\pi}, n \in \{0, 1, 2, ...\}$

(b) $\pm\sqrt{n\pi}, n \in \{1, 2, ...\}$

(c) $\pi/2 + 2n\pi$, $n \in \{....., -2, -1, 0, 1, 2, ...\}$

(d) $2n\pi$, $n \in \{...., -2, -1, 0, 1, 2, ...\}$

Q.13 Let $f(x) = \frac{\alpha x}{x+1}$, $x \ne -1$. Then, for what value of α is $f[f(x)] = x$?

(a) $\sqrt{2}$ (b) $-\sqrt{2}$

(c) 1 (d) -1

Q.14 Let $g(x) = 1 + x - [x]$ and $f(x) = \begin{cases} -1, & x < 0 \\ 0, & x = 0 \\ 1, & x > 0 \end{cases}$,

then for all x, $f[g(x)]$ is equal to

(a) x (b) 1

(c) f (x) (d) g(x)

Q.15 If $g\{f(x)\} = |\sin x|$ and $f\{g(x)\} = (\sin\sqrt{x})^2$, then

(a) $f(x) = \sin^2 x$, $g(x) = \sqrt{x}$
(b) $f(x) = \sin x$, $g(x) = |x|$
(c) $f(x) = x^2$, $g(x) = \sin\sqrt{x}$
(d) f and g cannot be determined

Q.16 If $f(x) = \cos(\log x)$, then $f(x)\cdot f(y) - \frac{1}{2}\left[f\left(\frac{x}{y}\right) + f(xy)\right]$ has the value

(a) -1 (b) $\frac{1}{2}$
(c) -2 (d) None of these

Q.17 Let $f(x) = |x - 1|$. Then,
(a) $f(x^2) = \{f(x)\}^2$ (b) $f(x + y) = f(x) + f(y)$
(c) $f(|x|) = |f(x)|$ (d) None of these

Q.18 The function $f : [0, 3] \to [1, 29]$, defined by $f(x) = 2x^3 - 15x^2 + 36x + 1$, is
(a) one-one and onto
(b) onto but not one-one
(c) one-one but not onto
(d) neither one-one nor onto

Q.19 $f(x) = \begin{cases} x, & \text{if } x \text{ is rational} \\ 0, & \text{if } x \text{ is irrational} \end{cases}$,

$g(x) = \begin{cases} 0, & \text{if } x \text{ is rational} \\ x, & \text{if } x \text{ is irrational} \end{cases}$. Then, $f - g$ is
(a) one-one and into
(b) neither one-one nor onto
(c) many one and onto
(d) one-one and onto

Q.20 If $f : [0, \infty) \to [0, \infty)$ and $f(x) = \frac{x}{1+x}$, then f is
(a) one-one and onto
(b) one-one but not onto
(c) onto but not one-one
(d) neither one-one nor onto

Q.21 Let function $f : R \to R$ be defined by $f(x) = 2x + \sin x$ for $x \in R$. Then, f is
(a) one-to-one and onto
(b) one-to-one but not onto
(c) onto but not one-to-one
(d) neither one-to-one nor onto

Q.22 Let $E = \{1, 2, 3, 4\}$ and $F = \{1, 2\}$. Then, the number of onto functions from E to F is
(a) 14 (b) 16
(c) 12 (d) 8

Q.23 If X and Y are two non-empty sets where $f : X \to Y$, is function is defined such that
$f(C) = \{f(x) : x \in C\}$ for $C \subseteq X$ and $f^{-1}(D) = \{x : f(x) \in D\}$ for $D \subseteq Y$,
for any $A \subseteq Y$ and $B \subseteq Y$, then
(a) $f^{-1}\{f(A)\} = A$
(b) $f^{-1}\{f(A)\} = A$, only if $f(X) = Y$
(c) $f\{f^{-1}(B)\} = B$, only if $B \subseteq f(x)$
(d) $f\{f^{-1}(B)\} = B$

Q.24 If $f(x) = \sin x + \cos x$, $g(x) = x^2 - 1$, then $g\{f(x)\}$ is invertible in the domain

(a) $\left[0, \frac{\pi}{2}\right]$ (b) $\left[-\frac{\pi}{4}, \frac{\pi}{4}\right]$
(c) $\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$ (d) $[0, \pi]$

Q.25 Suppose $f(x) = (x + 1)^2$ for $x \geq -1$. If g(x) is the function whose graph is reflection of the graph of $f(x)$ with respect to the line $y = x$, then $g(x)$ equals:

(a) $-\sqrt{x} - 1, x \geq 0$ (b) $\frac{1}{(x+1)^2}, x > -1$
(c) $\sqrt{x+1}, x \geq -1$ (d) $\sqrt{x} - 1, x \geq 0$

Q.26 If $f : [1, \infty) \to [2, \infty)$ is given by $f(x) = x + \frac{1}{x}$, then $f^{-1}(x)$ equals

(a) $\frac{x + \sqrt{x^2 - 4}}{2}$ (b) $\frac{x}{1 + x^2}$
(c) $\frac{x - \sqrt{x^2 - 4}}{2}$ (d) $1 + \sqrt{x^2 - 4}$

Q.27 If the function $f : [1, \infty) \to [1, \infty)$ is defined by $f(x) = 2^{x(x-1)}$, then $f^{-1}(x)$ is

(a) $\left(\frac{1}{2}\right)^{x(x-1)}$ (b) $\frac{1}{2}(1 + \sqrt{1 + 4\log_2 x})$
(c) $\frac{1}{2}(1 - \sqrt{1 + 4\log_2 x})$ (d) not defined

Q.28 If $f(x) = 3x - 5$, then $f^{-1}(x)$

(a) is given by $\frac{1}{3x - 5}$
(b) is given by $\frac{x + 5}{3}$
(c) does not exist because f is not one-one
(d) does not exist because f is not onto

Q.29 Which of the following functions is periodic?
(a) $f(x) = x - [x]$, where $[x]$ denotes the greatest integer less than or equal to the real number x
(b) $f(x) = \sin(1/x)$ for $x \neq 0$, $f(0) = 0$
(c) $f(x) = x \cos x$
(d) None of the above

SECTION-C
(School / Board pattern)

Q.1 Let D be the domain of the real valued function defined by $f(x)=\sqrt{25-x^2}$ Then, write D.

Q.2 If $A=\{a,b,c,d\}$ and the function $f=\{(a,b),(b,d),(c,a),(d,c)\}$, write f^{-1}

Q.3 Is $g=\{(1,1),(2,3),(3,5),(4,7)\}$ a function? If g is described by $g(x)=\alpha x+\beta$, then what value should be assigned to α and β?

Q.4 Let the function $f:R\to R$ be defined by $f(x)=\cos x,\forall x\in R.$ Show that f is neither one-one nor onto.

Q.5 Let $f:R\to R$ be the function defined by $f(x)=\frac{1}{2-\cos x},\forall x\in R.$ Then, Find the range of F.

Q.6 Let n be a fixed positive integer. Define a relation R in Z as follows $\forall a,b\in Z$, aRb if and only if a-b is divisible by n. Show that R is an equivalence relation.

Q.7 Let R be relation defined on the set of natural number N as follows, $R=\{(x,y):x\in N,y\in N,2x+y=41\}$. find the domain and range of the realtion R. Also verify whether R is reflexive, symmetric and transitive.

Q.8 Let $A=[-1,1]$, then, discuss whether the following functions defined on A are one onto or bijective.

(i) $f(x)=\frac{x}{2}$ (ii) $g(x)=|x|$ (iii) $h(x)=x|x|$ (iv) $k(x)=x^2$

Q.9 Each of the following defines a relation of N
(i) x is greater than y, x, y $\in$ N.
(ii) $x+y=10,x,y\in N$
(iii) xy is square of an integer $x,y\in N.$
(iv) $x+4y=10,x,y\in N$
Determine which of the above relations are reflexive, symmetric and transitive.

Q.10. Let $A=\{1,2,3,.....,9\}$ and R be the relation in $A\times A$ defined by $(a,b)R(c,d)$ if a+d=b+c for (a,b), (c,d) in A × A. Prove that R is an equivalence relation and also obtain the equivalent class [(2,5)].

Q.11 Let * be the binary operation defined on Q. Find which of the following binary operations are commutative

(i) a * b = a - b, $\forall$ a, b $\in$ Q (ii) $a*b=a^2+b^2,\forall a,b\in Q$

(iii) $a*b=a+ab,\forall a,b\in Q$ (iv) $a*b=(a-b)^2,\forall a,b\in Q$

Q.12 If * this is binary operation defined on R by a * B $=1+ab,\forall a,b\in R.$ Then, the operation * is
(i) commutative but not associtative. (ii) associative but not commutative.
(iii) neither commutative nor associative. (iv) both commutative and associtatve.

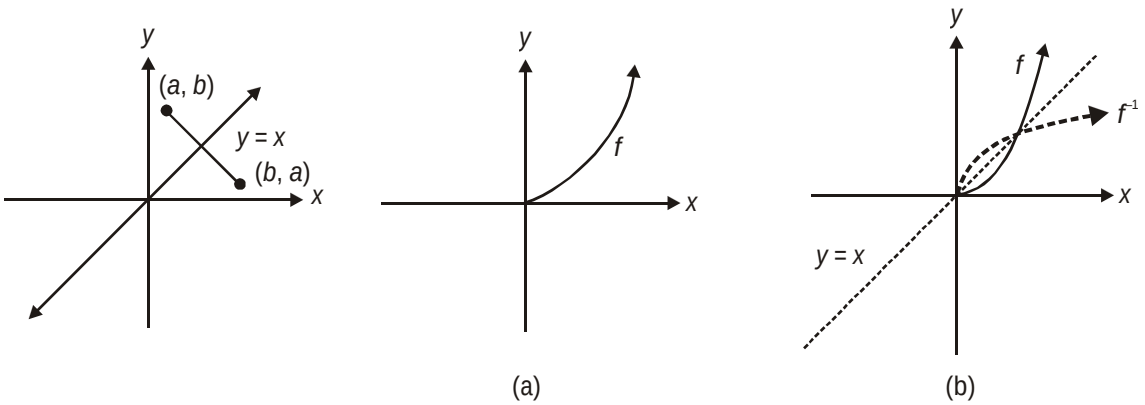
CHAPTER 2

INVERSE TRIGONOMETRIC FUNCTIONS

1. INVERSE FUNCTIONS

The inverse of a function f is denoted by f^{-1} (read f inverse or the inverse of). If (a, b) is an ordered pair of f, then (b, a) is an ordered pair of f^{-1}. The domain and range of f^{-1} are the range and domain, respectively, of f.

Do not confuse the -1 in f^{-1} with a negative exponent. The symbol f^{-1} does not mean $1/f^1$; f^{-1} refers to the inverse function of function f. Graphically, two functions that are inverses of each other are mirror images with reference to the line $y = x$, because ordered pairs (a, b) and (b, a) are mirror images with respect to the line $y = x$ as illustrated in Figure.



(a) (b)

Therefore, if the graph of a function f is known, as in Figure (a), then the graph of f^{-1} can be determined by reflecting f across the line $y = x$ (Figure (b)).

Another useful way of viewing inverse functions is in terms of composition. Basically, inverse functions undo each other, which can be more formally stated as follows: If f and g are inverses of each other, then

(a) $(f \circ g)(x) = f(g(x)) = x$ for all x in the domain of g; and

(b) $(g \circ f)(x) = g(f(x)) = x$ for all x in the domain of f.

This relationship of inverse functions can be used to verify whether two functions are indeed inverses of each other.

FINDING INVERSE FUNCTIONS

The idea of inverse functions undoing each other provides the basis for an informal approach to finding the inverse of a function. Consider the function $f(x) = 2x + 1$.

To each x this function assigns twice x plus 1. To undo this function, we could subtract 1 and divide by 2. So, the inverse should be $f^{-1}(x) = \frac{x-1}{2}$

Now let's verify that f and f^{-1} are inverses of each other.

$$(f \circ f^{-1})(x) = f(f^{-1}(x)) = f\left(\frac{x-1}{2}\right) + 1 = x \text{ and } (f^{-1} \circ f)(x) = f^{-1}(f^{-1}(x)) = f\left(\frac{x-1}{2}\right) + 1 = x$$

Thus, the inverse of f is given by, $f^{-1}(x) = \frac{x-1}{2}$

Let's consider another example of finding an inverse function by the undoing process.

Illustration 1 : Find the inverse of $f(x) = 2011x - 2012$.

Solution : To each x, the function f assigns 2011 times x minus 2012. To undo this we can add 2012 and then divide by 2011. So, the inverse should be $f^{-1}(x) = \frac{x+2012}{2011}$

To verify that f and f^{-1} are inverses we can show that

$$(f \circ f^{-1})(x) = f(f^{-1}(x)) = f\left(\frac{x+2012}{2011}\right) = 2011\left(\frac{x+2012}{2011}\right) - 2012 = x \text{ and}$$

$$(f^{-1} \circ f)(x) = f^{-1}(f(x)) = f^{-1}(2011x - 2012) = \frac{2011x - 2012 + 2012}{2011} = x$$

Thus, f and f^{-1} are inverses and we can write $f^{-1}(x) = \frac{x+2012}{2011}$

The following is a more formal technique for finding the inverse of a function.

1. Let $y = f(x)$.
2. Interchange x and y.
3. Solve the resulting equation for y in terms of x.
4. $f^{-1}(x)$ is determined by the final equation.

Illustration 2 : Find the inverse of $f(x) = 2010x - 2012$.

Solution : First, let $y = f(x)$, so the equation becomes $y = 2010x - 2012$.
Interchange x and y to produce $x = 2010y - 2012$.
Now solve the equation for y in terms of x.

$$x = 2010y - 2012 \quad \text{or} \quad x + 2012 = 2010y \quad \text{or} \quad \frac{x+2012}{2010} = y$$

Thus, the inverse of $f(x) = 2010x - 2012$ is $f^{-1}(x) = \frac{x+2012}{2010}$

Illustration 3 : Find the inverse of $f(x) = \frac{2}{3}x + \frac{3}{5}$

Solution : Let $y = f(x)$, so the equation becomes $y = \frac{2}{3}x + \frac{3}{5}$

Interchange x and y to produce $x = \frac{2}{3}y + \frac{3}{5}$

Now solve the equation for y in terms of x.

$$x = \frac{2}{3}y + \frac{3}{5}$$

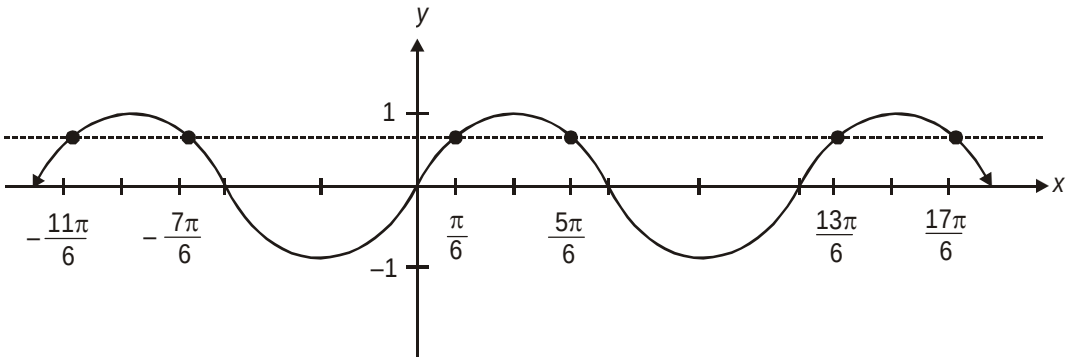
$$15(x) = 15\left(\frac{2}{3}y + \frac{3}{5}\right) \Rightarrow 15x = 10y + 9$$

$$15x - 9 = 10y \Rightarrow \frac{15x - 9}{10} = y$$

Thus, the inverse of $f(x) = \frac{2}{3}x + \frac{3}{5}$ is $f^{-1}(x) = \frac{15x - 9}{10}$.

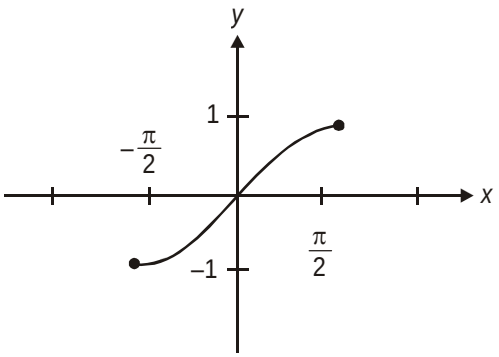
2. INVERSE TRIGONOMETRIC FUNCTIONS

It is evident that the sine function over the domain of all real numbers is not a one-to-one function. For example, suppose that we consider the solutions for $\sin x = \frac{1}{2}$. Certainly, $\frac{\pi}{6}$ is a solution, but there are infinitely many more solutions, such as $\frac{5\pi}{6}, \frac{13\pi}{6}, \frac{17\pi}{6}, -\frac{7\pi}{6}, -\frac{11\pi}{6}$, etc. as indicated in figure.



However we can form a one-to-one function from the sine function, and not eliminate any values from its range, by restricting the domain to the interval $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$. Therefore, we have a new function by

the equation $y = \sin x$ with a domain of $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$ and



a range of $-1 \le y \le 1$. (figure)

INVERSE SINE FUNCTION

The inverse sine function is defined by $y = \sin^{-1} x$ if and only if $x = \sin y$

where $-1 \le x \le 1$ and $-\frac{\pi}{2} \le y \le \frac{\pi}{2}$

In definition, the equation $y = \sin^{-1} x$ can be read as y is the angle whose sine is x. Therefore, $y = \sin^{-1}\frac{1}{2}$ means by y is the angle, between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$, inclusive, whose sin is $\frac{1}{2}$, thus $y = \frac{\pi}{6}$. (The angle could also be expressed as 30°).

Principal value of $\sin^{-1} x$ itself its value as inverse functions are consider as function only for a particular interval. So their is no need to define separate principal value.

Illustration 4 : Evaluate $\cos\left(\sin^{-1}\left(-\frac{1}{2}\right)\right)$

Solution : The expression $\cos\left(\sin^{-1}\left(-\frac{1}{2}\right)\right)$ means the cosine of the angle between $(-(\pi/2)$ and $\pi/2$, inclusive, whose sine is $-1/2$. We know that the angle between $-(\pi/2)$ and $\pi/2$, inclusive, whose sine is

$-1/2$, is

$-(\pi/6)$, then $\cos(-(\pi/6)) = \sqrt{3}/2$.

Therefore, $\cos\left(\sin^{-1}\left(-\frac{1}{2}\right)\right) = \frac{\sqrt{3}}{2}$.

INVERSE COSINE FUNCTION

The other trigonometric functions can also be used to introduce inverse functions. In each case, a restriction needs to be place on the original domain to create a one-to-one function that contains the entire range of the original function. Then a corresponding inverse function can be defined.

By restricting the domain of the cosine function to real numbers between 0 and π, inclusive, a one-to-one function with a range between -1 and 1, inclusive, is obtained. Then the following definition creates the inverse cosine function.

The inverse cosine function or arc cosine function is defined by

$y = \cos^{-1} x = \arccos x$ if and only if $x = \cos y$, where $-1 \le x \le 1$ and $0 \le y \le \pi$

particular interval. So their is no need to define separate principal value.

Illustration 5 : Solve $y = \cos^{-1}(-(\sqrt{3}/2))$ for y, where $0 \le y \le \pi$.

Solution : The expression cpsparticular interval. So their is no need to define separate principal value.

$\cos^{-1}(-(\sqrt{3}/2))$ can be interpreted as the angle whose cosine is $-(\sqrt{3}/2)$. We know that $y = 5\pi/6$.

Illustration 6 : Evaluate $\sin(\cos^{-1} 1/2)$

Solution : The expression $\sin(\cos^{-1} 1/2)$ means the sine of the angle, between 0 and π, inclusive, whose cosine is 1/2. We know that $\pi/3$ is the angle whose cosine is 1/2 and we know that $\sin(\pi/3) = \sqrt{}/2$. Therefore, $\sin(\cos^{-1} 1/2) = \sqrt{3}/2$.

INVERSE TANGENT FUNCTION

By restricting the domain of the tangent function to real numbers between $-(\pi/2)$ and $\pi/2$, $(-(\pi/2)$ and $\pi/2$ are not included since the tangent is undefined at those values) a one-to-one function is obtained. Therefore, the inverse tangent function.

The inverse tangent function or arctangent function is defined by

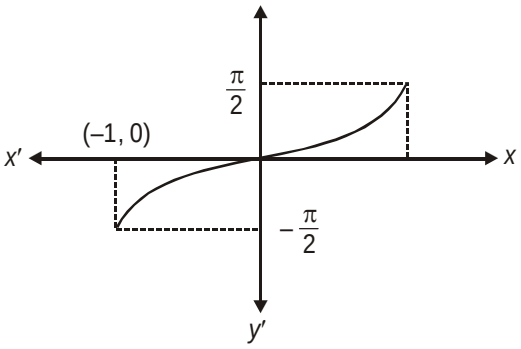
$y = \tan^{-1}x = \arctan x$ if an only if $x = \tan y$ where $-\infty < x < \infty$ and $-(\pi/2) < y < \pi/2$

Illustration 7 : Solve $y = \tan^{-1}(-(\sqrt{3}/3))$ for y, where $-90° < y < 90°$.

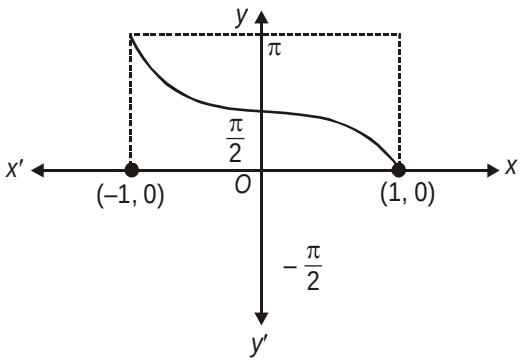
Solution : The expression $\tan^{-1}(-(\sqrt{3}/3))$ can be interpreted as the angle between $-90°$ and $90°$, whose tangent is $-(\sqrt{3}/3)$.

We know that $y = -30°$

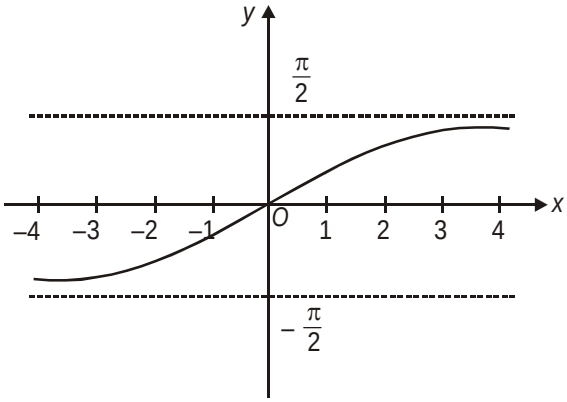
3. GRAPHS OF BASIC INVERSE TRIGONOMETRIC FUNCTIONS



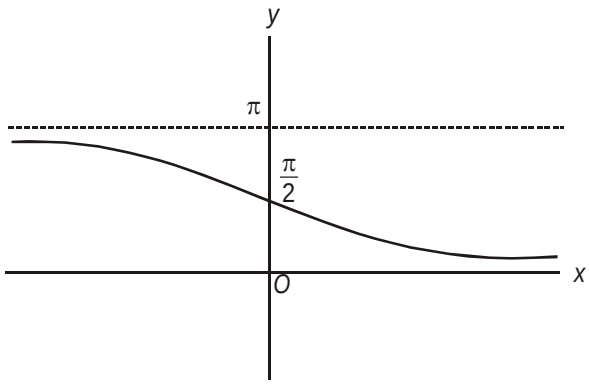
Graph of $y = \sin^{-1} x$



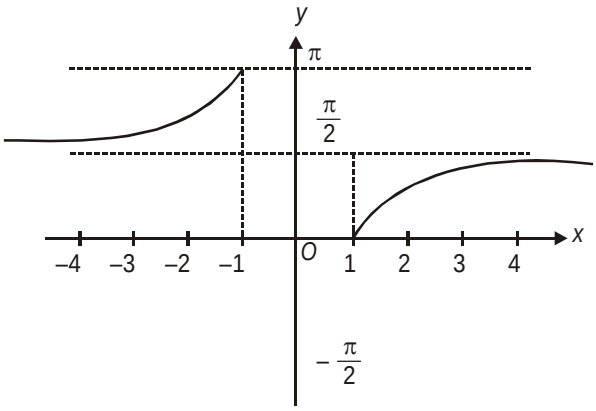
Graph of $y = \cos^{-1} x$



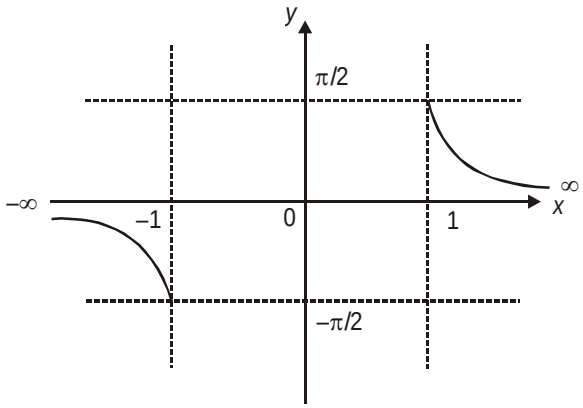
Graph of $y = \tan^{-1} x$



Graph of $y = \cot^{-1} x$



Graph of $y = \sec^{-1} x$



Graph of $y = \operatorname{cosec}^{-1} x$

DOMAIN AND RANGE OF INVERSE TRIGONOMETRIC FUNCTIONS :

Function	Domain	Range
$y = \sin^{-1} x$	$[-1, 1]$	$\left[-\frac{\pi}{2}, \frac{\pi}{2}\right]$
$y = \cos^{-1} x$	$[-1, 1]$	$[0, \pi]$
$y = \tan^{-1} x$	$(-\infty, \infty)$	$\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$
$y = \cot^{-1} x$	$(-\infty, \infty)$	$(0, \pi)$
$y = \sec^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$\left[0, \frac{\pi}{2}\right) \cup \left(\frac{\pi}{2}, \pi\right]$
$y = \operatorname{cosec}^{-1} x$	$(-\infty, -1] \cup [1, \infty)$	$\left[-\frac{\pi}{2}, 0\right) \cup \left(0, \frac{\pi}{2}\right]$

(a) 1st quadrant is common to the range of all the inverse functions.

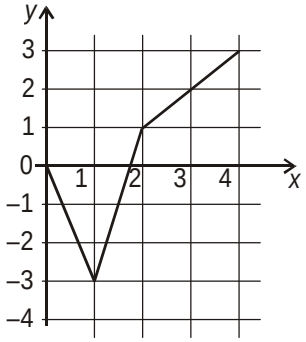
(b) 3rd quadrant is not used in inverse functions

(c) 4th quadrant is used in the clockwise direction i.e., $-\frac{\pi}{2} \le y \le 0$

(d) No inverse function is periodic.

Student Task

Q.1 Given is a partial graph of an even periodic function *f* whose period is 9. If [ ] = G.I.F. then the value of the expression.



$f(-3) + 2|f(-1)| + \left[f\left(\frac{7}{8}\right) \right] + f(0) + \text{arc cos}(f(-2)) + f(-7) + f(20)$

Q.2 Identify the pair(s) of functions which are identical. Also plot the graphs in each case.

(a) $y = \tan(\cos^{-1} x)$; $y = \frac{\sqrt{1-x^2}}{x}$ (b) $y = \tan(\cot^{-1} x)$; $y = \frac{1}{x}$

(c) $y = \sin(\text{arc tan } x)$; $y = \frac{x}{\sqrt{1+x^2}}$ (d) $y = \cos(\text{arc tan } x)$; $y = \sin(\text{arc cot } x)$

Q.3 Find the domain and range of the following functions. (Read the symbols [ ] = G.I.F., { } = F.P.F.)

(i) $f(x) = \cot^{-1}(2x - x^2)$ (ii) $f(x) = \sec^{-1}(\log_3 \tan x + \log_{\tan x} 3)$

(iii) $f(x) = \cos^{-1}\left(\frac{\sqrt{2x^2+1}}{x^2+1}\right)$ (iv) $f(x) = \tan^{-1}\left(\log_{4/5}\left(5x^2 - 8x + 4\right)\right)$

ANSWERS

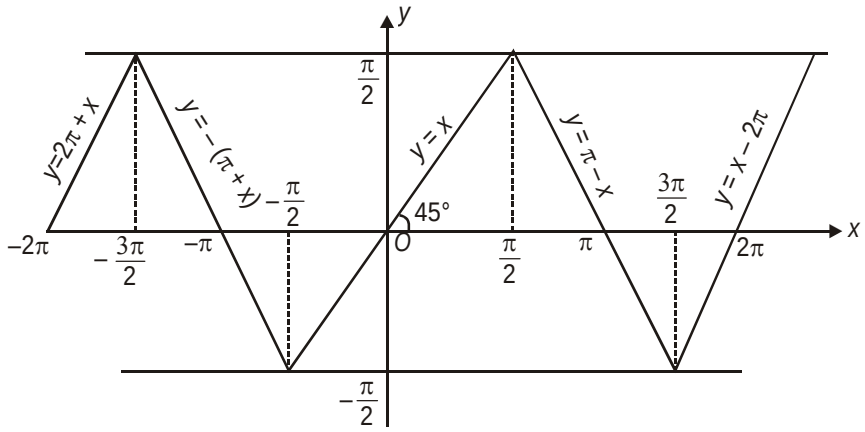
1. 8
2. (a), (b), (c) and (d) all are identical.
3. (i) $x \in R$; $R : \left[\frac{\pi}{4}, \pi\right)$ (ii) $D : x \in \left(n\pi, n\pi + \frac{\pi}{2}\right) - \left\{x \mid x = n\pi + \frac{\pi}{4}\right\}$ $R : \left[\frac{\pi}{3}, \frac{2\pi}{3}\right] = \left[\frac{\pi}{2}\right]$

(iii) $D : x \in R$ $R : \left[0, \frac{\pi}{2}\right)$, (iv) $D : x \in R$ $R : \left(-\frac{\pi}{2}, \frac{\pi}{4}\right]$

4. PROPERTIES OF INVERSE TRIGONOMETRIC FUNCTIONS

PROPERTY 1

(i) $\sin^{-1}(\sin x) = x$, provided that $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$

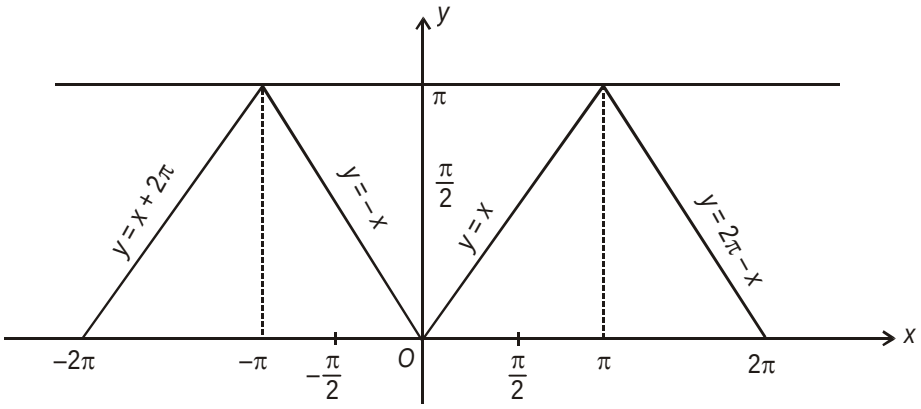


Let sin $x = y$. Then, for any θ satisfying $-\frac{\pi}{2} \le x \le \frac{\pi}{2}$

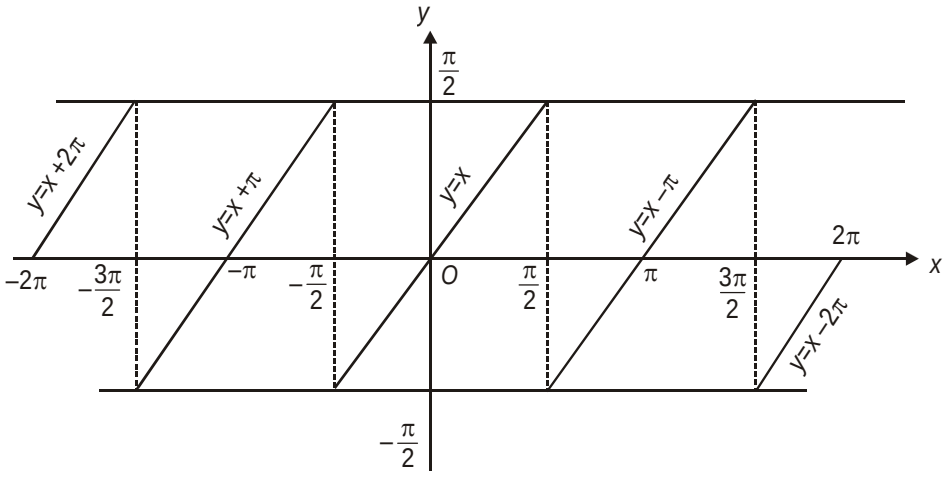
$\sin x = y \Rightarrow x = \sin^{-1} y$

$x = \sin^{-1} (\sin x)$ $[\because y = \sin x]$

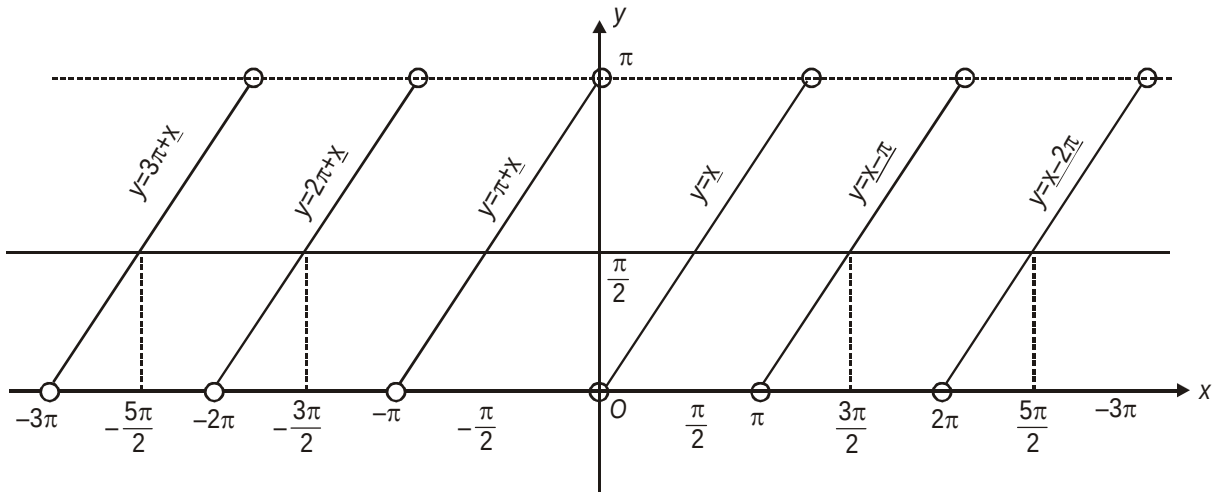
(ii) $\cos^{-1} (\cos x) = x$, provided that $0 \le x \le \pi$



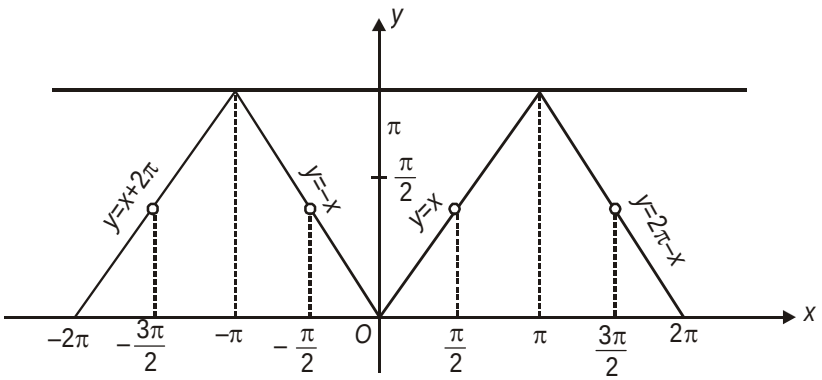
(iii) $\tan^{-1} (\tan x) = x$, provided that $-\frac{\pi}{2} < x < \frac{\pi}{2}$



(iv) $\cot^{-1}(\cot x) = x$, provided that $0 < x < \pi$



(v) $\sec^{-1}(\sec x) = x$, provided that $0 \leq x < \frac{\pi}{2}$ or $\frac{\pi}{2} < x \leq \pi$



(vi) $\text{cosec}^{-1}(\text{cosec } x) = x$, provided that $-\frac{\pi}{2} \leq x < 0$ or $0 < x \leq \frac{\pi}{2}$

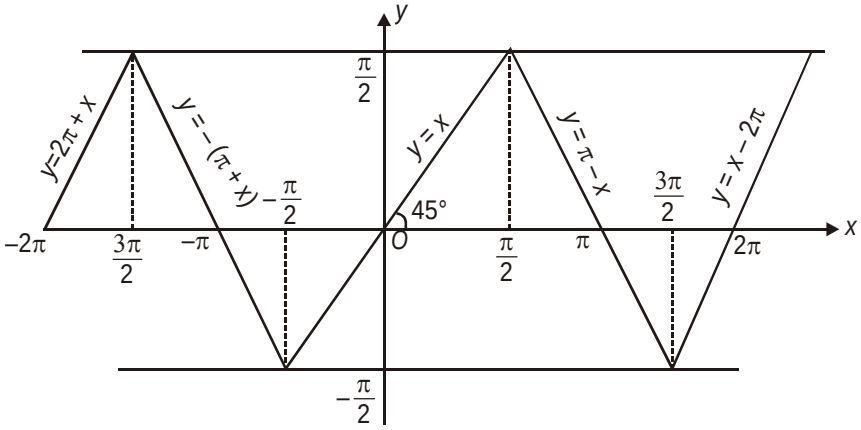


Illustration 8 : Find the value of $\cos^{-1}\cos\left(\frac{7\pi}{6}\right)$

Solution : $\cos^{-1}\left(\cos\frac{7\pi}{6}\right) \neq \frac{7\pi}{6}$ [because $\frac{7\pi}{6}$ does not lie between 0 and π]

Now, $\cos^{-1}\left(\cos\frac{7\pi}{6}\right) = \cos^{-1}\left[\cos\left(2\pi - \frac{5\pi}{6}\right)\right]$ $\left(\because \frac{7\pi}{6} = 2\pi - \frac{5\pi}{6}\right)$

$= \cos^{-1}\left(\cos\frac{5\pi}{6}\right) = \frac{5\pi}{6}$ $[\because \cos(2\pi - \theta) = \cos\theta]$

Illustration 9 : The value of the expression

$$\sin^{-1}\left(\sin\frac{22\pi}{7}\right)+\cos^{-1}\left(\cos\frac{5\pi}{3}\right)+\tan^{-1}\left(\tan\frac{5\pi}{7}\right)+\sin^{-1}(\cos 2) \text{ is –}$$

(a) $\frac{17\pi}{42}-2$ (b) –2 (c) $\frac{-\pi}{21}-2$ (d) None of these

Solution : (a) $\sin^{-1}\sin\left(\frac{22\pi}{7}\right)=\sin^{-1}\sin\left(3\pi+\frac{\pi}{7}\right)=-\frac{\pi}{7}$,

$$\cos^{-1}\cos\left(\frac{5\pi}{3}\right)=\cos^{-1}\cos\left(2\pi-\frac{\pi}{3}\right)=\frac{\pi}{3}$$

$\tan^{-1}\tan\left(\frac{5\pi}{7}\right)=\tan^{-1}\tan\left(\frac{5\pi}{7}-\pi\right)=-\frac{2\pi}{7}$, $\sin^{-1}\cos(2)=\frac{\pi}{2}-\cos^{-1}\cos 2=\frac{\pi}{2}-2$

$\therefore$ Required value = $-\frac{\pi}{7}+\frac{\pi}{3}-\frac{2\pi}{7}+\frac{\pi}{2}-2 = \frac{(-18+35)\pi}{42}-2=\frac{17\pi}{42}-2$

Illustration 10 : $\cos^{-1} x$ is equal to –

(a) $\sin^{-1}\sqrt{1-x^2}$ if – 1 < x < 1 (b) $-\sin^{-1}\sqrt{1-x^2}$ if –1 < x < 0

(c) $\sin^{-1}\sqrt{1-x^2}$ if – 1 < x < 0 (d) $\sin^{-1}\sqrt{1-x^2}$ if 0 < x < 1

Solution : (d) Let $\cos^{-1} x = \theta$ then $x = \cos\theta$ and $0 \le \theta \le \pi$

$$\therefore \quad \sin^{-1}\sqrt{1-x^2} = \sin(\sin\theta) = \begin{cases} \theta & \text{if } 0\le\theta\le\frac{\pi}{2} \\ \pi-\theta & \text{if } \frac{\pi}{2}<\theta\le\pi \end{cases}$$

$$= \begin{cases} \cos^{-1} x & \text{if } 0\le x\le 1 \\ \pi-\cos^{-1} x & \text{if } -1\le x<0 \end{cases}$$

$\therefore$ $\cos^{-1} x = \sin^{-1}\sqrt{1-x^2}$ if 0 < x < 1 is true.

Illustration 11 : $\cos^{-1} x$ is equal to –

(a) $-\tan^{-1}\frac{\sqrt{1-x^2}}{x}$ if – 1 < x < 0 (b) $\tan^{-1}\frac{\sqrt{1-x^2}}{x}$ if –1 < x < 0

(c) $-\tan^{-1}\frac{\sqrt{1-x^2}}{x}$ if 0 < x < 1 (d) $\pi+\tan^{-1}\frac{\sqrt{1-x^2}}{x}$ if –1 < x < 0

Solution : (d) Let $\cos^{-1} x = \theta$ then $x = \cos\theta$ and $0 \le \theta \le \pi$

$$\therefore \quad \tan^{-1}\frac{\sqrt{1-x^2}}{x}=\tan^{-1}(\tan\theta) = \begin{cases} \theta & , 0\le\theta\le\frac{\pi}{2} \\ \theta-\pi & , \frac{\pi}{2}<\theta\le\pi \end{cases} =$$

$$\begin{cases} \cos^{-1} x & , 0<x\le 1 \\ -\pi+\cos^{-1} x & , -1\le x<0 \end{cases}$$

i.e., $\cos^{-1} x=\pi+\tan^{-1}\frac{\sqrt{1-x^2}}{x}, -1<x<0$ is correct.

Student Task

Q.1 Solve : $\cos^{-1} x = \tan^{-1} \frac{\sqrt{1-x^2}}{x}$

Q.2 Solve : $\sec^{-1} x = \tan^{-1} \sqrt{x^2 - 1}$

Q.3 Solve : $\sin^{-1} x = \pi - \cot^{-1} \frac{\sqrt{1-x^2}}{x}$

Q.4 Solve : $\sec^{-1} x = \pi - \tan^{-1} \sqrt{x^2 - 1}$

Q.5 Solve : $\cos^{-1} \frac{1-x^2}{1+x^2} = 2\tan^{-1} x$

ANSWERS :

1. $x \in (0, 1]$ 2. $x \in [1, \infty)$ 3. $x \in [-1, 0)$
4. $x \in (-\infty, -1]$ 5. $x \in [0, \infty)$

PROPERTY 2

(i) $\sin(\sin^{-1} x) = x$, Provided that $-1 \leq x \leq 1$

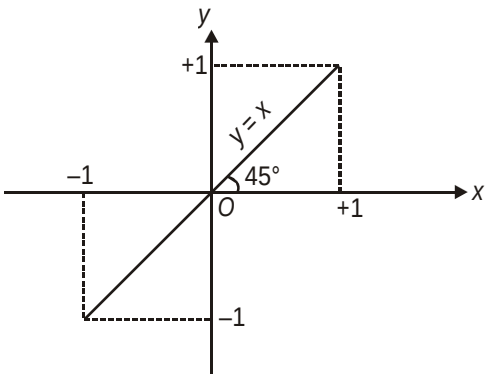
Let $\sin \theta = x$ ..... (1)

$\theta = \sin^{-1} x$ ..... (2)

substituting the value of θ in (1) from (2), $\sin(\sin^{-1} x) = x$,

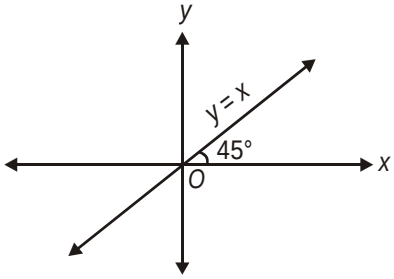
Here x is a number and will have no unit.

(ii) $\cos(\cos^{-1} x) = x$, Provided that $-1 \leq x \leq 1$



(iii) $\tan(\tan^{-1} x) = x$, Provided that $-\infty < x < \infty$

(iv) $\cot(\cot^{-1} x) = x$, Provided that $-\infty < x < \infty$



(v) $\sec(\sec^{-1} x) = x$,

Provided that $-\infty < x \leq 1$ or $1 \leq x < \infty$

(vi) $\operatorname{cosec}(\operatorname{cosec}^{-1} x) = x$,

Provided that $-\infty < x \leq -1$ or $1 \leq x < \infty$

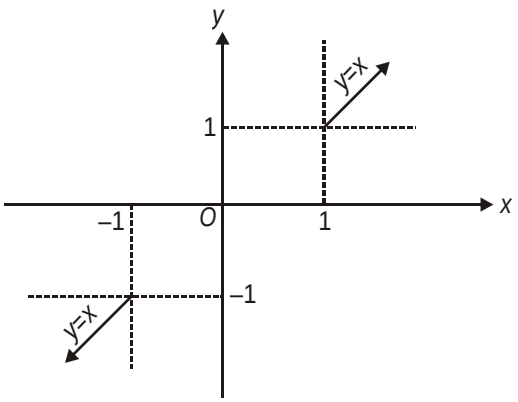


Illustration 12 : $\sin(\cot^{-1} x) = ?$

(a) $\sqrt{1+x^2}$ (b) x (c) $(1+x^2)^{-3/2}$ (d) $(1+x^2)^{-1/2}$

Solution : (d) Let $\cot^{-1} x = \theta \Rightarrow x = \cot\theta$

Now $\text{cosec}\,\theta = \sqrt{1+\cot^2\theta} = \sqrt{1+x^2}$

$$\therefore \quad \sin\theta = \frac{1}{\text{cosec}\,\theta} = \frac{1}{\sqrt{1+x^2}} \Rightarrow \theta = \sin^{-1} = \frac{1}{\sqrt{1+x^2}}$$

Hence $\sin(\cot^{-1} x) = \sin\left(\sin^{-1}\frac{1}{\sqrt{1+x^2}}\right) = \frac{1}{\sqrt{1+x^2}} = (1+x^2)^{-1/2}$

Illustration 13 : $\cot\left[\cos^{-1}\left(\frac{7}{25}\right)\right] =$

(a) 25/24 (b) 25/7 (c) 24/25 (d) None of these

Solution : (d) $\cot\left[\cos^{-1}\left(\frac{7}{25}\right)\right] = \cot\left[\cot^{-1}\left(\frac{7}{24}\right)\right] = \frac{7}{24}$

Student Task

Q.1 Solve : $\sin \sin^{-1} x = |x|$

Q.2 Solve : $\sin \sin^{-1} 2x = 2x + 2011$

Q.3 Solve : $\sin \sin^{-1} 2013\,x = \sin 2013\,x$

ANSWERS :

1. $x \in [0, 1]$ 2. $x = \phi$ 3. $x = 0$

PROPERTY : 3

(i) $\sin^{-1}(-x) = -\sin^{-1} x$ Provided that $-1 \leq x \leq 1$

Let $\sin^{-1}(-x) = \theta$

or $-x = \sin\theta. \Rightarrow x = -\sin\theta$ or $x = \sin(-\theta)$

or $-\theta = \sin^{-1} x$ or $\theta = -\sin^{-1} x$

$\sin^{-1}(-x) = -\sin^{-1} x$

(ii) $\cos^{-1}(-x) = \pi - \cos^{-1} x$ Provided that $-1 \leq x \leq 1$

Let $\cos^{-1}(-x) = q$

or $-x = \cos\theta \Rightarrow x = -\cos\theta$

or $x = \sin(\pi - \theta)$ or $\cos^{-1} x = \pi - \theta$ or $\cos^{-1} x = \pi - \cos^{-1}(-x)$
or $\cos^{-1}(-x) = \pi - \cos^{-1} x$

(iii) $\tan^{-1}(-x) = -\tan^{-1} x$ Provided that $-\infty < x < \infty$
(iv) $\cot^{-1}(-x) = \pi - \cot^{-1} x$ Provided that $-\infty < x < \infty$
(v) $\sec^{-1}(-x) = \pi - \sec^{-1} x$ Provided that $-\infty < x \leq 1$ or $1 \leq x < \infty$
(vi) $\operatorname{cosec}^{-1}(-x) = -\operatorname{cosec}^{-1} x$ Provided that $-\infty < x \leq -1$ or $1 \leq x < \infty$

Hence $\sin(\cot^{-1} x) = \sin\left(\sin^{-1}\frac{1}{\sqrt{1+x^2}}\right) = \frac{1}{\sqrt{1+x^2}} = (1+x^2)^{-1/2}$

Illustration 14 : Find the value of $\cos\left[\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)+\frac{\pi}{6}\right]$.

Solution : $\cos\left[\cos^{-1}\left(-\frac{\sqrt{3}}{2}\right)+\frac{\pi}{6}\right] = \cos\left[\pi-\cos^{-1}\left(\frac{\sqrt{3}}{2}\right)+\frac{\pi}{6}\right]$

$[\because \cos^{-1}(-x) = \pi - \cos^{-1} x]$

$= \cos\left(\pi - \frac{\pi}{6} + \frac{\pi}{6}\right) = \cos\pi = -1$

Illustration 15 : Find the value of $\cos^{-1}(-1)$
Solution : $\cos^{-1}(-1) = \pi - \cos^{-1}(1) = \pi - 0 = \pi$

PROPERTY 4

(i) $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$ $x \in [-1,1]$

Let $\sin^{-1} x = \theta \Rightarrow x = \sin\theta \Rightarrow x = \cos\left(\frac{\pi}{2} - \theta\right)$

or $\cos^{-1} x = \frac{\pi}{2} - \theta$ or $\cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x$ or $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$

(ii) $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$ $x \in R$

(iii) $\sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}$ $x \in (-\infty, -1] \cup [1, \infty]$

Illustration 16 : Find the value of $\sin\{\sin^{-1}\frac{1}{2} + \cos^{-1}\frac{1}{2}\}$

Solution : $\sin\{\sin^{-1}\frac{1}{2} + \cos^{-1}\frac{1}{2}\} = \sin\left(\frac{\pi}{2}\right) = 1$ $\left(\because \sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}\right)$

Illustration 17 : If $\sin^{-1} x - \cos^{-1} x = \frac{\pi}{6}$, then find the value of x

Solution : $\sin^{-1} x - \cos^{-1} x = \frac{\pi}{2} \Rightarrow \frac{\pi}{2} - \cos^{-1} x - \cos^{-1} x = \frac{\pi}{6}$

$\Rightarrow 2\cos^{-1}x = \frac{\pi}{2}-\frac{\pi}{6}=\frac{2\pi}{6}=\frac{\pi}{3} \Rightarrow \cos^{-1}x=\frac{\pi}{6}$ $\quad \therefore x = \cos\frac{\pi}{6}=\frac{\sqrt{3}}{2}$

Illustration 18 : If $\sin^{-1}x+\sin^{-1}y=\frac{2\pi}{3}$, then find the value of $\sin^{-1}x - \cos^{-1}y$

Solution : We have, $\sin^{-1}x+\sin^{-1}y=\frac{2\pi}{3} \Rightarrow \left(\frac{\pi}{2}-\cos^{-1}x\right)+\left(\frac{\pi}{2}-\cos^{-1}y\right)=\frac{2\pi}{3}$

$\Rightarrow \pi-(\cos^{-1}x+\cos^{-1}y)=\frac{2\pi}{3} \Rightarrow \cos^{-1}x+\cos^{-1}y=\pi-\frac{2\pi}{3}=\frac{\pi}{3}$

Illustration 19 : If $(\sin^{-1}x)^2 + (\cos^{-1}x)^2 = \frac{5\pi^2}{8}$, then find the value of x.

Solution : We have, $(\sin^{-1}x)^2 + (\cos^{-1}x)^2 = \frac{5\pi^2}{8}$

$\Rightarrow \quad (\sin^{-1}x)^2+\left(\frac{\pi}{2}-\sin^{-1}x\right)^2=\frac{5\pi^2}{8} \Rightarrow 2(\sin^{-1}x)^2=\pi\sin^{-1}x-\frac{3\pi^2}{8}=0$

$\Rightarrow \quad \sin^{-1}x=\frac{\pi\pm\sqrt{4\pi^2}}{4} \Rightarrow \sin^{-1}x=\frac{3\pi}{4} \text{ or } -\frac{\pi}{4}$

$\Rightarrow \quad x=\sin\frac{3\pi}{4} \text{ or } \sin\left(-\frac{\pi}{4}\right) \quad \therefore \quad x=\frac{1}{\sqrt{2}} \text{ or } -\frac{1}{\sqrt{2}}$

Student Task

Q.1 Consider the equation $(\sin^{-1}x)^3 + (\cos^{-1}x)^3 = 2p\pi^3$. Find the value(s) of parameter 'p' so that the given equation has the real solution(s).

Q.2 Find the greatest value of n if $\cot^{-1}\frac{n}{\pi}>\frac{\pi}{6}$, $n\in N$

Q.3 If $\cot^{-1}\left(\frac{n^2-10n+21.6}{\pi}\right)>\frac{\pi}{6}$, $n \in N$, then find the least value of n.

ANSWERS :

1. $p\in\left[\frac{1}{64},\frac{7}{16}\right]$ **2. 5** **3. 3**

PROPERTY 5 : Conversion Property

(i) $\sin^{-1}x = \text{cosec}^{-1}\left(\frac{1}{x}\right)$; $-1 \le x \le 1$ and $\text{cosec}^{-1}x = \sin^{-1}\left(\frac{1}{x}\right)$, $x \in R - (-1, 1)$

Let $\sin^{-1}x = y \Rightarrow x = \sin y \Rightarrow \text{cosec } y = \left(\frac{1}{x}\right) \Rightarrow y = \text{cosec}^{-1}\left(\frac{1}{x}\right)$

$\Rightarrow \quad \sin^{-1}x = \text{cosec}^{-1}\left(\frac{1}{x}\right)$

(ii) $\cos^{-1} x = \sec^{-1}\left(\frac{1}{x}\right)$ $-1 \le x \le 1$ and $\sec^{-1} x = \cos^{-1}\left(\frac{1}{x}\right), x \in R - (-1, 1)$

(iii) $\tan^{-1} x = \begin{cases} \cot^{-1}\left(\frac{1}{x}\right), & x > 0 \\ -\pi + \cot^{-1}\left(\frac{1}{x}\right), & x < 0 \end{cases}$

Illustration 20 : If $\sin^{-1} x - \cos^{-1} x = \frac{\pi}{6}$, then find the value of x.

Solution : $\sin^{-1} x - \cos^{-1} x = \frac{\pi}{6} \Rightarrow \frac{\pi}{2} - \cos^{-1} x - \cos^{-1} x = \frac{\pi}{6}$

$\Rightarrow \quad 2\cos^{-1} x = \frac{\pi}{2} - \frac{\pi}{6} = \frac{2\pi}{6} = \frac{\pi}{3} \Rightarrow \cos^{-1} x = \frac{\pi}{6} \quad \therefore x = \cos\frac{\pi}{6} = \frac{\sqrt{3}}{2}$

PROPERTY 6 :

(i) $\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$, if $x > 0, y = 0$ and $xy < 1$, $\frac{x+y}{1-xy}$ is positive and therefore $\tan^{-1}\left(\frac{x+y}{1-xy}\right)$ is between 0° and 90°.

If however, $xy > 1$ then $\frac{x+y}{1-xy}$ is negative therefore according to our definition $\tan^{-1}\left(\frac{x+y}{1-xy}\right)$ is a negative angle. We know that $\tan(\pi + \gamma) = \tan\gamma$, the formula should be $\tan^{-1} x + \tan^{-1} y = \pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right)$

Similarly, if $x < 0, y < 0$ and $xy > 1$ then formula should be

$\tan^{-1} x + \tan^{-1} y = -\pi + \tan^{-1}\left(\frac{x+y}{1-xy}\right)$

Let $\tan^{-1} x = \theta_1$ and $\tan^{-1} y = \theta_2$ ..... (1)
then, $\tan\theta_1 = x$ and $\tan\theta_2 = y$

so, $\tan(\theta_1 + \theta_2) = \frac{\tan\theta_1 + \tan\theta_2}{1 - \tan\theta_1 . \tan\theta_2}$

or $\tan(\theta_1 + \theta_2) = \frac{x+y}{1-xy}$ or $\theta_1 + \theta_2 = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$

or $\tan^{-1} x + \tan^{-1} y = \tan^{-1}\left(\frac{x+y}{1-xy}\right)$ [By (1)]

(ii) $\tan^{-1} x - \tan^{-1} y = \begin{cases} \tan^{-1}\left(\frac{x-y}{1+xy}\right), & \text{if } xy > -1 \\ \pi + \tan^{-1}\left(\frac{x-y}{1+xy}\right), & \text{if } x > 0, y < 0 \text{ and } xy < -1 \\ -\pi + \tan^{-1}\left(\frac{x-y}{1+xy}\right), & \text{if } x < 0, y > 0 \text{ and } xy < -1 \end{cases}$

(iii) $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \tan^{-1}\left[\frac{x+y+z-xyz}{1-xy-yz-zx}\right]$

If $x > 0, y > 0, z > 0$ and $xy + yz + zx < 1$

(a) If $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \pi$ then $x + y + z = xyz$

(b) If $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z = \frac{\pi}{2}$ then $xy + yz + zx = 1$

Note :- If $x_1, x_2, x_3, \ldots\ldots x_n \in R$, then $\tan^{-1} x_1 + \tan^{-1} x_2 + \ldots + \tan^{-1} x_n = \tan^{-1}\left(\frac{S_1 - S_2 + S_5 - S_7 + \ldots}{1 - S_2 + S_4 - S_6 + \ldots}\right)$

where S_k denotes the sum of the products of $x_1, x_2, \ldots x_n$ taken k at a time.

Illustration 21 : Prove that $2\tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{2} + 2\tan^{-1}\frac{1}{8} = \frac{\pi}{4}$

Solution : $2\tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{2} + 2\tan^{-1}\frac{1}{8} = 2\tan^{-1}\frac{\frac{1}{5}+\frac{1}{8}}{1-\frac{1}{5}.\frac{1}{8}} + \tan^{-1}\frac{1}{7}$

$= 2\tan^{-1}\frac{1}{3} + \tan^{-1}\frac{1}{7} = \tan^{-1}\frac{2.\frac{1}{3}}{1-\frac{1}{9}} + \tan^{-1}\frac{1}{7}$

$= \tan^{-1}\frac{3}{4} + \tan^{-1}\frac{1}{7} = \tan^{-1}\frac{\frac{3}{4}+\frac{1}{7}}{1-\frac{3}{4}.\frac{1}{7}} = \tan^{-1}\frac{25}{25} = \tan^{-1} 1 = 45° = \frac{\pi}{4}$

Illustration 22 : Prove that

$$\tan^{-1}\sqrt{\left\{\frac{a(a+b+c)}{bc}\right\}} + \tan^{-1}\sqrt{\left\{\frac{b(a+b+c)}{ac}\right\}} + \tan^{-1}\sqrt{\left\{\frac{c(a+b+c)}{ab}\right\}} = \pi$$

Solution : L.H.S. $= \tan^{-1}\frac{\sqrt{\left\{\frac{a(a+b+c)}{bc}\right\}} + \left\{\frac{b(a+b+c)}{ac}\right\}}{1-\sqrt{\left\{\frac{a(a+b+c)}{bc}\right\}}.\left\{\frac{b(a+b+c)}{ac}\right\}} + \tan^{-1}\sqrt{\left\{\frac{c(a+b+c)}{ab}\right\}}$

$= \tan^{-1}\frac{\sqrt{\{(a+b+c)\}}}{-\sqrt{\{ab\}}(a+b)} + \tan^{-1}\sqrt{\left\{\frac{c(a+b+c)}{ab}\right\}}$ =

$\tan^{-1}\left[\sqrt{\frac{(a+b+c)c}{ab}}\right] + \tan^{-1}\left[-\sqrt{\left[\frac{c(a+b+c)}{ab}\right]}\right]$

$$= \pi - \tan^{-1}\left[\sqrt{\frac{c(a+b+c)}{ab}}\right] + \tan^{-1}\left[-\sqrt{\left[\frac{c(a+b+c)}{ab}\right]}\right] = \pi$$

Illustration 23

To solve : $\tan^{-1} 2x + \tan^{-1} 3x = \frac{\pi}{4}$

Solution : $\tan^{-1}\frac{2x+3x}{1-6x^2} = \frac{\pi}{4}$ or $\frac{5x}{1-6x^2} = \tan\frac{\pi}{4} = 1$

or $6x^2 + 5x - 1 = 0$ or $(6x - 1)(x + 1) = 0$

$\therefore$ $x = \frac{1}{6}$ and -1. But -1 inadmissible, hence $x = \frac{1}{6}$

Illustration 24 : To solve : $\tan^{-1}(x+1) + \tan^{-1}(x-1) = \tan^{-1}\frac{8}{31}$

Solution : $\tan^{-1}\frac{x+1+x-1}{1-(x+1)(x-1)} = \tan^{-1}\frac{8}{31}$ or $\frac{2x}{1-(x^2-1)} = \frac{8}{31}$

or $4(2 - x^2) = 31x$ or $4x^2 + 31x - 8 = 0$ or $(4x - 1)(x + 8) = 0$

$\therefore$ $x = \frac{1}{4}$ and $x = -8$

Illustration 25 : If $\tan^{-1} 2$ and $\tan^{-1} 3$ be two angles of a triangle, then find the third angle.

Solution : $A + B = \tan^{-1} 2 + \tan^{-1} 3 = \tan^{-1}\frac{2+3}{1-2.3} + \pi$ ($\because ab > 11$)

$= \pi + \tan^{-1}(-1) = \pi - \frac{\pi}{4}$

Hence the third angle $= \pi - (A + B) = \pi - \left(\pi - \frac{\pi}{4}\right) = \frac{\pi}{4}$

Illustration 26 : Solve the equation $\tan^{-1}\frac{x+1}{x-1} + \tan^{-1}\frac{x-1}{x} = \tan^{-1}(-7)$

Solution : Taking the tangents of both sides of the equation, we have

$$\frac{\tan\left[\tan^{-1}\frac{x+1}{x-1}\right] + \tan\left[\tan^{-1}\frac{x-1}{x}\right]}{1 - \tan\left[\tan^{-1}\frac{x+1}{x-1}\right]\tan\left[\tan^{-1}\frac{x-1}{x}\right]} = \tan\{\tan^{-1}(-7)\} = -7$$

i.e., $\frac{\frac{x+1}{x-1} + \frac{x-1}{x}}{1 - \frac{x+1}{x-1}\frac{x-1}{x}} = -7$ i.e., $\frac{2x^2 - x + 1}{1 - x} = -7$ so that $x = 2$

This value makes the left-hand side of the given equation positive, so that there is no value of x strictly satisfying the given equation.
The value x = 2 is a solution of the equation

$$\tan^{-1}\frac{x+1}{x-1} + \tan^{-1}\frac{x-1}{x} = \pi + \tan^{-1}(-7)$$

PROPERTY-7 :

(i) $\sin^{-1} x + \sin^{-1} y = \begin{cases} \sin^{-1}\{x\sqrt{1-y^2} + y\sqrt{1-x^2}\}, & \text{if } -1 \le x, y \le 1 \text{ and } x^2+y^2 \le 1 \\ & \text{or if } xy < 0 \text{ and } x^2+y^2 > 1 \\ \pi - \sin^{-1}\{x\sqrt{1-y^2} + y\sqrt{1-x^2}\}, & \text{if } 0 < x, y \le 1 \text{ and } x^2+y^2 > 1 \\ -\pi - \sin^{-1}\{x\sqrt{1-y^2} + y\sqrt{1-x^2}\}, & \text{if } -1 \le x, y < 0 \text{ and } x^2+y^2 > 1 \end{cases}$

(ii) $\sin^{-1} x - \sin^{-1} y = \begin{cases} \sin^{-1}\{x\sqrt{1-y^2} - y\sqrt{1-x^2}\}, & \text{if } -1 \le x, y \le 1 \text{ and } x^2+y^2 \le 1 \\ & \text{or if } xy > 0 \text{ and } x^2+y^2 > 1 \\ \pi - \sin^{-1}\{x\sqrt{1-y^2} - y\sqrt{1-x^2}\}, & \text{if } 0 < x \le 1, -1 \le y \le 0 \text{ and } x^2+y^2 > 1 \\ -\pi - \sin^{-1}\{x\sqrt{1-y^2} - y\sqrt{1-x^2}\}, & \text{if } -1 \le x < 0, 0 < y \le 1 \text{ and } x^2+y^2 > 1 \end{cases}$

Let $\sin^{-1} x = \theta_1$ and $\sin^{-1} y = \theta_2$ ..... (1)
then $\sin\theta_1 = x$ and $\sin\theta_2 = y$
so $\sin(\theta_1 \pm \theta_2) = \sin\theta_1 . \cos\theta_2 \pm \cos\theta_1 . \sin\theta_2$

or $\sin(\theta_1 \pm \theta_2) = x\sqrt{1-y^2} \pm y\sqrt{1-x^2}$

or $\sin^{-1} x \pm \sin^{-1} y = \sin^{-1}[x\sqrt{1-y^2} \pm y\sqrt{1-x^2}]$

If $x > 0$, $y > 0$ and $x^2 + y^2 > 1$, then the $\sin^{-1} x + \sin^{-1} y$ may be an angle more than $\pi/2$ while

$\sin^{-1}(x\sqrt{1-y^2} + y\sqrt{1-x^2})$ is an angle between $-\frac{\pi}{2}$ and $\frac{\pi}{2}$

$\sin^{-1} x \pm \sin^{-1} y = \pi - \sin^{-1}(x\sqrt{1-y^2} \pm y\sqrt{1-x^2})$

Illustration 27 : If $\sin^{-1} 6x + \sin^{-1} 6\sqrt{3}\, x = -\frac{\pi}{2}$, then find the value of x

Solution : $\sin^{-1}\ 6\sqrt{3}\ x = -\frac{\pi}{2} - \sin^{-1} 6x = -(\sin^{-1} 1 + \sin^{-1} 6x) = -\sin^{-1}(1.\sqrt{1-(6x)^2} - 6x.\sqrt{1-1})$

$\Rightarrow \quad -\sin^{-1}\sqrt{1-36x^2} = \sin^{-1}(-\sqrt{1-36x^2})$

$\Rightarrow \quad 6\sqrt{3}x = -\sqrt{1-36x^2} \quad \Rightarrow 108\ x^2 = 1 - 36\ x^2 \Rightarrow 144\ x^2 = 1 \Rightarrow$

$x^2 = \frac{1}{144} \Rightarrow x = \pm\frac{1}{12}$

But only $x = -\frac{1}{12}$ satisfies the equation

Illustration 28 : Prove that $\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3} = \sin^{-1}\frac{1}{\sqrt{5}} + \cot^{-1} 3 = 45°$

Solution : $\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{3} = \tan^{-1}\frac{\frac{1}{2}+\frac{1}{3}}{1-\frac{1}{2}.\frac{1}{3}} = \tan^{-1} 1 = 45°$

Let $\cot^{-1} 3 = \theta$ or $\cot\theta = 3$

$\therefore \quad \sin\theta = \frac{1}{\sqrt{10}} \quad \therefore \quad \theta = \sin^{-1}\left(\frac{1}{\sqrt{10}}\right)$

$\therefore \quad \sin^{-1}\frac{1}{\sqrt{5}} + \cot^{-1} 3 = \sin^{-1}\frac{1}{\sqrt{5}} + \sin^{-1}\frac{1}{\sqrt{10}}$

$= \sin^{-1}\left\{\frac{1}{\sqrt{5}}.\frac{3}{\sqrt{10}} + \frac{1}{\sqrt{10}}.\frac{2}{\sqrt{5}}\right\} = \sin^{-1}\frac{5}{\sqrt{10}} = \sin^{-1}\frac{1}{\sqrt{2}} = 45°$

PROPERTY 8 :

(i) $\cos^{-1} x + \cos^{-1} y = \begin{cases} \cos^{-1}\{xy - \sqrt{1-x^2}\sqrt{1-y^2}\}, & \text{if } -1 \le x, y \le 1 \text{ and } x + y \ge 0 \\ 2\pi - \cos^{-1}\{xy - \sqrt{1-x^2}\sqrt{1-y^2}\}, & \text{if } -1 \le x, y \le 1 \text{ and } x + y \le 0 \end{cases}$

(ii) $\cos^{-1} x - \cos^{-1} y = \begin{cases} \cos^{-1}\{xy + \sqrt{1-x^2}\sqrt{1-y^2}\}, & \text{if } -1 \le x, y \le 1 \text{ and } x \le y \\ -\cos^{-1}\{xy + \sqrt{1-x^2}\sqrt{1-y^2}\}, & \text{if } -1 \le y \le 0,\ 0 < x \le 1 \text{ and } x \ge y \end{cases}$

Let $\cos^{-1} x = \theta_1$ and $\cos^{-1} y = \theta_2$
then $\cos\theta_1 = x$ and $\cos\theta_2 = y$ ..... (1)
so, $\cos(\theta_1 \pm \theta_2) = \cos\theta_1.\cos\theta_2 \mp \sin\theta_1.\sin\theta_2$

or $\cos(\theta_1 \pm \theta_2) = \cos\theta_1.\cos\theta_2 \mp \sqrt{1-\cos^2\theta_1}\sqrt{1-\cos^2\theta_2}$

or $\cos(\theta_1 \pm \theta_2) = (xy \mp \sqrt{1-x^2}\sqrt{1-y^2})$ [By (1)]

or $\theta_1 \pm \theta_2 = \cos^{-1}(xy \mp \sqrt{1-x^2}\sqrt{1-y^2})$

or $\cos^{-1} x_1 \pm \cos^{-1} x_2 = \cos^{-1}(xy \mp \sqrt{1-x^2}\sqrt{1-y^2})$

Illustration 29 : $\sec^{-1}\frac{x}{a} - \sec^{-1}\frac{x}{b} = \sec^{-1} b - \sec^{-1} a$, then find the value of x.

Solution : $\cos^{-1}\frac{a}{x} + \cos^{-1}\frac{1}{a} = \cos^{-1}\frac{1}{b} + \cos^{-1}\frac{b}{x}$

$\therefore \quad \frac{a}{x}.\frac{1}{a} - \sqrt{\left(1-\frac{a^2}{x^2}\right)}\sqrt{\left(1-\frac{1}{a^2}\right)} = \frac{1}{b}.\frac{b}{x} - \sqrt{\left(1-\frac{b^2}{x^2}\right)}\sqrt{\left(1-\frac{1}{b^2}\right)}$

or $b^2(x^2 - a^2)(a^2 - 1) = a^2(x^2 - b^2)(b^2 - 1)$
or $x^2(a^2 - b^2) = a^2 b^2 (a^2 - b^2) \quad \therefore \quad x = ab$

Illustration 30 : $\sin^{-1}\left(\frac{3}{5}\right) + \cos^{-1}\left(\frac{12}{13}\right) = \cos^{-1}\left(\frac{33}{65}\right)$

Solution : LHS = $\sin^{-1}\left(\frac{3}{5}\right) + \cos^{-1}\left(\frac{12}{13}\right)$ or $\cos^{-1}\left(\frac{4}{5}\right) + \cos^{-1}\left(\frac{12}{13}\right)$

[$\because \cos^{-1} x = \sin^{-1}\sqrt{1-x^2}$

or $\cos^{-1}\left[\frac{4}{5}\times\frac{12}{13}-\sqrt{1-\frac{16}{25}}\sqrt{1-\frac{155}{169}}\right]$ or $\cos^{-1}\left[\frac{4}{5}\times\frac{12}{13}-\sqrt{\frac{9}{25}}\sqrt{\frac{25}{169}}\right]$

or $\cos^{-1}\left[\frac{48}{65}-\frac{15}{65}\right]$ or $\cos^{-1}\left[\frac{33}{65}\right]$ = RHS

Student Task

Q.1 If $\sin^{-1}x + \sin^{-1}y + \sin^{-1}z = \pi$ and $x^4 + y^4 + z^4 + 4x^2y^2z^2 = \lambda(x^2y^2 + y^2z^2 + z^2x^2)$ then find the value of λ.

Q.2 Evaluate : $\tan^{-1}\sqrt{\frac{a}{bc}(a+b+c)} + \tan^{-1}\sqrt{\frac{b}{ac}(a+b+c)} + \tan^{-1}\sqrt{\frac{c}{ab}(a+b+c)}$

Q.3 Find the number of solution(s) of equation $\cos^{-1}\left(\sqrt{6}x\right)+\cos^{-1}\left(3\sqrt{3}x^2\right)=\pi/2$.

ANSWERS :

1. 2 2. π 3. 1

PROPERTY 9 :

(i) $2\sin^{-1}x = \begin{cases} \sin^{-1}(2x\sqrt{1-x^2}) & , \quad if\ -\frac{1}{\sqrt{2}} \le x \le \frac{1}{\sqrt{2}} \\ \pi-\sin^{-1}(2x\sqrt{1-x^2}) & , \quad if\ \frac{1}{\sqrt{2}} \le x \le 1 \\ -\pi-\sin^{-1}(2x\sqrt{1-x^2}) & , \quad if\ -1 \le x \le -\frac{1}{\sqrt{2}} \end{cases}$

Let $\sin^{-1}x = \theta$ then $\sin\theta = x$

so, $\sin 2\theta = 2\sin\theta\cos\theta$ or $\sin 2\theta = 2x\sqrt{1-x^2}$ $[\because \cos\theta = \sqrt{1-\sin^2\theta}]$

or $2\theta = \sin^{-1}(2x\sqrt{1-x^2})$ or $2\sin^{-1}x = \sin^{-1}(2x\sqrt{1-x^2})$

(ii) $3\sin^{-1}x = \begin{cases} \sin^{-1}(3x-4x^3), & if\ -\frac{1}{2} \le x \le \frac{1}{2} \\ \pi-\sin^{-1}(3x-4x^3), & if\ \frac{1}{2} < x \le 1 \\ -\pi-\sin^{-1}(3x-4x^3), & if\ -1 \le x < -\frac{1}{2} \end{cases}$

Let $\sin^{-1}x = \theta$ then $x = \sin\theta$

Now, $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$ or $\sin 3\theta = 3x - 4x^2$

or $3\theta = \sin^{-1}(3x - 4x^3)$ or $3\sin^{-1}x = \sin^{-1}(3x - 4x^3)$

(iii) $2\cos^{-1}x = \begin{cases} \cos^{-1}(2x^2-1) & , \quad if\ 0 \le x \le 1 \\ 2\pi-\cos^{-1}(2x^2-1), & if\ -1 \le x \le 0 \end{cases}$

Let $\cos^{-1}x = \theta$ then $x = \cos\theta$

so, $\cos 2\theta = 2\cos^2\theta - 1$ or $\cos 2\theta = 2x^2 - 1$ or $2\theta = \cos^{-1}(2x^2 - 1)$ or $2\cos^{-1}x = \cos^{-1}(2x^2 - 1)$

(iv) $3\cos^{-1}x = \begin{cases} \cos^{-1}(4x^3-3x) & , \quad if\ \frac{1}{2} \le x \le 1 \\ 2\pi-\cos^{-1}(4x^3-3x) & , \quad if\ -\frac{1}{2} \le x \le \frac{1}{2} \\ 2\pi+\cos^{-1}(4x^3-3x) & , \quad if\ -1 \le x - \frac{1}{2} \end{cases}$

Let $\cos^{-1} x = \theta$ then $x = \cos\theta$

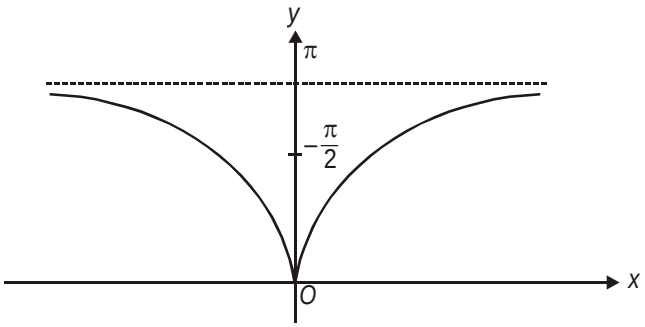
Now, $\cos 3\theta = 4\cos^3\theta - 3\cos\theta$ or $\cos 3\theta = 4x^3 - 3x$

or $3\theta = \cos^{-1}(4x^3 - 3x)$ or $3\cos^{-1} x = \cos^{-1}(4x^3 - 3x)$

(v) $2\tan^{-1} x = \begin{cases} \tan^{-1}\left(\dfrac{2x}{1-x^2}\right) = \sin^{-1}\left(\dfrac{2x}{1+x^2}\right), & \text{if } -1 < x < 1 \\ \pi + \tan^{-1}\left(\dfrac{2x}{1-x^2}\right) = \pi - \sin^{-1}\left(\dfrac{2x}{1+x^2}\right) & \text{if } x > 1 \\ -\pi + \tan^{-1}\left(\dfrac{2x}{1-x^2}\right) = -\pi - \sin^{-1}\left(\dfrac{2x}{1+x^2}\right) & \text{if } x < -1 \end{cases}$

$2\tan^{-1} x = \begin{cases} \cos^{-1}\left(\dfrac{1-x^2}{1+x^2}\right), & \text{if } 0 \le x < \infty \\ -\cos^{-1}\left(\dfrac{1-x^2}{1+x^2}\right), & \text{if } -\infty < x \le 0 \end{cases}$

Graph of $y = \cos^{-1}\dfrac{1-x^2}{1+x^2}$



Let $\tan^{-1} x = \theta$ then $x = \tan\theta$

Now, $\tan 2\theta = \dfrac{2\tan\theta}{1-\tan^2\theta} \Rightarrow \tan 2\theta = \dfrac{2x}{1-x^2}$ or $2\theta = \tan^{-1}\left(\dfrac{2x}{1-x^2}\right)$ or $2\tan^{-1} x = \tan\left(\dfrac{2x}{1-x^2}\right)$

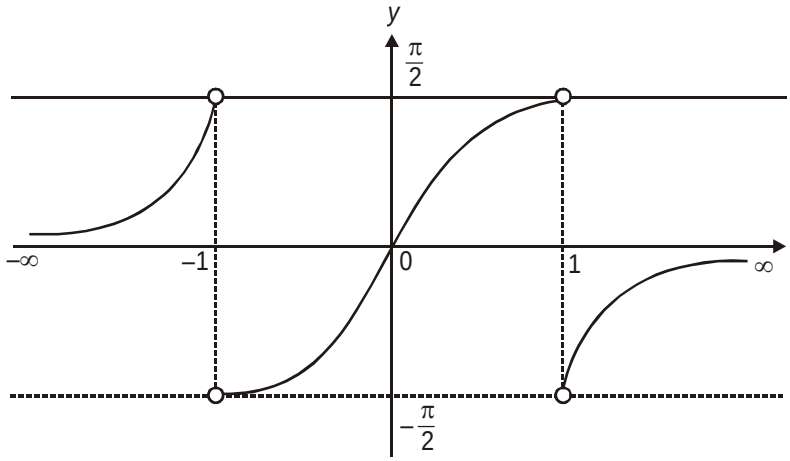
Again, $\sin 2\theta = \dfrac{2\tan\theta}{1+\tan^2\theta}$ or $\sin 2\theta = \dfrac{2x}{1+x^2} \Rightarrow 2\theta = \sin^{-1}\left(\dfrac{2x}{1+x^2}\right)$

$\Rightarrow 2\tan^{-1} x = \sin^{-1}\left(\dfrac{2x}{1+x^2}\right)$ and $\cos 2\theta = \dfrac{1-\tan^2\theta}{1+\tan^2\theta}$

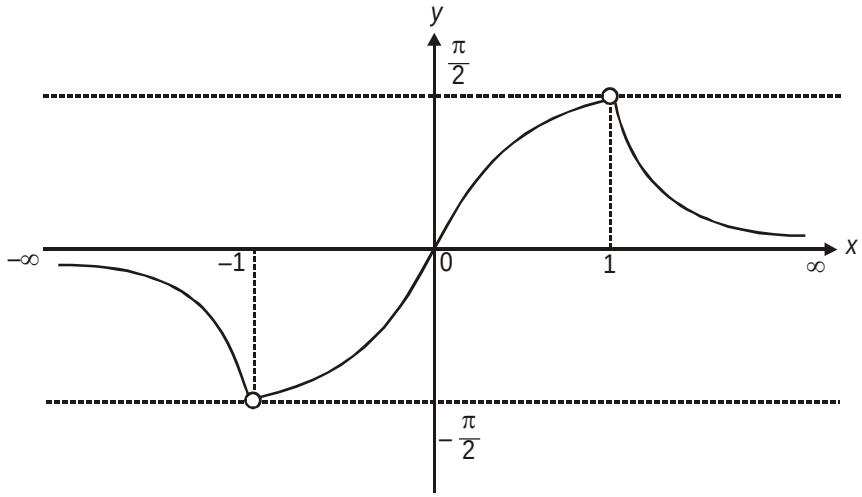
or $\cos 2\theta = \dfrac{1-x^2}{1+x^2}$ or $2\theta = \cos^{-1}\left(\dfrac{1-x^2}{1+x^2}\right)$

$2\tan^{-1} x = \cos^{-1}\left(\dfrac{1-x^2}{1+x^2}\right)$

Graph of $y = \tan^{-1}\dfrac{2x}{1-x^2}$



Graph of $y = \sin^{-1}\dfrac{2x}{1+x^2}$



(vi) $3\tan^{-1}x = \begin{cases} \tan^{-1}\left(\dfrac{3x-x^3}{1-3x^2}\right), & \text{if } -\dfrac{1}{\sqrt{3}} < x < \dfrac{1}{\sqrt{3}} \\ \pi + \tan^{-1}\left(\dfrac{3x-x^3}{1-3x^2}\right), & \text{if } x > \dfrac{1}{\sqrt{3}} \\ -\pi + \tan^{-1}\left(\dfrac{3x-x^3}{1-3x^2}\right), & \text{if } x < -\dfrac{1}{\sqrt{3}} \end{cases}$

Let $\tan^{-1}x = \theta$ then $x = \tan\theta$

Now, $\tan 3\theta = \dfrac{3\tan\theta - \tan^3\theta}{1-3\tan^2\theta}$ or $\tan 3\theta = \dfrac{3x-x^3}{1-3x^2}$ or $3\theta = \tan^{-1}\left[\dfrac{3x-x^3}{1-3x^2}\right]$

or $3\tan^{-1}x = \tan^{-1}\left(\dfrac{3x-x^3}{1-3x^2}\right)$

Illustration 31 : $\sin^{-1}\left(\dfrac{4x}{x^2+4}\right) + 2\tan^{-1}\left(-\dfrac{x}{2}\right)$ is independent of x then –

(a) $x \in [3, 4]$ (b) $x \in [-2, 2]$ (c) $x \in [-1, 1]$ (d) $x \in [1, \infty]$

Solution : (b) $\sin^{-1}\left(\frac{4x}{x^2+4}\right)+2\tan^{-1}\left(-\frac{x}{2}\right)$

$= \sin^{-1}\left(\frac{2.(x/2)}{(x/2)^2+1}\right)-2\tan^{-1}\frac{x}{2}=2\tan^{-1}\frac{x}{2}-2\tan^{-1}\frac{x}{2}=0$

Hence, $\left|\frac{x}{2}\right|\le 1$; $|x|\le 2 \Rightarrow -2\le x\le 2$

Illustration 32 : Prove that $3\tan^{-1}\frac{1}{4}+\tan^{-1}\frac{1}{20}=\frac{1}{4}\pi-\tan^{-1}\frac{1}{1985}$

Solution : Since $\tan 3\alpha=\frac{3\tan\alpha-\tan^3\alpha}{1-3\tan^2\alpha}$

$\therefore\ 3\tan^{-1}\frac{1}{4}=\tan^{-1}\left(\frac{3\left(\frac{1}{4}\right)-\left(\frac{1}{4}\right)^3}{1-3\left(\frac{1}{4}\right)^2}\right)$ or $3\tan^{-1}\frac{1}{4}=\tan^{-1}\frac{47}{52}$

$\therefore\ 3\tan^{-1}\frac{1}{4}+\tan^{-1}\frac{1}{20}=\tan^{-1}\frac{47}{52}+\tan^{-1}\frac{1}{20}\ =\ \tan^{-1}\frac{\frac{47}{52}+\frac{1}{20}}{1-\frac{47}{52}.\frac{1}{20}}=\tan^{-1}\frac{992}{993}$

and $\frac{1}{4}\pi-\tan^{-1}\frac{1}{1985}=\tan^{-1}\frac{1-\frac{1}{1985}}{1+\frac{1}{1985}}=\tan^{-1}\frac{1984}{1986}=\tan^{-1}\frac{992}{993}$

Illustration 33 : To solve $2\tan^{-1}(\cos x)=\tan^{-1}(2\operatorname{cosec} x)$

Solution : $\tan^{-1}\frac{2\cos x}{1-\cos^2 x}=\tan^{-1}(2\operatorname{cosec} x)$ or $\frac{2\cos x}{\sin x}=\frac{2}{\sin x}$ or $\sin x(\sin x-\cos x)=0$

Hence, either $\sin x=0 \Rightarrow x=n\pi$.

or $\sin x-\cos x=0 \Rightarrow x=\frac{\pi}{4}=n\pi+\frac{\pi}{4}$

Illustration 34 : Find the value of x for $\sin^{-1}\left(\frac{2a}{1+a^2}\right)-\cos^{-1}\left(\frac{1-b^2}{1+b^2}\right)=\tan^{-1}\left(\frac{2x}{1-x^2}\right)$.

Solution : Put $a=\tan\theta$ $\therefore \frac{2a}{1+a^2}=\frac{2\tan\theta}{1+\tan^2\theta}=\sin 2\theta$ $\therefore \sin^{-1}\left(\frac{2a}{1+a^2}\right)=\sin^{-1}(\sin 2\theta)\Rightarrow 2\theta$

$=2\tan^{-1}a$

Put $b=\tan\phi$

$\therefore\ \frac{1-b^2}{1+b^2}=\frac{1-\tan^2\phi}{1+\tan^2\phi}=\cos 2\phi$ $\therefore\ \cos^{-1}\frac{1-b^2}{1+b^2}=2\phi=2\tan^{-1}b$

$\therefore$ L.H.S. = $2(\tan^{-1}a - \tan^{-1}b) = 2\tan^{-1}\frac{a-b}{1+ab}$

Put $x = \tan\alpha$ $\therefore$ $\frac{2x}{1-x^2} = \frac{2\tan\alpha}{1-\tan^2\alpha} = \tan 2\alpha$

$\therefore$ R.H.S. = $\tan^{-1}\frac{2x}{1-x^2} = 2a = 2\tan^{-1}x$ L.H.S. = R.H.S.

$\Rightarrow$ $2\tan^{-1}\frac{a-b}{1+ab} = 2\tan^{-1}x \Rightarrow x = \frac{a-b}{1+ab}$

Illustration 35 : Find the value of $\tan\left[2\tan^{-1}\frac{1}{5} - \frac{\pi}{4}\right]$

Solution : $2\tan^{-1}\frac{1}{5} = \tan^{-1}\frac{1}{5} + \tan^{-1}\frac{1}{5} = \tan^{-1}\frac{\frac{1}{5}+\frac{1}{5}}{1-\frac{1}{5}.\frac{1}{5}} = \tan^{-1}\frac{2/5}{24/25} = \tan^{-1}\frac{5}{12}$

$$\tan\left(2\tan^{-1}\frac{1}{5} - \frac{\pi}{4}\right) = \tan\left(\tan^{-1}\frac{5}{12} - \frac{\pi}{4}\right) = \frac{\tan\left(\tan^{-1}\frac{5}{12}\right) - \tan\frac{\pi}{4}}{1+\tan\left(\tan^{-1}\frac{5}{12}\right)\tan\frac{\pi}{4}}$$

$$= \frac{\frac{5}{12}-1}{1+\frac{5}{12}.1} = \frac{-\frac{7}{12}}{\frac{17}{12}} = -\frac{7}{17}$$

PROPERTY-10

(i) $\sin^{-1}x = \cos^{-1}\sqrt{1-x^2} = \tan^{-1}\frac{x}{\sqrt{1-x^2}} = \cot^{-1}\frac{\sqrt{1-x^2}}{x} = \sec^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right) = \text{cosec}^{-1}\left(\frac{1}{x}\right)$

Let $\sin^{-1}x = \theta$ then $x = \sin\theta$

Now, $\cos\theta = \sqrt{1-\sin^2\theta} \Rightarrow \cos\theta = \sqrt{1-x^2}$ or $\theta = \cos^{-1}\sqrt{1-x^2}$

or $\sin^{-1}x = \cos^{-1}\sqrt{1-x^2} = \sec^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)$ $[\because \theta = \sin^{-1}x]$

Again, $\tan\theta = \frac{\sin\theta}{\cos\theta} = \frac{\sin\theta}{\sqrt{1-\sin^2\theta}}$

$$\tan\theta = \frac{x}{\sqrt{1-x^2}} \Rightarrow \theta = \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$$

or $\sin^{-1}x = \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right)$ or $\cot^{-1}x = \left(\frac{\sqrt{1-x^2}}{x}\right)$ $\left[\because \tan^{-1}x = \cot^{-1}\left(\frac{1}{x}\right)\right]$

Hence, $\sin^{-1}x = \cos^{-1}\sqrt{1-x^2} = \sec^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)$

$= \tan^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right) = \cot^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right) = \operatorname{cosec}^{-1}\left(\frac{1}{x}\right)$

(ii) $\cos^{-1}x = \sin^{-1}\sqrt{1-x^2} = \tan^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right) = \cot^{-1}\left(\frac{x}{\sqrt{1-x^2}}\right) = \sec^{-1}\frac{1}{x} = \operatorname{cosec}^{-1}\left(\frac{1}{\sqrt{1-x^2}}\right)$

(iii) $\tan^{-1}x = \sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) = \cos^{-1}\left(\frac{1}{\sqrt{1+x^2}}\right) = \cot^{-1}\left(\frac{1}{x}\right) = \sec^{-1}\sqrt{1+x^2} = \operatorname{cosec}^{-1}\left(\frac{\sqrt{1+x^2}}{x}\right)$

Illustration 36 : Find the value of $\sin^{-1}(\cot^{-1}(\tan\cos^{-1}x)$.

Solution : Put $\cos^{-1}x = \theta \;\therefore\; x = \cos\theta$

$\therefore \quad \tan\theta = \frac{\sqrt{1-x^2}}{x} \qquad \therefore \quad \tan(\cos^{-1}x) = \tan\theta = \frac{\sqrt{1-x^2}}{x}$

$\therefore \quad \cot^{-1}(\tan(\cos^{-1}x)) = \cot^{-1}\left(\frac{\sqrt{1-x^2}}{x}\right) = \phi$ (say)

$\therefore \quad \cot\phi = \frac{\sqrt{1-x^2}}{x} \qquad \therefore \quad \sin\phi = \frac{x}{1} \;\therefore\; \sin(\cot^{-1}(\tan(\cos^{-1}x)) = \sin\phi = x$

Student Task

Q.1 $\cot^{-1}(-\sqrt{3})$ equals

(a) $-\pi/6$ (b) $5\pi/6$ (c) $\pi/3$ (d) $2\pi/3$

Q.2 $\tan\left(90° - \cot^{-1}\frac{1}{3}\right)$ equals–

(a) 3 (b) 2/3 (c) 1/3 (d) $1/\sqrt{10}$

Q.3 $\cos^{-1}\left[\cos\left(-\frac{17}{15}\pi\right)\right]$ is equal to –

(a) $-\frac{17\pi}{15}$ (b) $\frac{17\pi}{15}$ (c) $\frac{2\pi}{15}$ (d) $\frac{13\pi}{15}$

Q.4 The value of $\tan\left[2\tan^{-1}\frac{1}{5} - \frac{\pi}{4}\right]$ is

(a) 0 (b) 1 (c) – 7/17 (d) None of these

Q.5 If $\sin^{-1}x + \sin^{-1}y = \frac{2\pi}{3}$, then $\cos^{-1}x + \cos^{-1}y$ equals

(a) $2\pi/3$ (b) $\pi/3$ (c) $\pi/6$ (d) π

Q.6 If $\tan^{-1} 2x + \tan^{-1} 3x = \frac{\pi}{4}$ then x equals–

(a) –1 (b) 1/6 (c) –1, 1/6 (d) None of these

Q.7 $\cos\left(\tan^{-1}\frac{3}{4}\right)$ equals–

(a) 4/5 (b) 3/4 (c) 4/3 (d) 5/3

Q.8 If $xy + yz + zx = 1$, then, $\tan^{-1} x + \tan^{-1} y + \tan^{-1} z$ equals–

(a) π (b) $\pi/2$ (c) 1 (d) None of these

Q.9 Let $f(x) = \sec^{-1} x + \tan^{-1} x$ Then $f(x)$ is real for

(a) $x \in [-1, 1]$ (b) $x \in R$ (c) $x \in (-\infty, -1] \cup [1, \infty)$ (d) None of these

Q.10 Which of the following is not the value of $2\tan^{-1} x$?

(a) $\tan^{-1}\frac{2x}{1+x^2}$ (b) $\tan^{-1}\frac{2x}{1-x^2}$ (c) $\cos^{-1}\frac{1-x^2}{1+x^2}$ (d) $\sin^{-1}\frac{2x}{1+x^2}$

Q.11 If $\sin^{-1} x = \pi/5$, then $\cos^{-1} x$ equals

(a) $\pi/10$ (b) $3\pi/10$ (c) $5\pi/4$ (d) $7\pi/4$

ANSWERS

1.	(b)	**2.**	(c)	**3.**	(d)	**4.**	(c)	**5.**	(b)	**6.**	(b)	**7.**	(a)
8.	(b)	**9.**	(c)	**10.**	(a)	**11.**	(b)						

5. INFINITE SERIES OF INVERSE TRIGONOMETRIC FUNCTIONS

$$\sin^{-1}(x) = x + \left(\frac{1}{2}\right)\frac{x^3}{3} + \left(\frac{1}{2}.\frac{3}{4}\right)\frac{x^5}{5} + \left(\frac{1}{2}\frac{3}{4}\frac{5}{6}\right)\frac{x^7}{7} + ... = \sum_{n=0}^{\infty}\left(\frac{(2n)!}{2^{2n}(n!)^2}\right)\frac{x^{2n+1}}{(2n+1)}; |x| \le 1$$

$$\cos^{-1} x = \frac{\pi}{2} - \sin^{-1} x = \frac{\pi}{2} - \left[x + \left(\frac{1}{2}\right)\frac{x^3}{3} + \left(\frac{1}{2}.\frac{3}{4}\right)\frac{x^5}{5} + \left(\frac{1}{2}\frac{3}{4}\frac{5}{6}\right)\frac{x^7}{7} + ...\right]$$

$$= \frac{\pi}{2} - \sum_{n=0}^{\infty}\left(\frac{(2n)!}{2^{2n}(n!)^2}\right)\frac{x^{2n+1}}{(2n+1)}; |x| \le 1$$

$$\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + ... = \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{(2n+1)}; |x| \le 1 \quad x \ne i, -i$$

$$\cot^{-1} x = \frac{\pi}{2} - \tan^{-1} x = \frac{\pi}{2} - \left(x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + ...\right) = \frac{\pi}{2} - \sum_{n=0}^{\infty}\frac{(-1)^n x^{2n+1}}{2n+1}, |x| \le 1 \quad x \ne i, -i$$

$$\sec^{-1} x = \cos^{-1}(x^{-1}) = \frac{\pi}{2} - \left[x^{-1} + \left(\frac{1}{2}\right)\frac{x^{-3}}{3} + \left(\frac{1}{2}.\frac{3}{4}\right)\frac{x^{-5}}{5} + \left(\frac{1}{2}\frac{3}{4}\frac{5}{6}\right)\frac{x^{-7}}{7} + ...\right]$$

$$= \sum_{n=0}^{\infty}\left(\frac{(2n)!}{2^{2n}(n!)^2}\right)\frac{x^{-(2n+1)}}{2x+1}, |x| \ge 1$$

$$\operatorname{cosec}^{-1}(x) = \sin^{-1}(x^{-1}) = x^{-1} + \left(\frac{1}{2}\right)\frac{x^{-3}}{3} + \left(\frac{1}{2}.\frac{3}{4}\right)\frac{x^{-5}}{5} + \left(\frac{1}{2}\frac{3}{4}\frac{5}{6}\right)\frac{x^{-7}}{7} + ...$$

$$= \sum_{n=0}^{\infty}\left(\frac{(2n)!}{2^{2n}(n!)^2}\right)\frac{x^{-(2n+1)}}{(2n+1)}; |x| \ge 1$$

Alternative to the power series for $\tan^{-1} x$ is

$$\tan^{-1}(x) = \cfrac{x}{1+\cfrac{x^2}{3+\cfrac{4x^2}{5+\cfrac{9x^2}{7+\cfrac{16x^2}{9+\cfrac{25x^2}{\ddots}}}}}}$$

LOGARITHMIC FORMS OF INVERSE TRIGONOMETRIC FUNCTIONS

$$\sin^{-1} x = -i\log(ix + \sqrt{1-x^2})$$

$$\cos^{-1} x = -i\log(x + \sqrt{x^2-1}) = \frac{\pi}{2} + i\log(ix + \sqrt{1-x^2}) = \frac{\pi}{2} - \sin^{-1} x$$

$$\tan^{-1} x = \frac{i}{2}(\log(1-ix) - \log(1+ix))$$

$$\cot^{-1} x = \frac{i}{2}\left(\log\left(1-\frac{i}{x}\right)\right) - \log\left(1+\frac{i}{x}\right)$$

$$\sec^{-1} x = -i\log\left(\sqrt{\frac{1}{x^2}-1} + \frac{1}{x}\right) = i\log\left(\sqrt{1-\frac{1}{x^2}} + \frac{i}{x}\right) + \frac{\pi}{2} = \frac{\pi}{2} - \operatorname{cosec}^{-1} x$$

$$\operatorname{cosec}^{-1} x = -i\log\left(\sqrt{1-\frac{1}{x^2}} + \frac{i}{x}\right)$$

USEFUL TIPS

1. $\tan^{-1} x = -\pi + \cot^{-1}\frac{1}{x} = \sin^{-1}\frac{x}{\sqrt{1+x^2}} = \cos^{-1}\frac{1}{\sqrt{1+x^2}}$; $x < 0$

2. $\cos^{-1} x = \sec^{-1}\frac{1}{x} = \pi - \sin^{-1}\sqrt{1-x^2} = \pi + \tan^{-1}\frac{\sqrt{1-x^2}}{x} = \cot^{-1}\frac{x}{\sqrt{1+x^2}}$; $-1 < x < 0$

3. $\cos^{-1}\frac{1}{2} + \cos^{-1}\left(-\frac{1}{7}\right) + \cos^{-1}\left(\frac{13}{14}\right) = \pi$

4. $\sin^{-1}\left(\frac{3}{5}\right) - \cos^{-1}\left(\frac{12}{13}\right) = \sin^{-1}\left(\frac{16}{65}\right)$

5. $2\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{1}{7}\right) = \tan^{-1}\left(\frac{31}{17}\right)$

6. $2\tan^{-1}\left(\frac{3}{2}\right) + \tan^{-1}\left(\frac{12}{5}\right) = \pi$

7. If $\sum_{i=1}^{2n} \sin^{-1} x_i = n\pi$ then value of $\sum_{i=1}^{2n} x_i$ is $2n$.

8. There is only 1 solution for the equation $\sin^{-1} x + \cos^{-1}(1-x) = \sin^{-1}(-x)$

9. If $\tan^{-1}\frac{1}{1+2} + \tan^{-1}\frac{1}{1+2.3} + \tan^{-1}\frac{1}{1+3.4} + ... + \tan^{-1}\frac{1}{1+n(n+1)} = \tan^{-1}\theta$ then $\theta = \frac{n}{n+2}$

10. Value of $\tan^{-1}\left(\frac{1}{2}\tan 2A\right) + \tan^{-1}(\cot A) + \tan^{-1}(\cot^3 A)$ between $0 < A < (\pi/4)$ is independent on A.

11. $\cos\tan^{-1}\sin\cot^{-1} x = \sqrt{\frac{x^2+1}{x^2+2}}$

MISCELLANEOUS EXAMPLES

Example 1 :

Find the sum $\sum_{k=1}^{n} \tan^{-1}\left[\frac{2k}{2+k^2+k^4}\right]$. Hence show that

$$\frac{\pi}{4} = \tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{2}{11}\right) + \tan^{-1}\left(\frac{3}{46}\right) + \tan^{-1}\left(\frac{4}{137}\right) + ...\text{upto infinity}$$

Solution :

We first try to express $\tan^{-1}\left[\frac{2k}{2+k^2+k^4}\right]$ in the form $\tan^{-1}\left[\frac{(x-y)}{(1+xy)}\right]$ as follows :

Let $x - y = 2k$ and $xy = 1 + k^2 + k^4$

$\Rightarrow x(x - 2k) = 1 + k^2 + k^4$

$\Rightarrow x^2 - 2kx + k^2 = 1 + 2k^2 + k^4 \quad \Rightarrow \quad (x-k)^2 = (k^2+1)^2$

$\Rightarrow x - k = k^2 + 1 \quad \Rightarrow \quad x = k^2 + k + 1$

$\Rightarrow y = x - 2k = k^2 - k + 1$

Hence $\sum_{k=1}^{n} \tan^{-1}\left[\frac{2k}{2+k^2+k^4}\right] = \sum_{k=1}^{n} \tan^{-1}\left[\frac{(k^2+k+1)-(k^2-k+1)}{1+(k^2+k-1)(k^2-k+1)}\right]$

$= \sum_{k=1}^{n} \tan^{-1}(k^2+k+1) - \tan^{-1}(k^2-k+1) \qquad (V_k - V_{k-1} \text{ form})$

$[\because (k^2+k+1)(k^2-k+1) = k^4+k^2+1 > -1]$

$= (\tan^{-1}3 - \tan^{-1}1) + (\tan^{-1}7 - \tan^{-1}3) + (\tan^{-1}13 - \tan^{-1}7)$

$+ ..... \tan^{-1}(n^2+n+1) - \tan^{-1}(n^2-n+1)$

[By substituting k = 1, 2, ...n]

$= \tan^{-1}(n^2+n+1) - \tan^{-1}1 = \tan^{-1}(n^2+n+1) - \frac{\pi}{4}$

Since $\sum_{k=1}^{n} \tan^{-1}\left[\frac{2k}{2+k^2+k^4}\right] = \tan^{-1}(n^2+n+1) - \frac{\pi}{4}$, for all $n \in N$

Taking limits as n tends to infinity we get

$$\tan^{-1}\left(\frac{1}{2}\right) + \tan^{-1}\left(\frac{2}{11}\right) + \tan^{-1}\left(\frac{3}{46}\right) + \ldots\ldots \text{ to infinity}$$

$= \lim_{n\to\infty}\left(\tan^{-1}(n^2+n+1) - \frac{\pi}{4}\right) = \tan^{-1}\infty - \frac{\pi}{4} = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4}$

Example 2 :

If $\sin^{-1}\left(x - \frac{x^2}{2} + \frac{x^3}{4} - \ldots\ldots\right) + \cos^{-1}\left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \ldots\ldots\right) = \frac{\pi}{2}$ for $0 < |x| < \sqrt{2}$, then find the value of x.

Solution :

We know that, $\sin^{-1}y + \cos^{-1}y = \frac{\pi}{2}, |y| \le 1$

$\therefore$ According to question $x - \frac{x^2}{2} + \frac{x^3}{3} - \ldots\ldots = x^2 - \frac{x^4}{2} + \frac{x^6}{4} \ldots\ldots$

$\Rightarrow \frac{x}{1+\frac{x}{2}} = \frac{x}{1+\frac{x^2}{2}}$ $(\because 0 < |x| < \sqrt{2})$ $\Rightarrow \frac{x}{2+x} = \frac{x^2}{2+x^2}$ $\Rightarrow 2x + x^3 = 2x^2 + x^3 \Rightarrow x = x^2$

But $x \neq 0$ hence $x = 1$

Example 3 :

Solve $\tan^{-1}2x + \tan^{-1}3x = n\pi + \frac{3\pi}{4}$

Solution :

We have $\tan^{-1}\frac{2x+3x}{1-6x^2} = n\pi + \frac{3\pi}{4}$

$\therefore \frac{2x+3x}{1-6x^2} = \tan\left(n\pi + \frac{3\pi}{4}\right) = -1$

$\therefore 6x^2 - 5x - 1 = 0$ or $(6x + 1)(x - 1) = 0$

$\therefore x = 1$, or $-\frac{1}{6}$

Example 4 :

$$\tan^{-1}\left(\frac{c_1x - y}{c_1y + x}\right) + \tan^{-1}\left(\frac{c_2 - c_1}{1 + c_2c_1}\right) + \tan^{-1}\left(\frac{c_3 - c_2}{1 + c_3c_2}\right) + \ldots\ldots + \tan^{-1}\left(\frac{1}{c_n}\right) \text{ equals}$$

(a) $\tan^{-1}\frac{y}{x}$ (b) $\tan^{-1}\frac{x}{y}$ (c) $-\tan^{-1}\frac{y}{x}$ (d) None of these

Solution : (b)

we have $\tan^{-1}\left(\frac{c_1x-y}{c_1y+x}\right)+\tan^{-1}\left(\frac{c_2-c_1}{1+c_2c_1}\right)+\tan^{-1}\left(\frac{c_3-c_2}{1+c_3c_2}\right)+.....+\tan^{-1}\left(\frac{1}{c_n}\right)$

$= \tan^{-1}\left(\frac{\frac{x}{y}-\frac{1}{c_1}}{1+\frac{x}{y}\frac{1}{c_1}}\right)+\tan^{-1}\left(\frac{\frac{1}{c_1}-\frac{1}{c_2}}{1+\frac{1}{c_1c_2}}\right)+\tan^{-1}\left(\frac{\frac{1}{c_2}-\frac{1}{c_3}}{1+\frac{1}{c_2c_3}}\right)+.....+\tan^{-1}\left(\frac{1}{c_n}\right)$

$=\tan^{-1}\left(\frac{x}{y}\right)-\tan^{-1}\left(\frac{1}{c_1}\right)+\tan^{-1}\left(\frac{1}{c_1}\right)-\tan^{-1}\left(\frac{1}{c_2}\right)+\tan^{-1}\left(\frac{1}{c_2}\right)-...-\tan^{-1}\left(\frac{1}{c_n}\right)+\tan^{-1}\left(\frac{1}{c_n}\right)$

$= \tan^{-1}\left(\frac{x}{y}\right)$

Example 5 :

Find the greater angle out of A and B if $A = 2\tan^{-1}(2\sqrt{2}-1)$ and $B = 3\sin^{-1}\left(\frac{1}{3}\right)+\sin^{-1}\left(\frac{3}{5}\right)$

Solution :

$A = 2\tan^{-1}(2\sqrt{2}-1) = 2\tan^{-1}(1.828)$

$\therefore\ A > 2\tan^{-1}\sqrt{3}\ [\sqrt{3} = 1.732 < 1.828] \Rightarrow A > \sqrt{3}$ ....(i)

We have $\sin^{-1}\frac{1}{3} < \sin^{-1}\frac{1}{2} = \frac{\pi}{6} \Rightarrow 3\sin^{-1}\frac{1}{3} < \frac{\pi}{2}$

Using $\sin 3\theta = 3\sin\theta - 4\sin^3\theta$

We have $\sin^{-1}\frac{1}{3} = \sin^{-1}\left(3\times\frac{1}{3}-4\left(\frac{1}{3}\right)^3\right)$

$= \sin^{-1}\left(\frac{23}{27}\right) = \sin^{-1}(0.852) \quad \therefore\ 3\sin^{-1}\frac{1}{3} < \sin^{-1}\left(\frac{\sqrt{3}}{2}\right)$ $\left(\because\ \frac{\sqrt{3}}{2} = 0.868 > 0.852\right)$

i.e. $3\sin^{-1}\frac{1}{3} < \frac{\pi}{3}$ ...(ii)

Also $\sin^{-1}\frac{3}{5} = \sin^{-1}(0.6) < \sin^{-1}\frac{\sqrt{3}}{2} = \frac{\pi}{3} \quad \therefore\ \sin^{-1}\frac{3}{5} < \frac{\pi}{3}$

$\therefore\ B = 3\sin^{-1}\frac{1}{3}+\sin\frac{3}{5} < \frac{\pi}{3}+\frac{\pi}{3} = \frac{2\pi}{3}$

$\therefore\ B < \frac{2\pi}{3}$ ....(iii)

By (i) and (iii), $A > B$

Example 6 :

If $\sin(\cot^{-1}(x+1)) = \cos(\tan^{-1}x)$ then find the value of x.

Solution :

$\sin[\cot^{-1}(x+1) = \sin\left(\sin^{-1}\frac{1}{\sqrt{x^2+2x+2}}\right) = \frac{1}{\sqrt{x^2+2x+2}}$

$$\cos(\tan^{-1}x) = \cos\left(\cos^{-1}\frac{1}{\sqrt{1+x^2}}\right) = \frac{1}{\sqrt{1+x^2}}$$

Hence $\frac{1}{\sqrt{x^2+2x+2}} = \frac{1}{\sqrt{1+x^2}} \Rightarrow x^2 + 2x + 2 = 1 + x^2 \Rightarrow x = -x^2 + 2x + 1/2$

Example 7 :

$\sin(\cot^{-1}\cos\tan^{-1}x)$ is equal to –

(a) $\sqrt{\frac{x^2+1}{x^2+2}}$ (b) $\sqrt{\frac{x+2}{x^2+1}}$ (c) $\sqrt{\frac{x^2-1}{x^2+2}}$ (d) $\sqrt{\frac{x^2+2}{x^2+1}}$

Solution : (a)

$\sin(\cot^{-1}\cos\tan^{-1}x) = \sin(\cot^{-1}\cos\theta)$, where $\theta = \tan^{-1}x$ or $x = \tan\theta$

$$= \sin\left\{\sin^{-1}\left(\frac{1}{\sqrt{1+\cos^2\theta}}\right)\right\} = \frac{1}{\sqrt{1+\cos^2\theta}} = \frac{\sec\theta}{\sqrt{1+\sec^2\theta}} = \frac{\sqrt{1+x^2}}{\sqrt{2+x^2}}$$

Example 8 :

If x_1, x_2, x_3, x_4 are roots of equation $x^4 - x^3\sin 2\beta + x^2\cos 2\beta - x\cos\beta - \sin\beta = 0$, then $\sum_{i=1}^{4}\tan^{-1}x_i$ equals

(a) β (b) $\pi/2 - \beta$ (c) $\pi - \beta$ (d) $-\beta$

Solution : (b)

$p_1 = \Sigma x_1 = \sin 2\beta$
$p_2 = \Sigma x_1 x_2 = \cos 2\beta$
$p_3 = \Sigma x_1 x_2 x_3 = \cos\beta$
$p_4 = \Sigma x_1 x_2 x_3 x_4 = -\sin\beta$

$$\tan^{-1}x_1 + \tan^{-1}x_2 + \tan^{-1}x_3 + \tan^{-1}x_4 = \tan^{-1}\frac{p_1 - p_3}{1 - p_2 + p_4} \quad \ldots(1)$$

$p_1 - p_3 = \sin 2\beta - \cos\beta = \cos\beta(2\sin\beta - 1)$
$1 - p_2 + p_4 = 1 - \cos 2\beta - \sin\beta = 2\sin^2\beta - \sin\beta = \sin\beta(2\sin\beta - 1)$

$$\therefore \tan^{-1}\frac{p_1 - p_3}{1 - p_2 + p_4} = \tan^{-1}\left(\frac{\cos\beta}{\sin\beta}\right) = \tan^{-1}(\cot\beta) = \tan^{-1}\left[\tan\left(\frac{\pi}{2} - \beta\right)\right] = \frac{\pi}{2} - \beta$$

Example 9 :

$\tan^{-1}(1/3) + \tan^{-1}(1/7) + \tan^{-1}(1/13) + \ldots + \tan^{-1}\left(\frac{1}{n^2+n+1}\right) + \ldots\infty$ is equal to

(a) 0 (b) $\pi/4$ (C) $\pi/2$ (d) $2\pi/3$

Solution : (b)

$$T_n = \tan^{-1}\left(\frac{1}{n^2+n+1}\right) = \tan^{-1}\left(\frac{(n+1)-n}{1+(n+1)n}\right) = \tan^{-1}(n+1) - \tan^{-1}n$$

$\Rightarrow T_1 = \tan^{-1}(1/3) = \tan^{-1}2 - \tan^{-1}1$
$T_2 = \tan^{-1}(1/7) = \tan^{-1}3 - \tan^{-1}2$
.....
$T_n = \tan^{-1}(n+1) - \tan^{-1}n$

On adding, $T_1 + T_2 + T_3 + \ldots + T_n = \tan^{-1}(n+1) - \tan^{-1}1 = \tan^{-1}\left(\frac{n}{n+2}\right)$

$$\therefore \quad \lim_{n\to\infty}(T_1+T_2+T_3+.....+T_n)=\lim_{n\to\infty}\tan^{-1}\left(\frac{1}{1+2/n}\right)=\tan^{-1}1=\frac{\pi}{4}$$

Example 10 :

If $\sin^{-1}\frac{x}{a}+\sin^{-1}\frac{y}{b}=\alpha$, then –

(a) $\frac{x^2}{a^2}-\frac{2xy}{ab}\cos\alpha+\frac{y^2}{b^2}=\sin^2\alpha$ (d) $\frac{x^2}{a^2}-\frac{2xy}{ab}\sin\alpha+\frac{y^2}{b^2}=\cos^2\alpha$

(c) $\frac{x^2}{a^2}+\frac{2xy}{ab}\cos\alpha+\frac{y^2}{b^2}=\sin^2\alpha$ (d) $\frac{x^2}{a^2}+\frac{2xy}{ab}\sin\alpha+\frac{y^2}{b^2}=\cos^2\alpha$

Solution : (c)

$$\cos\alpha=\sqrt{1-\frac{x^2}{a^2}}\sqrt{1-\frac{y^2}{b^2}}-\frac{xy}{ab}.$$

Squaring after rearranging, $\left(1-\frac{x^2}{a^2}\right)\left(1-\frac{y^2}{b^2}\right)=\cos^2\alpha+2\frac{xy}{ab}\cos\alpha+\frac{x^2y^2}{a^2b^2}$

or $\frac{x^2}{a^2}+\frac{2xy}{ab}\cos\alpha+\frac{y^2}{b^2}=\sin^2\alpha$

Example 11 :

$\cot^{-1}(\sqrt{\cos\alpha})-\tan^{-1}(\sqrt{\cos\alpha})=x$, then sin x equals

(a) $\tan^2\left(\frac{\alpha}{2}\right)$ (B) $\cot^2\left(\frac{\alpha}{2}\right)$ (C) $\tan\alpha$ (d) $\cot\left(\frac{\alpha}{2}\right)$

Solution : (a)

$$\tan^{-1}\left[\frac{1}{\sqrt{\cos\alpha}}\right]-\tan^{-1}\left[\sqrt{\cos\alpha}\right]=x \Rightarrow \tan^{-1}\left[\frac{\frac{1}{\sqrt{\cos\alpha}}-\sqrt{\cos\alpha}}{1+\frac{\sqrt{\cos\alpha}}{\sqrt{\cos\alpha}}}\right]=x \Rightarrow \tan x=\frac{1-\cos\alpha}{2\sqrt{\cos\alpha}}$$

$$\therefore \quad \sin x=\frac{1-\cos\alpha}{1+\cos\alpha}=\frac{2\sin^2(\alpha/2)}{2\cos^2(\alpha/2)}=\tan^2\left(\frac{\alpha}{2}\right)$$

Example 12 :

$\tan^{-1}\left(\frac{1}{4}\right)+\tan^{-1}\left(\frac{2}{9}\right)$ equal to

(a) $\frac{1}{2}\cos^{-1}\left(\frac{3}{5}\right)$ (b) $\frac{1}{2}\sin^{-1}\left(\frac{3}{5}\right)$ (c) $\frac{1}{2}\tan^{-1}\left(\frac{3}{5}\right)$ (d) $\tan^{-1}\left(\frac{1}{2}\right)$

Solution : (d)

$$\tan^{-1}\left(\frac{1}{4}\right)+\tan^{-1}\left(\frac{2}{9}\right)=\tan^{-1}\left[\frac{\frac{1}{4}+\frac{2}{9}}{1-\frac{1}{4}\times\frac{2}{9}}\right]=\tan^{-1}\left[\frac{1}{2}\right]$$

Example 13 :

The trigonometric equation $\sin^{-1}x = 2\sin^{-1}a$, has a solution for

(a) $\frac{1}{2} < |a| < \frac{1}{\sqrt{2}}$ (b) All real values of a

(c) $|a| \le \frac{1}{\sqrt{2}}$ (d) $|a| \ge \frac{1}{\sqrt{2}}$

Solution : (c)

$$\sin^{-1} x = 2\sin^{-1} a$$

$$-\frac{\pi}{2} \le \sin^{-1} x \le \frac{\pi}{2} \; ; \quad -\frac{\pi}{2} \le 2\sin^{-1} a \le \frac{\pi}{2} \; ; \quad -\frac{\pi}{4} \le \sin^{-1} a \le \frac{\pi}{4}$$

$$\sin\left(-\frac{\pi}{4}\right) \le a \le \sin\frac{\pi}{4} \quad ; \quad -\frac{1}{\sqrt{2}} \le a \le \frac{1}{\sqrt{2}} \; ; \; |a| \le \frac{1}{\sqrt{2}}$$

Example 14 :

Find the positive integral solutions x and y of the equation.

$$\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) + \cos^{-1}\frac{y}{\sqrt{1+y^2}} = \tan^{-1}(3)$$

Solution :

We have, $\sin^{-1}\left(\frac{x}{\sqrt{1+x^2}}\right) + \cos^{-1}\frac{y}{\sqrt{1+y^2}} = \tan^{-1}(3)$

$$\tan^{-1} x + \tan^{-1}\frac{1}{y} = \tan^{-1} 3 \; ; \qquad \tan^{-1} x - \tan^{-1} 3 = \tan^{-1}\frac{1}{y}$$

$$\tan^{-1} 3 - \tan^{-1} x = \tan^{-1}\frac{1}{y} \Rightarrow \tan^{-1}\frac{3-x}{1+3x} = \tan^{-1}\frac{1}{y}$$

since x and y are positive integers $\Rightarrow x = 1,2 \Rightarrow x = 1, y = 1$

$x = 2, y = 7$

Example 15 :

If $\sin^{-1}\left(\frac{x}{5}\right) + \operatorname{cosec}^{-1}\left(\frac{5}{4}\right) = \frac{\pi}{2}$ then find the value of x.

Solution :

$$\sin^{-1}\left(\frac{x}{5}\right) + \operatorname{cosec}^{-1}\left(\frac{5}{4}\right) = \frac{\pi}{2} \qquad \text{..... (1)}$$

Let $\theta = \operatorname{cosec}^{-1}\left(\frac{5}{4}\right); \; \operatorname{cosec}\theta = \frac{5}{4}$

$$\therefore \sin\theta = \frac{1}{\operatorname{cosec}\theta} = \frac{4}{5} \Rightarrow \theta = \sin^{-1}\left(\frac{4}{5}\right) \qquad \text{..... (2)}$$

We know that $\sin\frac{\pi}{2} = 1 \quad \therefore \frac{\pi}{2} = \sin^{-1}(1)$ ..... (3)

From eq. (1), $\sin^{-1}\left(\frac{x}{5}\right) = \frac{\pi}{2} - \operatorname{cosec}^{-1}\left(\frac{5}{4}\right) = \frac{\pi}{2} - \theta$

Using Eq. (2) and eq. (3)

$$\sin^{-1}\left(\frac{x}{5}\right)=\sin^{-1}(1)-\sin^{-1}\left(\frac{4}{5}\right)=\sin^{-1}\left(\sqrt{1-\frac{16}{25}}-\frac{4}{5}\sqrt{1-1}\right)$$ (Using Formula of $\sin^{-1}A-\sin^{-1}B$)

$$\Rightarrow \sin^{-1}\frac{x}{5}=\sin^{-1}\left(\frac{3}{5}-0\right)\Rightarrow \sin^{-1}\frac{x}{5}=\sin^{-1}\frac{3}{5}\Rightarrow x=3$$

Example 16 :

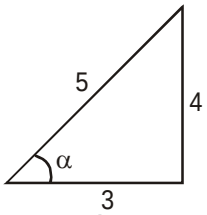
Prove that : $\sin^{-1}\frac{4}{5}+\cos^{-1}\frac{12}{13}+\sin^{-1}\frac{16}{65}=\frac{\pi}{2}$

Solution :

We may write this identity in the form

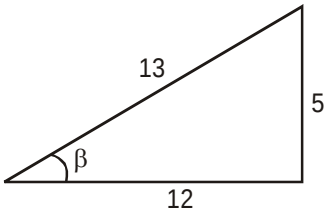
$$\sin^{-1}\frac{4}{5}+\cos^{-1}\frac{12}{13}=\frac{\pi}{2}-\sin^{-1}\frac{16}{65}=\cos^{-1}\frac{16}{65}$$

Let $\alpha=\sin^{-1}\frac{4}{5}$, so that $\sin\alpha=\frac{4}{5}$ and $\beta=\cos^{-1}\frac{12}{13}$ so that $\cos\beta=\frac{12}{13}$



We have to express $\alpha + \beta$ as an inverse cosine.
Now $\cos(\alpha+\beta)=\cos\alpha\cos\beta-\sin\alpha\sin\beta$, whence by reading off the values of the functions from the figures in the margin, we have, $\cos(\alpha+\beta)=\frac{3}{5}.\frac{12}{13}-\frac{4}{5}.\frac{5}{13}=\frac{16}{65}$

$$\therefore \quad \alpha+\beta=\cos^{-1}\frac{16}{65}$$



Example 17 :

Solve : $\sin^{-1}x+\sin^{-1}(1-x)=\cos^{-1}x$

Solution :

By transposition, $\sin^{-1}(1-x)=\cos^{-1}x-\sin^{-1}x$
Let $\cos^{-1}x=a$, and $\sin^{-1}x=b$, then $\sin^{-1}(1-x)=\alpha-\beta$
$\therefore\ 1-x=\sin^{-1}(\alpha-\beta)=\sin\alpha\cos\beta-\cos\alpha\sin\beta$

But $\cos\alpha=x$ and therefore $\sin\alpha=\sqrt{1-x^2}$ Also $\sin\beta=x$, and therefore $\cos\beta=\sqrt{1-x^2}$
$\therefore\ 1-x=(1-x^2)-x^2=1-2x^2\ \therefore\ 2x^2-x=0$ whence $x=0$, or $1/2$

Example 18 :

Solve : **(i)** $\sin^{-1}x+\sin^{-1}y=\frac{2\pi}{3},\cos^{-1}x+\cos^{-1}y=\frac{\pi}{3}$

(ii) $\tan^{-1}(x-1)+\tan^{-1}x+\tan^{-1}(x+1)=\tan^{-1}3x$

Solution :

(i) $\sin^{-1}x+\sin^{-1}y=\frac{2\pi}{3}$ or $\frac{\pi}{2}-\cos^{-1}x+\frac{\pi}{2}-\cos^{-1}y=\frac{2\pi}{3}$

$\Rightarrow \cos^{-1}x+\cos^{-1}y=\frac{\pi}{3}=\cos^{-1}x-\cos^{-1}y$

$\Rightarrow \cos^{-1}y=0\Rightarrow y=1$ and $\cos^{-1}x=\frac{\pi}{3}\Rightarrow x=\frac{1}{2}$

(ii) The given equation can be written as

$$\tan^{-1}(x+1)+\tan^{-1}(x-1)=\tan^{-1}3x-\tan^{-1}x$$

$\Rightarrow \tan^{-1}\frac{(x+1)\tan^{-1}(x-1)}{1-(x^2-1)}=\tan^{-1}\frac{3x-x}{1+3x^2}$

$\Rightarrow \frac{2x}{2-x^2}=\frac{2x}{1+3x^2} \Rightarrow x=0$ or $x=\pm\frac{1}{2}$

Example 19 :

The inequality $\sin^{-1}x > \cos^{-1}x$ holds –

(a) for all values of x (b) for $x\in\left(\frac{1}{\sqrt{2}},1\right]$ (c) for $x\in\left(0,\frac{1}{\sqrt{2}}\right)$ (d) for $x \in (-1, 0)$

Solution : (b)

The given inverse functions are defined for $x \in [-1, 1]$. However if $x < 0$, $\sin^{-1}x$ is negative and $\cos^{-1}x$ is positive.

If $0 \le x \le 1$, both are positive. Since function is monotonically increasing.

Therefore $\sin(\sin^{-1}x) > \sin(\cos^{-1}x)$ i.e., $x>\sqrt{1-x^2}$

i.e., $2x^2 > 1$ or $x>\frac{1}{\sqrt{2}}$ i.e., $x\in\left(\frac{1}{\sqrt{2}},1\right]$

Example 20 :

Find the greatest and least values of $(\sin^{-1}x)^3 + (\cos^{-1}x)^3$

Solution :

We have, $(\sin^{-1}x)^3 + (\cos^{-1}x)^3 = (\sin^{-1}x + \cos^{-1}x)^3 - 3\sin^{-1}x\cos^{-1}x(\sin^{-1}x + \cos^{-1}x)$

$= \frac{\pi^3}{8} - 3(\sin^{-1}x\cos^{-1}x)\frac{\pi}{2} = \frac{\pi^3}{8}-\frac{3\pi}{2}\sin^{-1}x\left(\frac{\pi}{2}-\sin^{-1}x\right)$

$= \frac{\pi^3}{8}-\frac{3\pi^2}{4}\sin^{-1}x+\frac{3\pi}{2}(\sin^{-1}x)^2 = \frac{\pi^3}{8}+\frac{3\pi}{4}\left[(\sin^{-1}x)^2-\frac{\pi}{2}\sin^{-1}x\right]$

$= \frac{\pi^3}{8}+\frac{3\pi}{2}\left[\left(\sin^{-1}x-\frac{\pi}{4}\right)^2\right]-\frac{3\pi^3}{32} = \frac{\pi^3}{32}+\frac{3\pi}{2}\left(\sin^{-1}x-\frac{\pi}{4}\right)^2$ so, the least value is $\frac{\pi^3}{32}$

Example 21 :

The number of solutions of $\sin^{-1}x+\sin^{-1}2x=\frac{\pi}{3}$ is –

(a) 0 (b) 1 (c) 2 (d) infinite

Solution : (b)

$$2x = \sin\left(\frac{\pi}{3} - \sin^{-1}x\right) = \frac{\sqrt{3}}{2}\sqrt{1-x^2} - \frac{1}{2}x$$

$5x = 3\sqrt{(1-x^2)}$ or $x^2 = \frac{3}{28}$; $x = \frac{1}{2}\sqrt{\frac{3}{7}}$ is the only solution.

Example 22 :

If $\phi = \tan^{-1}\frac{x\sqrt{3}}{2k-x}$ and $\theta = \sin^{-1}\frac{2x-k}{2\sqrt{x^2-xk+k^2}}$, $x > 0$, $k > 0$, find the value of

$\cos 2(\phi-\theta) + \sin 2(\phi-\theta)$

Solution :

$$\theta = \sin^{-1}\frac{2x-k}{2\sqrt{x^2-xk+k^2}} = \tan^{-1}\frac{2x-k}{k\sqrt{3}}$$

and $\phi = \tan^{-1}\frac{x\sqrt{3}}{2k-x}$ give $\tan(\theta-\phi) = \dfrac{\frac{x\sqrt{3}}{2k-x} - \frac{2x-k}{k\sqrt{3}}}{1+\frac{x(2x-k)}{k(2k-x)}} = \frac{1}{\sqrt{3}}$

$$\cos 2(\phi-\theta) + \sin 2(\phi-\theta) = \frac{1-\tan^2(\phi-\theta)}{1+\tan^2(\phi-\theta)} + \frac{2\tan(\phi-\theta)}{1+\tan^2(\phi-\theta)} = \frac{1+\sqrt{3}}{2}$$

Example 23 :

Prove that $\cos^{-1}\left[1 - \frac{1-\cos x}{12\cos x + 13}\right] = \pi - 2\cot^{-1}\left(\frac{1}{5}\tan\frac{x}{2}\right)$

Solution :

$$1 - \frac{1-\cos x}{12\cos x+13} = \frac{13\cos x + 12}{12\cos x + 13} = \frac{26\cos^2\frac{x}{2} - 1}{24\cos^2\frac{x}{2} + 1} = \frac{25\cos^2\frac{x}{2} - \sin^2\frac{x}{2}}{25\cos^2\frac{x}{2} + \sin^2\frac{x}{2}}$$

$$= \frac{1 - \frac{1}{25}\tan^2\frac{x}{2}}{1 + \frac{1}{25}\tan^2\frac{x}{2}} = \cos\left[2\tan^{-1}\left(\frac{1}{5}\tan\frac{x}{2}\right)\right]$$

$$\therefore \quad \cos^{-1}\left[1 - \frac{1-\cos x}{12\cos x+13}\right] = 2\tan^{-1}\left(\frac{1}{5}\tan\frac{x}{2}\right) = \pi - 2\cot^{-1}\left(\frac{1}{5}\tan\frac{x}{2}\right)$$

Example 24 :

If x_1, x_2, x_3, x_4 are the roots of the equation $x^4 - (\sin 2\beta)x^3 + (\cos 2\beta)x^2 - (\cos\beta)x - \sin\beta = 0$

prove that $\tan^{-1}x_1 + \tan^{-1}x_2 + \tan^{-1}x_3 + \tan^{-1}x_4 = n\pi + \frac{\pi}{2} - \beta$.

Solution :

The sums of the products of the roots taken one, two, three and four at a time are given by

$\Sigma x_1 = -(-\sin 2\beta)$, $\Sigma x_1x_2 = (-1)^2\cos 2\beta$, $\Sigma x_1x_2x_3 = (-1)^3(-\cos\beta)$ and $\Sigma x_1x_2x_3x_4 = (-1)^4(-\sin\beta)$

$$\tan(\tan^{-1}x_1+\tan^{-1}x_2+\tan^{-1}x_3+\tan^{-1}x_4)=\frac{\Sigma x_1-\Sigma x_1x_2x_3x_4}{1-\Sigma x_1x_2+x_1x_2x_3x_4}=\frac{\sin2\beta-\cos\beta}{1-\cos2\beta-\sin\beta}=\cot\beta$$

$$\sum_1^4\tan^{-1}x_1=n\pi+\frac{\pi}{2}-\beta,\ n=0,1,2,\ldots\ldots\ldots\ldots$$

Example 25 :

If $\sin^{-1}x>\cos^{-1}x$ then

(a) $x\in\left(-1,-\frac{1}{\sqrt{2}}\right)$ (b) $x\in\left(0,\frac{1}{\sqrt{2}}\right)$ (c) $x\in\left(\frac{1}{\sqrt{2}},1\right)$ (d) $x\in\left(-\frac{1}{\sqrt{2}},0\right)$

Solution : (c)

We have $\sin^{-1}x>\cos^{-1}x$

$\Rightarrow$ $\sin^{-1}x>\pi/2-\sin^{-1}x \Rightarrow 2\sin^{-1}x>\pi/2 \Rightarrow \sin^{-1}x>\pi/4$

Also $\sin(\sin^{-1}x)>\sin(\pi/4)$

$\Rightarrow$ $x>\frac{1}{\sqrt{2}}\Rightarrow x\in\left(\frac{1}{\sqrt{2}},1\right]$ since $-1\le x\le 1$

Example 26 :

Prove that the equation $\sin[\cot^{-1}(\cos(\tan^{-1}x))]=\frac{a}{b}$ has solutions provided $\frac{1}{\sqrt{2}}\le\left|\frac{a}{b}\right|<1$ and find the solutions.

Solution :

$$\sin[\cot^{-1}(\cos(\tan^{-1}x))]=\sin\left[\cot^{-1}\left(\cos\left(\cos^{-1}\frac{1}{\sqrt{1+x^2}}\right)\right)\right]$$

$$=\sin\left[\cot^{-1}\frac{1}{\sqrt{1+x^2}}\right]=\sin\left[\sin^{-1}\frac{\sqrt{1+x^2}}{\sqrt{2+x^2}}\right]=\sqrt{\frac{1+x^2}{2+x^2}}$$

The given equation reduces to $x^2=\frac{2a^2-b^2}{b^2-a^2}$

As $|\sin\theta|<1,|a|<|b|$ **and real values of x exist provided**

$2a^2\ge b^2$ **i.e.,** $\frac{1}{\sqrt{2}}\le\left|\frac{a}{b}\right|<1$ **and then** $x=\pm\sqrt{\frac{2a^2-b^2}{b^2-a^2}}$

Example 27 :

Solve the equation $\theta=\tan^{-1}(2\tan^2\theta)-\frac{1}{2}\sin^{-1}\frac{3\sin2\theta}{5+4\cos2\theta}$

Solution :

With $t=\tan\theta,\sin2\theta=\frac{2t}{1+t^2},\cos2\theta=\frac{1-t^2}{1+t^2}$

and $\frac{3\sin2\theta}{5+4\cos2\theta}=\frac{6t}{9+t^2}$

$\therefore$ $\sin^{-1}\frac{3\sin2\theta}{5+4\cos2\theta}=\tan^{-1}\frac{6t}{9-t^2}=2\tan^{-1}\frac{t}{3}$

The given equation can be written as $\theta = \tan^{-1}(2\tan^2\theta) - \tan^{-1}\left(\frac{1}{3}\tan\theta\right)$ **i.e.,**

$$\tan\theta = \frac{2\tan^2\theta - \frac{1}{3}\tan\theta}{1 + \frac{2}{3}\tan^3\theta}$$

$\therefore$ tam $\theta = 0$ or $2\tan^3\theta - 6\tan\theta + 4 = 0$ i.e. $\tan\theta = 1, 1, -2$

Example 28 :

Find the value of $\text{cosec}^{-1}\sqrt{5} + \text{cosec}^{-1}\sqrt{65} + \text{cosec}^{-1}\sqrt{325} + ... + ...$ ad infinitum.

Solution :

The given series can be written

$\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{8} + \tan^{-1}\frac{1}{18} + ... + \tan^{-1}\frac{1}{2n^2} + ...$ ad infinitum.

$\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{8} = \tan^{-1}\frac{2}{3}, \tan^{-1}\frac{2}{3} + \tan^{-1}\frac{1}{18} = \tan^{-1}\frac{3}{4}$.

In general, $\tan^{-1}\frac{n-1}{n} + \tan^{-1}\frac{1}{2n^2} = \tan^{-1}\frac{n(2n^2-2n+1)}{2n^3-n+1}$

and $2n^3 - n + 1 = (2n^2 - 2n + 1)(n + 1)$ so that, by induction,

$\tan^{-1}\frac{1}{2} + \tan^{-1}\frac{1}{8} + ... + \tan^{-1}\frac{1}{2n^2} = \tan^{-1}\frac{n}{n+1}$ for all $n \in N$.

As $n \to \infty$, $\tan^{-1}\frac{n}{n+1} \to \tan^{-1}1 = \frac{\pi}{4}$

Example 29 :

Prove that $\cos\left(\frac{1}{3}\sin^{-1}a\right)$ has six values and that the product of the six values is $-\frac{1}{16}(1-a^2)$.

Solution :

If $\theta = \sin^{-1}a$ and $t = \cos\frac{\theta}{3}$ then $\sin\theta = a$ and $\cos^2\theta = 1 - a^2$,

$\cos\theta = \cos 3.\frac{\theta}{3} = 4t^3 - 3t$ and t satisfies the polynomial equation of 6th degree

$$16t^6 - 24t^4 + 9t^2 - 1 + a^2 = 0$$

This has 6 roots and accordingly $\cos\left(\frac{1}{3}\sin^{-1}a\right)$ has six values whose product is $\frac{a^2-1}{16}$

Example 30 :

The number of real solution of (x, y) where $|y| = \sin x$, $y = \cos^{-1}(\cos x)$, $-2\pi \le x \le 2\pi$ is –

(a) 2 (b) 1 (c) 3 (d) 4

Solution : (c)

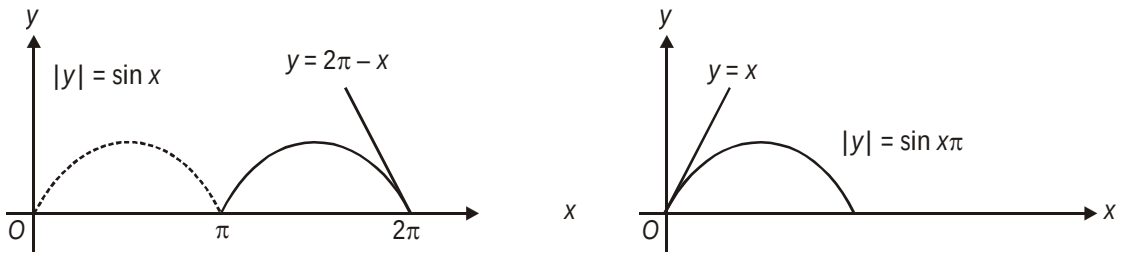
In $[\pi, 2\pi]$ In $[0, \pi]$, $|y| = \sin x$

$|y| = \sin x$, $y = \cos^{-1}\cos(2\pi - x) = 2\pi - x$

$y = \cos^{-1}\cos(x) = x$

One solution : $x = 2\pi$, $y = 0$

One solution : $x = 0$, $y = 0$

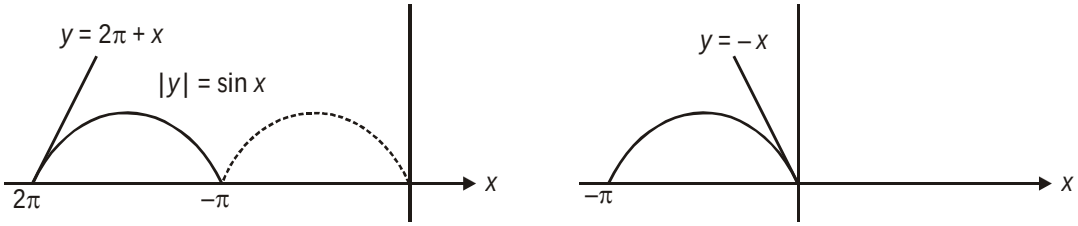


In $[-2\pi, -\pi]$ In $[-\pi, 0]$, $|y| = \sin x$

$|y| = \sin x$, $y = \cos^{-1}\{\cos(2\pi + x) = 2\pi + x$ $y = \cos^{-1}(\cos(-x) = -x$

One solution : $x = -2\pi$, $y = 0$

One solution : $x = 0$, $y = 0$



$\therefore$ There are 3 solutions (0, 0), $(2\pi, 0)$, $(-2\pi, 0)$.

Example 31 :

Find the value of $\cos^{-1} x + \cos^{-1}\left(\frac{x}{2} + \frac{\sqrt{3-3x^2}}{2}\right)$ where $\left(\frac{1}{2} \le x \le 1\right)$

Solution :

Let $x = \cos\theta$, where $0 \le \theta \le \frac{\pi}{3}$ because $\frac{1}{2} \le x \le 1$.

Given expression is = $\cos^{-1}\cos\theta + \cos^{-1}\left[\frac{1}{2}\cos\theta + \frac{\sqrt{3}}{2}\sin\theta\right]$

$= \theta + \cos^{-1}\cos\left(\theta - \frac{\pi}{3}\right)$; $0 \le \theta \le \frac{\pi}{3} \Rightarrow \frac{-\pi}{3} \le \theta - \frac{\pi}{3} \le 0$ $= \theta + \cos^{-1}\cos\left(\frac{\pi}{3} - \theta\right) = \theta + \frac{\pi}{3} - \theta = \frac{\pi}{3}$

Example 32 :

Solve for x : $\tan^{-1}(1 + x) + \tan^{-1} x + \tan^{-1}(x - 1) = \tan^{-1} 3$

Solution :

$\tan^{-1}(1 + x) + \tan^{-1}(x - 1) = \tan^{-1} 3 - \tan^{-1} x$

$\tan^{-1}\left(\frac{1+x+x-1}{1-x^2+1}\right) = \tan^{-1}\left(\frac{3-x}{1+3x}\right)$, when $x^2 - 1 < 1$ and $3x < 1$

$\Rightarrow \frac{2x}{2-x^2} = \frac{3-x}{1+3x}$, when $-\sqrt{2} < x < \frac{1}{3}$ $\Rightarrow$ $2x(1 + 3x) = (3 - x)(2 - x^2)$. when $-\sqrt{2} < x < \frac{1}{3}$

$\Rightarrow$ $2x + 6x^2 = 6 - 2x - 3x^2 + x^3$ $\Rightarrow$ $x^3 - 9x^2 - 4x + 6 = 0$

$\Rightarrow$ $(x + 1)(x^2 - 10x + 6) = 0$, when $-\sqrt{2} < x < \frac{1}{3}$

$\Rightarrow$ $x = -1$ and neglecting $x^2 - 10x + 6 = 0$ as its root does not $\in \left(-\sqrt{2}, \frac{1}{3}\right)$ $\therefore$ $x = -1$

Example 33 :

If $p > q > 0$ and $pr < -1 < qr$, then prove that

$$\tan^{-1}\frac{p-q}{1+pq}+\tan^{-1}\frac{q-r}{1+qr}+\tan^{-1}\frac{r-p}{1+rp}=\pi$$

Solution :

Since $p, q > 0$, therefore for $pq > 0$, $\tan^{-1}\frac{p-q}{1+pq}=\tan^{-1}p-\tan^{-1}q$ .... (1)

Since $qr > -1$, $\tan^{-1}\frac{q-r}{1+qr}=\tan^{-1}q-\tan^{-1}r$ .... (2)

Since $pr < -1$ and $r < 0$ $\tan^{-1}\frac{r-p}{1+rp}=\pi+\tan^{-1}r-\tan^{-1}p$ .... (3)

On adding (1), (2) and (3) we get

$$\tan^{-1}\frac{p-q}{1+pq}+\tan^{-1}\frac{q-r}{1+qr}+\tan^{-1}\frac{r-p}{1+rp}=\pi$$

Example 34 :

If $\tan^{-1}y = 4\tan^{-1}x$ $(|x|<\tan\frac{\pi}{8})$, find y as an algebraic function of x and hence prove that $\tan\frac{\pi}{8}$ is a root of the equation
$x^4 - 6x^2 + 1 = 0$.

Solution :

We have $\tan^{-1}y = 4\tan^{-1}x = 2\tan^{-1}\frac{2x}{1-x^2}$ (as $|x| < 1$)

$$= \tan^{-1}\frac{\frac{4x}{(1-x^2)}}{1-\frac{4x^2}{(1-x^2)^2}}=\tan^{-1}\frac{4x(1-x^2)}{x^4-6x^2+1} \quad \left(\text{as } \left|\frac{2x}{1-x^2}\right| < 1\right)$$

$$\Rightarrow\ y=\frac{4x(1-x^2)}{x^4-6x^2+1}$$

If $x=\tan\frac{\pi}{8}\Rightarrow\tan^{-1}y=4\tan^{-1}x=\frac{\pi}{2}\Rightarrow y=\infty\Rightarrow x^4-6x^2+1=0$

Example 35 :

If $\sin^{-1}\left(\frac{2p}{1+p^2}\right)-\cos^{-1}\left(\frac{1-q^2}{1+q^2}\right)=\tan^{-1}\frac{2x}{1-x^2}$, prove that $x=\frac{p-q}{1+pq}$, where $p, q \in (0, 1)$

Solution :

L.H.S. $= \sin^{-1}\left(\frac{2p}{1+p^2}\right)-\cos^{-1}\left(\frac{1-q^2}{1+q^2}\right) = 2\tan^{-1}p - 2\tan^{-1}q = 2[\tan^{-1}p - \tan^{-1}q] =$

$$2\tan^{-1}\left(\frac{p-q}{1+pq}\right)$$

$= 2\tan^{-1}x$ = R.H.S. where $x=\left(\frac{p-q}{1+pq}\right)$

Example 36 :

Show that $(\sin^{-1}x)^3+(\cos^{-1}x)^3=\alpha\pi^3$ has no real solutions for α < 1/32.

Solution :

$$\text{LHS} = (\sin^{-1}x+\cos^{-1}x)((\sin^{-1}x)^2-(\cos^{-1}x)(\sin^{-1}x)+(\cos^{-1}x)^2)=\alpha\pi^3$$

$$\Rightarrow \frac{\pi}{2}((\sin^{-1}x)^2-(\cos^{-1}x)(\sin^{-1}x)+(\cos^{-1}x)^2)=\alpha\pi^3$$

$$\Rightarrow \frac{\pi}{2}\left(\frac{\pi^2}{4}-3\cos^{-1}x\sin^{-1}x\right)=\alpha\pi^3 \Rightarrow \frac{\pi^2}{8}-\frac{3}{2}.\frac{\pi}{2}\sin^{-1}x+\frac{3}{2}(\sin^{-1}x)^2=\alpha\pi^2$$

$\Rightarrow 12y^2-6\pi y+\pi^2(1-8\alpha)=0$, where y = $\sin^{-1}$ x ..... (1)

In order to eq. (1) to have real roots we must have

$$D\geq 0\Rightarrow 36\pi^2-4.12\pi^2(1-8\alpha)\geq 0\Rightarrow 3-4+32\alpha\geq 0\Rightarrow \alpha\geq\frac{1}{32}$$

Example 37 :

Solve the equation $\sin(2\cos^{-1}(\cot(2\tan^{-1}x)))=0$

Solution :

$$\sin(2\cos^{-1}(\cot(2\tan^{-1}x)))=0$$

$$\Rightarrow 2\cos^{-1}(\cot(2\tan^{-1}x))=n\pi,\ n\in I$$

$$\Rightarrow \cos^{-1}(\cot(2\tan^{-1}x))=\frac{n\pi}{2},\ n\in I$$

Since we are interested only in principal values $\Rightarrow$ n = 0, 1, 2

$$\therefore \cos^{-1}(\cot(2\tan^{-1}x))=0,\pi/2,\pi$$

$$\Rightarrow \cot(2\tan^{-1}x)=1,0,-1 \Rightarrow 2\tan^{-1}x=m\pi+\frac{\pi}{4},m\pi+\frac{\pi}{2},m\pi-\frac{\pi}{4}$$

Hence again we are interested in principal values.

i.e. $-\pi<m\pi+\frac{\pi}{4}<\pi\Rightarrow m=0,-1\Rightarrow 2\tan^{-1}x=\frac{\pi}{4}$ and $-\frac{3\pi}{4}$ and correspondingly

$$x=\tan\left(\frac{\pi}{8}\right),\tan\left(\frac{-3\pi}{8}\right)$$

Similarly, $-\pi<m\pi-\frac{\pi}{4}<\pi\Rightarrow m=0,1 \Rightarrow x=\tan\left(\frac{-\pi}{8}\right),\tan\left(\frac{3\pi}{8}\right)$

$$-\pi<m\pi+\frac{\pi}{2}<\pi\Rightarrow m=0,-1 \Rightarrow x=\tan\left(\pm\frac{\pi}{4}\right)$$

Hence the results are $x=\pm\tan\left(\frac{\pi}{4}\right),\ \pm\tan\left(\frac{\pi}{8}\right),\ \pm\tan\left(\frac{3\pi}{8}\right)$

UNSOLVED EXERCISE

Section-A

(Straight objective type with single option correct)

Topic: 1 Domain and Range Based Questions

Q.1 Solution set of $[\sin^1 x] > [\cos^{-1}x]$, where [.] denotes the greater integer function is

(a) [sin 1, 1] (b) $\left[\frac{1}{\sqrt{2}},1\right]$

(c) (cos 1, sin 1) (d) None of these

Q.2 All possible values of p and q for which

$$\cos^{-1}\sqrt{p}+\cos^{-1}\sqrt{1-p}+\cos^{-1}\sqrt{1-q}=\frac{3\pi}{4}$$

is satisfied is

(a) $p>1,q=\frac{1}{2}$ (b) $q>1,p=\frac{1}{2}$

(c) $0\le p\le 1,q=\frac{1}{2}$ (d) None of these

Q.3 The set of values of k for which $x^2 - kx + \sin^{-1}(\sin 4) > 0$ for all real x is

(a) ϕ (b) (–2, 2)

(c) R (d) None of these

Q.4 Which of the following is the solution set of the equation

$$2\cos^{-1}(x)=\cot^{-1}\left(\frac{2x^2-1}{2x\sqrt{1-x^2}}\right)?$$

(a) (0, 1) (b) (–1, 1) – {0)

(c) (–1, 0) (d) [–1, 1]

Q.5 Domain of $f(x)=\sqrt{\sin^{-1}(2x)+\frac{\pi}{6}}$ is –

(a) $\left(-\frac{1}{2},\frac{1}{2}\right]$ (b) $\left[-\frac{1}{4},\frac{3}{2}\right)$

(c) $\left[-\frac{1}{4},\frac{1}{4}\right]$ (d) $\left[-\frac{1}{4},\frac{1}{2}\right]$

TOPIC: 2 PROPERTY OF INVERSE BASED QUESTIONS

Q.1 If $\sin^{-1}\left(x-\frac{x^2}{2}+\frac{x^3}{4}-....\right)+\cos^{-1}\left(x^2-\frac{x^4}{2}+\frac{x^6}{4}-...\right)=\frac{\pi}{2}$

For $0 < |x| < \sqrt{2}$, then x equals

(a) $\frac{1}{2}$ (b) 1

(c) $-\frac{1}{2}$ (d) –1

Q.2 The number of real solutions of $\tan^{-1}\sqrt{x(x+1)}+\sin^{-1}\sqrt{x^2+x+1}=\frac{\pi}{2}$ is

(a) Zero (b) One

(c) Two (d) Infinite

Q.3 The inequality $\sin^{-1}\sin(5) > x^2 - 4x$ holds if

(a) $x=2-\sqrt{9-2\pi}$

(b) $x=2+\sqrt{9-2\pi}$

(c) $x\in(2-\sqrt{9-2\pi},2+\sqrt{9-2\pi})$

(d) $x>2+\sqrt{9-2\pi}$

Q.4 If $\sin^{-1}x + \sin^{-1}(1-x) = \cos^{-1}x$, then x will be equal to

(a) 1, – 1 (b) 1, 0

(c) $0,\frac{1}{2}$ (d) None of these

Q.5 If $A = 2\tan^{-1}(2\sqrt{2}-1)$ and $B = 3\sin^{-1}\frac{1}{3}+\sin^{-1}\frac{3}{5}$ then

(a) $A > B$ (b) $A < B$

(c) $A = B$ (d) None of these

Q.6 Let $f(x) = \sin^{-1}x + \cos^{-1}x$ then

(a) $f\left(-\frac{1}{2}\right)=\frac{\pi}{2}$

(b) $f(-2)=\frac{\pi}{2}$

(c) $f\left(\frac{1}{1+\pi^2}\right)=\pi,k\in R$

(d) $f(k^2-2k+3)=\frac{\pi}{2},k\in R$

Q.7 The number of real solutions of (x, y), where $|y| = \sin x$ and $y = \cos^{-1}(\cos x)$, $-2\pi \le x \le 2\pi$ is

(a) 2 (b) 1

(c) 3 (d) 4

Q.8 If m = number of positive integral solutions of equation $\tan^{-1}x + \cot^{-1}y = \tan^{-1}3$ and for positive integral value of n, $(1 + ax)^n = 1 + 8x + 24x^2 + ...$ then which of the following is incorrect

(a) $m - a = 2$ (b) $m + a = 4$
(c) $2m - a = 2$ (d) $a^m = 4$

Q.9 If the mapping $f(x) = ax + b$, $a > 0$ maps $[-1, 1]$ onto $[0, 2]$, then $\cot[\cot^{-1}7 + \cot^{-1}8 + \cot^{-1}18] =$
(a) $f(1)$ (b) $f(0)$
(c) $f(2)$ (d) $f(-1)$

Q.10 If $\sum_{i=1}^{n}\sin^{-1}x_i = \frac{n\pi}{2}$ then $\sum_{i=1}^{n}x_i$ is
(a) n (b) $2n$
(c) $\frac{n(n+1)}{2}$ (d) None of these

Q.11 The greatest and least value of $(\sin^{-1}x)^3 + (\cos^{-1}x)^3$ is
(a) $-\frac{\pi}{2}, \frac{\pi}{2}$ (b) $\frac{-\pi^3}{8}, \frac{\pi^3}{8}$
(c) $\frac{\pi^3}{3}, \frac{7\pi^3}{8}$ (d) None of these

Q.12 Considering the principal value of inverse trigonometric functions, the solutions of the equation $2(\text{arc sin } x)^2 - (\text{arc sin } x) - 6 = 0$ are $x =$ .....
(a) $\sin\frac{7\pi}{11}$ (b) $-\sin\frac{21\pi}{11}$
(c) a and b both (d) None of these

Q.13 The value of the expression $\cos^{-1}[\cos\{2\cot^{-1}(\sqrt{2}-1)\}]$ is
(a) $\frac{\pi}{2}$ (b) $\frac{\pi}{4}$
(c) $\frac{3\pi}{4}$ (d) None of these

Q.14 The value of the angle $\tan^{-1}(\tan65° - 2\tan40°)$ in degrees is equal to
(a) $-20°$ (b) $20°$
(c) $40°$ (d) $25°$

Q.15 If $f(x) = 2x + 1$ and $g(x) = \frac{x-1}{2}$ for all real x, then $(fog)^{-1}\left(\frac{1}{x}\right)$ is equal to
(a) x (b) $\frac{1}{x}$
(c) $-x$ (d) $-\frac{1}{x}$

Q.16 Which one of the following quantities is negative
(a) $\tan(\cos^{-1}(\cos5))$ (b) $\cot(\sin^{-1}(\sin4))$
(c) $\cos(\tan^{-1}(\tan4))$ (d) $\sin(\cot^{-1}(\cot4))$

Q.17 If $\sin^{-1}x = \pi/5$, $x \in (-1, 1)$, then $\cos^{-1}x$ equals
(a) $3\pi/10$ (b) $5\pi/10$
(c) $-3\pi/10$ (d) $9\pi/10$

Q.18 The number of real solution of $\tan^{-1}\sqrt{x(x+1)} + \sin^{-1}\sqrt{x^2+x+1} = \frac{\pi}{2}$ is –
(a) zero (b) One
(c) Two (d) Infinite

Q.19 If $0 < x < 1$, then $\sqrt{1+x^2}\left[\left\{x\cos(\cot^{-1}x) + \sin(\cot^{-1}x)\right\}^2\right]^{1/2}$
(a) $\frac{x}{\sqrt{1+x^2}}$ (b) x
(c) $x\sqrt{1+x^2}$ (d) $\sqrt{1+x^2}$

TOPIC: 3 SUM AND DIFFERENENCE FORMULAE BASED QUESTIONS

Q.1 The solution of $\tan^{-1}2x + \tan^{-1}3x + \frac{\pi}{4}$ is
(a) $\frac{1}{6}$ (b) -1
(c) $\left(\frac{1}{6}, -1\right)$ (d) None of these

Q.2 $\tan^{-1}\left(\frac{1}{3}\right) + \tan^{-1}\left(\frac{1}{7}\right) + \tan^{-1}\left(\frac{1}{13}\right) + ... + \tan^{-1}\left(\frac{1}{n^2+n+1}\right)$ + ... to ∞ is equal to
(a) $\frac{2\pi}{3}$ (b) 0
(c) $\frac{\pi}{2}$ (d) $\frac{\pi}{4}$

Q.3 The value of $\cos(2\cos^{-1}x + \sin^{-1}x)$ at $x = \frac{1}{5}$ is
(a) 1 (b) 3
(c) 0 (d) $\frac{-2\sqrt{6}}{5}$

Q.4 If $\sin^{-1} x + \sin^{-1} y + \sin^{-1} z = \frac{3\pi}{2}$ and $f(1) = 2$, $f(p + q) = f(p)\, f(q)$ for all p, q then

$$x^{f(1)} + y^{f(2)} + z^{f(3)} - \frac{x+y+z}{x^{f(1)} + y^{f(2)} + z^{f(3)}}$$

(a) 0 (b) 1
(c) 2 (d) 3

Q.5 If $\cos^{-1} x + \cos^{-1} y + \cos^{-1} z = 3\pi$ then $xy + yz + zx$ = ...
(a) 0 (b) 3
(c) 1 (d) None of these

Q.6 If $2\tan^{-1} x + \sin^{-1} \frac{2x}{1+x^2}$ is independent of x then x lies in the interval
(a) $[1, \infty)$ (b) $[-1, 1]$
(c) $(-\infty, -1]$ (d) None of these

Q.7 The value of $\tan^{-1} 1 + \tan^{-1} 2 + \tan^{-1} 3$ is
(a) 0 (b) 1
(c) π (d) $-\pi$

Q.8 If $a_1, a_2, a_3, \ldots, a_n$ form an A.P. with common difference d ($a_i > 0$, $d > 0$), then the value of

$$\tan^{-1}\frac{d}{1+a_1a_2} + \tan^{-1}\frac{d}{1+a_2a_3} + \ldots + \tan^{-1}\frac{d}{1+a_{n-1}a_n}$$

is
(a) $n \tan^{-1} a_1$ (b) $n \tan^{-1} a_n$
(c) $\tan^{-1} a_1 - \tan^{-1} a_n$ (d) None of these

Q.9 If we consider only the principal values of the inverse trigonometric functions, then the value of

$$\tan\left(\cos^{-1}\frac{1}{5\sqrt{2}} - \sin^{-1}\frac{4}{\sqrt{17}}\right) \text{ is}$$

(a) $\sqrt{29}/3$ (b) 29/3
(c) $\sqrt{3}/29$ (d) 3/29

SECTION-B
(Previous Years Questions)

Q.1 The number of real solution of $\tan^{-1}\sqrt{x(x+1)} + \sin^{-1}\sqrt{x^2+x+1} = \frac{\pi}{2}$ is
(a) zero (b) one
(c) two (d) infinite

Q.2 If $\alpha = 3\sin^{-1}\left(\frac{6}{11}\right)$ and $\beta = 3\cos^{-1}\left(\frac{4}{9}\right)$, where the inverse trigonometric functions take only the principal values, then the correct option(s) is/are
(a) $\cos\beta > 0$ (b) $\sin\beta < 0$
(c) $\cos(\alpha + \beta) > 0$ (d) $\cos\alpha < 0$

Q.3 If $\sin^{-1}\left(x - \frac{x^2}{2} + \frac{x^3}{4} - \ldots\right) + \cos^{-1}\left(x^2 - \frac{x^4}{2} + \frac{x^6}{4} - \ldots\right) = \frac{\pi}{2}$, for $0 < |x| < \sqrt{2}$, then x equals
(a) $\frac{1}{2}$ (b) 1
(c) $-\frac{1}{2}$ (d) -1

Q.4 The principal value of $\sin^{-1}\left(\sin\frac{2\pi}{3}\right)$ is
(a) $-\frac{2\pi}{3}$ (b) $\frac{2\pi}{3}$
(c) $\frac{\pi}{3}$ (d) $\frac{5\pi}{3}$

Q.5 If $\tan^{-1} y = \tan^{-1} x + \tan^{-1}\left(\frac{2x}{1-x^2}\right)$, where $|x| < \frac{1}{\sqrt{3}}$. Then, the value of y is
(a) $\frac{3x - x^3}{1 - 3x^2}$ (b) $\frac{3x + x^2}{1 - 3x^2}$
(c) $\frac{3x - x^2}{1 + 3x^2}$ (d) $\frac{3x + x^3}{1 + 3x^2}$

Q.6 The value of $\cot\left\{\sum_{n=1}^{23}\cot^{-1}\left(1 + \sum_{k=1}^{n} 2k\right)\right\}$ is
(a) $\frac{23}{25}$ (b) $\frac{25}{23}$
(c) $\frac{23}{24}$ (d) $\frac{24}{23}$

Q.7 If x, y and z are in AP and $\tan^{-1} x$ $\tan^{-1} y$ and $\tan^{-1} z$ are also in AP, then
(a) x = y = z (b) 2x = 3y = 6z
(c) 6x = 3y = 2z (d) 6x = 4y = 3z

Q.8 If $0 < x < 1$, then $\sqrt{1+x^2}$ [{x cos ($\cot^{-1} x$) + sin ($\cot^{-1} x$)}^{2} – 1]$^{1/2}$ is equal to
(a) $\frac{x}{\sqrt{1+x^2}}$ (b) x
(c) $x\sqrt{1+x^2}$ (d) $\sqrt{1+x^2}$

Q.9 The value of x for which $\sin[\cot^{-1}(1+x)] = \cos(\tan^{-1}x)$ is

(a) $\frac{1}{2}$ (b) 1

(c) 0 (d) $-\frac{1}{2}$

Q.10 The value of $\tan\left[\cos^{-1}\left(\frac{4}{5}\right)+\tan^{-1}\left(\frac{2}{3}\right)\right]$ is

(a) $\frac{6}{17}$ (b) $\frac{17}{6}$

(c) $\frac{16}{7}$ (d) None of these

SECTION-C
(School / Board pattern)

Q.1 Find the value of $\tan^{-1}\left(\tan\frac{5\pi}{6}\right)+\cos^{-1}\left(\cos\frac{13\pi}{6}\right)$

Q.2 Find the value of $\tan^{-1}\left(-\frac{1}{\sqrt{3}}\right)+\cot^{-1}\left(\frac{1}{\sqrt{3}}\right)+\tan^{-1}\left[\sin\left(\frac{-\pi}{2}\right)\right]$

Q.3 Show that $2\tan^{-1}(-3)=\frac{-\pi}{2}+\tan^{-1}\left(\frac{-4}{3}\right)$.

Q.4 find the real solution of $\tan^{-1}\sqrt{x(x+1)}+\sin^{-1}\sqrt{x^2+x+1}=\frac{\pi}{2}$.

Q.5 If $2\tan^{-1}(\cos\theta)=\tan^{-1}(2\operatorname{cosec}\theta)$, then show that $\theta=\frac{\pi}{4}$, where n is any integer.

Q.6 Show that $\cos\left(2\tan^{-1}\frac{1}{7}\right)=\sin\left(4\tan^{-1}\frac{1}{3}\right)$

Q.7 Solve the equation $\cos\left(\tan^{-1}x\right)=\sin\left(\cot^{-1}\frac{3}{4}\right)$.

Q.8 Find the simplified form of $\cos^{-1}\left(\frac{3}{4}\cos x+\frac{4}{5}\sin x\right)$, where $x\in\left[\frac{-3\pi}{4},\frac{\pi}{4}\right]$.

Q.9 Show that $\sin^{-1}\frac{5}{13}+\cos^{-1}\frac{3}{5}=\tan^{-1}\frac{63}{16}$.

Q.10 Find the value of $4\tan^{-1}\frac{1}{5}-\tan^{-1}\frac{1}{239}$.

Q.11 Show that $\tan\left(\frac{1}{2}\sin^{-1}\frac{3}{4}\right)=\frac{4-\sqrt{7}}{3}$ and justify why the other value $\frac{4+\sqrt{7}}{3}$ is ignored?

Q.12 If $a_1, a_2, a_3, \ldots\ldots, a_n$ is an arithmetic progression with common difference d, then evaluate the following expression.

$$\tan\left[\tan^{-1}\left(\frac{d}{1+a_1a_2}\right)+\tan^{-1}\left(\frac{d}{1+a_2a_3}\right)+\tan^{-1}\left(\frac{d}{1+a_3a_4}\right)+\ldots\ldots+\tan^{-1}\left(\frac{d}{1+a_{n-1}a_n}\right)\right]$$

CHAPTER 3 MATRICES & DETERMINANTS

1. MATRICES

INTRODUCTION

Matrices are one of the most powerful tools in mathematics. The evolution of concept of matrices is the result of an attempt to obtain compact and simple methods of solving system of linear equations. Many physical operations such as magnification, rotation and reflection through a plane can be represented mathematically by matrices.

The number $a_1b_2 - a_2b_1$ which determines uniqueness of solution is associated with the matrix

$A = \begin{bmatrix} a_1 & b_1 \\ a_2 & b_2 \end{bmatrix}$ and is called the determinant of A or det A. Determinants have wide applications in Engineering, Science, Economics, Social Science, etc.

Determinants can be used in finding the area of a triangle, adjoint and inverse of a square matrix, consistency and inconsistency of system of linear equations and solution of linear equations in two or three variables.

MATRIX

Definition : A rectangular arrangement of numbers in rows and columns, is called a Matrix. This arrangement is enclosed by small ( ) or big [ ] bracket. A matrix is represented by capital letters A, B, C etc. and its element are by small letters a, b, c, x, y etc.

Ex. $A = \begin{bmatrix} a & b & c \\ d & e & f \end{bmatrix}$, $B = \begin{bmatrix} -1 & 3 & 2 \\ 2 & 9 & 4 \end{bmatrix}$, $C = \begin{bmatrix} 3 & 4 \\ 5 & 6 \end{bmatrix}$, $D = [\ell]$, $E = \begin{bmatrix} a \\ g \\ h \end{bmatrix}$, $F = [1, 4, 9]$

Order of a matrix : A matrix which has m rows and n columns is called a matrix of order $m \times n$. A matrix A of order $m \times n$ is usually written in the following manner –

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} & \ldots a_{1j} & \ldots a_{1n} \\ a_{21} & a_{22} & a_{23} & \ldots a_{2j} & \ldots a_{2n} \\ \ldots\ldots & \ldots\ldots & \ldots\ldots & \ldots\ldots & \ldots\ldots \\ a_{i1} & a_{i2} & a_{i3} & \ldots a_{ij} & \ldots a_{in} \\ \ldots\ldots & \ldots\ldots & \ldots\ldots & \ldots\ldots & \ldots\ldots \\ a_{m1} & a_{m2} & a_{m3} & \ldots a_{mj} & \ldots a_{mn} \end{bmatrix} \quad \text{or} \quad A = [a_{ij}]_{m \times n} \text{ where } \begin{matrix} i = 1,2,\ldots.m \\ j = 1,2,\ldots.n \end{matrix}$$

Here a_{ij} denotes the element of ith row and j^{th} column. Example : Order of matrix $\begin{bmatrix} 1 & -2 \\ 3 & 5 \\ 4 & -7 \end{bmatrix}$ is 3×2

2. DIFFERENT TYPES OF MATRICES

(i) Square Matrix: A matrix for which the number of rows is equal to the number of columns (each equal to n) is called a square matrix of order n.

Ex. $A = \begin{bmatrix} 1 & 0 & -2 & 3 \\ 2 & -1 & 4 & 5 \\ 3 & 2 & 4 & -1 \\ 1 & 0 & 0 & 2 \end{bmatrix}$ is a square matrix of order 4.

(ii) **Null Matrix:** The matrix whose all elements are zero is called null matrix or zero matrix. It is usually denoted by O.

(iii) **Identity Matrix :** A square matrix in which all the elements along the main diagonal (elements of the from a_{ii}) are unity is called an identity matrix or a unit matrix. An identity matrix of order n is denoted by I_n.

Ex. $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(iv) **Scalar matrix :** A matrix whose diagonal elements are all equal and other entries are zero, is called a scalar matrix

Ex. $A = \begin{bmatrix} k & 0 & 0 \\ 0 & k & 0 \\ 0 & 0 & k \end{bmatrix}$

(v) **Triangular Matrix:** A square matrix whose elements above the main diagonal or below the main diagonal are all zero is called a triangular matrix.

(a) $[a_{ij}]_{n \times n}$ is said to be upper triangular matrix if $i > j \Rightarrow a_{ij} = 0$,

(b) $[a_{ij}]_{n \times n}$ is said to be lower triangular matrix if $i < j \Rightarrow a_{ij} = 0$.

Ex. $\begin{bmatrix} 1 & 2 & -1 & 3 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 3 & -1 \\ 0 & 0 & 0 & -3 \end{bmatrix}$ is an upper triangular matrix.

(vi) **Diagonal Matrix:** A square matrix of any order with zero elements every where, except on the main diagonal, is called a diagonal matrix.

Ex. $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & -1 \end{bmatrix}$ is a diagonal matrix of order 3.

(vii) **Row Matrix:** A matrix which has only one row and n columns is called a row matrix of length n.

Ex. $[2 \; -1 \; 3 \; 0]_{1\times 4}$ is a row matrix of length 4.

(viii) **Column Matrix:** A matrix which has only one column and m rows is called a column matrix of length m.

Ex. $x = \begin{bmatrix} 1 \\ 3 \\ -2 \\ -1 \end{bmatrix}$ is a column matrix of length 4.

(ix) **Sub Matrix:** A matrix obtained by omitting some rows or some columns or both of a given matrix A is called a sub matrix of A.

Ex. If $A = \begin{bmatrix} 2 & 0 & 4 \\ 5 & 6 & 8 \\ 3 & -2 & 29 \end{bmatrix}$, then $\begin{bmatrix} 2 & 0 \\ 5 & 6 \end{bmatrix}$ is a submatrix of A which is obtained by omitting third row and third column of A.

(x) Singleton Matrix : If in a Matrix there is only one element then it is called Singleton Matrix. Thus $A = [a_{ij}]_{m\times n}$ is *a* Singleton Matrix if $m = n = 1$

Ex. [*b*], [6], [-*a*], [–9] are Singleton Matrices.

(xi) Equal Matrix :

Two Matrix *A* and *B* are said to be equal Matrix if they are of same order and their corresponding elements are equal.

Ex. If $A = \begin{bmatrix} -1 & 6 & 9 \\ -6 & 3 & 11 \end{bmatrix}$ and $B = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{bmatrix}$ are equal Matrix then

$a_1 = -1$, $a_2 = -6$, $b_1 = 6$, $b_2 = 3$, $c_1 = 9$, $c_2 = 11$

(xii) Singular Matrix :

Matrix *A* is said to be singular matrix if its determinant $|A| = 0$, otherwise non-singular matrix i.e.

If $\det |A| = 0 \Rightarrow$ singular

and $\det |A| \neq 0 \Rightarrow$ non-singular

(xiii) Trace of a matrix : The sum of the elements of a square matrix *A* lying along the principal diagonal is called the trace of *A* i.e. *tr*(*A*).

Thus if $A = [a_{ij}]_{n\times n}$

Then $tr(A) = \sum_{i=1}^{n} a_{ij} = a_{11} + a_{22} + ... + a_{nn}$

Properties of trace of a matrix : Let $A = [a_{ij}]_{n\times n}$ and $B = [b_{ij}]_{n\times n}$ and λ be a scalar.

(i) $tr(\lambda A) = \lambda\, tr(A)$

(ii) $tr(A + B) = tr(A) + tr(B)$

(iii) $tr(AB) = tr(BA)$

3. ADDITION AND SUBTRACTION OF MATRICES :

If $A = [a_{ij}]_{m\times n}$ and $B = [b_{ij}]_{m\times n}$ are two matrices of the same order then their sum $A + B$ is a matrix whose each element is the sum of corresponding element.

i.e. $A + B = [a_{ij} + b_{ij}]_{m\times n}$

Ex. If $A = \begin{bmatrix} 3 & 5 & 7 \\ 2 & 4 & 6 \end{bmatrix}$ and $B = \begin{bmatrix} 2 & 3 & 4 \\ 1 & 6 & 9 \end{bmatrix}$ then $A + B = \begin{bmatrix} 3+2 & 5+3 & 7+4 \\ 2+1 & 4+6 & 6+9 \end{bmatrix} = \begin{bmatrix} 5 & 8 & 11 \\ 3 & 10 & 15 \end{bmatrix}$

Similarly their subtraction $A - B$ is defined as

$A - B = [a_{ij} - b_{ij}]_{m\times n}$ i.e. in above example $A - B = \begin{bmatrix} 3-2 & 5-3 & 7-4 \\ 2-1 & 4-6 & 6-9 \end{bmatrix} = \begin{bmatrix} 1 & 2 & 3 \\ 1 & -2 & -3 \end{bmatrix}$

Matrix addition and substraction can be possible only when matrices are of same order.

Properties of Matrics addition : If *A*, *B* and *C* are Matrices of same order, then –

(i) Commutative Law : $A + B = B + A$

(ii) Associative Law : $(A + B) + C = A + (B + C)$

(iii) $A + O = O + A = A$, where O is zero matrix which is additive identity of the matrix.

(iv) $A + (-A) = 0 = (-A) + A$, where $(-A)$ is obtained by changing the sign of every element of *A* which is additive inverse of the matrix.

(v) Cancellation Law : $\left.\begin{matrix} A+B=A+C \\ B+A=C+A \end{matrix}\right\} \Rightarrow B = C$

(vi) $tr(A \pm B) = tr(A) \pm tr(B)$

MULTIPLICATION OF A MATRIX BY A SCALAR :

If $A = [a_{ij}]_{m \times n}$ and λ is a scalar, then $lA = [\lambda a_{ij}]_{m \times n}$

Ex. $A = \begin{bmatrix} 2 & 1 & 3 \\ 2 & 5 & -1 \\ 0 & 1 & 2 \end{bmatrix} \Rightarrow -3A = \begin{bmatrix} -6 & -3 & -9 \\ -6 & -15 & 3 \\ 0 & -3 & -6 \end{bmatrix}$.

Properties of scalar multiplication :

If $A = [a_{ij}]$ and $B = [b_{ij}]$ be two matrices of the same order, say $m \times n$, and k and ℓ are scalars, then

(i) $k(A + B) = kA + kB$, $(k + \ell)A = kA + \ell A$

(ii) $k(A + B) = k([a_{ij}] + [b_{ij}]) = k[a_{ij} + b_{ij}] = [k(a_{ij} + b_{ij})] = [(k a_{ij}) + (k b_{ij})]$
$= [k a_{ij}] + [k b_{ij}] = k[a_{ij}] + k[b_{ij}] = kA + kB$

(iii) $(k + \ell) A = (k + \ell)[a_{ij}] = [k a_{ij}] + [\ell a_{ij}] = k[a_{ij}] + \ell [a_{ij}] = kA + \ell A$

4. MULTIPLICATION OF TWO MATRICES :

Let $A = [a_{ij}]_{m \times n}$ and $B = [b_{jk}]_{n \times p}$ be two matrices such that the number of columns in A is equal to number of rows in B. Then the $m \times p$ matrix $C = [c_{ik}]_{m \times p}$, where $c_{ik} = \sum_{j=1}^{n} a_{ij} . b_{jk}$ (where i = 1, 2, 3 ... m, k = 1, 2, 3 ... p), is called the product of the matrices A and B. We have

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \dots & \dots & \dots & \dots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{bmatrix}_{m \times n} \begin{bmatrix} b_{11} & b_{12} & \dots & b_{1p} \\ b_{21} & b_{22} & \dots & b_{2p} \\ \dots & \dots & \dots & \dots \\ b_{n1} & b_{n2} & \dots & b_{np} \end{bmatrix}_{n \times p} = \begin{bmatrix} c_{11} & c_{12} & \dots & c_{1p} \\ c_{12} & c_{22} & \dots & c_{2p} \\ \dots & \dots & \dots & \dots \\ c_{m1} & c_{m2} & \dots & c_{mp} \end{bmatrix}_{m \times p},$$

where $C_{ij} = a_{i1} b_{1j} + a_{i2} b_{2j} + \dots + a_{in} b_{nj} = \sum_{k=1}^{n} a_{ik} b_{kj}$

Properties of matrix multiplication :

(i) Matrix multiplication is associative
i.e., $(AB)C = A(BC)$, A, B and C are $m \times n$, $n \times p$ and $p \times q$ matrices respectively.

(ii) Multiplication of matrices is distributive over addition of matrices :
i.e., $A(B + C) = AB + AC$

(iii) Existence of multiplicative identity of square matrices : If A is a square matrix of order n and I_n is the identity matrix of order n, then $AI_n = I_n A = A$.

(iv) Whenever AB and BA both exist, it is not necessary that $AB = BA$.

(v) The product of two matrices can be a zero matrix while neither of them is a zero matrix.

Ex. If $A = \begin{bmatrix} 0 & 1 \\ 0 & 0 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 0 & 0 \end{bmatrix}$ then $AB = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$, while neither A nor B is a null matrix.

(vi) In the case of matrix multiplication of $AB = 0$, then it doesn't necessarily imply that $A = 0$ or $B = 0$ or $BA = 0$.

Illustration 1 : If ω is an imaginary cube root of unity, show that

$$\begin{bmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{bmatrix} + \begin{bmatrix} \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \\ 1 & \omega & \omega^2 \end{bmatrix} + \begin{bmatrix} \omega^2 & 1 & \omega \\ 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \end{bmatrix}$$ is a null matrix.

Solution : $\begin{bmatrix} 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \end{bmatrix} + \begin{bmatrix} \omega & \omega^2 & 1 \\ \omega^2 & 1 & \omega \\ 1 & \omega & \omega^2 \end{bmatrix} + \begin{bmatrix} \omega^2 & 1 & \omega \\ 1 & \omega & \omega^2 \\ \omega & \omega^2 & 1 \end{bmatrix}$

$= \begin{bmatrix} 1+\omega+\omega^2 & 1+\omega+\omega^2 & 1+\omega+\omega^2 \\ 1+\omega+\omega^2 & 1+\omega+\omega^2 & 1+\omega+\omega^2 \\ 1+\omega+\omega^2 & 1+\omega+\omega^2 & 1+\omega+\omega^2 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$, which is a null matrix.

Illustration 2 : If $A = \begin{bmatrix} 1 & 3 \\ 3 & 2 \\ 2 & 5 \end{bmatrix}$ and $B = \begin{bmatrix} -1 & -2 \\ 0 & 5 \\ 3 & 1 \end{bmatrix}$ and $A + B - D = 0$ (zero matrix), find the matrix D

Solution : Let $D = \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix}$

$\therefore \quad A+B-C = \begin{bmatrix} 1 & 3 \\ 3 & 2 \\ 2 & 5 \end{bmatrix} + \begin{bmatrix} -1 & -2 \\ 0 & 5 \\ 3 & 1 \end{bmatrix} - \begin{bmatrix} a & b \\ c & d \\ e & f \end{bmatrix} \Rightarrow \begin{bmatrix} 1-1-a & 3-2-b \\ 3+0-c & 2+5-d \\ 2+3-e & 5+1-f \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}$

$\Rightarrow \quad -a = 0 \Rightarrow a = 0,\ 1 - b = 0 \Rightarrow b = 1,\ 3 - c = 0 \Rightarrow c = 3,\ 7 - d = 0 \Rightarrow d = 7,$
$5 - e = 0 \Rightarrow e = 5,\ 6 - f = 0 \Rightarrow f = 6,$

$\therefore \quad D = \begin{bmatrix} 0 & 1 \\ 3 & 7 \\ 5 & 6 \end{bmatrix}$

Illustration 3 : A matrix $A = [a_{ij}]$ of order 2×3 whose elements are such that $a_{ij} = i + j$ is -

(a) $\begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$ (b) $\begin{bmatrix} 2 & 3 & 4 \\ 5 & 4 & 3 \end{bmatrix}$ (c) $\begin{bmatrix} 2 & 3 & 4 \\ 5 & 5 & 4 \end{bmatrix}$ (d) None of these

Solution : (a) a_{ij} is the element of ith row and j^{th} column of matrix A

$\therefore \quad a_{11} = 1 + 1 = 2,\ a_{12} = 1 + 2 = 3,\ a_{13} = 1 + 3 = 4$
$a_{21} = 2 + 1 = 3.\ a_{22} = 2 + 2 = 4,\ a_{23} = 2 + 3 = 5$

$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \end{bmatrix} = \begin{bmatrix} 2 & 3 & 4 \\ 3 & 4 & 5 \end{bmatrix}$

Illustration 4 : If $A = \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$ and $A^2 - 4A - nI = 0$, then find the value of n.

Solution : $A^2 = \begin{bmatrix} 5 & -4 \\ -4 & 5 \end{bmatrix}$, $4A = \begin{bmatrix} 8 & -4 \\ -4 & 8 \end{bmatrix}$, $nI = \begin{bmatrix} n & 0 \\ 0 & n \end{bmatrix} \Rightarrow A^2 - 4A - nI$

$$= \begin{bmatrix} 5-8-n & -4+4-0 \\ -4+4-0 & 5-8-n \end{bmatrix} = \begin{bmatrix} -3-n & 0 \\ 0 & -3-n \end{bmatrix}$$

$\therefore \quad A^2 - 4A - nI = 0$

$$\Rightarrow \quad \begin{bmatrix} -3-n & 0 \\ 0 & -3-n \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \Rightarrow -3-n = 0 \Rightarrow n = -3$$

Illustration 5 : If matrix $P = \begin{bmatrix} 0 & -\tan(\theta/2) \\ \tan\theta/2 & 0 \end{bmatrix}$ then $(I-P)\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ is equal to -

(A) P (B) $P + I$ (C) I (D) None of these

Solution : (B) $I - P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} - \begin{bmatrix} 0 & -\tan(\theta/2) \\ \tan(\theta/2) & 0 \end{bmatrix} = \begin{bmatrix} 1 & \tan(\theta/2) \\ -\tan\theta/2 & 1 \end{bmatrix}$

$$\therefore \quad (I-P)\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix} = \begin{bmatrix} 1 & \tan(\theta/2) \\ -\tan\theta/2 & 1 \end{bmatrix} \cdot \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta + \tan(\theta/2)\sin\theta & -\sin\theta + \tan(\theta/2)\cos\theta \\ -\tan(\theta/2)\cos\theta + \sin\theta & \tan(\theta/2)\sin\theta + \cos\theta \end{bmatrix}$$

$$= \begin{bmatrix} 1-2\sin^2(\theta/2)+2\sin^2(\theta/2) & -2\sin(\theta/2)\cos(\theta/2)+\tan(\theta/2)(2\cos^2(\theta/2)-1) \\ -\tan(\theta/2)(2\cos^2\theta/2-1)+2\sin(\theta/2)\cos(\theta/2) & \tan(\theta/2)(2\sin(\theta/2)\cos(\theta/2))+(1-2\sin^2(\theta/2)) \end{bmatrix}$$

$$= \begin{bmatrix} 1 & -\tan(\theta/2) \\ \tan\theta/2 & 1 \end{bmatrix} = I + P$$

Student Task

Q.1 If $E(\theta) = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$, then value of $E(\alpha).E(\beta)$ is -

(a) $E(0^o)$ (b) $E(90^o)$ (c) $E(\alpha + \beta)$ (d) $E(\alpha - \beta)$

Q.2 If X and Y two matrices are such that $X - Y = \begin{bmatrix} 3 & 2 \\ -1 & 0 \end{bmatrix}$ and $X + Y = \begin{bmatrix} 1 & -2 \\ 3 & 4 \end{bmatrix}$ then Y matrices is

(a) $\begin{bmatrix} 2 & 0 \\ 1 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} -1 & -2 \\ 3 & 4 \end{bmatrix}$ (c) $\begin{bmatrix} -1 & -2 \\ 2 & 2 \end{bmatrix}$ (d) None of these

Q.3 If $X = \begin{bmatrix} 1 & a \\ 0 & 1 \end{bmatrix}$ and $3X - \begin{bmatrix} 2 & 3 \\ 0 & 2 \end{bmatrix} = \begin{bmatrix} 1 & 3 \\ 0 & 1 \end{bmatrix}$ then a is equal to -

(a) 1 (b) 2 (c) 0 (d) –2

Q.4 If $A = \begin{bmatrix} 1 & -3 & 2 \\ 2 & k & 5 \\ 4 & 2 & 1 \end{bmatrix}$ is a singular matrix, then k is equal to -

(a) –1 (b) 8 (c) 4 (d) 6

Q.5 If $[1 \ x \ 2]\begin{bmatrix}2 & 3 & 1\\0 & 4 & 2\\0 & 3 & 2\end{bmatrix}\begin{bmatrix}x\\1\\-1\end{bmatrix} = 0$, then the value of x is –

(a) –1 (b) 0 (c) 1 (d) 2

ANSWERS

(1) (c), (2) (c), (3) (b), (4) (d), (5) (a)

TRANSPOSE OF A MATRIX

The matrix obtained from any given matrix A, by interchanging rows and columns, is called the transpose of A and is denoted by A' or A^T .

Ex. If $A=\begin{bmatrix}1 & -2\\4 & 5\\7 & 8\end{bmatrix}_{3\times 2}$, then $A'=\begin{bmatrix}1 & 4 & 7\\-2 & 5 & 8\end{bmatrix}_{2\times 3}$

Properties of Transpose of a matrix

(i) $(A')'=A$

(ii) $(A+B)'=A'+B'$

(iii) $(\alpha A)'=\alpha A'$, α being any scalar

(iv) $(AB)'=B'A'$

CONJUGATE OF A MATRIX :

The matrix obtained from any given matrix A, containing complex number as its elements, on replacing its elements by the corresponding conjugate complex numbers is called the conjugate of A and is denoted by $\bar{A}$.

Ex. $A=\begin{bmatrix}1+2i & 2-3i & 3+4i\\4-5i & 5+6i & 6-7i\\8 & 7+8i & 7\end{bmatrix}$, then $\bar{A}=\begin{bmatrix}1-2i & 2+3i & 3-4i\\4+5i & 5-6i & 6+7i\\8 & 7-8i & 7\end{bmatrix}$

Properties of Conjugate of a matrix :

(i) $\overline{(\bar{A})}=A$ (ii) $\overline{(A+B)}=\bar{A}+\bar{B}$

(iii) $\overline{(\alpha A)}=\bar{\alpha}\bar{A}$, α being any scalar (iv) $\overline{(AB)}=\bar{A}\bar{B}$

Transpose conjugate of a matrix : The transpose of the conjugate of a matrix A is called transposed conjugate of A and is denoted by A^θ. The conjugate of the transpose of A is the same as the transpose of the conjugate of A i.e., $\overline{(A')}=\overline{(A)}'=A^\theta$

If $A = [a_{ij}]_{m\times n}$, then $A^\theta=[b_{ji}]_{n\times m}$, where $b_{ji} = \bar{a}_{ij}$

i.e., the $(j, i)^{th}$ element of A^θ = the conjugate of $(i, j)^{th}$ element of A.

Ex. If $A=\begin{bmatrix}1+2i & 2-3i & 3+4i\\4-5i & 5+6i & 6-7i\\8 & 7+8i & 7\end{bmatrix}$, then $A^\theta=\begin{bmatrix}1-2i & 4+5i & 8\\2+3i & 5-6i & 7-8i\\3-4i & 6+7i & 7\end{bmatrix}$

Properties of transpose conjugate of a matrix :

(i) $(A^\theta)^\theta = A$ (ii) $(A+B)^\theta = A^\theta + B^\theta$

(iii) $(kA)^\theta = \bar{k}A^\theta$, k being any scalar (iv) $(AB)^\theta = B^\theta A^\theta$

5. SYMMETRIC AND SKEW-SYMMETRIC MATRIX

Symmetric Matrix : A square matrix $A = [a_{ij}]$ is called symmetric matrix if $a_{ij} = a_{ji}$ for all i, j or $A^T = A$

Ex. $\begin{bmatrix} a & h & g \\ h & b & f \\ g & f & c \end{bmatrix}$

(i) Every unit matrix and square zero matrix are symmetric matrices.

(ii) Maximum number of different element in a symmetric matrix is $\frac{n(n+1)}{2}$.

Skew -Symmetric Matrix : A square matrix $A = [a_{ij}]$ is called skew-symmetric matrix. if $a_{ij} = -a_{ji}$ for all i, j or $A^T = -A$

Ex. $\begin{bmatrix} 0 & h & g \\ -h & 0 & f \\ -g & -f & 0 \end{bmatrix}$

(i) All Principal diagonal elements of a skew-symmetric matrix are always zero because for any diagonal element
$a_{ii} = -a_{ii} \Rightarrow a_{ii} = 0$

(ii) Trace of a skew symmetric matrix is always 0

Properties of Symmetric and skew-symmetric matrices

(i) If A is a square matrix, then $A + A^T, AA^T, A^TA$ are symmetric matrices while $A - A^T$ is skew-symmetric matrices.

(ii) If A is a symmetric Matrix, then $-A$, KA, A^T, A^n, A^{-1}, $B^T AB$ are also symmetric matrices where $n \in N$, $K \in R$ and B is a square matrix of order that of A.

(iii) If A is a skew symmetric matrix, then :
(a) A^{2n} is a symmetric matrix for $n \in N$
(b) A^{2n+1} is a skew-symmetric matrices for $n \in N$
(c) kA is also skew -symmetric matrix where k $\in R$
(d) $B^T AB$ is also skew - symmetric matrix where B is a square matrix of order that of A.

(iv) If A, B are two symmetric matrices, then –
(a) $A \pm B$, $AB + BA$ are also symmetric matrices.
(b) $AB - BA$ is a skew-symmetric matrix
(c) AB is a symmetric matrix when $AB = BA$.

(v) If A, B are two skew-symmetric matrices, then –
(a) $A \pm B$, $AB - BA$ are skew-symmetric matrices
(b) $AB + BA$ is a symmetric matrix

(vi) Every square matrix A can uniquely be expressed as sum of a symmetric and skew symmetric matrix i.e.

$$A = \left[\frac{1}{2}(A + A^T)\right] + \left[\frac{1}{2}(A - A^T)\right]$$

SOME SPECIAL TYPE OF MATRIX

1. Hermitian and Skew-Hermitian Matrix

A square matrix $A = [a_{ij}]$ is said to be Hermitian if $a_{ij} = \bar{a}_{ji} \ \forall i, j$ i.e., $A = A^{\theta}$.

e.g., $\begin{bmatrix} a & b+ic \\ b-ic & d \end{bmatrix}$, $\begin{bmatrix} 3 & 3-4i & 5+2i \\ 3+4i & 5 & -2+i \\ 5-2i & -2-i & 2 \end{bmatrix}$ are Hermitian matrices.

If A is a Hermitian matrix then $\bar{a}_{ii} = a_{ii} \Rightarrow a_{ii}$ is real $\forall i$. Thus every diagonal element of a Hermitian Matrix must be real.

A square matrix A = [a_{ij}] is said to be a skew-Hermitian if $a_{ij} = -\bar{a}_{ji}, \forall i, j$ i.e., $A^{\theta} = -A$.

Ex. $\begin{bmatrix} 0 & -2+i \\ 2+i & 0 \end{bmatrix}, \begin{bmatrix} 3i & -3+2i & -1-i \\ 3+2i & -2i & -2-4i \\ 1-i & 2-4i & 0 \end{bmatrix}$ are skew-Hermitian matrices.

If A is a skew-Hermitian matrix, then $a_{ij} = -\bar{a}_{ij}$ i.e., a_{ij} must be purely imaginary or zero.

2. Orthogonal matrix : A square matrix A is said to be orthogonal, if $AA' = A'A = I$, where I is a unit matrix.

(i) If A is orthogonal, then A' is also orthogonal.

(ii) If A and B are orthogonal matrices then AB and BA are also orthogonal matrices.

3. Unitary matrix : A square matrix A is called unitary matrix if $AA^{\theta} = A^{\theta}A = I$.

4. Idempotent matrix: A square matrix A is called idempotent provided it satisfies the relation $A^2 = A$.

Ex. : The matrix $A = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$ is idempotent as

$$A^2 = \begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}\begin{bmatrix} 2 & -2 & -4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix} = \begin{bmatrix} 2 & -2 & 4 \\ -1 & 3 & 4 \\ 1 & -2 & -3 \end{bmatrix}$$

5. Involutary matrix : A matrix A such that $A^2 = I$, is called involutary matrix.

6. Nilpotent matrix : A square matrix A is called a nilpotent matrix, if there exists a positive integer m such that $A^m = 0$. If m is the least positive integer such that $A^m = 0$, then m is called the index of the nilpotent matrix A.

7. ELEMENTARY OPERATION (TRANSFORMATION) OF A MATRIX :

(i) The interchange of any two rows or two columns : Symbolically the interchange of ith and j^{th} rows is denoted by $R_i \leftrightarrow R_j$ and interchange of ith and j^{th} column is denoted by $C_i \leftrightarrow C_j$.

For example, applying $R_1 \leftrightarrow R_2$ to $\begin{bmatrix} 1 & 2 & 1 \\ -1 & \sqrt{3} & 1 \\ 5 & 6 & 7 \end{bmatrix}$, we get $\begin{bmatrix} -1 & \sqrt{3} & 1 \\ 1 & 2 & 1 \\ 5 & 6 & 7 \end{bmatrix}$

(ii) The multiplication of the elements of any row or column by a non zero number : Symbolically, the multiplication of each element of the ith row by k, where $k \neq 0$ is denoted by $R_i \to k R_i$.

The corresponding column operation is denoted by $C_i \to kC_i$

For example, applying $C_3 \to \frac{1}{7}C_3$, to $B = \begin{bmatrix} 1 & 2 & 1 \\ -1 & \sqrt{3} & 1 \end{bmatrix}$ we get $\begin{bmatrix} 1 & 2 & 1/7 \\ -1 & \sqrt{3} & 1/7 \end{bmatrix}$

(iii) The addition to the elements of any row or column, the corresponding elements of any other row or column multiplied by any non zero number.

Symbolically, the addition to the elements of ith row, the corresponding elements of jth row multiplied by k is denoted by

$R_i \to R_i + kR_j$.

The corresponding column operation is denoted by $C_i \to C_i + kC_j$.

For example, applying $R_2 \to R_2 - 2R_1$, to $C = \begin{bmatrix} 1 & 2 \\ 2 & -1 \end{bmatrix}$, we get $\begin{bmatrix} 1 & 2 \\ 0 & -5 \end{bmatrix}$

DETERMINANT OF A MATRIX

If $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ be a square matrix, then its determinant, denoted by $|A|$ or det (A) is

defined as $|A| = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$

Properties of the Determinant of a matrix :

(i) $|A|$ exists $\Leftrightarrow A$ is a square matrix

(ii) $|AB| = |A|\ |B|$

(iii) $|A^T| = |A|$

(iv) $|kA| = k^n\ |A|$, if A is a square matrix of order n.

(v) If A and B are square matrices of same order then $|AB| = |BA|$

(vi) If A is a skew symmetric matrix of odd order then $|A| = 0$

(vii) If A = diag $(a_1, a_2 ..... a_n)$ then $|A| = a_1 a_2 ... a_n$

(viii) $|A|^n = |A^n|$, $n \in N$.

To know how to calculate value of determinant, please sea 'value of a determinant' in determinant section.

8. ADJOINT OF A MATRIX

If every element of a square matrix A be replaced by its cofactor in $|A|$, then the transpose of the matrix so obtained is called the adjoint of matrix A and it is denoted by adj A.

Thus if $A = [a_{ij}]$ be a square matrix and F_{ij} be the cofactor of a_{ij} in $|A|$, then Adj. $A = [F^{ij}]^T$ (To know how to find minor and cofactor please see 'minor & cofactor' in determinant section)

Hence if $A = \begin{bmatrix} a_{11} & a_{12} & ...a_{1n} \\ a_{21} & a_{22} & ...a_{2n} \\ a_{n1} & a_{n2} & ...a_{nn} \end{bmatrix}$, then Adj. $A = \begin{bmatrix} F^{11} & F^{12} & ...F^{1n} \\ F^{21} & F^{22} & ...F^{2n} \\ F^{n1} & F^{n2} & ...F^{nn} \end{bmatrix}^T$

Ex. if $A = \begin{bmatrix} 9 & -4 \\ -2 & 3 \end{bmatrix}$ then adj $A = \begin{bmatrix} 3 & 2 \\ 4 & 9 \end{bmatrix}^T = \begin{bmatrix} 3 & 4 \\ 2 & 9 \end{bmatrix}$

Properties of adjoint matrix : If A, B are square matrices of order n and I_n is corresponding unit matrix, then

(i) A (adj. A) = $|A|\ I_n$ = (adj A) A. (Thus A (adj A) is always a scalar matrix)

(ii) $|\text{adj } A| = |A|^{n-1}$

(iii) adj (adj A) = $|A|^{n-2} A$

(iv) $|\text{adj (adj } A)| = |A|^{(n-1)^2}$

(v) adj (A^T) = (adj $A)^T$

(vi) adj (AB) = (adj B) (adj A)

(vii) adj (A^m) = (adj $A)^m$, $m \in N$

(viii) adj $(kA) = k^{n-1}$ (adj. A), $k \in R$

(ix) adj $(I_n) = I_n$

(x) adj 0 = 0

(xi) A is symmetric $\Rightarrow$ adj A is also symmetric

(xii) A is diagonal $\Rightarrow$ adj A is also diagonal

(xiii) A is triangular $\Rightarrow$ adj A is also triangular

(xiv) A is singular $\Rightarrow$ $|\text{adj } A| = 0$

9. INVERSE OF A MATRIX

If A and B are two matrices such that $AB = I = BA$ then B is called the inverse of A and it is denoted by A^{-1}, thus

$A^{-1} = B \Leftrightarrow AB = I = BA$

To find inverse matrix of a given matrix A we use following formula

$A^{-1} = \dfrac{adj.A}{|A|}$. Thus A^{-1} exists $\Leftrightarrow |A| \neq 0$

(i) Matrix A is called invertible if A^{-1} exists. (ii) Inverse of a matrix is unique.

Properties of Inverse Matrix : Let A and B are two invertible matrices of the same order, then

(i) $(A^T)^{-1} = (A^{-1})^T$

(ii) $(AB)^{-1} = B^{-1}A^{-1}$

(iii) $(A^k)^{-1} = (A^{-1})^k$, $k \in N$

(iv) adj (A^{-1}) = (adj $A)^{-1}$

(v) $(A^{-1})^{-1} = A$

(vi) $|\ A^{-1}\ | = \dfrac{1}{|A|} = |A|^{-1}$

(vii) If A = diag $(a_1, a_2, ..., a_n)$, then A^{-1} = diag $(a_1^{-1}, a_2^{-1}, ...a_n^{-1})$

(viii) A is symmetric matrix $\Rightarrow$ A^{-1} is symmetric matrix.

(ix) A is triangular matrix and $|A| \neq 0 \Rightarrow A^{-1}$ is a triangular matrix.

(x) A is scalar matrix $\Rightarrow A^{-1}$ is scalar matrix

(xi) A is diagonal matrix $\Rightarrow$ A^{-1} is diagonal matrix

(xii) $AB = AC \Rightarrow B = C$, iff $|A| \neq 0$.

PERIOD OF A MATRIX

If for any Matrix A, $A^{k+1} = A$ then k is called period of Matrix (where k is a least positive integer)
Ex. If $A^3 = A$, $A^5 = A$, $A^7 = A$,.....then it is a periodic matrix and $A^{2+1} = A$ so its period is = 2

SUBMATRIX

Let A be $m \times n$ matrix, then a matrix obtained by leaving some rows or columns or both of A is called a sub matrix of A

Ex. if $A' = \begin{bmatrix} 2 & 1 & 0 \\ 3 & 2 & 2 \\ 2 & 5 & 3 \end{bmatrix}$ and $\begin{bmatrix} 2 & 2 \\ 5 & 3 \end{bmatrix}$ are sub matrices of Matrix $A = \begin{bmatrix} 2 & 1 & 0 & -1 \\ 3 & 2 & 2 & 4 \\ 2 & 5 & 3 & 1 \end{bmatrix}$

10. RANK OF A MATRIX

A number r is said to be the rank of a m x n matrix A if (a) Every square sub matrix of order $(r+1)$ or more is singular and (b) There exists at least one square submatrix of order r which is non-singular.

Thus, the rank of matrix is the order of the highest order non-singular sub matrix.

To find the rank of matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 3 & 4 & 5 \end{bmatrix}$. We have $|A| = 0$, therefore $r(A)$ is less then 3, we observe that $\begin{bmatrix} 5 & 6 \\ 4 & 5 \end{bmatrix}$ is a non-singular square sub matrix of order 2. Hence $r(A) = 2$.

(i) The rank of the null matrix is not defined and the rank of every non null matrix is greater than or equal to one.

(ii) The rank of matrix is same as the rank of its transpose i.e. $r(A) = r(A^T)$

(iii) Elementary transformation do not alter the rank of matrix.

APPLICATION OF MATRICES

Matrix instead of being a quantity is an operator to be used for transformation of vectors or variables. The addition and multiplication of matrices are compositions of operators instead of scalar quantities.

(A) Geometrical applications

(i) Reflection : If the point (x, y) is reflected in y-axis the new coordinates are

$x' = -x, y' = y$ .....(i)

These equations can be put in the form $x' = -1x + 0y \; y' = 0x + 1y$

In terms of matrices, they can be written as $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

Hence the reflection in y-axis is obtained by pre-multiplying the column vector $\begin{bmatrix} x \\ y \end{bmatrix}$ with $\begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$.

Similarly the reflection in x-axis is obtained by pre-multiplying with $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$.

(ii) Reflection about the line $y = x$: If we reflect the point (x, y) in the line $y = x$, y is changed into x and x into y, so that the new coordinates are : $y' = x, x' = y$

and so $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

Similarly the reflection of (x, y) in the line $x + y = 0$ is obtained by $x' = -y, y' = -x$

or $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} 0 & -1 \\ -1 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix}$

(iii) Rotation about the origin : If the coordinates of a point P are (x, y), O being the origin and the line OP is rotated about O as centre through an angle α in the anti-clockwise direction, the new coordinates of P are

$x' = r\cos(\theta+\alpha) = r\cos\theta\cos\alpha - r\sin\theta\sin\alpha = x\cos\alpha - y\sin\alpha$

and $y' = r\sin(\theta+\alpha) = r\sin\theta\cos\alpha + r\cos\theta\sin\alpha = y\cos\alpha + x\sin\alpha$

so that $\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix}$

where (r, θ) are the polar coordinates of $(x - y)$. If the rotation is in clockwise direction, we put $-\alpha$ for α.

11. SYSTEM OF LINEAR SIMULTANEOUS EQUATIONS

Consider the system of linear non-homogeneous simultaneous equations in three unknowns x, y and z, given by

$a_1x + b_1y + c_1z = d_1$, $a_2x + b_2y + c_2z = d_2$ and $a_3x + b_3y + c_3z = d_3$.

Let $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$,

Let $|A| = \Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$, $\Delta_x = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$, obtained on replacing first column of Δ by B.

Similarly let $\Delta_y = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$ and $\Delta_z = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$.

It can be shown that $AX = B$, $x\Delta = \Delta_x, y.\Delta = \Delta_y, z\Delta = \Delta_z$.

Illustration 6 : If $A = \begin{bmatrix} 2 & -1 \\ -7 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 4 & 1 \\ 7 & 2 \end{bmatrix}$ then find the value of B^TA^T.

Solution : (A) $B^TA^T = \begin{bmatrix} 4 & 7 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} 2 & -7 \\ -1 & 4 \end{bmatrix} = \begin{bmatrix} 8-7 & -28+28 \\ 2-2 & -7+8 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Illustration 7 : $\begin{bmatrix} 1 & -\tan\theta/2 \\ \tan\theta/2 & 1 \end{bmatrix}\begin{bmatrix} 1 & \tan\theta/2 \\ -\tan\theta/2 & 1 \end{bmatrix}^{-1}$ is equal to -

(a) $\begin{bmatrix} \sin\theta & -\cos\theta \\ \cos\theta & \sin\theta \end{bmatrix}$ (b) $\begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix}$

(c) $\begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$ (d) None of these

Solution : (c) $\begin{bmatrix} 1 & \tan\theta/2 \\ -\tan\theta/2 & 1 \end{bmatrix}^{-1} = \frac{1}{\sec^2\theta/2}\begin{bmatrix} 1 & -\tan\theta/2 \\ \tan\theta/2 & 1 \end{bmatrix}$

$\therefore$ Product $\frac{1}{\sec^2\theta/2}\begin{bmatrix} 1 & -\tan\theta/2 \\ \tan\theta/2 & 1 \end{bmatrix}\begin{bmatrix} 1 & -\tan\theta/2 \\ \tan\theta/2 & 1 \end{bmatrix}$

$= \frac{1}{\sec^2\theta/2}\begin{bmatrix} 1-\tan^2\theta/2 & -2\tan\theta/2 \\ 2\tan\theta/2 & 1-\tan^2\theta/2 \end{bmatrix}$

$= \frac{1}{\sec^2\theta/2}\begin{bmatrix} 1-\tan^2\theta/2 & -2\tan\theta/2 \\ 2\tan\theta/2 & 1-\tan^2\theta/2 \end{bmatrix}$

$= \begin{bmatrix} \cos^2\theta/2-\sin^2\theta/2 & -2\sin\theta/2\cos\theta/2 \\ 2\sin\theta/2\cos\theta/2 & \cos^2\theta/2-\sin^2\theta/2 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$

Illustration 8 : If $A = \begin{bmatrix} \ell_1 & m_1 & n_1 \\ \ell_2 & m_2 & n_2 \\ \ell_3 & m_3 & n_3 \end{bmatrix}$, where (ℓ_1, m_1, n_1), (ℓ_2, m_2, n_2) and (ℓ_3, m_3, n_3) are the direction of cosine of three mutually perpendicular straight lines, then prove that $AA' = I$.

Solution : Since (ℓ_1, m_1, n_1), (ℓ_2, m_2, n_2) and (ℓ_3, m_3, n_3) are the direction cosines of three mutually perpendicular straight lines $\ell_1^2+m_1^2+n_1^2=1$, $\ell_2^2+m_2^2+n_2^2=1$ and $\ell_3^2+n_3^2+m_3^2=1$

and $\ell_1\ell_2 + m_1m_2 + n_1n_2 = \ell_2\ell_3 + m_2m_3 + n_2n_3 = \ell_3\ell_1 + m_3m_1 + n_3n_1 = 0$

We have $A = \begin{bmatrix} \ell_1 & m_1 & n_1 \\ \ell_2 & m_2 & n_2 \\ \ell_3 & m_3 & n_3 \end{bmatrix} \Rightarrow A' = \begin{bmatrix} \ell_1 & \ell_2 & \ell_3 \\ m_1 & m_2 & m_3 \\ n_1 & n_2 & n_3 \end{bmatrix}$.

$AA' = \begin{bmatrix} \ell_1 & \ell_2 & \ell_3 \\ m_1 & m_2 & m_3 \\ n_1 & n_2 & n_3 \end{bmatrix}\begin{bmatrix} \ell_1 & m_1 & n_1 \\ \ell_2 & m_2 & n_2 \\ \ell_3 & m_3 & n_3 \end{bmatrix} \Rightarrow \begin{bmatrix} \Sigma\ell_1^2 & \Sigma\ell_1m_1 & \Sigma\ell_1n_1 \\ \Sigma\ell_1n_1 & \Sigma m_1^2 & \Sigma m_1n_1 \\ \Sigma n_1\ell_1 & \Sigma m_1n_1 & \Sigma n_1^2 \end{bmatrix} \Rightarrow$

$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I$

Illustration 9 : If $A = \begin{bmatrix} -1 & 7 \\ 2 & 3 \end{bmatrix}$, then skew-symmetric part of A is –

(a) $\begin{bmatrix} -1 & 7 \\ 2 & 3 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & -5/2 \\ 5/2 & 0 \end{bmatrix}$ (c) $\begin{bmatrix} -1 & -9/2 \\ 9/2 & 3 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 5/2 \\ -5/2 & 0 \end{bmatrix}$

Solution : (d) Let $A = B + C$, where $B = \frac{1}{2}(A + A^T)$ and $C = \frac{1}{2}(A - A^T)$ are respectively symmetric and skew-symmetric parts of A.

Now, $C = \frac{1}{2}\left\{\begin{bmatrix}-1 & 7\\ 2 & 3\end{bmatrix} - \begin{bmatrix}-1 & 2\\ 7 & 3\end{bmatrix}\right\} = \frac{1}{2}\begin{bmatrix}0 & 5\\ -5 & 0\end{bmatrix} = \begin{bmatrix}0 & 5/2\\ -5/2 & 0\end{bmatrix}$

Illustration 10 : Show that the matrix $\begin{bmatrix}1 & 1 & 3\\ 5 & 2 & 6\\ -2 & -1 & -3\end{bmatrix}$ is a nilpotent matrix of index 3.

Solution : Let $A = \begin{bmatrix}1 & 1 & 3\\ 5 & 2 & 6\\ -2 & -1 & -3\end{bmatrix} \Rightarrow A^2 = \begin{bmatrix}0 & 0 & 0\\ 3 & 3 & 9\\ -1 & -1 & -3\end{bmatrix}$

$\therefore \quad A^3 = A^2.A = \begin{bmatrix}0 & 0 & 0\\ 3 & 3 & 9\\ -1 & -1 & -3\end{bmatrix} \times \begin{bmatrix}1 & 1 & 3\\ 5 & 2 & 6\\ -2 & -1 & -3\end{bmatrix} = \begin{bmatrix}0 & 0 & 0\\ 0 & 0 & 0\\ 0 & 0 & 0\end{bmatrix} \Rightarrow A^3 = 0$ i.e., A is a nilpotent matrix of index 3.

Illustration 11 : If $A = \begin{bmatrix}2 & 0 & 0\\ 2 & 2 & 0\\ 2 & 2 & 2\end{bmatrix}$, then adj (adj A) is equal to -

(a) $8\begin{bmatrix}1 & 0 & 0\\ 1 & 1 & 0\\ 1 & 1 & 1\end{bmatrix}$ (b) $16\begin{bmatrix}1 & 0 & 0\\ 1 & 1 & 0\\ 1 & 1 & 1\end{bmatrix}$ (c) $64\begin{bmatrix}1 & 0 & 0\\ 1 & 1 & 0\\ 1 & 1 & 1\end{bmatrix}$ (d) none of these

Solution : (b) $|A| = \begin{bmatrix}2 & 0 & 0\\ 2 & 2 & 0\\ 2 & 2 & 2\end{bmatrix} = (2)(2)(2) = 8$

Now adj (adj A) $= |A|^{3-2}A = 8\begin{bmatrix}2 & 0 & 0\\ 2 & 2 & 0\\ 2 & 2 & 2\end{bmatrix} = 16\begin{bmatrix}1 & 0 & 0\\ 1 & 1 & 0\\ 1 & 1 & 1\end{bmatrix}$

Illustration 12 : If $A = \begin{pmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{pmatrix}$, then show that $A^2 - 4A - 5I = 0$, where I and 0 are the unit matrix and the null matrix of order 3 respectively. Use this result to find A^{-1} .

Solution : Given $A = \begin{pmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{pmatrix}$ $\therefore \quad A^2 = \begin{pmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{pmatrix} \times \begin{pmatrix}1 & 2 & 2\\ 2 & 1 & 2\\ 2 & 2 & 1\end{pmatrix} = \begin{pmatrix}9 & 8 & 8\\ 8 & 9 & 8\\ 8 & 8 & 9\end{pmatrix}$

$$\therefore \quad A^2 - 4A - 5I = \begin{pmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{pmatrix} - 4\begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} - 5\begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 9 & 8 & 8 \\ 8 & 9 & 8 \\ 8 & 8 & 9 \end{pmatrix} - \begin{pmatrix} 4 & 8 & 8 \\ 8 & 4 & 8 \\ 8 & 8 & 4 \end{pmatrix} - \begin{pmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 5 \end{pmatrix} = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$\therefore \quad A^2 - 4A - 5I = 0 \quad$ or $\quad 5I = A^2 - 4A$

on multiplying by A^{-1}, we get

$$5A^{-1} = A - 4I = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} - \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix} = \begin{pmatrix} 1 & 2 & 2 \\ 2 & 1 & 2 \\ 2 & 2 & 1 \end{pmatrix} - \begin{pmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{pmatrix}$$

$$\therefore \quad A^{-1} = \frac{1}{5}\begin{pmatrix} -3 & 2 & 2 \\ 2 & -3 & 2 \\ 2 & 2 & -3 \end{pmatrix} = \begin{pmatrix} -3/5 & 2/5 & 2/5 \\ 2/5 & -3/5 & 2/5 \\ 2/5 & 2/5 & -3/5 \end{pmatrix}$$

Illustration 13 : Solve the system of equations, $x + 2y + 3z = 1$; $2x + 3y + 2z = 2$; $3x + 3y + 4z = 1$ with the help of matrix inversion.

Solution : The given system of equations in the matrix form can be written as

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix}\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \Rightarrow AX = B$$

where $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 3 & 2 \\ 3 & 3 & 4 \end{bmatrix}, X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}$.

Now $|A| = 1(12 - 6) - 2(8 - 6) + 3(6 - 9) = 6 - 4 - 9 = -7 \neq 0$.
Hence the given system has unique solution.

Let C be the matrix of cofactors of elements in $|A|$. Then $C = \begin{bmatrix} C_{11} & C_{12} & C_{13} \\ C_{21} & C_{22} & C_{23} \\ C_{31} & C_{32} & C_{33} \end{bmatrix}$

Here

$$C_{11} = \begin{vmatrix} 3 & 2 \\ 3 & 4 \end{vmatrix} = 6, C_{12} = \begin{vmatrix} 2 & 2 \\ 3 & 4 \end{vmatrix} = -2, C_{13} = \begin{vmatrix} 2 & 3 \\ 3 & 3 \end{vmatrix} = -3, C_{21} = -\begin{vmatrix} 2 & 3 \\ 3 & 4 \end{vmatrix} = 1, C_{22} = \begin{vmatrix} 1 & 3 \\ 3 & 4 \end{vmatrix} = -5, C_{23} = -\begin{vmatrix} 1 & 2 \\ 3 & 3 \end{vmatrix} = 3,$$

$$C_{31} = \begin{vmatrix} 2 & 3 \\ 3 & 2 \end{vmatrix} = -5, C_{32} = -\begin{vmatrix} 1 & 3 \\ 2 & 2 \end{vmatrix} = 4, C_{33} = \begin{vmatrix} 1 & 2 \\ 2 & 3 \end{vmatrix} = -1$$

$$\therefore \quad C = \begin{bmatrix} 6 & -2 & -3 \\ 1 & -5 & 3 \\ -5 & 4 & -1 \end{bmatrix} \quad \Rightarrow \text{Adj } A = C' = \begin{bmatrix} 6 & 1 & -5 \\ -2 & -5 & 4 \\ -3 & 3 & -1 \end{bmatrix}$$

$$\therefore \quad A^{-1} = \frac{Adj\,A}{|A|} = -\frac{1}{7}\begin{bmatrix} 6 & 1 & -5 \\ -2 & -5 & 4 \\ -3 & 3 & -1 \end{bmatrix} = \begin{bmatrix} -6/7 & -1/7 & 5/7 \\ 2/7 & 5/7 & -4/7 \\ 3/7 & -3/7 & 1/7 \end{bmatrix}$$

$$\therefore \quad A^{-1}B = \begin{bmatrix} -6/7 & -1/7 & 5/7 \\ 2/7 & 5/7 & -4/7 \\ 3/7 & -3/7 & 1/7 \end{bmatrix}\begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix} \Rightarrow \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -3/7 \\ 8/7 \\ -2/7 \end{bmatrix} \qquad (\because\ A^{-1}B = X)$$

$\therefore \quad x = -3/7,\ y = 8/7,\ z = -2/7$

12. MATRICES AT A GLANCE

(i) A matrix is an ordered rectangular array of numbers or functions.

(ii) A matrix having m rows and n columns is called a matrix of order $m \times n$.

(iii) If $A = [a_{ij}]_{m\times n}$ and $B = [b_{jk}]_{n\times p}$, then $AB = C = [C_{ik}]_{m\times p}$, where $c_{ik} = \sum_{j=1}^{n} a_{ij}b_{jk}$.

(iv) If $A = [a_{ij}]_{m\times n}$, then A' or $A^T = [a_{ji}]_{n\times m}$

(v) (a) $(A')' = A$ (b) $(kA)' = kA'$, (c) $(A + B)' = A' + B'$, (d) $(AB)' = B'A'$

(vi) Elementary operations of a matrix are :

(a) $R_i \leftrightarrow R_j$ or $C_i \leftrightarrow C_j$

(b) $R_i \to kR_i$ or $C_i \to kC_i$

(c) $R_i \to R_i + kR_j$ or $C_i \to C_i + kC_j$

(vii) If A and B are two square matrices such that $AB = BA = I$, then B is the inverse matrix of A and is denoted by A^{-1} and A is the inverse of B.

(viii) Inverse of a square matrix, if it exists, is unique.

(ix) Determinant of a matrix $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$ is given by (expanding along R_1)

$$|A| = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = a_1\begin{vmatrix} b_2 & c_2 \\ b_3 & c_3 \end{vmatrix} - b_1\begin{vmatrix} a_2 & c_2 \\ a_3 & c_3 \end{vmatrix} + c_1\begin{vmatrix} a_2 & b_2 \\ a_3 & b_3 \end{vmatrix}$$

(x) If elements of a row or a column in a determinant can be expressed as sum of two or more elements, then the given determinant can be expressed as sum of two or more determinants.

(xi) If to each element of a row or a column of a determinant the equimultiples of corresponding elements of other rows or columns are added, then value of determinant remains same.

(xii) Area of a triangle with vertices (x_1, y_1), (x_2, y_2) and (x_3, y_3) is given by $\Delta = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$

(xiii) Minor of an element a_{ij} of the determinant of matrix A is the determinant obtained by deleting i^{th} row and j^{th} column and denoted by M_{ij}.

(xiv) Cofactor of a_{ij} is given by $A_{ij} = (-1)^{i+j} M_{ij}$.

(xv) A square matrix A has inverse if and only if A is non-singular.

(xvi) $A^{-1} = \frac{1}{|A|}(adj\ A)$

(xvii) If $a_1x + b_1y + c_1z = d_1$, $a_2x + b_2y + c_2z = d_2$, $a_3x + b_3y + c_3z = d_3$, then these equations can be written as $AX = B$, where $A = \begin{bmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{bmatrix}$, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$ and $B = \begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}$

(xviii) Unique solution of equation $AX = B$ is given by $X = A^{-1}B$, where $|A| \neq 0$.

(xix) A system of equation is consistent or inconsistent according as its solution exists or not.

(xx) For a square matrix A in matrix equation $AX = B$:

(*a*) $|A| \neq 0$, there exists unique solution.

(*b*) $|A| = 0$ and (adj A) $B \neq 0$, then there exists no solution.

(c) $|A| = 0$ and (adj A) $B = 0$, then system may or may not be consistent.

Student Task

Q.1 (adj A^T) – (adj $A)^T$ equals

(a) $2\,|A|$ (b) $2\,|A|I$ (c) zero matrix (d) unit matrix

Q.2 If $A = \begin{bmatrix} 1 & -5 & 7 \\ 0 & 7 & 9 \\ 11 & 8 & 9 \end{bmatrix}$, then the value of tr A is

(a) 17 (b) 25 (c) 3 (d) 12

Q.3 If $A = \begin{bmatrix} ab & b^2 \\ -a^2 & -ab \end{bmatrix}$, then A is –

(a) Involutory matrix (b) Idempotent matrix

(c) Nilpotent matrix (d) None of these

Q.4 If A and B are square matrices of same orders, then adj (AB) equals –

(a) adj A. adj B (b) adj B. adj A (c) adj A + adj B (d) adj A – adj B

Q.5 If $A = \begin{bmatrix} 1 & 2 \\ -4 & -1 \end{bmatrix}$, then A^{-1} equals

(a) $\frac{1}{7}\begin{bmatrix} 1 & 2 \\ -4 & -1 \end{bmatrix}$ (b) $\begin{bmatrix} -1 & -2 \\ 4 & 1 \end{bmatrix}$ (c) $\frac{1}{9}\begin{bmatrix} 1 & 2 \\ 4 & 1 \end{bmatrix}$ (d) $\frac{1}{7}\begin{bmatrix} -1 & -2 \\ 4 & 1 \end{bmatrix}$

Q.6 If $A = \begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$ then $A^2 - 5A + 6I$ equals –

(a) $\begin{bmatrix} 1 & -1 & -5 \\ -1 & -1 & 4 \\ -3 & -10 & 4 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & -1 & -3 \\ -1 & -1 & 10 \\ -5 & 4 & 4 \end{bmatrix}$ (c) 0 (d) I

Q.7 If $A = \begin{bmatrix} 0 & -i \\ i & 0 \end{bmatrix}$, B = $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$, C = $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$, then which of the following statement is true

(a) $AB = BA$ (b) $AB = -BA$ (c) $A^2 = BC$ (d) $A^2 = B + C$

Q.8 If $A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$, then AA^T equals –

(a) $\begin{bmatrix} \cos 2\theta & -\sin 2\theta \\ \sin 2\theta & \cos 2\theta \end{bmatrix}$ (b) $\begin{bmatrix} \cos^2\theta & \sin^2\theta \\ \sin^2\theta & \cos^2\theta \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

ANSWERS

1. (c)	2. (a)	3. (c)	4. (b)	5. (d)
6. (b)	7. (b)	8. (c)		

13. DETERMINANTS

Definition : A determinant is a special kind of symbol used into determine certain properties of systems of equations and functions. Many complicated expressions can be easily handled if they are expressed as determinants. In other words, an expression expressed in equal number of rows and column and put between two vertical lines is named as determinant of that expression Ex. ,

$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}$, $\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ are the determinants of second and third order respectively.

Obviously in every determinant, the number of rows and columns are equal and this number is called the order of that determinant.

VALUE OF A DETERMINANT

The expression which has been expressed in a determinant form is called the value of that determinant.

To find the value of a third order determinant

Let $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$

To find its value we expand it by any row or column as the sum of three determinants of order 2. If we expand it by first row then

$$\Delta = (-1)^{1+1} a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} + (-1)^{1+2} a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + (-1)^{1+3} a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

$$= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix} = a_{11}(a_{22}a_{23} - a_{32}a_{23}) - a_{12}(a_{21}a_{33} - a_{31}a_{23}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

A determinant is called a triangular determinant if its every element above or below the diagonal is zero.

For example $\begin{vmatrix} a & 0 & 0 \\ b & c & 0 \\ d & e & f \end{vmatrix}$ is a triangular determinant. In particular when all the elements except

diagonal elements are zero, then it is called a diagonal determinant.

For example $\begin{vmatrix} a & 0 & 0 \\ 0 & b & 0 \\ 0 & 0 & c \end{vmatrix}$ is a diagonal determinant.

We generally use R_1, R_2, R_3,..... to denote first, second, third.... row and C_1, C_2, C_3 ..... to denote first, second, third ..... column of a determinant.

MINOR & COFACTOR

Minor : The Determinant that is left by cancelling the row and column intersecting at a particular element is called the minor of that element.

If $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ then Minor of a_{11} is $M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$, Similarly $M_{12} = \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$

Using this concept the value of Determinant can be

$\Delta = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}$

or $\Delta = -a_{21}M_{21} + a_{22}M_{22} - a_{23}M_{23}$

or $\Delta = a_{31}M_{31} - a_{32}M_{32} + a_{33}M_{33}$

Cofactor : The cofactor of an element a_{ij} is denoted by F_{ij} and is equal to $(-1)^{i+j} M_{ij}$ where M is a minor of element a_{ij}

if $\Delta = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$ then $F_{11} = (-1)^{1+1} M_{11} = M_{11} = \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix}$

$F_{12} = (-1)^{1+2} M_{12} = -M_{12} = -\begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix}$

(i) The sum of products of the element of any row with their corresponding cofactor is equal to the value of determinant i.e.

$\Delta = a_{11}F_{11} + a_{12}F_{12} + a_{13}F_{13}$

(ii) The sum of the product of element of any row with corresponding cofactor of another row is equal to zero i.e.

$a_{11}F_{21} + a_{12}F_{22} + a_{13}F_{23} = 0$

(iii) If order of a determinant (D) is 'n' then the value of the determinant formed by replacing every element by its cofactor is Δ^{n-1}.

14. PROPERTIES OF DETERMINANT

Property 1 : The value of determinant remains unchanged, if the rows and the column are interchanged.

This is always denoted by ' and is also called transpose

Ex. $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and $D' = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$

Then $D' = D$, D and D' are transpose of each other

Since the determinant remains unchanged when rows and columns are interchanged, it is obvious that any theorem which is true for 'rows' must also be true for 'Columns'.

Property 2 : If any two rows (or columns) of a determinant be inter changed, the determinant is unaltered in numerical value, but is changed in sign only.

Ex. $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and $D' = \begin{vmatrix} a_2 & b_2 & c_2 \\ a_1 & b_1 & c_1 \\ a_3 & b_3 & c_3 \end{vmatrix}$ then $D' = -D$

Property 3 : If a determinant has two rows (or columns) identical, then its value is zero.

Ex. Let $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \end{vmatrix}$ then, $D = 0$

Property 4 : If all the elements of any row (or column) be multiplied by the same number, then the value of determinant is multiplied by that number.

Ex. $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and $D' = \begin{vmatrix} ka_1 & kb_1 & kc_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ then $D' = k\,D$

Property 5 : If each elements of any row (or column) can be expressed as a sum of two terms, then the determinant can be expressed as the sum of the determinants

Ex. $\begin{vmatrix} a_1 + x & b_1 + y & c_1 + z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} + \begin{vmatrix} x & y & z \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$

Property 6 : The value of a determinant is not altered by adding to the elements of any row (or column) the same multiples of the corresponding elements of any other row (or column)

Ex. $D = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$ and $D' = \begin{vmatrix} a_1 + ma_2 & b_1 + mb_2 & c_1 + mc_2 \\ a_2 & b_2 & c_2 \\ a_3 - na_1 & b_3 - nb_1 & c_3 - nc_1 \end{vmatrix}$ then $D' = D$

It should be noted that while applying Property 6 at least one row (or column) must remain unchanged

Property 7 : If $\Delta = f(x)$ and $f(a) = 0$ then $(x\text{-}a)$ is a factor of Δ.

Ex. $D = \begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}$ If we replace a by b then $D = \begin{vmatrix} 1 & 1 & 1 \\ b & b & c \\ b^2 & b^2 & c^2 \end{vmatrix} = 0 \Rightarrow (a - b)$ is a factor of D

15. SYMMETRIC & SKEW SYMMETRIC DETERMINANT

Symmetric determinant : A determinant is called symmetric determinant if for its every element.

$a_{ij} = a_{ji} \quad \forall\ i, j$

Ex. $\begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$

Skew Symmetric determinant : A determinant is called skew symmetric determinant if for its every element $a_{ij} = -a_{ji} \forall\ i,\ j$

Ex. $\begin{vmatrix} 0 & 3 & -1 \\ -3 & 0 & 5 \\ 1 & -5 & 0 \end{vmatrix}$

(i) Every diagonal element of a skew symmetric determinant is always zero.

(ii) The value of a skew symmetric determinant of even order is always a perfect square and that of odd order is always zero.

MULTIPLICATION OF TWO DETERMINANTS

Multiplication of two second order determinants is defined as follows

$$\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \times \begin{vmatrix} \ell_1 & m_1 \\ \ell_2 & m_2 \end{vmatrix} = \begin{vmatrix} a_1\ell_1 + b_1\ell_2 & a_1 m_1 + b_1 m_2 \\ a_2\ell_1 + b_2\ell_2 & a_2 m_1 + b_2 m_2 \end{vmatrix}$$

Multiplication of two third order determinants is defined as follows

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} \times \begin{vmatrix} \ell_1 & m_1 & n_1 \\ \ell_2 & m_2 & n_2 \\ \ell_3 & m_3 & n_3 \end{vmatrix} = \begin{vmatrix} a_1\ell_1 + b_1\ell_2 + c_1\ell_3 & a_1 m_1 + b_1 m_2 + c_1 m_3 & a_1 n_1 + b_1 n_2 + c_1 n_3 \\ a_2\ell_1 + b_2\ell_2 + c_2\ell_3 & a_2 m_1 + b_2 m_2 + c_2 m_3 & a_2 n_1 + b_2 n_2 + c_2 n_3 \\ a_3\ell_1 + b_3\ell_2 + c_3\ell_3 & a_3 m_1 + b_3 m_2 + c_3 m_3 & a_3 n_1 + b_3 n_2 + c_3 n_3 \end{vmatrix}$$

In above case the order of determinant is same, if the order is different then for their multiplication first of all they should be expressed in the same order.

SUMMATION OF DETERMINANTS

Let $\Delta = \begin{vmatrix} f(r) & a & \ell \\ g(r) & b & m \\ h(r) & c & n \end{vmatrix}$ where a, b, c, ℓ, m and n are constants, independent of r. Then

$$\sum_{r=1}^{n} \Delta_r = \begin{vmatrix} \sum_{r=1}^{n} f(r) & a & \ell \\ \sum_{r=1}^{n} g(r) & b & m \\ \sum_{r=1}^{n} h(r) & c & n \end{vmatrix}$$

Here function of r can be the elements of only one row or one column.

LIMIT OF A DETERMINANT

$$\text{Let } \Delta(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ \ell(x) & m(x) & n(x) \\ u(x) & v(x) & w(x) \end{vmatrix}, \text{ then } \lim_{x\to a} \Delta(x) = \begin{vmatrix} \lim_{x\to a} f(x) & \lim_{x\to a} g(x) & \lim_{x\to a} h(x) \\ \lim_{x\to a} \ell(x) & \lim_{x\to a} m(x) & \lim_{x\to a} n(x) \\ \lim_{x\to a} u(x) & \lim_{x\to a} v(x) & \lim_{x\to a} w(x) \end{vmatrix},$$

provided each of nine limiting values exist finitely.

DIFFERENTIATION OF DETERMINANTS

$$\text{Let } \Delta(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ \ell(x) & m(x) & n(x) \\ u(x) & v(x) & w(x) \end{vmatrix},$$

$$\text{then } \Delta'(x) = \begin{vmatrix} f'(x) & g'(x) & h'(x) \\ \ell(x) & m(x) & n(x) \\ u(x) & v(x) & w(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ \ell'(x) & m'(x) & n'(x) \\ u(x) & v(x) & w(x) \end{vmatrix} + \begin{vmatrix} f(x) & g(x) & h(x) \\ \ell(x) & m(x) & n(x) \\ u'(x) & v'(x) & w'(x) \end{vmatrix}$$

DETERMINANTS INVOLVING INTEGRATIONS

Let $\Delta(x)=\begin{vmatrix} f(x) & g(x) & h(x) \\ a & b & c \\ \ell & m & n \end{vmatrix}$ where a, b, c, ℓ, m and are constants.

$$\Rightarrow \int_a^b \Delta(x)\,dx = \begin{vmatrix} \int_a^b f(x)\,dx & \int_a^b g(x)\,dx & \int_a^b h(x)\,dx \\ a & b & c \\ \ell & m & n \end{vmatrix}$$

Illustration 14 : Let $f(x) = \begin{vmatrix} x^3 & \sin x & \cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$, where p is a constant. Then $\frac{d^3}{dx^3}[f(x)]$ at $x = 0$ is –

(a) p (b) $p + p^2$ (c) $p + p^3$ (d) independent of p

Solution : (d) : $\frac{d}{dx}f(x)=\begin{vmatrix} 3x^2 & \cos x & -\sin x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}+0+0$; $\frac{d^2}{dx^2}f(x)=\begin{vmatrix} 6x & -\sin x & -\cos x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$;

$$\frac{d^3}{dx^3}f(x)=\begin{vmatrix} 6 & -\cos x & \sin x \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}$$

$\frac{d^3}{dx^3}f(x)$ at $x = 0$ is $\begin{vmatrix} 6 & -1 & 0 \\ 6 & -1 & 0 \\ p & p^2 & p^3 \end{vmatrix}=0$ i.e. independent of p.

Illustration 15 : Let $f(x) = \begin{vmatrix} \sec x & \cos x & \sec^2 x+\cot x\operatorname{cosec} x \\ \cos^2 x & \cos^2 x & \operatorname{cosec}^2 x \\ 1 & \cos^2 x & \cos^2 x \end{vmatrix}$. Prove that

$$\int_0^{\pi/2} f(x)\,dx = -\left(\frac{\pi}{4}+\frac{8}{15}\right).$$

Solution : Operate $R_1 \to R_1 - \sec x\, R_3$

$$\Rightarrow f(x) = \begin{vmatrix} 0 & 0 & \sec^2 x+\cot x\operatorname{cosec} x-\cos x \\ \cos^2 x & \cos^2 x & \operatorname{cosec}^2 x \\ 1 & \cos^2 x & \cos^2 x \end{vmatrix}$$

$= (\sec^2 x + \cot x \operatorname{cosec} x - \cos x)(\cos^4 x - \cos^2 x)$

$= \left(1 + \frac{\cos^3 x}{\sin^2 x} - \cos^3 x\right)(\cos^2 x - 1)$

$= -\sin^2 x \frac{\sin^2 x + \cos^3 x - \cos^3 x \sin^2 x}{\sin^2 x} = -(\sin^2 x + \cos^5 x)$

$$\int_0^{\pi/2} f(x)\,dx = -\int_0^{\pi/2} (\sin^2 x + \cos^5 x)\,dx = -\left(\frac{1}{2}.\frac{\pi}{2} + \frac{4.2}{5.3.1}\right) = -\left(\frac{\pi}{4} + \frac{8}{15}\right)$$

APPLICATIONS OF DETERMINANT

1. Area of triangle : The area of a triangle whose vertices are (x_1, y_1), (x_2, y_2) and (x_3, y_3), is given by the expression $\frac{1}{2}[x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)]$

Now this expression can be written in the form of a determinant as

$$\Delta = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix} \quad \text{..... (1)}$$

(i) Since area is a positive quantity, we always take the absolute value of the determinant in (1).
(ii) If area is given, use both positive and negative values of the determinant for calculation.
(iii) The area of the triangle formed by three collinear points is zero.

2. Solution of system of (Non-Homogeneous) linear equations: Consider three linear simultaneous equation in 'x', 'y', 'z'

$a_1x + b_1y + c_1z = d_1$ ..... (i)
$a_2x + b_2y + c_2z = d_2$ .... (ii)
$a_3x + b_3y + c_3z = d_3$ .... (iii)

and if $\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$, $\Delta_1 = \begin{vmatrix} d_1 & b_1 & c_1 \\ d_2 & b_2 & c_2 \\ d_3 & b_3 & c_3 \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} a_1 & d_1 & c_1 \\ a_2 & d_2 & c_2 \\ a_3 & d_3 & c_3 \end{vmatrix}$, $\Delta_3 = \begin{vmatrix} a_1 & b_1 & d_1 \\ a_2 & b_2 & d_2 \\ a_3 & b_3 & d_3 \end{vmatrix}$

then using Cramer's rule of determinant we get

$$\frac{x}{\Delta_1} = \frac{y}{\Delta_2} = \frac{z}{\Delta_3} = \frac{1}{\Delta} \quad \text{i. e. } x = \frac{\Delta_1}{\Delta}, y = \frac{\Delta_2}{\Delta}, z = \frac{\Delta_3}{\Delta}$$

Case- I : If $\Delta \neq 0$ then $x = \frac{\Delta_1}{\Delta}, y = \frac{\Delta_2}{\Delta}, z = \frac{\Delta_3}{\Delta}$

$\therefore$ The system is consistent and has unique solutions

Case-II : If $\Delta = 0$ and

(i) If at least one of $\Delta_1, \Delta_2, \Delta_3$ is not zero then the system of equations a inconsistent i.e. has no solution.
(ii) If $\Delta_1 = \Delta_2 = \Delta_3 = 0$ or $\Delta_1, \Delta_2, \Delta_3$ are all zero, then the number of solutions of the system of equations is infinite.

The above can be shown diagrammatically as follows –

3. Solution of system of Homogeneous linear equation:

$a_1x + b_1y + c_1z = 0$ $a_2x + b_2y + c_2z = 0$ $a_3x + b_3y + c_3z = 0$

$$\Delta = \begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix}$$

(i) If $\Delta = 0$, equation has a non-trivial solution(non-zero solution) (i.e. at least one of the x, y, z is different from zero).
(ii) If $\Delta \neq 0$ then the only solution of the above system of equation is x = 0, y = 0 and z = 0 and it called trivial solution or zero solution.

Illustration 16 : Given $2x - y + 2z = 2$, $x - 2y - z = -4$, $x + y + \lambda z = 4$. Then the value of λ such that the given system of equation has no solution, is –

(a) 3 (b) 1 (c) 0 (d) –3

Solution : (a) For no solution $\Delta = 0$

$$\begin{vmatrix} 2 & -1 & 2 \\ 1 & -2 & -1 \\ 1 & 1 & \lambda \end{vmatrix} = 0 \Rightarrow \lambda = 3$$

Illustration 17 : If x, y, z are not all zero and if ax + by + cz = 0, bz + cy + az = 0, cx + ay + bz = 0. Prove that x : y : z = 1 : 1 : 1 or 1 : ω, ω^2 or 1 : ω^2 : ω, where ω is the complex cube roots of unity.

Solution : For non-trival solution $\begin{vmatrix} a & b & c \\ b & c & a \\ c & a & b \end{vmatrix} = 0$

$= (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca) = 0$ ..... (1)

$= (a + b + c)(a + \omega b + \omega^2 c)(a + \omega^2 b + \omega c) = 0$ ..... (2)

Now, by using cramer rule find x, y, z and using $(a + b + c) = 0$,

$(a + \omega b + \omega^2 c) = 0$ and $(a + \omega^2 b + \omega c) = 0$, prove the desired result.

Student Task

Q.1 Solve the following system of equations by determinant method.

(i) $x + y + z = 6$, $x + 2y + 3z = 14$, $x + 4y + 7z = 30$

(ii) $x + y + z = 6$, $x - y + z = 2$, $2x + y - z = 1$

(iii) $x + y + z = 3$, $x + 2y + 3z = 4$, $2x + 3y + 4z = 8$.

Q.2 Investigate for what values of λ, μ the equations $x + y + z = 6$; $x + 2y + 3z = 10$; $x + 2y + \lambda z = \mu$ have

(i) no solution **(ii)** unique solution

(iii) infinitely many solutions

Q.3 The existence of unique solution of the system $x + y + z = b$, $2x + 3y - z = 6$, $5x - y + az = 10$ depends on–

(a) b only (b) a only (c) a and b (d) Neither a nor b

Q.4 Given the system of equations $px + y + z = 1$, $x + py + z = p$, $x + y + pz = p^2$, then for what value of p does this system have no solution –

(a) 0 (b) (1, –2) (c) 1 (d) (0, 1)

Q.5 The system of equations : $\alpha x + y + z = \alpha - 1$; $x + \alpha y + z = \alpha - 1$; $x + y + \alpha z = \alpha - 1$ has no solution, if α is

(a) 1 (b) not – 2 (c) either – 2 or 1 (d) – 2

ANSWERS

1. (i) $x = k - 2$, $y = 8 - k$, $z = k$, $k \in R$. (ii) $x = 1$, $y = 2$, $z = 3$ (iii) no solution
2. (i) $\lambda = 3$, $\mu \neq 10$ (ii) $\lambda \neq 3$, $\mu \in R$ (iii) $\lambda = 3$, $\mu = 10$
3. (b) 4. (c) 5. (d)

EXERCISE

Q.1 If, $E=\begin{bmatrix}0&1&0\\0&0&1\\0&0&0\end{bmatrix}$ and $F=\begin{bmatrix}0&0&0\\1&0&0\\0&1&0\end{bmatrix}$ calculate the matrix product EF & FE and show that $E^2F + FE^2 = E$.

Q.2 Find the number of 2 × 2 matrix satisfying

(i) a_{ij} is 1 or –1 ; (ii) $a_{11}^2+a_{12}^2=a_{21}^2+a_{22}^2=2$; (iii) $a_{11}\,a_{21} + a_{12}\,a_{22} = 0$

Q.3 Find the value of x and y that satisfy the equations $\begin{bmatrix}3&-2\\3&0\\2&4\end{bmatrix}\begin{bmatrix}y&y\\x&x\end{bmatrix}=\begin{bmatrix}3&3\\3y&3y\\10&10\end{bmatrix}$

Q.4 Let $A=\begin{bmatrix}a&b\\c&d\end{bmatrix}$ and $B=\begin{bmatrix}p\\q\end{bmatrix}\neq\begin{bmatrix}0\\0\end{bmatrix}$. Such that $AB = B$ and $a + d = 5050$. Find the value of $(ad – bc)$.

Q.5 Define $A=\begin{bmatrix}0&1\\3&0\end{bmatrix}$. Find a vertical vector V such that $(A^8 + A^6 + A^4 + A^2 + I)V = \begin{bmatrix}0\\11\end{bmatrix}$

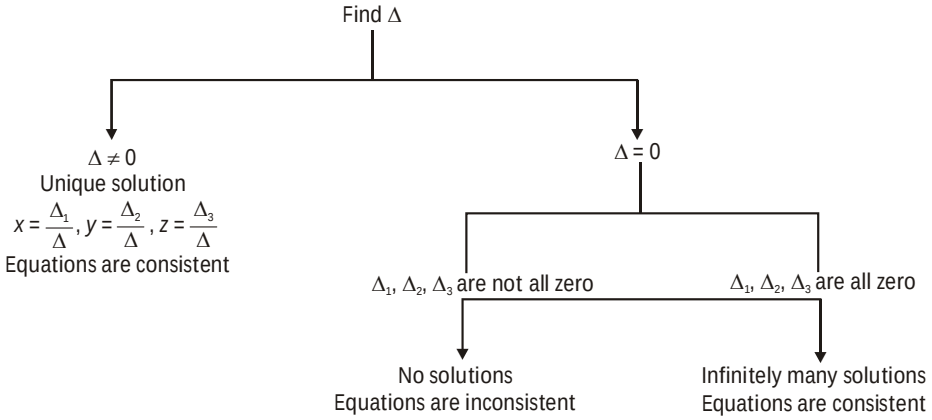
(where I is the 2 × 2 identity matrix).

Q.6 If, $A=\begin{bmatrix}1&0&2\\0&2&1\\2&0&3\end{bmatrix}$, then show that the matrix A is a root of the polynomial $f(x) = x^3 – 6x^2 + 7x + 2$.

Q.7 For a non zero λ, use induction to prove that :

(a) $\begin{bmatrix}\lambda&1&0\\0&\lambda&1\\0&0&\lambda\end{bmatrix}^n=\begin{bmatrix}\lambda^n&n\lambda^{n-1}&\frac{n(n-1)}{2}\lambda^{n-2}\\0&\lambda^n&n\lambda^{n-1}\\0&0&\lambda^n\end{bmatrix}$ for every $n \in N$.

(b) If, $A=\begin{bmatrix}0&1\\0&0\end{bmatrix}$, then $(aI + bA)^n = a^nI + na^{n-1}\,bA$, where I is a unit matrix of order 2, $\forall\ n \in N$.



Q.8. If the matrices $A=\begin{bmatrix}1 & 2\\ 3 & 4\end{bmatrix}$ and $B=\begin{bmatrix}a & b\\ c & d\end{bmatrix}$

(a, b, c, d not all simultaneously zero) commute, find the value of $\frac{d-b}{a+c-b}$. Also show that the matrix which commutes with A is of the form $\begin{bmatrix}\alpha-\beta & 2\beta/3\\ \beta & \alpha\end{bmatrix}$.

Q.9 If $\begin{bmatrix}a & b\\ c & 1-a\end{bmatrix}$ is an idempotent matrix. Find the value of $f(a)$, where $f(x) = x - x^2$, when $bc = 1/4$. Hence otherwise evaluate a.

Q.10 If the matrix A is involutary, show that $\frac{1}{2}(I + A)$ and $\frac{1}{2}(I - A)$ are idempotent and $\frac{1}{2}(I + A)\times\frac{1}{2}(I - A) = O$.

Q.11 Show that the matrix $A=\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}$ can be decomposed as a sum of a unit and a nilpotent matrix.

Hence evaluate the matrix $\begin{bmatrix}1 & 0\\ 2 & 1\end{bmatrix}^{2007}$.

Q.12 Given matrices $A=\begin{bmatrix}1 & x & 1\\ x & 2 & y\\ 1 & y & 3\end{bmatrix}; B=\begin{bmatrix}3 & -3 & z\\ -3 & 2 & -3\\ z & -3 & 1\end{bmatrix}$

Obtain x, y and z if the matrix AB is symmetric.

Q.13 Let X be the solution set of the equation $A^x = I$, where $A=\begin{bmatrix}0 & 1 & -1\\ 4 & -3 & 4\\ 3 & -3 & 4\end{bmatrix}$ and I is the corresponding unit matrix and $x \subseteq N$ then find the minimum value of $\Sigma(\cos^x\theta + \sin^x\theta)$, $\theta \in R$.

Q.14 $A=\begin{pmatrix}3 & a & -1\\ 2 & 5 & c\\ b & 8 & 2\end{pmatrix}$ is Symmetric and $B=\begin{pmatrix}d & 3 & a\\ b-a & e & -2b-c\\ -2 & 6 & -f\end{pmatrix}$ is Skew Symmetric, then find AB. Is AB a symmetric, Skew Symmetric or neither of them. Justify your answer.

Q.15 Express the matrix $\begin{bmatrix}1 & 2 & 5\\ 2 & 3 & -6\\ -1 & 0 & 4\end{bmatrix}$ as a sum of a lower triangular matrix & an upper triangular matrix with zero in its leading diagonal. Also Express the matrix as a sum of a symmetric & a skew symmetric matrix.

Q.16 A is a square matrix of order n.
ℓ = maximum number of distinct entries if A is a triangular matrix
m = maximum number of distinct entries if A is a diagonal matrix
p = minimum number of zeroes if A is a triangular matrix
if $\ell + 5 = p + 2m$, find the order of the matrix.

Q.17 If A is an idempotent non zero matrix and I is an identity matrix of the same order, find the value of n, $n \in N$, such that $(A + I)^n = I + 127\,A$.

Q.18 Consider the two matrices A and B where $A=\begin{bmatrix}1 & 2\\ 4 & 3\end{bmatrix}; B=\begin{bmatrix}5\\ -3\end{bmatrix}$. If $n(A)$ denotes the number of elements in A such that $n(XY) = 0$, when the two matrices X and Y are not conformable for multiplication. If $C = (AB)(B¢A)$; $D=(B¢A)(AB)$ then, find the value of $\left(\frac{n(C)\left(|D|^2 + n(D)\right)}{n(A) - n(B)}\right)$.

Answer Key :

1. $EF=\begin{bmatrix}1 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 0\end{bmatrix}, FE=\begin{bmatrix}0 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{bmatrix}$ 2. 8 3. $x=\frac{3}{2}, y=2$ 4. 5049

5. $V=\begin{bmatrix}0\\ \frac{1}{11}\end{bmatrix}$ 8. 1 9. $f(a) = 1/4$, $a = 1/2$ 11. $\begin{bmatrix}1 & 0\\ 4014 & 1\end{bmatrix}$

12. $\left(-\frac{4\sqrt{2}}{3}, \frac{2}{3}, 2\sqrt{2}\right), \left(\frac{4\sqrt{2}}{3}, \frac{2}{3}, -2\sqrt{2}\right), (3, 3, -1)$ 13. 2

14. AB is neither symmetric nor skew symmetric

15. $\begin{bmatrix}1 & 0 & 0\\ 2 & 3 & 0\\ -1 & 0 & 4\end{bmatrix}+\begin{bmatrix}0 & 2 & 5\\ 0 & 0 & -6\\ 0 & 0 & 0\end{bmatrix}; \begin{bmatrix}1 & 2 & 2\\ 2 & 3 & -3\\ 2 & -3 & 4\end{bmatrix}+\begin{bmatrix}0 & 0 & 3\\ 0 & 0 & -3\\ -3 & 3 & 0\end{bmatrix}$

16. 4 17. $n = 7$ 18. 650

3. MISCELLANEOUS EXAMPLES

Example 1 :

If α, β are the roots of the equation $x^2 - 2x + 4 = 0$, find the value of $\begin{vmatrix}\alpha+\beta & \alpha^2+\beta^2 & \alpha^3+\beta^3\\ \alpha^2+\beta^2 & \alpha^3+\beta^3 & \alpha^4+\beta^4\\ \alpha^3+\beta^3 & \alpha^4+\beta^4 & \alpha^5+\beta^5\end{vmatrix}$

Solution :

The two roots of the equation are $1 \pm i\sqrt{3}$ so that we can take

$\alpha = 1 + i\sqrt{3} = 2\left(\frac{1}{2} + i\frac{\sqrt{3}}{2}\right) = 2e^{i\pi/3}$ and $\beta = 1 - i\sqrt{3} = 2\left(\frac{1}{2} - i\frac{\sqrt{3}}{2}\right) = 2e^{-i\pi/3}$

$\alpha + \beta = 2$, $\alpha^2 + \beta^2 = 2^2\left(e^{i2\pi/3} + e^{-i2\pi/3}\right) = 2^2 . 2\cos\frac{2\pi}{3} = -2^2$,

$\alpha^3 + \beta^3 = 2^3 \left(e^{i\pi} + e^{-i\pi}\right) = -2^4$, $\alpha^4 + \beta^4 = 2^4 \left(e^{i4\pi/3} + e^{-i4\pi/3}\right) = -2^4$

and $\alpha^5 + \beta^5 = 2^5 \left(e^{i5\pi/3} + e^{-i5\pi/3}\right) = 2^5$

$$\text{Determinant} = \begin{vmatrix} 2 & -2^2 & -2^4 \\ -2^2 & -2^4 & -2^4 \\ -2^4 & -2^4 & 2^5 \end{vmatrix} = -2^8 \begin{vmatrix} 1 & -2 & -2^2 \\ 1 & 2^2 & 2 \\ -1 & -1 & 1 \end{vmatrix} = -2^8 \begin{vmatrix} 1 & -2 & -2^2 \\ 0 & 6 & 6 \\ 0 & -3 & -3 \end{vmatrix} = 0.$$

Example 2 :

If x, y, z are not all zero and $ax + a^3y + (a^4 - 1)\, z = 0$,
$bx + b^3y + (b^4 - 1)\, z = cx + c^3y + (c^4 - 1)z = 0$,
where no two of a, b, c are equal, show that abc (ab + bc + ca) = a + b + c.

Solution :

Since the system of equations has a non-trivial solution

$$\begin{vmatrix} a & a^3 & a^4 - 1 \\ b & b^3 & b^4 - 1 \\ c & c^3 & c^4 - 1 \end{vmatrix} = \begin{vmatrix} a & a^3 & a^4 \\ b & b^3 & b^4 \\ c & c^3 & c^4 \end{vmatrix} - \begin{vmatrix} a & a^3 & 1 \\ b & b^3 & 1 \\ c & c^3 & 1 \end{vmatrix} = 0$$

$$\begin{vmatrix} a & a^3 & a^4 \\ b & b^3 & b^4 \\ c & c^3 & c^4 \end{vmatrix} = abc \begin{vmatrix} 1 & a^2 & a^3 \\ 1 & b^2 & b^3 \\ 1 & c^2 & c^3 \end{vmatrix}$$

$$= abc \begin{vmatrix} 1 & a^2 & a^3 \\ 0 & b^2 - a^2 & b^3 - a^3 \\ 0 & c^2 - a^2 & c^3 - a^3 \end{vmatrix} = abc\,(a - b)\,(b - c)\,(c - a)\,(ab + bc + ca)$$

$$\begin{vmatrix} a & a^3 & 1 \\ b & b^3 & 1 \\ c & c^3 & 1 \end{vmatrix} = \begin{vmatrix} a - c & a^3 - c^3 & 0 \\ b - c & b^3 - c^3 & 0 \\ c & c^3 & 1 \end{vmatrix} = (a - b)\,(b - c)\,(c - a)\,(a + b + c)$$

$\therefore$ $abc\ (a - b)\,(b - c)\,(c - a)\,(ab + bc + ca) = (a - b)\,(b - c)\,(c - a)\,(a + b + c)$

$\Rightarrow$ abc (ab + bc + ca) = a + b + c

Example 3 :

If a, b, c are positive numbers, prove that $\begin{vmatrix} a & b & c \\ a - 2b & b - 2c & c - 2a \\ 2b + c & 2c + a & 2a + b \end{vmatrix} = 0$ implies $a = b = c$.

Solution :

Adding the second and third columns to the first and taking out the common factor, we have the given

$$\text{determinant} = (a+b+c)\begin{vmatrix} 1 & b & c \\ -1 & b-2c & c-2a \\ 3 & 2c+a & 2a+b \end{vmatrix} = (a+b+c)\begin{vmatrix} 1 & b & c \\ 0 & 2(b-c) & 2(c-a) \\ 0 & 3b-4c+a & 3c-4a+b \end{vmatrix}$$

after the operations

$R_3 + 3R_2 \to R_3$

$R_2 + R_1 \to R_2$

$$= 2(a+b+c)\begin{vmatrix} b-c & c-a \\ 3b-4c+a & 3c-4a+b \end{vmatrix}$$

$$= 2(a+b+c)\begin{vmatrix} b-c & c-a \\ a-c & b-a \end{vmatrix}$$ after the operation $R_2 - 3R_1 \to R_2$

$$= 2(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$$

As $a + b + c \neq 0$, the given condition implies

$a^2 + b^2 + c^2 - ab - bc - ca \equiv \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2] = 0$ giving $a = b = c$.

Example 4 :

If $\omega \neq 1$ is a cube root of unity, show that $\begin{vmatrix} 1 & \omega & \omega^2 & \omega^3 \\ \omega & \omega^2 & \omega^3 & 1 \\ \omega^2 & \omega^3 & 1 & \omega \\ \omega^3 & 1 & \omega & \omega^2 \end{vmatrix} = \pm i\,3\sqrt{3}$.

Solution :

As $1 + \omega + \omega^2 = 0$ and $\omega^3 = 1$, adding R_2, R_3, R_4 to R_1 the given determinant

$$= \begin{vmatrix} 1 & 1 & 1 & 1 \\ \omega & \omega^2 & 1 & 1 \\ \omega^2 & 1 & 1 & \omega \\ 1 & 1 & \omega & \omega^2 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 0 & 0 \\ \omega & \omega^2-\omega & 1-\omega & 1-\omega \\ \omega^2 & 1-\omega^2 & 1-\omega^2 & \omega-\omega^2 \\ 1 & 0 & \omega-1 & \omega^2-1 \end{vmatrix}$$

$$= (1-\omega)^3\begin{vmatrix} -\omega & 1 & 1 \\ 1+\omega & 1+\omega & \omega \\ 0 & -1 & -1-\omega \end{vmatrix} = (1-\omega)^3\begin{vmatrix} -\omega & 0 & -\omega \\ -\omega^2 & -\omega^2 & \omega \\ 0 & -1 & \omega^2 \end{vmatrix}$$

$$= -\omega^2(1-\omega)^3\begin{vmatrix} 1 & 0 & 1 \\ -\omega & -\omega & 1 \\ 0 & -1 & \omega^2 \end{vmatrix} = -\omega^2(1-\omega)^3\begin{vmatrix} 1 & 0 & 0 \\ -\omega & -\omega & 1+\omega \\ 0 & -1 & \omega^2 \end{vmatrix}$$

$$= 3(\omega-\omega^2) = \pm 3\left[\frac{-1+i\sqrt{3}}{2} - \frac{-1-i\sqrt{3}}{2}\right] = \pm\, i\,3\sqrt{3}.$$

Example 5 :

If x, y, z such that $2x + 2\sqrt{3}y - 15z = 0$, $5\sqrt{3}x - 10y + 3\sqrt{3}z = 2\sqrt{3}$ and $x - 2\sqrt{3}y + 6z = 0$ are respectively the A.M., G.M. and the reciprocal of the H.M. between two numbers show that the sum of squares of the two numbers is divisible by 10.

Solution :

$$\text{Determinant of Coefficients} = \begin{vmatrix} 2 & 2\sqrt{3} & -15 \\ 5\sqrt{3} & -10 & 3\sqrt{3} \\ 1 & -2\sqrt{3} & 6 \end{vmatrix} = 54$$

By Cramer's rule or otherwise $x = 2$, $y = \sqrt{3}$ and $z = \frac{2}{3}$.

As $\frac{1+3}{2} = 2$, $\sqrt{1.3} = \sqrt{3}$ and $\left(\frac{2.1.3}{1+3}\right)^{-1} = \frac{2}{3}$, the numbers are 1 and 3.

The sum of their square is 10.

Example 6 :

$$\text{Solve : } \begin{vmatrix} 2x^2+x-1 & 3x^2+4x+1 & 4x^2+3x-1 \\ 4x^2+2x-1 & 6x^2+8x-2 & 8x^2+4x-3 \\ 6x^2+3x-1 & 9x^2+12x-5 & 12x^2+5x-6 \end{vmatrix} = 0.$$

Solution :

After the operations $R_3 + R_1 - 2R_2 \to R_3$ and $R_2 - 2R_1 \to R_2$, we get

$$\text{Determinant} = (x+1)\begin{vmatrix} 2x-1 & 3x+1 \\ 1 & -4 \end{vmatrix} = 0 \; ; \; x = -1 \text{ and } x = \frac{3}{11} \text{ are the two solutions.}$$

Example 7 :

Find the values of a and b for which the equations $x + y + z = 3$, $x + 3y + 2z = 6$ and $x + ay + 3z = b$ have

(i) a unique solution and

(ii) no solution

Solution :

$x + y + z = 3$

$x + 3y + 2z = 6$

$x + ay + 3z = b$

$$\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 3 & 2 \\ 1 & a & 3 \end{vmatrix} = 5 - a$$

If $\Delta \neq 0$ i.e., $a \neq 5$, the solution is unique .

Case I : $a \neq 5$ and b any real number gives unique solution

Case II : If $a = 5$, then $\Delta = 0$

Now, we can choose λ μ, r such that

$$\left.\begin{aligned} \lambda + \mu + r &= 0 \\ \lambda + 3\mu + 5r &= 0 \\ \lambda + 2\mu + 3r &= 0 \end{aligned}\right\}$$

These equation are satisfied by $\lambda = 1$, $\mu = -2$, $r = 1$

Hence, for consistency $3\lambda + 6\mu + br = 0$

i.e., $3 - 12 + b = 0$ i.e., $b = 9$

$\therefore$ If $a = 5$, $B = 9$, equations are consistent and have infinite solution.

If $a = 5$, $b \neq 9$, The system has no solution.

Example 8 :

Prove that $\begin{vmatrix} yz-x^2 & zx-y^2 & xy-z^2 \\ zx-y^2 & xy-z^2 & yz-x^2 \\ xy-z^2 & yz-x^2 & zx-y^2 \end{vmatrix}$ is divisible by $(x + y + z)^2$. Find the quotient.

Solution :

$$\text{Determinant} = \frac{1}{x}\begin{vmatrix} xyz-x^3 & zx^2-xy^2 & x^2y-z^2x \\ zx-y^2 & xy-z^2 & yz-x^2 \\ xy-z^2 & yz-x^2 & zx-y^2 \end{vmatrix}$$

$$= \frac{1}{x}\begin{vmatrix} 3xyz-x^3-y^3-z^3 & 0 & 0 \\ zx-y^2 & xy-z^2 & yz-x^2 \\ xy-z^2 & yz-x^2 & zx-y^2 \end{vmatrix}$$ after the operations $R_1 + yR_2 + zR_3 \to R_1$

$= (3\,xyz - x^3 - y^3 - z^3)^2 = (x + y + z)^2\,(xy + yz + zx - x^2 - y^2 - z^2)^2$

The quotient is $(xy + yz + zx - x^2 - y^2 - z^2)^2$.

Example 9 :

If $A = \begin{bmatrix} 0 & -\tan\alpha/2 \\ \tan\alpha/2 & 0 \end{bmatrix}$ and I is a 2×2 unit matrix, then prove that $I + A = (I - A)\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \sin\alpha \end{bmatrix}$

Solution :

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \text{ and given } A = \begin{bmatrix} 0 & -\tan\alpha/2 \\ \tan\alpha/2 & 0 \end{bmatrix}$$

$$\therefore \quad I + A = \begin{bmatrix} 1 & -\tan\alpha/2 \\ \tan\alpha/2 & 1 \end{bmatrix} \qquad \ldots (1)$$

$$\text{R.H.S.} = (I - A)\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix} = \begin{bmatrix} 1 & \tan\alpha/2 \\ -\tan\alpha/2 & 1 \end{bmatrix}\begin{bmatrix} \cos\alpha & -\sin\alpha \\ \sin\alpha & \cos\alpha \end{bmatrix}$$

$$= \begin{bmatrix} 1 & \tan\alpha/2 \\ -\tan\alpha/2 & 1 \end{bmatrix}\begin{bmatrix} \frac{1-\tan^2\alpha/2}{1+\tan^2\alpha/2} & -\frac{2\tan\alpha/2}{1+\tan^2\alpha/2} \\ \frac{2\tan\alpha/2}{1+\tan^2\alpha/2} & \frac{1-\tan^2\alpha/2}{1+\tan^2\alpha/2} \end{bmatrix}$$

$$= \begin{bmatrix} \frac{1-\tan^2\alpha/2}{1+\tan^2\alpha/2} + \frac{2\tan^2\alpha/2}{1+\tan^2\alpha/2} & -\frac{2\tan\alpha/2}{1+\tan^2\alpha/2} + \frac{\tan\alpha/2(1-\tan^2\alpha/2)}{1+\tan^2\alpha/2} \\ \frac{-\tan\alpha/2(1-\tan^2\alpha/2}{1+\tan^2\alpha/2} + \frac{2\tan\alpha/2}{1+\tan^2\alpha/2} & \frac{2\tan^2\alpha/2}{1+\tan^2\alpha/2} + \frac{1-\tan^2\alpha/2}{1+\tan^2\alpha/2} \end{bmatrix}$$

$$= \begin{bmatrix} \dfrac{(1+\tan^2\alpha/2)}{(1+\tan^2\alpha/2} & \dfrac{-\tan\alpha/2(1+\tan^2\alpha/2)}{(1+\tan^2\alpha/2} \\ \dfrac{\tan\alpha/2(1+\tan^2\alpha/2)}{(1+\tan^2\alpha/2)} & \dfrac{(1+\tan^2\alpha/2)}{(1+\tan^2\alpha/2)} \end{bmatrix} = \begin{bmatrix} 1 & -\tan\alpha/2 \\ \tan\alpha/2 & 1 \end{bmatrix}$$

$= I + A$ = L.H.S. {from (1)}

Example 10 :

Verify that $A = \frac{1}{3}\begin{pmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & -1 \end{pmatrix}$ is an orthogonal matrix.

Solution :

$$A = \frac{1}{3}\begin{pmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & -1 \end{pmatrix} \quad \therefore \quad A' = \frac{1}{3}\begin{pmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ 2 & 2 & -1 \end{pmatrix}$$

$$\therefore \quad AA' = \frac{1}{3}\begin{pmatrix} 1 & -2 & 2 \\ -2 & 1 & 2 \\ -2 & -2 & -1 \end{pmatrix} \times \frac{1}{3}\begin{pmatrix} 1 & -2 & -2 \\ -2 & 1 & -2 \\ 2 & 2 & -1 \end{pmatrix} = \frac{1}{9}\begin{pmatrix} 1+4+4 & -2-2+4 & -2+4-2 \\ -2-2+4 & 4+1+4 & 4-2-2 \\ -2+4-2 & 4-2-2 & 4+4+1 \end{pmatrix}$$

$$= \frac{1}{9}\begin{pmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = I.$$ Hence A is an orthogonal matrix.

Example 11 :

If $f = \begin{vmatrix} (b+c)^2 & a^2 & a^2 \\ b^2 & (c+a)^2 & b^2 \\ c^2 & c^2 & (a+b)^2 \end{vmatrix}$ and $g = a + b + c$, then find fg^{-3}.

Solution :

With the operations $C_1 - C_2 \to C_1$ and $C_2 - C_3 \to C_2$

$$f = (a+b+c)^2 \begin{vmatrix} b+c-a & 0 & a^2 \\ b-c-a & c+a-b & b^2 \\ 0 & c-a-b & (a+b)^2 \end{vmatrix}$$

$$= 2(a+b+c)^2 \begin{vmatrix} b+c-a & 0 & a^2 \\ b-c-a & c+a-b & b^2 \\ a-b & -a & ab \end{vmatrix}$$ after the operation $R_3 - R_1 - R_2 \to R_3$

$$= 2(a+b+c)^2 \begin{vmatrix} c & -a & a(a+b) \\ -c & c-b & b(a+b) \\ a-b & -a & ab \end{vmatrix}$$ after the operation $R_1 + R_3 \to R_1$, $R_2 + R_3 \to R_2$

$$= 2\,(a+b+c)^2 \left(\begin{vmatrix} 0 & -a & a(a+b) \\ 0 & c-b & b(a+b) \\ a-b & -a & ab \end{vmatrix} + \begin{vmatrix} c & -a & a(a+b) \\ -c & c-b & b(a+b) \\ 0 & -a & ab \end{vmatrix} \right)$$

$$= 2\,(a+b+c)^2 \left(a(a^2-b^2) \begin{vmatrix} -1 & 1 \\ c-b & b \end{vmatrix} + c \begin{vmatrix} c-b-a & (a+b)^2 \\ -a & ab \end{vmatrix} \right)$$

$= 2\,abc\,(a+b+c)^3$, after simplification.

Hence $fg^{-3} = 2\,abc$.

Example 12 :

If a, b, c are positive numbers, show that the determinant $\begin{vmatrix} bc-a^2 & ca-b^2 & ab-c^2 \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ca-b^2 \end{vmatrix}$ vanishes iff a = b = c.

Solution :

The given determinant $= \frac{1}{a} \begin{vmatrix} abc-a^3 & ca^2-ab^2 & a^2b-c^2a \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ca-b^2 \end{vmatrix}$

$$= \frac{1}{a} \begin{vmatrix} 3abc-a^3-b^3-ab^2 & 0 & 0 \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ca-b^2 \end{vmatrix}$$

after the operation, $R_1 \leftarrow R_1 + bR_2 + cR_3$

$$= \left(3abc-a^3-b^3-c^3\right)^2 = \frac{1}{4}(a+b+c)^2 \left[(a-b)^2+(b-c)^2+(c-a)^2\right]$$

The determinant vanishes iff $a = b = c$.

Example 13 :

If $A = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix}$, show that $A^3 = pI + qA + rA^2$

Solution :

We have $A^2 = AA = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix} \times \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ p & q & r \\ pr & p+qr & q+r^2 \end{bmatrix}$

$$\therefore \quad A^3 = A^2.A = \begin{bmatrix} 0 & 0 & 1 \\ p & q & r \\ pr & p+qr & q+r^2 \end{bmatrix} \times \begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ p & q & r \end{bmatrix} = \begin{bmatrix} p & q & r \\ pr & p+qr & q+r^2 \\ pq+r^2p & pr+q^2+qr^2 & p+2qr+r^3 \end{bmatrix}$$

$$\text{and } pI + qA + rA^2 = p\begin{bmatrix}1&0&0\\0&1&0\\0&0&1\end{bmatrix}+q\begin{bmatrix}0&1&0\\0&0&1\\p&q&r\end{bmatrix}+r\begin{bmatrix}0&0&1\\p&q&r\\pr&p+qr&q+r^2\end{bmatrix}$$

$$=\begin{bmatrix}p&0&0\\0&p&0\\0&0&p\end{bmatrix}+\begin{bmatrix}0&q&0\\0&0&q\\pq&q^2&qr\end{bmatrix}+\begin{bmatrix}0&0&r\\pr&qr&r^2\\pr^2&pr+qr^2&qr+r^3\end{bmatrix}$$

$$=\begin{bmatrix}p+0+0&0+q+0&0+0+r\\0+0+pr&p+0+qr&0+q+r^2\\0+pq+pr^2&0+q^2+pr+qr^2&p+qr+qr+r^3\end{bmatrix}$$

$$=\begin{bmatrix}p&q&r\\pr&p+qr&q+r^2\\pq+pr^2&q^2+pr+qr^2&p+2qr+r^3\end{bmatrix}$$

Example 14 :

Show that $\begin{vmatrix}x+1&\omega&\omega^2\\\omega&x+\omega^2&1\\\omega^2&1&x+\omega\end{vmatrix} = x^3$, where ω is a complex cube root of unity.

Solution :

$$\text{Let } \Delta = \begin{vmatrix}x+1&\omega&\omega^2\\\omega&x+\omega^2&1\\\omega^2&1&x+\omega\end{vmatrix} = \begin{vmatrix}x&x&x\\\omega&x+\omega^2&1\\\omega^2&1&x+\omega\end{vmatrix}, \text{ by } R_1 : R_1 + R_2 + R_3$$

$$= x\begin{vmatrix}1&1&1\\\omega&x+\omega^2&1\\\omega^2&1&x+\omega\end{vmatrix}$$

Now in $\begin{vmatrix}1&1&1\\\omega&(x+\omega^2)&1\\\omega^2&1&(x+\omega)\end{vmatrix}$ if we put $x = 0$, we get

$$\begin{vmatrix}1&1&1\\\omega&\omega^2&1\\\omega^2&1&\omega\end{vmatrix} = \frac{1}{\omega}\begin{vmatrix}1&1&1\\\omega^2&\omega^3&\omega\\\omega^2&1&\omega\end{vmatrix} = \frac{1}{\omega}\begin{vmatrix}1&1&1\\\omega^2&1&\omega\\\omega^2&1&\omega\end{vmatrix} = 0$$

$\therefore$ x is a factor of $\begin{vmatrix} 1 & 1 & 1 \\ \omega & x+\omega^2 & 1 \\ \omega^2 & 1 & x+\omega \end{vmatrix}$ $\therefore$ we can write $\Delta = x^2(x + h)$

Equating the coefficient of x^2 on both sides, we have

$1 + \omega + \omega^2 = h \Rightarrow h = 0$ $\therefore$ $\Delta = x^3$.

Example 15 : Solve the equation $\begin{vmatrix} 4x & 6x+2 & 8x+1 \\ 6x+2 & 9x+3 & 12x \\ 8x+1 & 12x & 16x+2 \end{vmatrix} = 0$

Solution :

Determinant = $\begin{vmatrix} 4x & 6x+2 & 8x+1 \\ 6x+2 & 9x+3 & 12x \\ -3 & -4 & 3 \end{vmatrix}$, $R_3 \leftarrow R_3 + R_1 - 2R_2$

$$= \begin{vmatrix} 12x+1 & 14x+3 & 50x+10 \\ 18x+2 & 21x+3 & 75x+9 \\ 0 & -1 & 0 \end{vmatrix} \begin{matrix} C_1 \leftarrow C_3 + C_1 \\ C_2 \leftarrow C_3 + C_2 \\ C_3 \leftarrow 4C_3 + 3C_2 \end{matrix} = -97x - 11$$

So that $x = \frac{-11}{97}$.

Example 16 :

Show that $\begin{vmatrix} 1 & \beta\gamma+\alpha\beta & (\beta\gamma-\alpha\delta)^2 \\ 1 & \gamma\alpha+\beta\delta & (\gamma\alpha-\beta\delta)^2 \\ 1 & \alpha\beta+\gamma\delta & (\alpha\beta-\gamma\delta)^2 \end{vmatrix} = (\alpha - \beta)(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)(\gamma - \delta)$.

Solution :

the given determinant = $\begin{vmatrix} 1 & \beta\gamma+\alpha\beta & (\beta\gamma+\alpha\delta)^2 \\ 1 & \gamma\alpha+\beta\delta & (\gamma\alpha+\beta\delta)^2 \\ 1 & \alpha\beta+\gamma\delta & (\alpha\beta+\gamma\delta)^2 \end{vmatrix}$ [By $C_3 : C_3 + 4\ \alpha\ \beta\ \gamma\ \delta\ C_1$]

$= \begin{vmatrix} 1 & a & a^2 \\ 1 & b & b^2 \\ 1 & c & c^2 \end{vmatrix}$ where, $a = \beta\gamma + \alpha\delta$ etc.

$= (b - c)(c - a)(a - b)$

$= (\gamma\alpha + \beta\delta - \alpha\beta - \gamma\delta)(\alpha\beta + \gamma\delta - \beta\gamma - \alpha\delta)(\beta\gamma + \alpha\delta - \alpha\delta - \gamma\alpha - \beta\delta)$

$= (\alpha - \delta)(\gamma - \beta)(\beta - \delta)(\alpha - \gamma)(\gamma - \delta)(\beta - \alpha) = (\alpha - \beta)(\alpha - \gamma)(\alpha - \delta)(\beta - \gamma)(\beta - \delta)(\gamma - \delta)$.

Example 17 :

Show that the system of equations $\alpha x - 3y + z = 0$, $x + \alpha y + 3z = 1$, $3x + y + 5z = 2$ has a unique solution for $\alpha = 2$ and find the solution. Show that the system is consistent for $\alpha = -1$ and that it has an infinite number of solutions. Find all the solutions in terms of a parameter.

Solution :

For $\alpha = 2$,

the determinant of coefficients = $\begin{vmatrix} 2 & -3 & 1 \\ 1 & 2 & 3 \\ 3 & 1 & 5 \end{vmatrix} = -3.$

By Cramer's rule, $\frac{x}{\begin{vmatrix} 0 & -3 & 1 \\ 1 & 2 & 3 \\ 2 & 1 & 5 \end{vmatrix}} = \frac{y}{\begin{vmatrix} 2 & 0 & 1 \\ 1 & 1 & 3 \\ 3 & 2 & 5 \end{vmatrix}} = \frac{z}{\begin{vmatrix} 2 & -3 & 0 \\ 1 & 2 & 1 \\ 3 & 1 & 2 \end{vmatrix}} = \frac{-1}{3} \Rightarrow x = 2, y = 1, z = -1.$

For $\alpha = -1$, the determinant = $\begin{vmatrix} -1 & -3 & 1 \\ 1 & -1 & 3 \\ 3 & 1 & 5 \end{vmatrix} = 0$, so that the system does not have any solution if the number of equations in the system does not reduce atleast by one. The system is then said to be consistent and it has an infinite number of solutions. Since equation (1) = 2 equation (2) – equation (3), the system reduces to only two independent equations (2) and (3) and therefore it is consistent. With $z = t$, $x - y = 1 - 3t$ and $3x + y = 2 - 5t$ giving the one-parameter solution $\left(\frac{3}{4} - 2t, -\frac{1}{4} + t, t\right), -\infty < t < \infty.$

Example 18 :

If α, β, γ are the roots of $x^3 - x^2 - 8 = 0$, find the value of $\begin{vmatrix} \gamma+\alpha & \alpha+\beta & \beta+\gamma \\ \alpha(\beta+\gamma) & \beta(\gamma+\alpha) & \gamma(\alpha+\beta) \\ \alpha^2+1 & \beta^2+1 & \gamma^2+1 \end{vmatrix}$

Solution :

As x is absent, $\alpha + \beta + \gamma = -(-1)$, $\alpha\beta + \beta\gamma + \gamma\alpha = 0$, $\alpha\beta\gamma = -(-8)$

Given determinant = $\begin{vmatrix} 1-\beta & 1-\gamma & 1-\alpha \\ -\beta\gamma & -\gamma\alpha & -\alpha\beta \\ \alpha^2+1 & \beta^2+1 & \gamma^2+1 \end{vmatrix} = \begin{vmatrix} 2 & 1-\gamma & 1-\alpha \\ 0 & -\gamma\alpha & -\alpha\beta \\ 4 & \beta^2+1 & \gamma^2+1 \end{vmatrix}$ with $C_1 \leftarrow C_1 + C_2 + C_3$

$= -2\alpha \begin{vmatrix} 1 & 1-\gamma & 1-\alpha \\ 0 & \gamma & \beta \\ 2 & \beta^2+1 & \gamma^2+1 \end{vmatrix} = -2\alpha \begin{vmatrix} 1 & 1-\gamma & 1-\alpha \\ 0 & \gamma & \beta \\ 0 & \beta^2-1+2\gamma & \gamma^2-1+2\alpha \end{vmatrix}$ by $R_3 \leftarrow R_3 - 2R_1$

$= -2\alpha [\gamma(\gamma^2 - 1 + 2\alpha) - \beta(\beta^2 - 1 + 2\gamma)] = 6\ \alpha\beta\gamma = 48.$

UNSOLVED EXERCISE

Section-A

(Straight objective type with single option correct)

Topic : 1 Types of Matrices, Addition, Subtraction and Transpose of a Matrix Based Questions

Q.1 If I is a unit matrix of order 10, then the determinant of equal to:
(a) 10 (b) 1
(c) 1/10 (d) 9

Q.2 If A and B are non-singular matrices, then:
(a) $(AB)^{-1} = A^{-1}B^{-1}$ (b) $AB = BA$
(c) $(AB)' = A'B'$ (d) $(AB)^{-1} = B^{-1}A^{-1}$

Q.3 If A and B are square matrices of the same order, then
(a) $(AB)' = A'B'$
(b) $(AB)' = B'A'$
(c) $AB = 0$; If $|A| = 0$ or $|B| = 0$
(d) $AB = 0$; if $A = I$ or $B = I$

Q.4 If A, B, C are three $n \times n$ matrices, then $(ABC)' =$
(a) $A'B'C'$ (b) $C'B'A'$
(c) $B'C'A'$ (d) $B'A'C'$

Q.5 If A and B are square matrices of order 3 such that $|A| = -1$, $|B| = 3$, then $|3AB| =$
(a) –9 (b) –81
(c) –27 (d) 81

Q.6 If A is non-singular matrix, then det (A^{-1}) is equal to

(a) $\det\left(\frac{1}{A^2}\right)$ (b) $\frac{1}{\det(A^2)}$

(c) $\det\left(\frac{1}{A}\right)$ (d) $\frac{1}{\det(A)}$

Q.7 If the matrix $A = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & i \\ -i & a \end{bmatrix}$, where $i = \sqrt{-1}$, is unitary, then a is equal to
(a) –2 (b) –1
(c) 0 (d) 1

Q.8 For each real $x : -1 < x < 1$. Let $A(x)$ be the matrix $(1 - x)^{-1}\begin{bmatrix} 1 & -x \\ -x & 1 \end{bmatrix}$ and $z = \frac{x+y}{1+xy}$, then
(a) $A(z) = A(x) + A(y)$ (b) $A(z) = A(x)[A(y)]^{-1}$
(c) $A(z) = A(x)A(y)$ (d) $A(z) = A(x) - A(y)$

Q.9 If A is involutary matrix and I is unit matrix of the same order then $(I - A)(I + A)$ is equal to
(a) O (b) A
(c) I (d) $2A$

Q.10 If A is a non-singular matrix then $|\text{adj } A|$ is equal to (where n is the order of the matrix)
(a) $|A|^n$ (b) $|A|^{n-1}$
(c) $|A|^{n-2}$ (d) $|A|^{(n-1)^2}$

Q.11 Matrix A is such that $A^2 = 2A - I$ when I is unit matrix then A^n is equal to
(a) $nA - (n - 1)I$ (b) $nA - I$ (c) $2^{n-1}A - I$ (d) None of these

Q.12 If $\begin{bmatrix} i & 0 \\ 3 & -i \end{bmatrix} + X = \begin{bmatrix} i & 2 \\ 3 & 4+i \end{bmatrix} - X$, then X is equal to

(a) $\begin{bmatrix} 0 & -1 \\ 3 & i \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 1 \\ 0 & 2+i \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 0 \\ 0 & 2-i \end{bmatrix}$ (d) $\begin{bmatrix} i & 2 \\ 0 & 2+i \end{bmatrix}$

Q.13 $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 2 & 3 \\ -1 & -2 & -3 \end{bmatrix}$, then A is a nilpotent matrix of index
(a) 2 (b) 3
(c) 4 (d) 5

Q.14 If $A = \frac{1}{2}\begin{pmatrix} -1 & -\sqrt{3} \\ \sqrt{3} & 1 \end{pmatrix}$, then $A^{-1} - A^2$ is equal to a
(a) null matrix (b) invertible matrix
(c) unit matrix (d) none of these

Q.15 The values of λ for which the matrix $A = \begin{pmatrix} \lambda & 0 & \lambda \\ \lambda & 0 & -\lambda \\ 0 & 1 & 0 \end{pmatrix}$ is orthogonal is
(a) ± 1 (b) $\pm \frac{1}{\sqrt{3}}$
(c) $\pm \frac{1}{2}$ (d) $\pm \frac{1}{\sqrt{2}}$

Q.16 The values of a for which the matrix A = $\begin{pmatrix} a & a^2-1 & -3 \\ a+1 & 2 & a^2+4 \\ -3 & 4a & -1 \end{pmatrix}$ is symmetric are

(a) – 1 (b) – 2
(c) 3 (d) 2

Q.17 If A = $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 3 & -1 & 1 \end{bmatrix}$ then $A^3 - 3A^2 - A + 9I$ is equal to

(a) O (b) I
(c) A (d) A^2

Q.18 If A = $\begin{bmatrix} 4 & x+2 \\ 2x-3 & x+1 \end{bmatrix}$ is an invertible matrix, then x cannot take value

(a) – 1 (b) 2
(c) 3 (d) none of these

Q.19 Let α, β, γ be three real number and A = $\begin{bmatrix} 1 & \cos(\beta-\alpha) & \cos(\gamma-\alpha) \\ \cos(\alpha-\beta) & 1 & \cos(\gamma-\beta) \\ \cos(\alpha-\gamma) & \cos(\beta-\gamma) & 1 \end{bmatrix}$ then

(a) A is singular (b) A is non-singular
(c) A is orthogonal (d) none of these

Q.20 Let A = $\begin{bmatrix} 1 & 2 & 1 \\ 0 & 1 & -1 \\ 3 & 1 & 1 \end{bmatrix}$. The sum of all values of λ for which there exists a column vector X $\neq 0$ such that AX = λX, is

(a) 0 (b) 1
(c) 2 (d) 3

Topic : 2 Properties of Determinants

Q.1 If a, b, c are in A.P. and $f(x)$ = $\begin{vmatrix} x+a & x^2+1 & 1 \\ x+b & 2x^2-1 & 1 \\ x+c & 3x^2-2 & 1 \end{vmatrix}$, then $f'(x)$ is

(a) 0 (b) 1
(c) $a + bc$ (d) $\dfrac{abc}{a+b+c}$

Q.2 Let $\{D_1, D_2, D_3, ..., D_n\}$ be the set of third order determinates that can be made with the distinct non-zero real numbers $a_1, a_2 ..... a_9$ (using all), then

(a) $\sum_{i=1}^{n} D_i = 1$ (b) $\sum_{i=1}^{n} D_i = 0$
(c) $D_i = D_j$, $\forall$ i, j (d) None of these

Q.3 If $x_i = a_i b_i c_i$, $i = 1, 2, 3$ are three-digit positive integer such that each x_i is a multiple of 19, then for some integer n, $\Delta = \begin{vmatrix} a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \\ c_1 & c_2 & c_3 \end{vmatrix}$ is given by

(a) $19n + 1$ (b) $19n + 2$
(c) $19n$ (d) $19n + 3$

Q.4 If a, b, c are the sides of a triangle ABC such that $\begin{vmatrix} a^2 & b^2 & c^2 \\ (a+1)^2 & (b+1)^2 & (c+1)^2 \\ (a-1)^2 & (b-1)^2 & (c-1)^2 \end{vmatrix} = 0$, then

(a) ΔABC is non-isosceles right-angled triangle
(b) ΔABC is an equilateral
(c) ΔABC is an acute angled triangle with no two angles being equal
(d) ΔABC is is an isosceles

Q.5 Let $\begin{vmatrix} x & 2 & x \\ x^2 & x & 6 \\ x & x & 6 \end{vmatrix} = Ax^4 + Bx^3 + Cx^2 + Dx + E$.

Then value of $5A + 4B + 3C + 2D + E$ is equal to

(a) Zero (b) –16
(c) 16 (d) –17

Q.6 If a, b, c are the sides of a triangle ABC and A, B, C are angles opposite to a, b, c and $\Delta = \begin{vmatrix} a^2 & b\sin A & c\sin A \\ b\sin A & 1 & \cos A \\ c\sin A & \cos A & 1 \end{vmatrix}$, then

(a) Δ = area of triangle
(b) Δ = perimeter of the triangle
(c) $\Delta = \Sigma a^2$
(d) None of these

Q.7 If $f(x)$ = $\begin{vmatrix} \sin x & \sec x & x^2-1 \\ \operatorname{cosec} x & x\sin x & \cos x \\ \tan x & x\tan x & x^2+1 \end{vmatrix}$

then $\int_{-\pi/3}^{\pi/3} f(x)dx =$

(a) 1 (b) $\pi/3 + 1$
(c) 0 (d) $\pi/3 - 1$

Q.8 If *P, Q* and *R* represent the angles of an acute angled triangle, no two of them being equal then the value of

$$\begin{vmatrix} 1 & 1+\sin P & \sin P(1+\sin P) \\ 1 & 1+\sin Q & \sin Q(1+\sin Q) \\ 1 & 1+\sin R & \sin R(1+\sin R) \end{vmatrix}$$ is

(a) 0
(b) Positive
(c) Negative
(d) Can not be determined

Q.9 If α, β are the roots of the equation $x^2 + bx + c = 0$, then

$$\begin{vmatrix} 3 & 1+\alpha+\beta & 1+\alpha^2+\beta^2 \\ 1+\alpha+\beta & 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 \\ 1+\alpha^2+\beta^2 & 1+\alpha^3+\beta^3 & 1+\alpha^4+\beta^4 \end{vmatrix}$$ is

(a) $(b + 1)^2(b^2 - 4c)$
(b) $(1 + b + c)^2(b^2 - 4c)$
(c) $(b - c)^2(b^2 - 4c)$
(d) $(1 - b - c)^2(b^2 - 4c)$

Q.10 $\begin{vmatrix} \text{cosec}\,\alpha & 1 & 0 \\ 1 & 2\text{cosec}\,\alpha & 1 \\ 0 & 0 & 2\text{cosec}\,\alpha \end{vmatrix}$ gives the value equivalent to

(a) $\frac{1}{2}(\tan^2\alpha + \cot^2\alpha)$
(b) $\frac{1}{2}(\tan^3\alpha + \cot^3\alpha)$
(c) $\frac{1}{2}\left(\tan^2\frac{\alpha}{2} + \cot^2\frac{\alpha}{2}\right)$
(d) $\frac{1}{2}\left(\tan^3\frac{\alpha}{2} + \cot^3\frac{\alpha}{2}\right)$

Q.11 If α, β and γ are three different numbers

$$\begin{vmatrix} \alpha & \alpha^3 & \alpha^4-1 \\ \beta & \beta^3 & \beta^4-1 \\ \gamma & \gamma^3 & \gamma^4-1 \end{vmatrix}$$ is true iff

(a) $\alpha + \beta + \gamma = \alpha\beta\gamma$
(b) $\alpha\beta + \beta\gamma + \gamma\alpha = \alpha\beta\gamma$
(c) $\alpha\beta\gamma(\alpha + \beta + \gamma) = \alpha\beta + \beta\gamma + \gamma\alpha$
(d) $\alpha\beta\gamma(\alpha\beta + \beta\gamma + \gamma\alpha) = \alpha + \beta + \gamma$

Q.12 If $\begin{vmatrix} x & y & \alpha x - y \\ y & z & \alpha y - z \\ 2 & 1 & 0 \end{vmatrix} = 0$ and $\alpha \neq \frac{1}{2}$, then *x, y, z* are in

(a) A.P. (b) G.P.
(c) H.P. (d) None of these

Q.13 If $\Delta_1 = \begin{vmatrix} x & b & b \\ a & x & b \\ a & a & x \end{vmatrix}$ and $\Delta_2 = \begin{vmatrix} x & b \\ a & x \end{vmatrix}$, then

(a) $\Delta_1 = 3\Delta_2^2$ (b) $\frac{d}{dx}(\Delta_1) = 3\Delta_2$
(c) $\frac{d}{dx}(\Delta_1) = 3\Delta_2^2$ (d) $\Delta_1 = 3(\Delta_2)^{3/2}$

Q.14 The value of the determinant

$$\begin{vmatrix} \sin^2\left(x+\frac{3\pi}{2}\right) & \sin^2\left(x+\frac{5\pi}{2}\right) & \sin^2\left(x+\frac{7\pi}{2}\right) \\ \sin\left(x+\frac{3\pi}{2}\right) & \sin\left(x+\frac{5\pi}{2}\right) & \sin\left(x+\frac{7\pi}{2}\right) \\ \sin\left(x-\frac{3\pi}{2}\right) & \sin\left(x-\frac{5\pi}{2}\right) & \sin\left(x-\frac{7\pi}{2}\right) \end{vmatrix}$$

is

(a) 1 (b) 2
(c) 3 (d) 0

Q.15 If *z* is a complex number and $a_1, a_2, a_3, b_1, b_2, b_3$ are all real then

$$\begin{vmatrix} a_1z + b_1\bar{z} & a_2z + b_2\bar{z} & a_3z + b_3\bar{z} \\ b_1z + a_1\bar{z} & b_2z + a_2\bar{z} & b_3z + a_3\bar{z} \\ b_1z + a_1 & b_2z + a_2 & b_3z + a_3 \end{vmatrix}$$ is equal to

(a) $(a_1a_2a_3 + b_1b_2b_3)|z|^2$ (b) $|z|^2$
(c) 3 (d) 0

Q.16 If α, β, γ are the roots of $px^3 + qx^2 + r = 0$ then the value of the determinant

$$\begin{vmatrix} \alpha\beta & \beta\gamma & \gamma\alpha \\ \beta\gamma & \gamma\alpha & \alpha\beta \\ \gamma\alpha & \alpha\beta & \beta\gamma \end{vmatrix}$$ is

(a) *pq* (b) *qr*
(c) 0 (d) *pr*

Q.17 If $[x]$ stands for the greatest integer less than or equal to x, then the value of $\begin{vmatrix} [e] & [\pi] & [\pi^2-6] \\ [\pi] & [\pi^2-6] & [e] \\ [\pi^2-6] & [e] & [\pi] \end{vmatrix}$ is

(a) –8 (b) 8
(c) 0 (d) 1

Topic : 3 Adjoint and Inverse of a Matrix Based Questions

Q.1 If $F(x) = \begin{bmatrix} \cos x & -\sin x & 0 \\ \sin x & \cos x & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $G(y) = \begin{bmatrix} \cos y & 0 & \sin y \\ 0 & 1 & 0 \\ -\sin y & 0 & \cos y \end{bmatrix}$ then $[F(x)G(y)]^{-1}$ is equal to

(a) $F(-x)F(-y)$ (b) $F(x^{-1})G(y^{-1})$
(c) $F(-y)F(-x)$ (d) $G(-y^{-1})F(-x^{-1})$

Q.2 If A is a $n \times n$ matrix, then adj (adj A) =
(a) $|A|^{n-1} A$ (b) $|A|^{n-2} A$
(c) $|A|^n n$ (d) None of these

Q.3 The inverse of the matrix $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ is

(a) $\begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

Q.4 Let $\begin{bmatrix} 1 & 0 & 0 \\ 5 & 2 & 0 \\ -1 & 6 & 1 \end{bmatrix}$, then the adjoint of A is

(a) $\begin{bmatrix} 2 & -5 & 32 \\ 0 & 1 & -6 \\ 0 & 0 & 2 \end{bmatrix}$ (b) $\begin{bmatrix} -1 & 0 & 0 \\ -5 & -2 & 0 \\ 1 & 6 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} -1 & 0 & 0 \\ -5 & -2 & 0 \\ 1 & 6 & -1 \end{bmatrix}$ (d) None of these

Q.5 If $A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}$, then A(adj A) =

(a) $\begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix}$ (b) $\begin{bmatrix} 0 & 10 \\ 10 & 0 \end{bmatrix}$

(c) $\begin{bmatrix} 10 & 1 \\ 1 & 10 \end{bmatrix}$ (d) None of these

Q.6 $\begin{bmatrix} 1 & 3 \\ 3 & 10 \end{bmatrix}^{-1} =$

(a) $\begin{bmatrix} 10 & 3 \\ 3 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 10 & -3 \\ -3 & 1 \end{bmatrix}$

(c) $\begin{bmatrix} 1 & 3 \\ 3 & 10 \end{bmatrix}$ (d) $\begin{bmatrix} -1 & -3 \\ -3 & -10 \end{bmatrix}$

Q.7 The matrix $\begin{pmatrix} 1 & a & 2 \\ 1 & 2 & 5 \\ 2 & 1 & 1 \end{pmatrix}$ is not invertible, if 'a' has the value

(a) 2 (b) 1
(c) 0 (d) –1

Q.8 If $A = \begin{bmatrix} 3 & -3 & 4 \\ 2 & -3 & 4 \\ 0 & -1 & 1 \end{bmatrix}$, then $A^{-1} =$
(a) A (b) A^2
(c) A^3 (d) A^4

Q.9 If A and B are non-singular square matrices of same order, then adj (AB) is equal to
(a) (adj A) (adj B) (b) (adj B) (adj A)
(c) (adj B^{-1})(adj A^{-1}) (d) (adj A^{-1})(adj B^{-1})

Topic : 4 Solving System of Equation Based Q1uestions

Q.1 If the equations $a_1x^2 + b_1x + c_1 = 0$ and $a_2x^2 + b_2x + c_2 = 0$ possess a common root, then $\begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \begin{vmatrix} b_1 & c_1 \\ b_2 & c_2 \end{vmatrix}$ is equal to

(a) $\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$ (b) $2\cdot\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}$

(c) $\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}^2$ (d) None of these

Q.2 If $[p]$ stands for the greatest integer less than or equal to p, then in order that the set of equations, $2x - 3y = 4$, $7x - 2y = 2$, $[3\pi]x - [2e]y = [4a]$ may be consistent is that a should lie in

(a) $\left[\frac{1}{2}, 1\right]$ (b) [2, 3]

(c) $\left[\frac{3}{2}, \frac{7}{4}\right)$ (d) $\left[1, \frac{3}{2}\right)$

Q.3 If t is real and $\lambda = \frac{t^2 - 3t + 4}{t^2 + 3t + 4}$, then the number of solutions of the system of equations $3x - y + 4z = 3$, $x + 2y - 3z = -2$, $6x + 5y + \lambda z = -3$ is

(a) One (b) Two
(c) Zero (d) Infinite

Q.4 The set of equations : $\lambda x - y + (\cos\theta)z = 0$; $3x + y + 2z = 0$; $(\cos\theta)x + y + 2z = 0$; $0 \le \theta < 2\pi$, has non-trivial solutions

(a) For no value of λ and θ

(b) For all values of λ and θ

(c) For all values of λ and only two values of θ

(d) For only one value of λ and all values of θ

Q.5 The set of homoegeneous equations:

$tx + (t + 1)y + (t - 1)z = 0$; $(t + 1)x + ty + (t + 2)z = 0$; $(t - 1)x + (t + 2)y + tz = 0$

has non-trivial solution for

(a) Three values of t (b) Two values of t
(c) One value of t (d) No value of t

Q.6 X, Y and Z are positive numbers greater than 10 such that Y and Z have respectively 1 and 0 at their unit's place and Δ is the determinant $\begin{vmatrix} X & 4 & 1 \\ Y & 0 & 1 \\ Z & 1 & 0 \end{vmatrix}$. If $(\Delta + 1)$ is divisible by 10 then X has at its unit's place

(a) 1 (b) 0
(c) 2 (d) None of these

Q.7 If $x^a y^b = e^m$, $x^c y^d = e^n$, $\Delta_1 = \begin{vmatrix} m & b \\ n & d \end{vmatrix}$, $\Delta_2 = \begin{vmatrix} a & m \\ c & n \end{vmatrix}$ and $\Delta_3 = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$ then the values of x and y are respectively

(a) $\frac{\Delta_1}{\Delta_3}$ and $\frac{\Delta_2}{\Delta_3}$

(b) $\frac{\Delta_2}{\Delta_1}$ and $\frac{\Delta_3}{\Delta_1}$

(c) $\log\left(\frac{\Delta_1}{\Delta_3}\right)$ and $\log\left(\frac{\Delta_2}{\Delta_3}\right)$

(d) e^{Δ_1/Δ_3} and e^{Δ_2/Δ_3}

Q.8 If $a \ne p$, $b \ne q$, $c \ne r$ and the system of equation

$px + by + cz = 0$

$ax + qy + cz = 0$

$ax + by + rz = 0$

has a non-zero solution, then value of $\frac{p+q}{q-z} + \frac{q+b}{q-b} + \frac{r+c}{r-c}$ is

(a) 2 (b) – 2
(c) – 1 (d) 1

Q.9 Number of real values of l for which the system of equations

$(\lambda + 3)x + (\lambda + 2)y + z = 0$

$3x + (\lambda + 3)y + z = 0$

$2x + 3y + z = 0$

has a non-trivial solutions is

(a) 0 (b) 1
(c) 2 (d) infinite

Q.10 Let $\Delta(x) = \begin{vmatrix} x^2 - 5x + 3 & 3x - 5 & 3 \\ 3x^2 + x + 9 & 6x + 1 & 9 \\ 7x^2 - 6x + 9 & 14x - 6 & 21 \end{vmatrix} = ax^3 + bx^2 + cx + d$ then a equals

(a) 1 (b) 2
(c) 3 (d) None of these

SECTION-B
(Previous Years Questions)

Q.1 If A and B are square matrices of equal degree, then which one is correct among the following?
(a) A + B = B + A (b) A + B = A – B
(c) A – B = B – A (d) AB = BA

Q.2 Let $P = \begin{bmatrix} 1 & 0 & 0 \\ 4 & 1 & 0 \\ 16 & 4 & 1 \end{bmatrix}$ and I be the identity matrix of order 3. If $Q = [q_{ij}]$ is a matrix, such that $P^{50} - Q = I$, then $\frac{q_{31} + q_{32}}{q_{21}}$ equals
(a) 52 (b) 103
(c) 201 (d) 205

Q.3 If $A = \begin{bmatrix} 1 & 2 & 2 \\ 2 & 1 & -2 \\ c & 2 & b \end{bmatrix}$ is a matrix satisfying the equation $AA^T = 9I$, where, I is 3 × 3 identity matrix, then the ordered pair (a, b) is equal to
(a) (2, – 1) (b) (– 2, 1)
(c) (2, 1) (d) (– 2, – 1)

Q.4 If $A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \\ 0 & -2 & 4 \end{bmatrix}$, $6A^{-1} = A^2 + cA + dI$, then (c, d) is
(a) (– 6, 11) (b) (– 11, 6)
(c) (11, 6) (d) (6, 11)

Q.5 If $P = \begin{bmatrix} \sqrt{3}/2 & 1/2 \\ -1/2 & \sqrt{3}/2 \end{bmatrix}$, $A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ and $Q = PAP^T$, then $P^TQ^{2005}P$ is
(a) $\begin{bmatrix} 1 & 2005 \\ 0 & 1 \end{bmatrix}$ (b) $\begin{bmatrix} 1 & 2005 \\ 2005 & 1 \end{bmatrix}$
(c) $\begin{bmatrix} 1 & 0 \\ 2005 & 1 \end{bmatrix}$ (d) $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

Q.6 If $A = \begin{bmatrix} \alpha & 0 \\ 1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 0 \\ 5 & 1 \end{bmatrix}$ then value of α for which $A^2 = B$, is
(a) 1 (b) – 1
(c) 4 (d) no real values

Q.7 Let ω be a complex number such that $2\omega + 1 = z$, where $z = \sqrt{-3}$. If $\begin{vmatrix} 1 & 1 & 1 \\ 1 & -\omega^2 - 1 & \omega^2 \\ 1 & \omega^2 & \omega^7 \end{vmatrix} = 3k$, then k is equal to
(a) – z (b) z
(c) – 1 (d) 1

Q.8 If $\alpha, \beta \neq 0$ and $f(n) = \alpha^n + \beta^n$ and $\begin{vmatrix} 3 & 1+f(1) & 1+f(2) \\ 1+f(1) & 1+f(2) & 1+f(3) \\ 1+f(2) & 1+f(3) & 1+f(4) \end{vmatrix} = K(1 - \alpha)^2(1 - \beta)^2(\alpha - \beta)^2$, then K is equal to
(a) $\alpha\beta$ (b) $\frac{1}{\alpha\beta}$
(c) 1 (d) – 1

Q.9 Let $P = [a_{ij}]$ be a 3 × 3 matrix and let $Q = [b_{ij}]$, where $b_{ij} = 2^{i+j} a_{ij}$ for $1 \le i, j \le 3$. If the determinant of P is 2, then the determinant of the matrix Q is
(a) 2^{10} (b) 2^{11}
(c) 2^{12} (d) 2^{13}

Q.10 If $A = \begin{bmatrix} \alpha & 2 \\ 2 & \alpha \end{bmatrix}$ and $|A^3| = 125$, then the value of α is
(a) ± 1 (b) ± 2
(c) ± 3 (d) ± 5

Q.11 The number of distinct real roots of $\begin{vmatrix} \sin x & \cos x & \cos x \\ \cos x & \sin x & \cos x \\ \cos x & \cos x & \sin x \end{vmatrix} = 0$ in the interval $-\frac{\pi}{4} \le x \le \frac{\pi}{4}$ is
(a) 0 (b) 2
(c) 1 (d) 3

Q.12 If $f(x) = \begin{vmatrix} 1 & x & x+1 \\ 2x & x(x-1) & (x+1)x \\ 3x(x-1) & x(x-1)(x-2) & (x+1)x(x-1) \end{vmatrix}$, then f (100) is equal to
(a) 0 (b) 1
(c) 100 (d) – 100

Q.13 The parameter on which the value of the

determinant $\begin{vmatrix} 1 & a & a^2 \\ \cos(p-d)x & \cos px & \cos(p+d)x \\ \sin(p-d)x & \sin px & \sin(p+d)x \end{vmatrix}$

does not depend upon, is

(a) a (b) p
(c) d (d) x

Q.14 The determinant $\begin{vmatrix} xp+y & x & y \\ yp+z & y & z \\ 0 & xp+y & yp+z \end{vmatrix} = 0$, if

(a) x, y, z are in AP
(b) x, y, z are in GP
(c) x, y, z are in HP
(d) xy, yz, zx are in AP

Q.15 Consider the set A of all determinants of order 3 with entries 0 or 1 only. Let B be the subset of A consisting of all determinants with value 1. Let C be the subset of A consisting of all determinants with value – 1. Then,

(a) C is empty
(b) B has as many elements as C
(c) $A = B \cup C$
(d) B has twice as many elements as C

Q.16 If $A = \begin{bmatrix} 2 & -3 \\ -4 & 1 \end{bmatrix}$, then adj $(3A^2 + 12A)$ is equal to

(a) $\begin{bmatrix} 72 & -84 \\ -63 & 51 \end{bmatrix}$ (b) $\begin{bmatrix} 51 & 63 \\ 84 & 72 \end{bmatrix}$

(c) $\begin{bmatrix} 51 & 84 \\ 63 & 72 \end{bmatrix}$ (d) $\begin{bmatrix} 72 & -63 \\ -84 & 51 \end{bmatrix}$

Q.17 If $A = \begin{bmatrix} 5a & -b \\ 3 & 2 \end{bmatrix}$ and A adj $A = AA^T$, then $5a + b$ is equal to

(a) – 1 (b) 5
(c) 4 (d) 13

Q.18 If A is a 3 × 3 non-singular matrix such that $AA^T = A^TA$ and $B = A^{-1}A^T$, then BB^T is equal to

(a) I + B (b) I
(c) B^{-1} (d) $(B^{-1})^T$

Q.19 If $P = \begin{bmatrix} 1 & \alpha & 3 \\ 1 & 3 & 3 \\ 2 & 4 & 4 \end{bmatrix}$ is the adjoint of a 3 × 3 matrix A and $|A| = 4$, then α is equal to

(a) 4 (b) 11
(c) 5 (d) 0

Q.20 If P is a 3 × 3 matrix such that $P^T = 2P + I$, where P^T is the transpose of P and I is the 3 × 3 identity matrix, then there exists a column matrix, $X = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \neq \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ such that

(a) $PX = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$ (b) PX = X

(c) PX = 2X (d) PX = – X

Q.21 Let $\omega \neq 1$ be a cube root of unity and S be the set of all non-singular matrices of the form $\begin{bmatrix} 1 & a & b \\ \omega & 1 & c \\ \omega^2 & \omega & 1 \end{bmatrix}$, where each of a, b and c is either ω or ω^2. Then, the number of distinct matrices in the set S is

(a) 2 (b) 6
(c) 4 (d) 8

Q.22 Let M and N be two 3 × 3 non-singular skew-symmetric matrices such that MN = NM. If P^T denotes the transpose of P, then $M^2N^2(M^TN)^{-1}(MN^{-1})^T$ is equal to

(a) M^2 (b) $-N^2$
(c) $-M^2$ (d) MN

Q.23 The system of linear equations $x + \lambda y - z = 0$; $\lambda x - y - z = 0$; $x + y - \lambda z = 0$ has a non-trivial solution for

(a) infinitely many values of λ
(b) exactly one value of λ
(c) exactly two values of λ
(d) exactly three values of λ

Q.24 The set of all values of λ for which the system of linear equation
$2x_1 - 2x_2 + x_3 = \lambda x_1$, $2x_1 - 3x_2 + 2x_3 = \lambda x_2$ and $-x_1 + 2x_2 = \lambda x_3$ has a non-trivial solution

(a) is an empty set
(b) is a singleton set
(c) contains two elements
(d) contains more than two elements

Q.25 The number of value of *k*, for which the system of equation
$(k + 1)x + 8y = 4y \Rightarrow kx + (k + 3)y = 3k - 1$
has no solution, is

(a) infinite (b) 1
(c) 2 (d) 3

Q.26 The number of 3 × 3 matrices A whose entries are either 0 or 1 and for which the system $A\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}$ has exactly two distinct solutions, is

(a) 0 (b) $29 - 1$
(c) 168 (d) 2

Q.27 Given, $2x - y + 2z = 2$, $x - 2y + z = -4$, $x + y + \lambda z = 4$ then the value of λ such that the given system of equations has no solution, is

(a) 3 (b) 1
(c) 0 (d) – 3

Q.28 The number of values of k for which the system of equations $(k + 1)x + 8y = 4k$ and $kx + (k + 3)y = 3k - 1$ has infinitely many solutions, is/are

(a) 0 (b) 1
(c) 2 (d) ∞

Q.29 If the system of equations $x + ay = 0$, $az + y = 0$ and $ax + z = 0$ has infinite solutions, then the value of *a* is

(a) – 1 (b) 1
(c) 0 (d) no real values

Q.30 If the system of equations $x - ky - z = 0$, $kx - y - z = 0$, $x + y - z = 0$ has a non-zero solution, then possible values of *k* are

(a) – 1, 2 (b) 1, 2
(c) 0, 1 (d) – 1, 1

Section-C
(School / Board pattern)

Q.1 Solve for x and y, $x\begin{bmatrix} 2 \\ 1 \end{bmatrix} + y\begin{bmatrix} 3 \\ 5 \end{bmatrix} + \begin{bmatrix} -8 \\ -11 \end{bmatrix} = 0$

Q.2 If X and Y are 2 × 2 matrices, then solve the following matrix equations for X and Y

$$2X + +3Y = \begin{bmatrix} 2 & 3 \\ 4 & 0 \end{bmatrix}, 3x + 2y = \begin{bmatrix} -2 & 2 \\ 1 & -5 \end{bmatrix}$$

Q.3 Give an example of matirces A,B and C, such that AB = AC, where A is non-zero but $B \neq C$.

Q.4 If $A = \begin{bmatrix} 1 & 2 \\ -2 & 1 \end{bmatrix}, B = \begin{bmatrix} 2 & 3 \\ 3 & -4 \end{bmatrix}$ and $C = \begin{bmatrix} 1 & 0 \\ -1 & 0 \end{bmatrix}$, verify

(i) $(AB)C = A(BC)$. (ii) $A(B + C) = AB + AC$.

Q.5 If $A = [2\ 1], B = \begin{bmatrix} 5 & 3 & 4 \\ 8 & 7 & 6 \end{bmatrix}$ and $C = \begin{bmatrix} -1 & 2 & 1 \\ 1 & 0 & 2 \end{bmatrix}$, then verify that $A(B + C) = (AB + AC)$.

Q.6 Find x, y and z, if $A = \begin{bmatrix} 0 & 2y & 2 \\ x & y & -z \\ x & -y & z \end{bmatrix}$ satisfies $A' = A^{-1}$.

Q.7 Using the properties find the value of $\begin{vmatrix} 3x & -x + y & -x + z \\ x - y & 3y & z - 1 \\ x - z & y - z & 3z \end{vmatrix}$

Q.8 Using the properties prove that $\begin{vmatrix} y + z & z & y \\ z & z + x & x \\ y & x & x + y \end{vmatrix} = 4xyz$

Q.9 Using matirc method, solve the systemof equations $3x+2y-2z=3$, $x+2y+3z=6$ and $2x-y+z=2$

Q.10 If $A=\begin{vmatrix} 2 & 2 & -4 \\ -4 & 2 & -4 \\ 2 & -1 & 5 \end{vmatrix}$ and $B=\begin{vmatrix} 1 & -1 & 0 \\ 2 & 3 & 4 \\ 0 & 1 & 2 \end{vmatrix}$, then find BA and use this to solve the system of equations $y+2z=7$, $x-y=3$ and $2x+3y+4z=17$

Q.11 Prove that $\begin{vmatrix} bc-a^2 & ca-b^2 & ab-c^2 \\ ca-b^2 & ab-c^2 & bc-a^2 \\ ab-c^2 & bc-a^2 & ca-b^2 \end{vmatrix}$ is divisible by $(a+b+c)$ and find the quotient.

Q.12 If $x+y+z=0$, then prove that $\begin{vmatrix} xa & yb & zc \\ yc & za & xb \\ zb & xc & ya \end{vmatrix} = xyz \begin{vmatrix} a & b & c \\ c & a & b \\ b & c & a \end{vmatrix}$.

CHAPTER 4

CONTINUITY & DIFFERENTIABILITY

1. LIMIT, CONTINUITY AND DIFFERENTIABILITY

INTRODUCTION

The key idea underlying the development of calculus is the concept of limit. Calculus is the mathematics of change. The immense practical power of calculus derives from its ability to describe and predict the behavior of changing quantities. Falling apples, orbiting spacecraft, growing populations, decaying radioactive materials, rising consumer prices : all can be modeled using the calculus of change. The calculus tool that makes all this possible is called the derivative function.

LIMITS OF A FUNCTION

Let y = f(x) be a function of x and for some particular value of x say x = a, the value of y is indeterminate, then we consider the value of the function at the points which are very near to 'a'. If these values tend to a definite unique number ℓ as x tends to 'a' (either from left or from right) then this unique number ℓ is called the limits of f(x) at x = a and we write it as

$$\lim_{x \to a} f(x) = \ell \, .$$

Meaning of '$x \to a$':

Let x be a variable and a be a constant. If x assumes values nearer and nearer to 'a' then we can say 'x tends to a' and we write

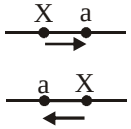
'$x \to a$'.

It should be noted that as $x \to a$ we have $x \neq a$.

(i) $x \neq a$

(ii) x assumes values nearer and nearer to 'a' and

(iii) we are not specifying any manner in which x should approach to a. x may



approach to 'a' from left or right as shown in figure.

EVALUATION OF LEFT HAND AND RIGHT HAND LIMITS :

The statement $x \to a^-$ means that x is tending to a from the left hand side, i.e. x is a number less than a but very close to a. Therefore $x \to a^-$ is equivalent to $x = a - h$ where $h > 0$ such that $h \to 0$. Similarly $x \to a^+$ is equivalent to $x = a + h$ where $h \to 0$. Thus, we have the following algorithms for finding left hand and right hand limits at $x = a$

1. Algorithm For Finding Left Hand Limit :

To evaluate LHL of $f(x)$ at $x = a$, i.e. $\lim_{x \to a^-} f(x)$ we proceed as follows :

(i) Write $\lim_{x \to a^-} f(x)$

(ii) Put $x = a - h$ and replace $x \to a^-$ by $h \to 0$ to obtain

$$\lim_{h \to 0} f(a - h).$$

(iii) Simplify $\lim_{h \to 0} f(a - h)$ by using the formula for the given function.

(iv) The value obtain in step (iii) is the LHL of $f(x)$ at $x = a$.

2. Algorithm For Finding Right Hand Limit :

To evaluate RHL of $f(x)$ at $x = a$ i.e. $\lim_{x\to a^+} f(x)$ we proceed as follows :

(i) Write $\lim_{x\to a^+} f(x)$

(ii) Put $x = a + h$ and replace $x \to a^+$ by $h \to 0$ to obtain $\lim_{h\to 0} f(a+h)$

(iii) Simplify $\lim_{h\to 0} f(a+h)$ by using the formula for the given function.

(iv) The value obtained in step (iii) is the RHL of $f(x)$ at $x = a$.

Existence of Limit : The limit of a function at some point exists only when its left-hand limit and right hand limit at that point exist and are equal. Thus

$\lim_{x\to a} f(x)$ exists $\Rightarrow \lim_{x\to a^-} f(x) = \lim_{x\to a^+} f(x) = \ell$

Where ℓ is called the limit of the function

DIFFERENCE BETWEEN THE VALUE OF A FUNCTION AT A POINT AND THE LIMIT AT A POINT

Let $f(x)$ be a function and let a be a point. Then, we have the following possibilities :

(i) $\lim_{x\to a} f(x)$ exists but $f(a)$ (the value of $f(x)$ at $x = a$) does not exist

***Illustration 1* :** Consider the function $f(x)$ defined by $f(x)\ \dfrac{x^2-9}{x-3}$

Clearly, this function is not defined at x = 3 i.e. f(3) does not exist, because it attains the form $\dfrac{0}{0}$. But, it can be easily seen that $\lim_{x\to 3^-} f(x) = \lim_{x\to 3^+} f(x) = 6$. So, $\lim_{x\to 3} f(x)$ exists.

Thus, the $\lim_{x\to 3} f(x)$ exists but the value $f(3)$ does not exist.

(ii) The value $f(a)$ exists but $\lim_{x\to a} f(x)$ does not exists

Illustration 2 : Consider the function $f(x) = [x]$ where [.] = G.I.F.

value of function at $x = 2$

$\Rightarrow \quad f(2) = [2] = 2$

LHL at $x = 2$

$\lim_{x\to 2^-} f(x) = \lim_{h\to 0} f(2-h) = \lim_{h\to 0} [2-h] = 1$

RHL at x = 2

$\lim_{x\to 2^+} f(x) = \lim_{h\to 0} f(2+h) = \lim_{h\to 0} [2+h] = 2$

Clearly RHL $\neq$ LHL so limit of function does not exist but value of function exist

(iii) $\lim_{x\to a} f(x)$ and $f(a)$ both exist but are unequal

Illustration 3 : Consider the function $f(x)$ defined by $f(x) = \begin{cases} \dfrac{x^2-4}{x-2}, & x \neq 2 \\ 3\ , & x = 2 \end{cases}$

It can be easily seen that $\lim_{x\to 2^-} f(x) = 4 = \lim_{x\to 2^+} f(x)$

So, $\lim_{x\to 2} f(x)$ exists and is equal to 4. Also, the value $f(2)$ exists and is equal to 3.

Thus $\lim_{x\to 2} f(x)$ and f(2) both exist but are unequal.

(iv) $\lim_{x\to a} f(x)$ and $f(a)$ both exist and are equal

Consider the function $f(x)$ defined by $f(x) = \begin{cases} \dfrac{x^2-4}{x-2}, & x \neq 2 \\ 4\ , & x = 2 \end{cases}$

For this function, it can be easily seen that $\lim_{x\to 2} f(x)$ and $f(2)$ both exist and are equal to 4.

2. LIMIT OF A SEQUENCE

The sequence, $x_1, x_2, \ldots\ldots, x_n, \ldots\ldots$ ..... (1)
is understood as a function $x_n = f(n)$ defined on the set of natural numbers $X = \{1, 2, \ldots\ldots\}$.
By analogy with the limit of a function at a point at infinity, we introduce here the concept of the limit of a sequence.
Namely, a number a is the limit of the sequence x_n (n = 1, 2, ...)

$$\lim_{n\to\infty} x_n = a$$

if for any $\varepsilon > 0$ there exists a number N, depending on ε, such that for all natural $n > N$ the following inequality is fulfilled

$$| x_n - a | < \varepsilon.$$

Illustration 4 : Let $x_1 = 0.9$, $x_2 = 0.99$, $x_3 = 0.999$, ... then prove that $\lim_{n\to\infty} x_n = 1$

Solution : We have $x_n = 1 - \dfrac{1}{10^n}$ (n = 1, 2, ...)

Let $\varepsilon > 0$ be an arbitrary number. Then $|x_n - 1| = \dfrac{1}{10^n} < \varepsilon$

if $n > \log\dfrac{1}{\varepsilon} = N$. Hence, $\lim_{n\to\infty} x_n = 1$

Student Task

Find $\lim_{x\to\infty} x_n$ if :

Q.1 $x_n = \dfrac{3n^2 + 5n + 4}{2 + n^2}$

Q.2 $x_n = \dfrac{1^2 + 2^2 + \ldots.. + n^2}{5n^3 + n + 1}$

Q.3 $x_n = \left(\dfrac{3n^2 + n - 2}{4n^2 + 2n + 7} \right)^3$

Q.4 $x_n = \sqrt[n]{5n}$

Q.5 $x_n = \left(\dfrac{2n^3}{2n^2 + 3} + \dfrac{1 - 5n^2}{5n + 1} \right)$

Q.6 $x_n = \sqrt[3]{n^2 - n^3} + n$

ANSWERS

1. 3 **2.** 1/15 **3.** 27/64 **4.** 1 **5.** 1/5 **6.** 1/4

3. BASIC LIMIT THEOREMS

Theorem 1 : If each of the terms of an algebraic sum of a finite number of functions has a limit as $x \to a$, then there exists a limit of this algebraic sum as $x \to a$ and it is equal to the same algebraic sum of the limits of the terms.

Corollary : A function can have only one limit as $x \to a$.

Indeed, if $f(x) \to A$ and $f(x) \to A'$ as $x \to a$

then, by virtue of theorem 1, we get

$f(x) - f(x) = 0 \to A - A'$ as $x \to a$.

Since the limit of a constant function is equal to the function itself and is unique, we have $A - A' = 0$, i.e. $A' = A$.

Note : It was assumed in the initial condition of the theorem that each of the functions has a limit and it was to be proved that their sum also has a limit. The converse is, generally speaking, not true : from the fact of existence of the limit of a sum it does not follow that each term of this sum has a limit. For instance, we have

$$\lim_{x \to \infty} (\sin^2 x + \cos^2 x) = 1, \text{ while } \lim_{x \to \infty} \sin^2 x \text{ and } \lim_{x \to \infty} \cos^2 x \text{ do not exist, and therefore, here}$$

$$\lim_{x \to \infty} (\sin^2 x + \cos^2 x) \neq \lim_{x \to \infty} \sin^2 x + \lim_{x \to \infty} \cos^2 x$$

Thus, the statement : the limit of a sum is equal to the sum of the limits of the terms is not rigorous.

A similar note should be borne in mind when considering the limit of a product (theorem 2) and the limit of a quotient (theorem 4).

Theorem 2 : If each of the factors of a product of a finite number of functions has a limit as $x \to a$, then the limit of the product as $x \to$ a is equal to the product of the limits of the factors.

Corollary 1 : A constant factor may be taken outside the limit sign. Indeed, if c is a constant function, then

$$\lim_{x \to a} [c\, f(x)] = \lim_{x \to a} c \lim_{x \to a} f(x) = c \lim_{x \to a} f(x)$$

Corollary 2 : If a function $f(x)$ has a limit as $x \to a$, then the limit of the nth power of the function as $x \to a$ (where n is a positive integer) is equal to the nth power of the limit of that function, i.e.

$$\lim_{x \to a} [f(x)]^n = \left[\lim_{x \to a} f(x) \right]^n$$

Illustration 5 : $\lim_{x\to\infty}\frac{(x+10)(x+20)^2(x+30)^3}{x^6} = \lim_{x\to\infty}\left[\left(1+\frac{10}{x}\right)\left(1+\frac{20}{x}\right)^2\left(1+\frac{30}{x}\right)^3\right]$

$= \lim_{x\to\infty}\left(1+\frac{10}{x}\right)\lim_{x\to\infty}\left(1+\frac{20}{x}\right)^2\lim_{x\to\infty}\left(1+\frac{30}{x}\right)^3 = 1\times1\times1 = 1$

Theorem 3 : If a function $f(x)$ has a non-zero limit as $x \to a$, then the limit of the inverse function $1/f(x)$ as $x \to a$ is equal to the inverse of the limit of the given function, i.e. $\lim_{x\to a}\frac{1}{f(x)} = \frac{1}{\lim_{x\to a} f(x)}$

Theorem 4 : If both the dividend $f(x)$ and the division $g(x)$ have limits as $x \to a$ and the limit of the divisor is non-zero, then the limit of their quotient (fraction) as $x \to a$ is equal to the quotient of the limits of the dividend (the numerator of the fraction) and the divisor (the denominator of the fraction), i.e. $\lim_{x\to a}\frac{f(x)}{g(x)} = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$

Illustration 6 : $\lim_{x\to1}\frac{x^2-1}{x^3-1} = \lim_{x\to1}\frac{(x-1)(x+1)}{(x-1)(x^2+x+1)} = \lim_{x\to1}\frac{(x+1)}{(x^2+x+1)} = \frac{\lim_{x\to1}(x+1)}{\lim_{x\to1}(x^2+x+1)} = \frac{2}{3}$

The following theorem is given without proof.

Theorem 5 : If a function $f(x)$ has a limit $x \to a$ and there exists $\sqrt[n]{f(x)}$ (where n is a natural number) both at the point a and in some of its neighbourhoods U_a, then

$$\lim_{x\to a}\sqrt[n]{f(x)} = \sqrt[n]{\lim_{x\to a} f(x)}$$

From above theorems you can remember :

Let $\underset{x\to a}{Lim}\, f(x) = \ell$ and $\underset{x\to a}{Lim}\, g(x) = m$. If ℓ and m are finite then :

(i) $\underset{x\to a}{Lim}\big(f(x) \pm g(x)\big) = \ell \pm m$.

(ii) $\underset{x\to a}{Lim}\big(f(x).g(x)\big) = \ell.m$.

(iii) $\underset{x\to a}{Lim}\frac{f(x)}{g(x)} = \frac{\ell}{m}$, provided $m \neq 0$.

(iv) $\underset{x\to a}{Lim}\, k\, f(x) = k\,\underset{x\to a}{Lim}\, f(x)$ = kℓ; where k is a constant.

(v) $\underset{x\to a}{Lim}\,[f(x) + k] = \underset{x\to a}{Lim}\, f(x) + k$, where k is a constant .

(vi) If $f(x) \le g(x)$, then $\underset{x\to a}{Lim}\, f(x) \le \underset{x\to a}{Lim}\, g(x)$.

Limit in Case of Composite Function

$\underset{x\to a}{Lim}\, f\big[g(x)\big] = f\left(\underset{x\to a}{Lim}\, g(x)\right)$; provided f is continuous at $x = \underset{x\to a}{Lim} g(x)$.

For example : $\underset{x\to a}{Lim}\, \ell n(f(x)) = \ell n\left\{\underset{x\to a}{Lim}\, f(x)\right\} = \ell n\, \ell \;\left(where \underset{x\to a}{Lim}\, f(x) = \ell,\ \ell > 0\right)$.

Student Task

Find the limits :

Q.1 $\lim\limits_{x\to 1} \dfrac{4x^5+9x+7}{3x^6+x^3+1}$

Q.2 $\lim\limits_{x\to 2} \dfrac{x^3+3x^2-9x-2}{x^3-x-6}$

Q.3 $\lim\limits_{x\to -1} \dfrac{x+1}{\sqrt{6x^2+3}+3x}$

Q.4 $\lim\limits_{x\to \infty} \left(\dfrac{x^3}{3x^2-4}-\dfrac{x^2}{3x+2}\right)$

Q.5 $\lim\limits_{x\to +\infty} \left(\sqrt{9x^2+1}-3x\right)$

Q.6 $\lim\limits_{x\to +\infty} \dfrac{2\sqrt{x}+3\sqrt[3]{x}+5\sqrt[5]{x}}{\sqrt{3x-2}+\sqrt[3]{2x-3}}$

ANSWERS

1. 4 2. 15/11 3. 1 4. 2/9 5. 0 6. $2/\sqrt{3}$

4. LIMITS AND INEQUALITIES : THE SQUEEZE PRINCIPLE

Algebraic limit rules don't help with limits like $\lim\limits_{x\to 0} x.\sin\left(\dfrac{1}{x}\right)$ and $\lim\limits_{x\to \infty} \dfrac{\sin x}{x}$

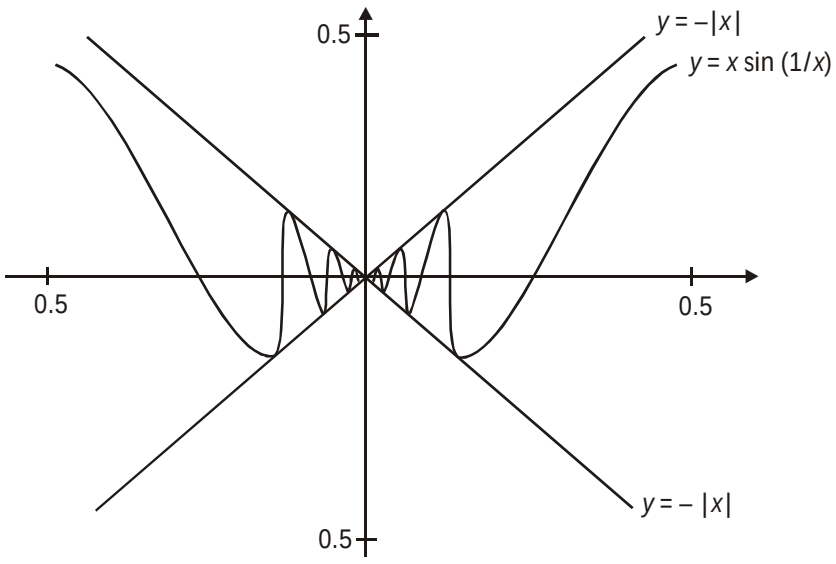
Squeezing inequalities

The idea is to use carefully-chosen inequalities to "squeeze" the function of interest between two simpler functions. For the first limit above, the key inequalities say that for any x,

$$-|x| \le x\sin\left(\frac{1}{x}\right) \le |x|$$

A graph shows convincingly that the inequality is valid:

Finding a limit by squeezing



As $x \to 0$ both the left and right sides of this three-term inequality tend to 0.

"Squeezed" between $|x|$ and $|x|$, $x \sin(1/x)$ has only one place to go :

$$x\sin\left(\frac{1}{x}\right) \to 0 \text{ as } x \to 0$$

There's a general theorem here. The graph above makes it reasonable; we omit the format proof.

Theorem : (The squeeze principle). Suppose that $f(x) \le g(x) \le h(x)$ for all x near $x = a$. If $f(x) \to L$ and $h(x) \to L$ as $x \to a$, then $g(x) \to L$ as $x \to a$.

Note :- The theorem is good for more than computation. It also guarantees that $\lim_{x\to a} g(x)$ exists. The squeeze principle is conceptually easy, the hard part, in practice, is to choose the "bounding functions" f and h properly.

Illustration 7 : Explain why $\lim_{x\to\infty} \frac{\sin x}{x} = 0$

Solution : The inequality $-\frac{1}{x} \le \frac{\sin x}{x} \le \frac{1}{x}$ holds for all $x > 0$. Because the left and right sides tends to 0 as $x \to \infty$ so, must the middle expression.

5. STANDARD METHODS OF EVALUATION OF LIMITS

When $x \to \infty$

In this case expression should be expressed as a function $1/x$ and then after removing indeterminate form, (If it is there) replace $1/x$ by 0.

Illustration 8 : $\lim_{x\to\infty} \frac{2x^2+3x}{3x^2+4} = ?$

Solution : $\lim_{x\to\infty} \frac{2x^2+3x}{3x^2+4} = \lim_{x\to\infty} \frac{2+(3/x)}{3+(4/x^2)} = \frac{2}{3}$

Illustration 9 : Find $\lim_{x\to\infty} \frac{1^2+2^2+3^2+.....+x^2}{x^3}$

Solution : $\lim_{x\to\infty} \frac{1^2+2^2+3^2+.....+x^2}{x^3} = \lim_{x\to\infty} \frac{x(x+1)(2x+1)}{6x^3} = \lim_{x\to\infty} \frac{(1+1/x)(2+1/x)}{6} = 1/3$

When $x \to a$, $a \in R$.

Factorisation method :

If $f(x)$ is of the form $\frac{h(x)}{g(x)}$ and of indeterminant form then this form is removed by factorising $g(x)$ and $h(x)$ and cancel the common factors, then put the value of x.

Illustration 10 : Find the value of $\lim_{x\to 64}\left(\frac{\sqrt{x}-8}{\sqrt[3]{x}-4}\right)$

Solution : $\lim_{x\to 64}\left(\frac{\sqrt{x}-8}{\sqrt[3]{x}-4}\right) \Rightarrow \lim_{y\to 2}\frac{y^3-8}{y^2-4}$ Here $y^6 = x$

$$\Rightarrow \lim_{y\to 2}\frac{(y-2)(y^2+2y+4)}{(y-2)(y+2)} = \frac{(y^2+2y+4)}{y+2} \Rightarrow \frac{4+4+4}{2+2} = 3$$

Illustration 11 : Evaluate $\lim_{x\to 1}\frac{x^3-x^2\log x+\log x-1}{x^2-1}$

Solution : The given limit = $\lim_{x\to 1}\frac{\left(x^3-1\right)-\left(x^2-1\right)\log x}{x^2-1}$

$$= \lim_{x\to 1}\frac{(x-1)\left(x^2+x+1\right)-(x-1)(x+1)\log x}{(x-1)(x+1)}$$

$$= \lim_{x\to 1}\frac{(x-1)\left[x^2+x+1-(x+1)\log x\right]}{(x-1)(x+1)} = \frac{1^2+1+1-(1+1)\log 1}{(1+1)} = \frac{3}{2}$$

Substitution Method :

For evaluating $\lim_{x\to a}\frac{g(x)}{h(x)}$, we follow the following steps-

(i) Put $x = a + h$, where h is small ($\neq 0$) as $x \to a$, $h \to 0$

(ii) Simplify numerator and denominator and cancel h throughout ($h \neq 0$)

(iii) Put $h = 0$, we get the required limit.

Illustration 12 : $\lim_{x\to 1}\frac{x^2-1}{x-1} = ?$

Solution : $\lim_{h\to 0}\frac{(1+h)^2-1}{(1+h)-1} = \lim_{h\to 0}\frac{h^2+2h}{h} = \lim_{h\to 0}(h+2) = 2$

Illustration 13 : Evaluate $\lim_{x\to -1^+}\frac{\sqrt{\pi}-\sqrt{\cos^{-1}x}}{\sqrt{x+1}}$

Solution :: Put $\cos^{-1}x = \theta \Rightarrow x = \cos\theta$

$\therefore$ Above limit $= \lim_{\theta\to\pi}\frac{\sqrt{\pi}-\sqrt{\theta}}{\sqrt{1+\cos\theta}} = \lim_{\theta\to\pi}\frac{\pi-\theta}{\sqrt{1+\cos\theta}}\cdot\frac{1}{\sqrt{\pi}+\sqrt{\theta}}$

$$\lim_{\theta\to\pi}\frac{\pi-\theta}{\sqrt{2\cos^2\theta/2}}\lim_{\theta\to\pi}\frac{1}{\sqrt{\pi}+\sqrt{\theta}} = \lim_{k\to 0}\frac{k}{\sqrt{2\sin^2\left(\frac{\pi}{2}-\frac{\theta}{2}\right)}}\cdot\frac{1}{2\sqrt{\pi}}$$

$$\text{(putting } \theta = \pi - k) = \lim_{k\to 0}\frac{k/2}{\frac{\sqrt{2}}{2}\sin\frac{k}{2}}\cdot\frac{1}{2\sqrt{\pi}} = \frac{1}{\sqrt{2\pi}}$$

Rationalisation Method

In this method we rationalise the factor containing the square root and simplify and we put the value of x.

Illustration 14 : Find $\lim_{x\to 5}\frac{1-\sqrt{x-4}}{x-5}$

Solution : $\lim_{x\to 5}\frac{1-\sqrt{x-4}}{x-5} \Rightarrow \lim_{x\to 5}\frac{1-\sqrt{x-4}}{x-5}\cdot\frac{1+\sqrt{x-4}}{1+\sqrt{x-4}}$

$$\Rightarrow \lim_{x\to 5}\frac{1-x+4}{(x-5)(1+\sqrt{x-4})} = \lim_{x\to 5}\frac{-(x-5)}{(x-5)(1+\sqrt{x-4})}$$

$$\Rightarrow \lim_{x\to 5}\frac{-1}{(1+\sqrt{x-4})} = \frac{-1}{(1+\sqrt{5-4})} = \frac{-1}{2}$$

Illustration 15 : Evaluate $\lim_{x\to\pm\infty} x\left(\sqrt{x^2+k}-x\right)$, $k > 0$

Solution : $\lim_{x\to\pm\infty} x\left(\sqrt{x^2+k}-x\right)\frac{\left(\sqrt{x^2+k}+x\right)}{\left(\sqrt{x^2+k}+x\right)}$ **=** $\lim_{x\to\pm\infty}\frac{x\left(x^2+k-x^2\right)}{\left(\sqrt{x^2+k}+x\right)}$ **=**

$$\lim_{x\to\pm\infty}\frac{xk}{\left(|x|\sqrt{\left(1+\frac{k}{x^2}\right)}\right)+x}$$

Here we have to consider two cases

(i) When $x\to\infty$; $|x| = x$

then the given limit = $\lim_{x\to\infty}\frac{xk}{x\sqrt{\left(1+\frac{k}{x^2}\right)}+x} = \lim_{x\to\infty}\frac{xk}{x\left(\sqrt{\left(1+\frac{k}{x^2}\right)}+1\right)} = \frac{k}{2}$

(ii) When $x\to-\infty$; $|x| = -x$ then we have $\lim_{x\to-\infty}\frac{xk}{-x\sqrt{\left(1+\frac{k}{x^2}\right)}+x}$

$$\lim_{x\to-\infty}\frac{xk}{x\left(-\sqrt{1+\frac{k}{x^2}}+1\right)} = \frac{k}{-1^-+1} = \frac{k}{0^-}\to-\infty$$

Expansion Method : If $x\to 0$ and there is atleast one function in the given expression which can be expanded then we express numerator and denominator in the ascending of x and remove the common factor there.

The following expansions of some standard functions are given.

(i) $e^x = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + ...$

(ii) $e^{-x} = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} + ...$

(iii) $\log(1+x) = x - \frac{x^2}{2} + \frac{x^3}{3} - .....$

(iv) $\log(1-x) = -x - \frac{x^2}{2} - \frac{x^3}{3} - \ldots\ldots$

(v) $a^x = 1 + (x \log a) + \frac{(x\log a)^2}{2!} + \frac{(x\log a)^3}{3!} + \ldots\ldots$

(vi) $\sin x = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \ldots\ldots$

(vii) $\cos x = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \ldots\ldots$

(viii) $\tan x = x + \frac{x^3}{3} + \frac{2}{15} x^5 + \ldots\ldots\ldots$

(ix) $\sin^{-1} x = x + \frac{x^3}{3!} + \frac{9x^5}{5!} + \ldots\ldots$

(x) $\cos^{-1} x = \frac{\pi}{2} - \left(x + \frac{x^3}{3!} + \frac{9x^5}{5!} + \ldots\ldots \right)$

(xi) $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \ldots\ldots\ldots$

(xii) $(1+x)^n = 1 + nx + \frac{n(n-1)}{2!} x^2 + \ldots\ldots$

Illustration 16 : $\lim_{x\to 0} \frac{e^x + \log\{(1-x)/e\}}{\tan x - x} = ?$

Solution : $\text{Limit} = \lim_{x\to 0} \frac{e^x + \log(1-x) - 1}{\tan x - x}$

$$= \lim_{x\to 0} \frac{\left(1 + x + \frac{x^2}{2!} + \ldots\ldots\right) - x - \frac{x^2}{2} - \frac{x^3}{3} \ldots\ldots - 1}{x + \frac{x^3}{3} + \frac{2x^5}{15} + \ldots\ldots - x} = -1/2$$

Illustration 17 : Evaluate $\lim_{x\to 0} \frac{2x - \sin^{-1} x}{2x + \tan^{-1} x}$.

Solution : $$\lim_{x\to 0} \frac{2x - \left(x + \frac{x^3}{6} + \ldots \right)}{2x + \left(x - \frac{x^3}{3} + \ldots \right)} = \frac{1}{3}$$

By application of standard results :

(i) $\lim_{x\to 0} \sin x = \lim_{x\to 0} \tan x = 0$.

(ii) $\lim_{x\to 0}\frac{\sin x}{x}=\lim_{x\to 0}\frac{x}{\sin x}=\lim_{x\to 0}\frac{\sin^{-1}x}{x}=\lim_{x\to 0}\frac{x}{\sin^{-1}x}=1$ (where x is measured in radian)

(iii) $\lim_{x\to 0}\cos x=\lim_{x\to 0}\left(\frac{1}{\cos x}\right)=1$

(iv) $\lim_{x\to 0}\frac{\tan x}{x}=\lim_{x\to 0}\frac{x}{\tan x}=\lim_{x\to 0}\frac{\tan^{-1}x}{x}=\lim_{x\to 0}\frac{x}{\tan^{-1}x}=1$ (where x is measured in radian)

(v) $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{x}=\lim_{x\to 0}(1+x)^{1/x}=e$, $\lim_{x\to\infty}\left(1+\frac{a}{x}\right)^{x}=\lim_{x\to\infty}\left(1+\frac{a}{x}\right)^{\frac{xa}{a}}=e^{a}$

(vi) If $\lim_{x\to a}f(x)=1$ and $\lim_{x\to a}\varphi(x)=\infty$ then, $\lim_{x\to a}f(x)^{\phi(x)}=\lim_{x\to a}[1+(f(x)-1)]^{\phi(x)}=e^{\lim_{x\to a}(f(x)-1).\phi(x)}$

(vii) If $\lim_{x\to a}f(x)$ exists and positive then $\lim_{x\to a}[f(x)]^{\varphi(x)}=e^{\lim_{x\to a}\varphi(x).\ell n[f(x)]}$

(viii) $\lim_{x\to 0}\frac{a^{x}-1}{x}=\log_{e}a$ (a > 0), $\lim_{x\to 0}\frac{e^{x}-1}{x}=1,,$

(ix) $\lim_{x\to a}\frac{x^{n}-a^{n}}{x-a}=n\ a^{n-1}$.

(x) $\lim_{x\to 0}\frac{\ln(1+x)}{x}=1$, $\lim_{x\to 0}\frac{\log_{a}(1+x)}{x}=\log_{a}e$

(xi) $\lim_{x\to 0}\frac{(1+x)^{n}-1}{x}=n$,

(xii) $\lim_{x\to\infty}\frac{\sin x}{x}=\lim_{x\to\infty}\frac{\cos x}{x}=0.$ (where x is measured in radian)

(xiii) $\lim_{x\to\infty}a^{x}=\begin{cases}0, & 0\le a<1\\ 1, & a=1\\ \infty, & a>1\\ \text{does not exist}, & a<0\end{cases}$

(xiv) $\lim_{x\to\infty}\frac{a_0x^{m}+a_1x^{m-1}+\ ....}{b_0x^{n}\ +\ b_1x^{n-1}+\ ...}=\begin{cases}\frac{a_0}{b_0}, & when\ m=n\\ 0, & when\ m<n\\ \infty, & when\ m>n\ \ and\ \frac{a_0}{b_0}>0\\ -\infty, & when\ m>n\ \ and\ \frac{a_0}{b_0}<0\end{cases}$

Illustration 18 : Evaluate $\lim_{x\to e}\frac{\ln x-1}{x-e}$.

Solution : Above limit = $\lim_{x\to e}\frac{\ln x/e}{e\left(\frac{x}{e}-1\right)} = \lim_{t\to 1}\frac{\ln t}{(t-1)e}$ (Putting $x/e = t$) Put $t = 1 + y$

$= \lim_{y\to 0}\frac{\ln(1+y)}{ey} = \frac{1}{e}$

Illustration 19 : Evaluate $\lim_{x\to 0}\left\{\tan\left(\frac{\pi}{4}+x\right)\right\}^{1/x}$

Solution : $\lim_{x\to 0}\left\{\tan\left(\frac{\pi}{4}+x\right)\right\}^{\frac{1}{x}} = \lim_{x\to 0}\left\{\frac{\tan\frac{\pi}{4}+\tan x}{1-\tan\frac{\pi}{4}\tan x}\right\}^{\frac{1}{x}}$

$= \lim_{x\to 0}\left\{\frac{1+\tan x}{1-\tan x}\right\}^{\frac{1}{x}} = \lim_{x\to 0}\left\{1+\frac{2\tan x}{1-\tan x}\right\}^{\frac{1}{x}}$

$= e^{\lim_{x\to 0}\left\{\frac{2\tan x}{1-\tan x}\right\}\frac{1}{x}} = e^{\lim_{x\to 0} 2\frac{\tan x}{x}\cdot\frac{1}{1-\tan x}} = e^2$

By Use of logarithm :

Illustration 20 **:** valuate $\lim_{x\to\infty} x^{1/x}$

Solution : Let $A = \lim_{x\to\infty} x^{1/x}$. Then,log A = $\lim_{x\to\infty}\frac{1}{x}\log x = 0 \Rightarrow A = e^0 = 1$.

Student Task

Evaluate the following limits :

Q.1 $\lim_{x\to\frac{\pi}{4}}\frac{1-\tan x}{1-\sqrt{2}\sin x}$.

Q.2 $\lim_{x\to 1}\left(\tan\frac{\pi x}{4}\right)^{\tan\frac{\pi x}{2}}$

Q.3 $\lim_{x\to 0}\frac{x\tan(2x)-2x\tan x}{(1-\cos 2x)^2}$

Q.4 $\lim_{x\to 0}\frac{(1+x)^5-1}{3x+5x^2}$.

Q.5 $\lim_{x\to 0}\frac{x}{\tan^{-1}2x}$

Q.6 $\lim_{x\to\frac{\pi}{2}}(\sec x)^{\cot x}$

ANSWERS

1. 2 2. e^{-1} 3. 1/2 4. 5/3 5. 1/2 6. 1

LEARNING OBJECTIVES

(i) A real valued function is continuous at a point in its domain if the limit of the function at that point equals the value of the function at that point. A function is continuous if it is continuous on the whole of its domain.

(ii) Sum, difference, product and quotient of continuous functions are continuous.

(iii) Every differentiable function is continuous, but the converse is not true.

(iv) Rolle's Theorem: If $f : [a, b] \to R$ is continuous on $[a, b]$ and differentiable on (a, b) such that $f(a) = f(b)$, then there exists some c in (a, b) such that $f'(c) = 0$.

(v) Mean value theorem: If $f : [a, b] \to R$ is continuous on $[a, b]$ and differentiable on (a, b). Then there exists some c in (a, b) such that $f'(c) = \dfrac{f(b) - f(a)}{b - a}$

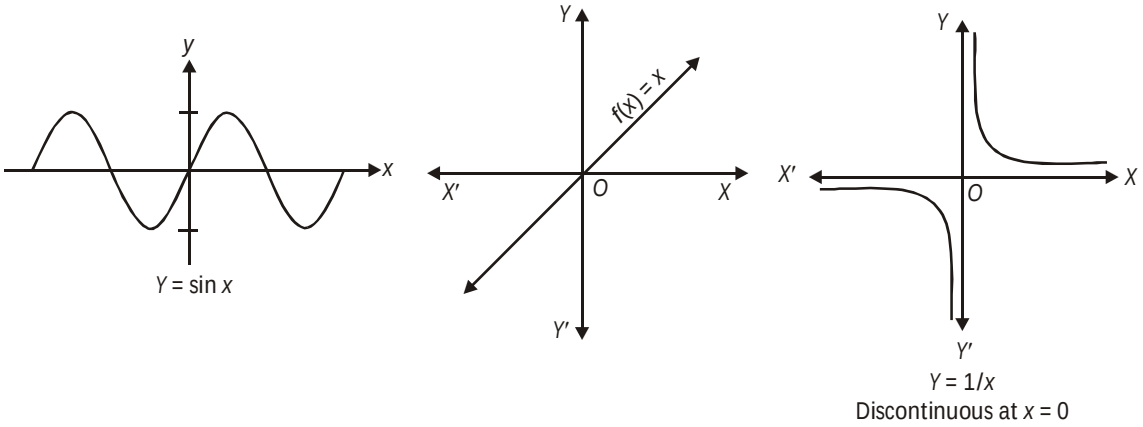
6. INTRODUCTION TO CONTINUITY

The word 'Continuous' means without any break or gap. If the graph of a function has no break or hole or gap or jump, then it is said to be continuous.

A function which is not continuous is called a discontinuous function.

In other words, if there is slight (finite) change in the value of a function by slightly changing the value of x then function is continuous, otherwise discontinuous, while studying graphs of functions, we see that graphs of function sin x, x, cos x, e^x etc. are continuous but greatest integer function $[x]$ has break at every integral point, so it is not continuous. Similarly, tan x, cot x, sec x, $1/x$ etc. are also discontinuous function.

Example :



(Continuous Function) Discontinuous at $x = 0$

For examining continuity of a function at a point , we find its limit and value at that point, If these two exist and are equal, then function is continuous at that point.

CONTINUITY OF A FUNCTION AT A POINT

A function $f(x)$ is said to be continuous at a point $x = a$ if (*i*) $f(a)$ exists

(ii) $\lim\limits_{x \to a} f(x)$ exists and finite so $\lim\limits_{x \to a^+} f(x) = \lim\limits_{x \to a^-} f(x)$

(iii) $\lim\limits_{x \to a} f(x) = f(a)$ or function $f(x)$ is continuous at $x = a$ if $\lim\limits_{x \to a^+} f(x) = \lim\limits_{x \to a^-} f(x) = f(a)$

i.e. If right hand limit at 'a' = left hand limit at 'a' = value of the function at 'a'.

CONTINUITY FROM LEFT AND RIGHT :

Function $f(x)$ is said to be

(i) Left continuous at $x = a$ if $\lim\limits_{x \to a^-} f(x) = f(a)$

(ii) Right continuous at $x = a$ if $\lim\limits_{x \to a^+} f(x) = f(a)$

Thus a function f(x) is continuous at a point x = a if it is left continuous as well as right continuous at $x = a$.

7. CONTINUITY OF A FUNCTION

Continuity On An Open Interval : A function $f(x)$ is said to be continuous on an open interval (a,b) if it is continuous at every point on the interval (a, b).

Continuity On A Closed Interval : A function $f(x)$ is said to be continuous on a closed interval $[a,b]$ if

(i) f is continuous on the open interval (a,b) (ii) $\lim_{x\to a^+} f(x) = f(a)$, and (iii) $\lim_{x\to b^-} f(x) = f(b)$

In other words, $f(x)$ is continuous on $[a,b]$ if it is continuous on (a,b) and it is continuous at a from the right and at b from the left.

CONTINUITY IN AN INTERVAL

(a) A function f is said to be continuous in (a, b) if f is continuous at each and every point $\in (a, b)$.

(b) A function f is said to be continuous in a closed interval $[a, b]$ if :

(i) f is continuous in the open interval and

(ii) f is right continuous at 'a' i.e. $\lim_{x\to a^+} f(x) = f(a)$ = a finite quantity.

(iii) f is left continuous at 'b' i.e. $\lim_{x\to b^-} f(x) = f(b)$ = a finite quantity

Note that a function f which is continuous in $[a, b]$ possesses the following properties :

(a) If $f(a)$ and $f(b)$ possess opposite sign, then there exists at least one solution of the equation $f(x)=0$ in the open interval (a, b).

(b) If K is any real number between $f(a)$ and $f(b)$, then there exists at least one solution of the equation $f(x) = K$ in the open interval (a, b).

SINGLE POINT CONTINUITY

Functions which are continuous only at one point are said to exhibit single point continuity

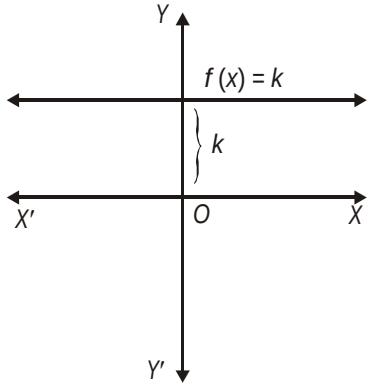
eg. $f(x)=\begin{cases} x \ \text{if} \ x\in Q \\ -x \ \text{if} \ x\notin Q \end{cases}$ and $g(x)=\begin{cases} x \ \text{if} \ x\in Q \\ 0 \ \text{if} \ x\notin Q \end{cases}$ are both continuous only at $x = 0$.

CONTINUITY OF BASIC ELEMENTARY FUNCTIONS

A function $f(x)$ is said to be continuous if it is continuous at each point of its domain.

Everywhere Continuous Function : A function $f(x)$ is said to be everywhere continuous if it is continuous on the entire real line $(-\infty, \infty)$. Some common type of functions that are continuous in their domain.

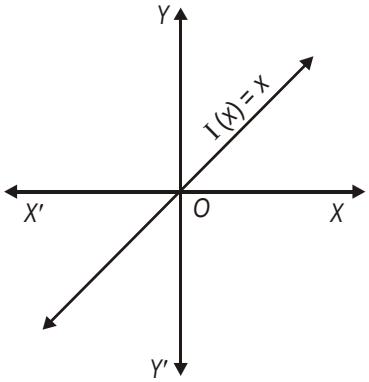
(i) Constant Function : Every constant function is



everywhere continuous. Let $f(x) = k$, where k is a constant, be a constant function.

Its graph is shown. It is evident from the graph that this function is everywhere continuous.

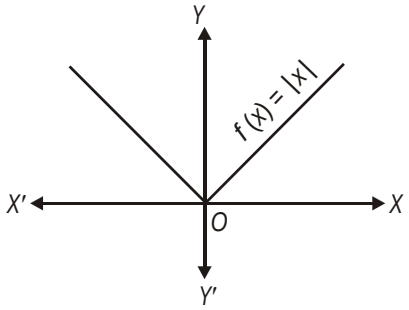
(ii) Identity Function : The identity function $I(x)$ is defined by $I(x) = x$ for all $x \in R$. This function is everywhere continuous as is evident from its graph.



(iii) Modulus Function :

The modulus function $f(x)$ is defined as

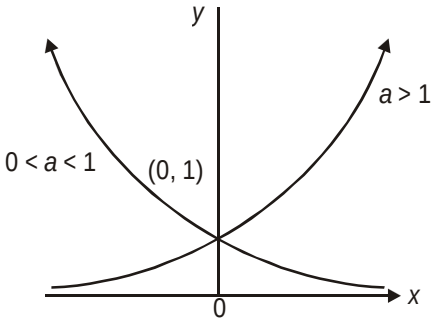
$f(x) = |x| = \begin{cases} x, \text{ if } x \geq 0 \\ -x, \text{if } x < 0 \end{cases}$ Clearly, the domain of f (x) is R and this function is everywhere continuous.



(iv) Power function :

$y = x^n$ (n is a natural number is continuous for any value of x.

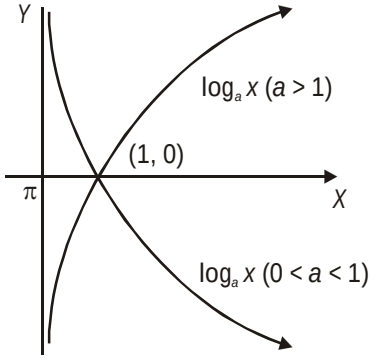
(v) Exponential function $y = a^x$ ($a > 0$) is continuous for any value of x.



(vi) Logarithmic Function :

If a is positive real number other than unity, then a function defined by $f(x) = \log_a x$ is called the logarithmic function.

Clearly its domain is the set of all positive real numbers and it is continuous on its domain.
It should be noted that it is not everywhere continuous



(vii) Polynomial Function :

A function of the form $f(x) = a_0 + a_1 x + a_2 x^2 + ...+ a_n x^n$, where $a_0, a_1, a_2,..., a_n \in R$ is called a polynomial function. This function is everywhere continuous.

(viii) Rational Function :

If $p(x)$ and $q(x)$ are two polynomials, then a function $f(x)$ of the form $f(x) = \frac{p(x)}{q(x)}$, $q(x) \neq 0$ is called a rational function. This function is continuous on its domain, i.e. it is everywhere continuous except at points where $q(x) = 0$.

(ix) Trigonometric function $y = \sin x$ is continuous for each value of x.

Indeed, assigning an increment Δx to the argument x and denoting by Δy the corresponding increment of the function y, we have $y + \Delta y = \sin(x + \Delta x)$,

hence, $\Delta y = \sin(x+\Delta x) - \sin x = 2\sin\frac{\Delta x}{2}.\cos\left(x+\frac{\Delta x}{2}\right)$.

For $\Delta x \neq 0$ we have $\left|\sin\frac{\Delta x}{2}\right| < \frac{|\Delta x|}{2}$, besides $\left|\cos\left(x+\frac{\Delta x}{2}\right)\right| \leq 1$

Therefore, $|\Delta y| < 2.\frac{|\Delta x|}{2}.1 = |\Delta x|$ i.e., $\lim_{\Delta x \to 0} \Delta y = 0$

Hence, the function sin x is continuous within the interval $(-\infty, +\infty)$. It is proved in the same way that

y = cos x is a continuous function within the interval $(-\infty, +\infty)$.

(x) Inverse trigonometric functions :

We know that the graph of the inverse of a function is the mirror image of the graph of the given function in the line $y = x$. Therefore, graphs of $\sin^{-1}x$, $\cos^{-1}x$,$\tan^{-1}x$,$\cot^{-1}x$, $\sec^{-1}x$ and $\text{cosec}^{-1}x$ are the mirror images of those of the corresponding trigonometric functions in the line $y = x$. Since trigonometric function are continuous in their domains, Therefore each of the inverse trigonometric functions is continuous in its domain.

8. BASIC THEOREMS ON CONTINUOUS FUNCTIONS

Theorem 1 : The sum of a finite number of continuous functions is a continuous function.

Indeed, if $f_1(x)$ and $f_2(x)$ are functions continuous on a set X and x_1 is any value from this set, then

$$\lim_{x \to x_1}[f_1(x) + f_2(x)] = \lim_{x \to x_1} f_1(x) + \lim_{x \to x_1} f_2(x) = f_1(x_1) + f_2(x_1),$$

i.e. the limit of the sum as $x \to x_1$ is equal to the value of this sum for $x = x_1$.

Consequently, the function $f_1(x) + f_2(x)$ is also continuous on the set X.

Theorem 2 : The product of a finite number of continuous functions is a continuous function.

This theorem is proved in an analogous way.

Corollary : The integral polynomial $P(x) = a_0 + a_1x + ... + a_nx^n$ is a continuous function.

Theorem 3 : The quotient obtained by the division of one continuous function by another continuous function is a function continuous at all points where the divisor is non-zero.

Corollary : The fractional rational function $R(x) = \frac{a_0 + a_1x + ... + a_nx^n}{b_0 + b_1x + ... + b_mx^m}$

is continuous everywhere except for those values of x there the denominator reduces to zero.

Theorem 4 : A continuous function of a continuous function is a continuous function. In other words, a composite function consisting of continuous functions is continuous.

Let x_1 be an arbitrary point belonging to the domain of definition of a composite function $f(\phi(x))$, where the function $u = f(x)$ is continuous at the point x_1, and the function $f(u)$ is continuous at the point $u_1 = f(x_1)$. On the basis of the strong property of permutability of a continuous function and its limit.

we have $$\lim_{x \to x_1} f(\phi(x)) = f\left(\lim_{x \to x_1} \phi(x)\right) = f(\phi(x_1))$$

i.e. the composite function $f(\phi(x))$ is continuous at the point x_1.

By virtue of Theorem 4, for instance, the functions $(\sin x)^2 = \sin^2 x$ and $\sin(x^2)$ are continuous, since the functions x^2 and sin x are continuous.

The functions we are going to consider further on are continuous everywhere, except, possibly, for some individual values of the argument.

For- instance, the function $\tan x = \frac{\sin x}{\cos x}$ by virtue of Theorem 3 of the present section, is continuous for all values of the argument x, except those for which $\cos x = 0$,

i.e. except for the values $x = (2k - 1)\frac{\pi}{2}$, where k is any integer.

Analogously, the function $\cot x = \frac{\cos x}{\sin x}$

is continuous for $\sin x \neq 0$, i.e. for $x \neq k\pi$ (k is an integer).

The theorem on the continuity of an inverse function is also true. We introduce it without proof.

Theorem 5 : If a function $y = f(x)$ is continuous and strictly monotone on the interval (a, b), then there exists a single-valued inverse function $x = \phi(y)$ defined on the interval $\langle f(a), f(b) \rangle$ the latter being continuous and monotone in the same sense.

By this theorem : the radical $\sqrt[n]{x}$ (n-natural), logarithmic function $\log_a x$ ($a > 0$, $a \neq 1$), principal values of the inverse trigonometric functions $\sin^{-1} x$, $\cos^{-1} x$, $\tan^{-1}x$, $\cot^{-1} x$ are continuous for any value of the argument x for which these functions are defined.

From above theorems we can conclude

(A) If f and g are two continuous functions on their common domain D, then

(i) f + g is continuous on D

(ii) $f - g$ is continuous on D

(iii) fg is continuous on D

(iv) αf is continuous on D where α is any real number.

(v) $\frac{f}{g}$ is continuous on $D - \{x ; g(x) \neq 0\}$

(vi) $\frac{1}{f}$ is continuous on $D - \{x ; f(x) \neq 0\}$

(B) The composition of two continuous functions is a continuous function.

(C) If f is continuous on its domain D, then $|f|$ is also continuous on D.

For examples :

(i) $e^{2x} + \sin x$ is a continuous function because it is the sum of two continuous function e^{2x} and $\sin x$.

(ii) $\sin (x^2 + 2)$ is a continuous function because it is the composite of two continuous functions $\sin x$ and $x^2 + 2$.

Note :- The product of one continuous and one discontinuous function may or may not be continuous. For example :

(i) $f(x) = x$ is continuous and $g(x) = \cos 1/x$ is discontinuous whereas their product $x \cos (1/x)$ is continuous

(ii) $f(x) = c$ is continuous and $g(x) = \sin 1/x$ is discontinuous whereas their product $c \sin (1/x)$ is discontinuous.

9. DISCONTINUOUS FUNCTIONS

A function is said to be a discontinuous function if it is discontinuous at at least one point in its domain.

The discontinuity may arise due to any of the following situations:

(i) $\lim\limits_{x \to a^+} f(x)$ or $\lim\limits_{x \to a^-} f(x)$ or both may not exist

(ii) $\lim\limits_{x \to a^+} f(x)$ as well as $\lim\limits_{x \to a^-} f(x)$ may exist, but are unequal

(iii) $\lim\limits_{x \to a^+} f(x)$ as well as $\lim\limits_{x \to a^-} f(x)$ both may exist, but either of the two or both may not be equal to f(a).

We classify the points of discontinuity according to various situations discussed above.

Removable Discontinuity: A function f is said to have removable discontinuity at $x = a$ if $\lim\limits_{x \to a^-} f(x) = \lim\limits_{x \to a^+} f(x)$ but their common value is not equal to $f(a)$. Such a discontinuity can be removed by assigning a suitable value to the function f at $x = a$.

Removable type of discontinuity can be further classified as :

(a) **Missing point discontinuity :** Where $\lim\limits_{x \to a} f(x)$ exists finitely but $f(a)$ is not defined. eg.

$f(x) = \dfrac{(1-x)(9-x^2)}{(1-x)}$ has a missing point discontinuity at $x = 1$, and $f(x) = \dfrac{\sin x}{x}$ has a missing point discontinuity at $x = 0$.

(b) **Isolated point discontinuity :** Where $\lim\limits_{x \to a} f(x)$ exists and $f(a)$ also exists but $\lim\limits_{x \to a} f(x) \neq f(a)$.

e.g. $f(x) = \dfrac{x^2 - 16}{x - 4}, x \neq 4$ & $f(4) = 9$ has an isolated point discontinuity t $x = 4$.

Similarly $f(x) = [x] + [-x] = \begin{cases} 0 \ \text{if} \ x \in \mathrm{I} \\ -1 \ \text{if} \ x \notin \mathrm{I} \end{cases}$ has an isolated point discontinuity at all $x \in \mathrm{I}$.

Non-removable type of discontinuity can be further classified as :

(a) Finite discontinuity e.g. $f(x) = x - [x]$ at all integral x, $f(x) = \tan^{-1}\dfrac{1}{x}$ at $x = 0$ and $f(x) = \dfrac{1}{1 + 2^{1/x}}$ at $x = 0$ (note that $f(0^+) = 0$, $f(0^-) = 1$).

(b) Infinite discontinuity e.g. $f(x)=\frac{1}{x-4}$ or $g(x)=\frac{1}{(x-4)^2}$ at $x = 4$, $f(x) = 2^{\tan x}$ at $x=\frac{\pi}{2}$ and

$f(x)=\frac{\cos x}{x}$ at $x = 0$.

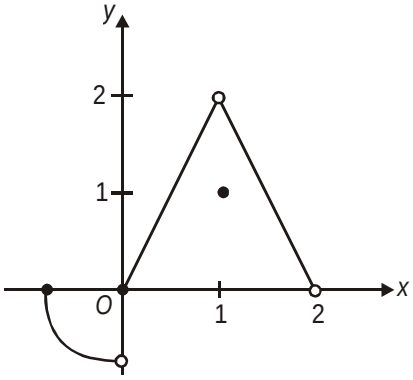
(c) Oscillatory discontinuity e.g. $f(x)=\sin\frac{1}{x}$ at $x = 0$.

In all these case the value of $f(a)$ of the function at $x = a$ (point of discontinuity) may or may not exist but $\lim_{x\to a} f(x)$ does not exist.

Note : From the adjacent graph note that

– f is continuous at $x = -1$

– f has isolated discontinuity at $x = 1$



– f has missing point discontinuity at $x = 2$

– f has non removable (finite type) discontinuity

Illustration 21 : Examine whether the function f given by $f(x) = x^2$ is continuous at $x = 0$.

Solution : First note that the function is defined at the given point $x = 0$ and its value is 0.

Then find the limit of the function at $x = 0$. Clearly $\lim_{x\to 0} f(x)=\lim_{x\to 0} x^2=0^2=0$

Thus $\lim_{x\to 0} f(x)=0=f(0)$. Hence, f is continuous at $x = 0$.

Illustration 22 : Determine all points at which the function $f(x)=\begin{cases} x^5-x^4+x^3+2x-1 & \text{if } x\le 1 \\ \frac{x^2+x+2}{x+1} & \text{if } x>1 \end{cases}$

is continuous.

Solution : First notice that $f(x)$ is the polynomial $x^5-x^4+x^3+2x-1$ when $x < 1$. Since polynomials are continuous everywhere,

$f(x)$ is continuous for $x < 1$. Now notice that $f(x)$ is the rational function $\frac{x^2+x+2}{x+1}$

when $x > 1$.

Since rational functions are continuous at every point where the denominator is not zero, $f(x)$ is certainly continuous for

$x > 1$. We need only check the point $x = 1$. We need to show that $\lim_{x\to 1} f(x) = f(1)$.

Since $f(x) = x^5 - x^4 + x^3 + 2x - 1$ when x > 1, we have,

$\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} (x^5 - x^4 + x^3 + 2x - 1) = (1)^5 - (1)^4 + (1)^3 + 2(1) - 1 = 2$

Since $f(x) = \frac{x^2 + x + 2}{x + 1}$ when x > 1, we have

$\lim_{x\to 1^+} f(x) = \lim_{x\to 1^+} \frac{x^2 + x + 2}{x + 1} = \frac{(1)^2 + (1) + 2}{1 + 1} = 2$

Thus $\lim_{x\to 1} f(x) = 2$. Since this also equals $f(1)$, f is continuous at $x = 1$. Therefore, f is continuous everywhere.

Illustration 23 : Is the function defined by $f(x) = |x|$, a continuous function ?

Solution : We may rewrite f as $f(x) = \begin{cases} -x, & \text{if } x < 0 \\ x, & \text{if } x \geq 0 \end{cases}$

We know that f is continuous at $x = 0$.
Let c be a real number such that $c < 0$. Then $f(c) = -c$.

Also, $\lim_{x\to c} f(x) = \lim_{x\to c} (-x) = -c$

Since $\lim_{x\to c} f(x) = f(c)$, f is continuous at all negative real numbers.

Now, let c be a real number such that $c > 0$. Then $f(c) = c$.

Also, $\lim_{x\to c} f(x) = \lim_{x\to c} x = c$.

Since $\lim_{x\to c} f(x) = f(c)$, f is continuous at all positive real numbers. Hence, f is continuous at all points.

Illustration 24 : Test the continuity of the function $f(x)$ at the origin : $f(x) = \begin{cases} \frac{|x|}{x}; & x \neq 0 \\ 1; & x = 0 \end{cases}$

Solution : We have (*LHL* at $x = 0$)

$= \lim_{x\to 0^-} f(x) = \lim_{h\to 0} f(0 - h) = \lim_{h\to 0} f(-h) = \lim_{h\to 0} \frac{|-h|}{-h} = \lim_{h\to 0} (-1) = -1$

and, (*RHL* at $x = 0$)

$= \lim_{x\to 0^+} f(x) = \lim_{h\to 0} f(0 + h)$

$= \lim_{h\to 0} f(h) = \lim_{h\to 0} \frac{|h|}{h} = \lim_{h\to 0} \frac{h}{h} = \lim_{h\to 0} 1 = 1$

Thus, we have $\lim_{x\to 0^-} f(x) \neq \lim_{x\to 0^+} f(x)$.

Hence f(x) is not continuous at the origin.

ALTERNATIVELY : We have,

$$f(x) = \begin{cases} \frac{|x|}{x}; & x \neq 0 \\ 1; & x = 0 \end{cases} = \begin{cases} \frac{x}{x} = 1 & , x > 0 \\ \frac{-x}{x} = -1 & , x < 0 \\ 1 & , x = 0 \end{cases} \qquad \left[\because |x| = \begin{cases} x, & x \geq 0 \\ -x, & x < 0 \end{cases} \right]$$

$\therefore$ (*LHL* at $x = 0$) $= \lim_{x \to 0^-} f(x) = \lim_{h \to 0} -1$

$\left[\because \ f(x) = -1 \ for \ x < 0 \ and \ x \to 0^- \right]$

$= -1$

(RHL at $x = 0$) $= \lim_{x \to 0^+} f(x) = \lim_{h \to 0} 1$

$\left[\because \ f(x) = 1 \ for \ x > 0 \ and \ x \to 0^+ \right]$

$= 1$

$\therefore \quad \lim_{x \to 0^-} f(x) \neq \lim_{x \to 0^+} f(x)$. Hence $f(x)$ is not continuous at the origin.

Illustration 25 : Find the point(s) of discontinuity of $y = \frac{1}{u^2 + u - 2}$, where $u = \frac{1}{x-1}$.

Solution : The function $u = f(x) = \frac{1}{x-1}$ is discontinuous at the point $x = 1$. . . . (i)

The function $y = g(x) = \frac{1}{u^2 + u - 2} = \frac{1}{(u+2)(u-1)}$ is discontinuous at $u = -2$ and $u = 1$.

when $u = -2$, $\frac{1}{x-1} = -2 \Rightarrow x = \frac{1}{2}$; $u = 1 \Rightarrow \frac{1}{x-1} = 1 \Rightarrow x = 2$

Hence, the composite function $y = g(f(x))$ is discontinuous at three points $x = 1/2$, 1, 2

Illustration 26 : Find all possible values of a and b so that f (x) is continuous for all $x \in R$ if

$$f(x) = \begin{cases} |ax + 3| & if \ x \leq -1 \\ |3x + a| & if \ -1 < x \leq 0 \\ \frac{b \sin 2x}{x} - 2b & if \ 0 < x < \pi \\ \cos^2 x - 3 & if \ x \geq \pi \end{cases}$$

Solution : $f(-1) = |3 - a|$; $f(-1^-) = |3 - a|$

$f(-1^+) = \lim_{h \to 0} |a + 3(-1 + h)| = |a - 3|$

also, $\left. \begin{matrix} f(0) = |a| = f(0^-) \\ f(0^+) = 2b - 2a = 0 \end{matrix} \right]$ Hence $|a| = 0 \Rightarrow a = 0$ again, $f(\pi) = -2$

$f(\pi^-) = \lim_{x \to \pi^-} \left(\frac{\sin 2x}{x} - 2 \right) b = -2b$. Hence $-2b = -2 \Rightarrow b = 1$

Illustration 27 : Show that the function $f(x) = \begin{cases} 2x+3; -3 \le x < -2 \\ x+1; -2 \le x < 0 \\ x+2; \ 0 \le x \le 1 \end{cases}$

is discontinuous at $x = 0$ and continuous at every other point in interval [–3, 1].

Solution :

Analytically : $f(x) = 2x + 3$ or $(x + 1)$ or $(x + 2)$ are polynomials of degree one, i.e. these are straight lines. Polynomials are always continuous in their domain. Hence, $f(x) = 2x + 3$ is continuous in

$-3 \le x < -2$; $f(x) = x + 1$ is continuous in $-2 < x < 0$ and $f(x) = x + 2$ in $0 < x < 0$. So the doubtful points where $f(x)$ may be discontinuous are those at which the definition of $f(x)$ changes.

Here, $x = -2$ and $x = 0$ are doubtful points. Now, let us test the continuity of $f(x)$ at these points.

At $x = -2$, LHL $= \lim_{x \to -2-h} f(x) = \lim_{x \to -2-h} 2x + 3 = \lim_{h \to 0} 2(-2-h) + 3 \lim_{h \to 0} -1 - 2h$

$= -1$

RHL $= \lim_{x \to -2+h} f(x) = \lim_{x \to -2+h} (x+1)1 = \lim_{x \to -2+h} (-2+h) + 1 = -1$

$f(-2) = -2 + 1 = -1$

LHL $= f(-2) =$ R.H.L.

$\therefore$ $f(x)$ is continuous at $x = -2$.

At $x = 0$, LHL $= \lim_{x \to 0-h} f(x) = \lim_{x \to 0-h} (x+1) = \lim_{h \to 0} (0-h+1) = 1$

RHL $= \lim_{x \to 0+h} f(x) = \lim_{x \to 0+h} (0+h+2) = 2$

$\therefore$ LHL $\neq$ R.H.L.

$\therefore$ $\lim_{x \to 0} f(x)$ does not exist and $f(x)$ is discontinuous at $x = 0$.

Student Task

Q.1 Let $f(x) = \begin{cases} \dfrac{1-\cos(4x)}{x^2} & , for\ x < 0 \\ a & , for\ x = 0 \\ \dfrac{\sqrt{x}}{(16+\sqrt{x})^{1/2} - 4} & , for\ x > 0 \end{cases}$; then find the value of 'a' for which $f(x)$ is continuous at $x = 0$.

Q.2 Find the value of $f(0)$ so that the function $f(x) = \dfrac{\sqrt{1+x} + \sqrt[3]{1+x}}{x}$ is continuous at $x = 0$.

Q.3 If $f(x) = -1 + |x - 1|$, $-1 \le x \le 3$ and $g(x) = 2 - |x + 1|$, $-2 \le x \le 2$, then calculate $f(g(x))$ and $g(f(x))$. Discuss the continuity of $f(g(x))$.

Q.4 If $f(x) = \begin{cases} x + \lambda & , x < 3 \\ 4 & , x = 3 \\ 3x - 5 & , x > 3 \end{cases}$ is continuous at $x = 3$ then find the value of λ

Q.5 For what value of 'k' is the function, $f(x) = \begin{cases} \frac{\sin 5x}{3x} & , \ x \neq 0 \\ k & , \ x = 0 \end{cases}$ continuous.

Q.6 If $f(x) = \begin{cases} \sin(1/x), & x \neq 0 \\ 0 & , \ x = 0 \end{cases}$, then it is discontinuous at

(a) $x = 0$ (b) All points (c) No point (d) None of these

ANSWERS

1. $a = 8$ 2. 5/6 3. Continuous everywhere
4. 1 5. 5/3 6. (a)

10. DIFFERENTIABILITY OF A FUNCTION IN A SET

Differentiability on an open interval : A function $f(x)$ defined on an open interval (a, b) is said to be differentiable or derivable in open interval (a, b) if it is differentiable at each point of (a, b).

Differentiability on a closed interval : A function $f(x)$ defined on $[a, b]$ is said to be differentiable or derivable at the end points a and b if it is differentiable from the right at a and from the left at b.

In other words $\lim\limits_{x \to a^+} \frac{f(x) - f(a)}{x - a}$ and $\lim\limits_{x \to b^-} \frac{f(x) - f(b)}{x - b}$ both exist.

If f is derivable in the open interval (a, b) and also at the end points a and b, then f is said to be derivable in the closed interval $[a, b]$

(i) Differentiable at every point of interval (a, b)
(ii) Right derivative exists at $x = a$
(iii) Left derivative exists at $x = b$.

Illustration 28 : Function $f(x) = \begin{cases} -x & if \quad x < 0 \\ x^2 & if \quad 0 \le x \le 1 \\ x^3 - x + 1 & if \quad x > 1 \end{cases}$ is differentiable at -

(A) $x = 0$ but not at $x = 1$ (B) $x = 1$ but not at x = 0
(C) $x = 0$ and $x = 1$ both (D) neither $x = 0$ nor $x = 1$

Solution : (B) **Differentiability at $x = 0$**

$$R\,[f'(0)] = \lim_{h \to 0} \frac{f(0+h) - f(0)}{h} = \lim_{h \to 0} \frac{(0+h)^2 - 0}{h} = \lim_{h \to 0} h = 0$$

$$L\,[f'(0)] = \lim_{h \to 0} \frac{f(0-h) - f(0)}{-h} = \lim_{h \to 0} \frac{-(0-h) - 0}{-h} = -1$$

$\therefore$ $R\,[f'(0)] \neq L\,[f'(0)]$

$\therefore$ $f(x)$ is not differentiable at $x = 0$

Differentiability at $x = 1$

$$R\,[f'(1)] = \lim_{h \to 0} \frac{f(1+h) - f(1)}{h} = \lim_{h \to 0} \frac{(1+h)^3 - (1+h) + 1 - 1}{h} = \lim_{h \to 0} \frac{2h + 3h^2 + h^3}{h}$$

$$L\,[f'(1)] = \lim_{h \to 0} \frac{f(1-h) - f(1)}{-h} = \lim_{h \to 0} \frac{f(1-h) - 1}{-h} = \lim_{h \to 0} \frac{-2h + h^2}{-h} = 2$$

Thus $R\,[f'(1)] = L\,f'(1)]$

$\because$ Function $f(x)$ is differentiable at $x = 1$

SOME STANDARD RESULTS ON DIFFERENTIABILITY

(i) Every polynomial function is differentiable at each $x \in R$.

(ii) The exponential function a^x , $a > 0$ is differentiable at each $x \in R$.

(iii) Every constant function is differentiable at each $x \in R$.

(iv) The logarithmic function is differentiable at each point in its domain.

(v) Trigonometric and inverse-trigonometric functions are differentiable in their domains.

(vi) The sum, difference, product and quotient of two differentiable functions is differentiable.

(vii) The composition of differentiable function is a differentiable function.

(viii) If a function is not differentiable but is continuous at a point, it geometrically implies there is a sharp corner or a kink at that point.

(ix) If $f(x)$ and $g(x)$ both are not differentiable at a point, then the sum function $f(x) + g(x)$ and the product function $f(x).\ g(x)$ can still be differentiable at that point.

11. RELATION BETWEEN THE CONTINUITY AND DIFFERENTIABILITY OF A FUNCTION

The function $y = f(x)$ ..... (1)

is called continuous at a point x if at this point $\lim_{\Delta x \to 0} \Delta y = 0$

function (1) is said to be differentiable at a point x if it has a derivative at this point, i.e. if there exists a finite limit. $\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = y'$ ..... (2)

There is a simple relation between these basic concepts of mathematical analysis.

Theorem : If a function is differentiable at some point, then the function is continuous at this point.

The converse is false : a continuous function can have no derivative.

Let the function $y = f(x)$ be differentiable at point x, i.e. equality (2) is fulfilled for this function.

We write the identity $\Delta y = \frac{\Delta y}{\Delta x}.\Delta x\ (\Delta x \neq 0)$

Hence, $\lim_{\Delta x \to 0} \Delta y = \lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x}.\ \lim_{\Delta x \to 0} \Delta x = y'.0 = 0$

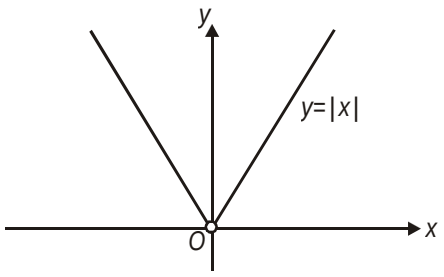
Consequently, the function $y = f(x)$ is continuous at the point x.

Corollary : If a function is discontinuous at some point, then it has no derivative at this point.

Continuous functions having no derivative at one point are exemplified by the function

$y = |x|$

This function is continuous for $x = 0$ but is not



differentiable for this value, since there exists no tangent line at the point $x = 0$ of its graph.

Mathematicians succeeded in constructing the graphs of some continuous functions differentiable at none of the points (*K*. Weierstrass and others).

The superb Russian mathematician N.I. Lobachevsky was the first to introduce a distinct differ-

ence between the notions of continuity and differentiability of a function.

Note that the derivative $y' = f'(x)$ of a continuous function $y = f(x)$ is not necessarility continuous.

If a function $f(x)$ has a continuous derivative $f'(x)$ on an interval $\langle a, b \rangle$, then the function is said to be smooth on this interval. A function f (x) whose derivative allows only a finite number of points of discontinuity of the first kind on a given interval $\langle a, b \rangle$ is termed piecewise smooth on this interval.

From above discussion we can conclude

(a) If $f'_+(a) = p$ & $f'_-(a) = q$ where p and q are finite then :

(i) $p = q \Rightarrow f$ is derivable at $x = a \Rightarrow f$ is continuous at $x = a$

(ii) $p \neq q \Rightarrow f$ is not derivable at $x = a$.

It is very important to note that f may be still continuous at $x = a$.

In short, for a function f :

Differentiability $\Rightarrow$ Continuity ; Continuity $\not\Rightarrow$ derivability

Not derivibaility $\not\Rightarrow$ discontinuous ; But discontinuity $\Rightarrow$ Non derivability

(b) If a function f is not differentiable but is continuous at $x = a$ it geometrically implies a sharp corner at $x = a$.

12. ALGEBRA OF A DIFFERENTIABLE FUNCTION

(i) If $f(x)$ and $g(x)$ are derivable at $x = a$ then the functions $f(x) + g(x)$, $f(x) - g(x)$, $f(x).g(x)$ will be derivable at $x = a$ and if $g(a) \neq 0$ then the function $f(x)/g(x)$ will also be derivable at $x = a$.

(ii) If $f(x)$ is differentiable at $x = a$ and $g(x)$ is not differentiable at $x = a$, then the product function $f(x)\,g(x)$ can still be differentiable at $x = a$. e.g., $f(x) = x$ and $g(x) = |x|$ at $x = 0$.

(iii) If $f(x)$ and $g(x)$ both are not differentiable at $x = a$, then the product function $f(x)\,g(x)$ can still be differentiable at $x = a$ i.e.,

$f(x) = |x|$ and $g(x) = -|x|$ at $x = 0$.

(iv) If $f(x)$ is differentiable at $x = a$ and $g(x)$ is not differentiable at $x = a$, then the sum function $f(x) + g(x)$ is not - differentiable at $x = a$.

(v) If $f(x)$ and $g(x)$ both are not - differentiable at $x = a$, then the sum function may be a differentiable function.

e.g. $f(x) = |x| + 1$ and $g(x) = -|x|$

(vi) If $f(x)$ is derivable at $x = a$ then it need not be true that $f'(x)$ is continuous at $x = a$.

e.g. $f(x) = \begin{cases} x^2 \sin\frac{1}{x} & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$.

(vii) A surprising result : Suppose that the function $f(x)$ and $g(x)$ defined in the interval (x_1, x_2) containing the point x_0, and if f is differentiable at $x = x_0$ with $f(x_0) = 0$ together with g is continuous as $x = x_0$ then the function $F(x) = f(x) \cdot g(x)$ is differentiable at $x = x_0$. e.g. $F(x) = \sin x \cdot x^{2/3}$ is differentiable at $x = 0$.

DIFFERENTIAL COEFFICIENT :

Let $y = f(x)$ be a continuous function of a variable quantity x, where x is independent and y is dependent variable quantity. Let δx be an arbitrary small change in the value of x and δy be the corresponding change in y then $\lim_{\delta x \to 0} \frac{\delta y}{\delta x}$ if it exists, is called the derivative or differential coefficient of y with respect to x and it is denoted by $\frac{dy}{dx}$. y', y_1 or Δy

The process of finding derivative of a function is called differentiation.

If we again differentiate (dy/dx) with respect to x then the new derivative so obtained is called second derivative of y with respect to x and it is denoted by $\left(\frac{d^2y}{dx^2}\right)$ or y'' or y_2 or $\Delta^2 y$. Similarly, we can find successive derivatives of y which may be denoted by $\frac{d^3y}{dx^3}, \frac{d^4y}{dx^4}, \ldots\ldots\ldots\ldots \frac{d^ny}{dx^n}, \ldots\ldots\ldots$

13. DERIVATIVE OF *f*(*x*) FROM THE FIRST PRINCIPLE/AB INITIO METHOD

If $f(x)$ is a derivable function then, $\underset{\delta x\to 0}{Lim}\frac{\delta y}{\delta x} = \underset{\delta x\to 0}{Lim}\frac{f(x+\delta x)-f(x)}{\delta x} = f'(x) = \frac{dy}{dx}$.

or simply $f'(x) = \underset{h\to 0}{Lim}\frac{f(x+h)-f(x)}{h}$.

The derivative of sin x : Let $y = \sin x$, where the argument x is expressed in radians.

We have $y + \Delta y = \sin(x + \Delta x)$. Hence, $\Delta y = \sin(x + \Delta x) - \sin x$

or $\Delta y = 2\sin\frac{\Delta x}{2}\cos\left(x + \frac{\Delta x}{2}\right)$

Dividing both sides of the last equality by Δx, we get

$$\frac{\Delta y}{\Delta x} = \frac{2\sin\frac{\Delta x}{2}}{\Delta x}\cos\left(x + \frac{\Delta x}{2}\right) \quad \text{or} \quad \frac{\Delta y}{\Delta x} = \frac{\sin\frac{\Delta x}{2}}{\frac{\Delta x}{2}}\cos\left(x + \frac{\Delta x}{2}\right)$$

Proceeding to the limit as $\Delta x \to 0$ and using the theorem on the limit of a product, we have

$$y' = \lim_{\Delta x\to 0}\frac{\Delta y}{\Delta x} = \lim_{\Delta x\to 0}\frac{\sin\frac{\Delta x}{2}}{\frac{\Delta x}{2}} . \lim_{\Delta x\to 0}\cos\left(x + \frac{\Delta x}{2}\right)$$

From the theorem on the limit of the ratio of the sine of an infinitely small arc to the arc itself it follows that

$$\lim_{\Delta x\to 0}\frac{\sin\frac{\Delta x}{2}}{\frac{\Delta x}{2}} = 1$$

Besides, by virtue of the continuity of the function cos x, we have $\left[\lim_{\Delta x\to 0}\cos\left(x + \frac{\Delta x}{2}\right)\right] = \cos x$

Consequently, $y' = 1 \cdot \cos x = \cos x$, i.e, $(\sin x)' = \cos x$

Thus, we obtain the following theorem : The derivative of sin x is equal to cos x.

Illustration 29 : Differentiate $\frac{1}{\sqrt{(x+a)}}$ from first principle.

Solution : Step I : Let $y = \frac{1}{\sqrt{(x+a)}} = (x+a)^{-1/2}$

Step II : Then $y + \Delta y = (x + \Delta x + a)^{-1/2}$

Step III : Then $\Delta y = [(x + \Delta x) + a]^{-1/2} - (x + a)^{-1/2} = (x + a)^{-1/2} \left[1 + \frac{\Delta x}{x + a}\right]^{-\frac{1}{2}} - (x + a)^{-1/2}$

$= (x + a)^{-1/2} \left[\left\{1 - \frac{1}{2}\left(\frac{\Delta x}{x + a}\right) + \frac{\left(-1/2\right)\left(-3/2\right)}{2!}\left(\frac{\Delta x}{x + a}\right)^2 .....\right\} - 1\right]$ (by the expansion of binomial theorem.)

$= (x + a)^{-1/2}\left[-\frac{\Delta x}{2(x + a)} + \frac{3(\Delta x)^2}{8(x + a)^2}.....\right]$

Step (IV) : $\frac{\Delta y}{\Delta x}\frac{\Delta y}{\Delta x} = (x + a)^{-1/2}\left[-\frac{1}{2(x + a)} + \frac{3(\Delta x)}{8(x + a)^2}.....\right]$

Step (V) : Taking limit as $\Delta x \to 0$

$$\lim_{\Delta x \to 0} \frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} (x + a)^{-1/2}\left[-\frac{1}{2(x + a)} + \frac{3(\Delta x)}{8(x + a)^2}.....\right]$$

$$\therefore \frac{dy}{dx} = (x + a)^{-1/2}\left[-\frac{1}{2(x + a)}\right] = -\frac{1}{2}(x + a)^{-1/2} = -\frac{1}{2(x + a)^{3/2}}$$

Illustration 30 : Differentiate $\sin^{-1}(e^x)$ from first principle.

Solution : Let $y = \sin^{-1}(e^x)$

Then $y + \Delta y = \sin^{-1}(e^{x + \Delta x})$ $\therefore$ $\sin(y + \Delta y) - \sin y = e^{x + \Delta x} - e^x$

$$\Rightarrow \quad 2\cos\left(y + \frac{\Delta y}{2}\right).\sin\left(\frac{\Delta y}{2}\right) = e^x(e^{\Delta x} - 1) \Rightarrow 2\cos\left(y + \frac{\Delta y}{2}\right).\frac{\sin\left(\frac{\Delta y}{2}\right)}{\Delta y}.\frac{\Delta y}{\Delta x}$$

$$= e^x\frac{\left(e^{\Delta x} - 1\right)}{\Delta r}$$

Now, when $\Delta x \to 0$; $e^{x + \Delta x} \to e^x$. Hence $y + \Delta y \to y$ i.e. $\Delta y \to 0$

Thus applying the limit $\Delta r \to 0$ and $\Delta y \to 0$, we have

$$\lim_{\Delta x \to 0}\frac{\Delta y}{\Delta x} = \lim_{\Delta x \to 0} e^x\left(\frac{e^{\Delta x} - 1}{\Delta x}\right).\lim_{\Delta x \to 0}\frac{1}{2\cos\left(y + \frac{\Delta y}{2}\right)\frac{\sin\left(\Delta y/2\right)}{\Delta y}}$$

$$\text{i.e.} \quad \frac{dy}{dx} = \lim_{\Delta x \to 0} e^x \frac{\left[\Delta x + \frac{(\Delta x)^2}{2!} + \frac{(\Delta x)^3}{3!} + ...to\ \infty\right]}{\Delta x} \times \lim_{\Delta x \to 0}\frac{1}{\cos\left(y + \frac{\Delta y}{2}\right)\frac{\sin\left(\frac{\Delta y}{2}\right)}{\left(\frac{\Delta y}{2}\right)}}$$

$$\text{i.e.} \quad \frac{dy}{dx} = e^x.\frac{1}{\cos y} = \frac{e^x}{\sqrt{1 - \sin^2 y}} = \frac{e^x}{\sqrt{1 - e^{2x}}}$$

14. THEOREMS ON DIFFERENTIATION

Theorem I : $\frac{d}{dx}[kf(x)] = k\, d/dx\,[f(x)]$, where k is a constant

Theorem II : $\frac{d}{dx}[f_1(x) \pm f_2(x) \pm f_3(x) \pm ...] = \frac{d}{dx}[f_1(x)] \pm \frac{d}{dx}[f_2(x)] \pm \frac{d}{dx}[f_3(x)]$ .....

Theorem III : $\frac{d}{dx}[f(x).\, g(x)] = f(x)\,\frac{d}{dx}[g(x)] + g(x)\,\frac{d}{dx}[f(x)]$

Theorem IV : $\frac{d}{dx}\left[\frac{f(x)}{g(x)}\right] = \frac{g(x)d/dx[f(x)] - f(x)d/dx[g(x)]}{[g(x)]^2}$

Theorem V : Derivative of the function of the function. If 'y' is a function of 't' and 't' is a function of 'x' then $\frac{dy}{dx} = \frac{dy}{dt}.\frac{dt}{dx}$

Theorem VI : Derivative of parametric equations if $x = f(t)$, $y = \psi(t)$ then $\frac{dy}{dx} = \frac{dy/dt}{dx/dt}$

Theorem VII : Derivative of a function with respect to another function If $f(x)$ and $g(x)$ are two functions of a variable x, then

$$\frac{d[f(x)]}{d[g(x)]} = \frac{d}{dx}f(x) \,/\, \frac{d}{dx}g(x)$$

Theorem VIII : $\frac{dy}{dx}.\frac{dx}{dy} = 1$

Theorem IX : The general power rule : If $y = f[g(x)] = [g(x)]^n$ then the chain rule yields

$$\frac{dy}{dx} = \frac{dy}{du}.\frac{du}{dx} = nu^{n-1}.\frac{d}{dx}g(x) = n[g(x)]^{n-1}.\frac{d}{dx}g(x).$$

If n is any real number and $g'(x)$ exists, then $\frac{d}{dx}[g(x)]^n = n[g(x)]^{n-1}.\frac{d}{dx}g(x)$

You might think of the general power rule as : power in front of function, reduce power of function by one, times derivative of function. A common error is to ignore the $\frac{d}{dx}g(x)$ term in the general power formula.

DIFFERENTIATION OF SOME STANDARD FUNCTIONS

(1) $\frac{d}{dx}(\text{constant}) = 0$

(2) $\frac{d}{dx}(kx) = k$; where k is constant

(3) $\frac{d}{dx}(x^n) = nx^{n-1}$

(4) $\frac{d}{dx}(e^x) = e^x$

(5) $\frac{d}{dx}(a^x) = a^x \log_e a$

(6) $\frac{d}{dx}(\log_e x) = 1/x$

(7) $\frac{d}{dx}(\log_a x) = \frac{1}{x\log_e a}$

(8) $\frac{d}{dx}(\sin x) = \cos x$

(9) $\frac{d}{dx}(\cos x) = -\sin x$

(10) $\frac{d}{dx}(\tan x) = \sec^2 x$

(11) $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$

(12) $\frac{d}{dx}(\sec x) = \sec x \tan x$

(13) $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$

(14) $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}, -1 < x < 1$

(15) $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}, -1 < x < 1$

(16) $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$

(17) $\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$

(18) $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}; |x| > 1$

(19) $\frac{d}{dx}(\operatorname{cosec}^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}; |x| > 1$

(20) $\frac{d}{dx}(\sinh x) = \cosh x$

(21) $\frac{d}{dx}(\cosh x) = \sinh x$

(22) $\frac{d}{dx}(\tanh x) = \operatorname{sech}^2 x$

(23) $\frac{d}{dx}(\coth x) = -\operatorname{cosec} h^2 x$

(24) $\frac{d}{dx}(\operatorname{sec} h\, x) = \operatorname{sech} x \tanh x$

(25) $\frac{d}{dx}(\operatorname{cosech} x) = -\operatorname{cosec} hx \coth x$

(26) $\frac{d}{dx}(\sin h^{-1} x) = \frac{1}{\sqrt{1+x^2}}$

(27) $\frac{d}{dx}(\cosh^{-1} x) = \frac{1}{\sqrt{x^2-1}}$

(28 $\frac{d}{dx}(\tanh^{-1} x) = \frac{1}{1+x^2}$

(29) $\frac{d}{dx}(\coth^{-1} x) = \frac{1}{x^2-1}$

(30) $\frac{d}{dx}(\operatorname{sech}^{-1} x) = -\frac{1}{x\sqrt{1-x^2}}$

(31) $\frac{d}{dx}(\operatorname{cosech}^{-1} x) = -\frac{1}{|x|\sqrt{x^2+1}}$

(32) $\frac{d}{dx}(e^{ax} \sin b\, x) = e^{ax}(a \sinh x + b \cos bx) = \sqrt{a^2+b^2}\ e^{ax} \sin(bx + \tan^{-1} b/a)$

(33) $\frac{d}{dx}(e^{ax} \cosh x) = e^{ax}(a \cos b\, x - b \sinh x) = \sqrt{a^2+b^2}\ e^{ax} \cos(bx + \tan^{-1} b/a)$

15. METHODS OF DIFFERENTIATION

Working rule for finding the derivative of implicit functions

First method :

(i) Differentiate every term of $f(x, y) = 0$ with respect to x.

(ii) Collect the coefficients of $\frac{dy}{dx}$ and obtain the value of $\frac{dy}{dx}$.

Second method : $f(x, y)$ = constant, then $\frac{dy}{dx} = \frac{\partial f / \partial x}{\partial f / \partial y}$ are partial differential coefficients of $f(x, y)$ with respect to and y respectively.

Note :- Partial differential coefficient $f(x, y)$ with respect to x means the ordinary differential coefficient of $f(x, y)$ with respect to x keeping y constant.

THE DERIVATIVE OF AN IMPLICIT FUNCTION

If the relation between the variables x and y is given by an equation containing both, and this equation is not immediately solvable for y, then y is called an implicit function of x. Implicit functions are given by $\phi(x, y) = 0$.

(i) In order to find dy/dx, in the case of implicit functions, we differentiate each term w.r.t. x regarding y as a functions of x and then collect terms in dy/dx together on one side to finally find dy/dx.

(ii) In answers of dy/dx in the case of implicit functions, both x and y are present.

DIFFERENTIATION OF LOGARITHMIC FUNCTIONS

If differentiation of an expression or an equation is done after taking log on both sides, then it is called logarithmic differentiation. This method is useful for the function having following forms - When base and power both are the functions of x i.e. the functions is of the form $[f(x)]^{g(x)}$.

$y = [f(x)]^{g(x)}$

$\log y = g(x) \log [f(x)]$

$$\frac{1}{y} \cdot \frac{dy}{dx} = \frac{d}{dx} [g(x). \log [f(x)]]$$

$$\frac{dy}{dx} = [f(x)]^{g(x)} \cdot \left\{\frac{d}{dx}\left[g(x)\log f(x)\right]\right\}$$

THE DERIVATIVE OF AN EXPONENTIAL FUNCTION

Let $y = a^x$, where $a > 0$

Then $\ln y = x \ln a$.

Taking the derivative of both sides with respect to x, we have

$$\frac{1}{y}y' = \ln a \quad \text{hence} \quad y' = y \ln a,$$

or, finally, $(a^x)' = a^x \ln a$ ..... (1)

Thus, the derivative of an exponential function is equal to the function itself multiplied by the natural logarithm of the base.

Using the chain rule, we can write the more general form of this as follows :

The derivative of $a^{f(x)}$: $\frac{d}{dx}(a^{f(x)}) = a^{f(x)} f'(x) \ln a$

THE DERIVATIVE OF A FUNCTION REPRESENTED PARAMETRICALLY :

It is sometimes convenient to represent the relation between the variables x and y be two equations

$x = \phi(t)$, $y = \psi(t)$. ..... (1)

where t is an auxiliary variable (parameter). This method is frequently applied in mechanics were the parameter t usually denotes time, and equations (1) represent the parametric equations of the path of a moving point $M(x, y)$.

Generally speaking, equations (1) define y as a composite function of x. Indeed, resolving the first equation of system (1) with respect to the parameter t (if it is possible), we have

$t = \theta(x)$,

where θ is a function inverse to the function ϕ. Hence, eliminating the parameter t from eq. (1), we get

$y = \phi(\theta(x))$ ..... (2)

Applying formula (2), we easily find the derivative y'_x as the derivative of a composite function. But in practice elimination of the parameter t frequently involves many difficulties and sometimes

turns out to be even impossible. Therefore, we give a rule for finding the derivative y'_x which does not require elimination of the parameter.

Theorem : If a function of an argument x is represented parametrically $x = \phi(t)$, $y = \psi(t)$, where the functions $\phi(t)$ and $\psi(t)$ are differentiable and $\phi'(t) \neq 0$, then the derivative of this function is

$$y'_x = \frac{y'_t}{x'_t} \qquad \text{..... (3)}$$

16. DIFFERENTIATION BY TRIGONOMETRICAL SUBSTITUTIONS

Sometimes it comes very easy to differentiate a function by using trigonometrical transformations. Usually this is done in case of inverse trigonometrical functions . This method saves a lot of energy and time.

Some suitable substitutions

Functions		**Substitutions**
(i)	$\sqrt{a^2 - x^2}$	$x = a \sin\theta$ or $a \cos\theta$
(ii)	$\sqrt{x^2 + a^2}$	$x = a \tan\theta$ or $a \cot\theta$
(iii)	$\sqrt{x^2 - a^2}$	$x = a \sec\theta$ or a cosec θ
(iv)	$\sqrt{\frac{a-x}{a+x}}$	$x = a \cos 2\theta$
(v)	$\sqrt{\frac{a^2-x^2}{a^2+x^2}}$	$x^2 = a^2 \cos 2\theta$
(vi)	$\sqrt{ax - x^2}$	$x = a \sin^2\theta$
(vii)	$\sqrt{\frac{x}{a+x}}$	$x = a \tan^2\theta$
(viii)	$\sqrt{\frac{x}{a-x}}$	$x = a \sin^2\theta$
(ix)	$\sqrt{(x-a)(x-b)}$	$x = a \sec^2\theta - b \tan^2\theta$
(x)	$\sqrt{(x-a)(b-x)}$	$x = a \cos^2\theta + b \sin^2\theta$

DIFFERENTIATION OF INFINITE SERIES

(i) If $y = \sqrt{f(x) + \sqrt{f(x) + \sqrt{f(x) + ...\infty}}} \quad \Rightarrow \quad y = \sqrt{f(x) + y} \quad \Rightarrow \quad y^2 = f(x) + y$

$$2y\frac{dy}{dx} = f'(x) + \frac{dy}{dx} \qquad \therefore \quad \frac{dy}{dx} = \frac{f'(x)}{2y-1}$$

(ii) If $y = f(x)^{f(x)^{f(x)^{....\infty}}}$ then $y = f(x)^y$.

$\therefore \quad \log y = y \log [f(x)]$

$$\frac{1}{y}\frac{dy}{dx} = \frac{y'.f'(x)}{f(x)} + \log f(x).\left(\frac{dy}{dx}\right)$$

$$\therefore \quad \frac{dy}{dx} = \frac{y^2 f'(x)}{f(x)[1 - y\log f(x)]}$$

Illustration 31 : Find the derivative of the function $y = \sin^3 x = (\sin x)^3$

Solution: We put $z = \sin x$, then $y = x^3$, Hence $z'_x = \cos x$ and $y'_z = 3z^2 = 3\sin^2 x$
Consequently, $y'_x = 3\sin^2 x \cos x$
If experienced, one may not write the intermediate variable z, introducing it mentally.

Illustration 32 : Find the derivative of the function $y = \sqrt{x^2 + 4x + 3}$

Solution : Using the formula for the derivative of a square root and applying the rule for differentiating a composite function, we have

$$y' = \frac{1}{2\sqrt{x^2+4x+3}}.(x^2+4x+3)' = \frac{1}{2\sqrt{x^2+4x+3}}.(2x+4) = \frac{x+2}{\sqrt{x^2+4x+3}}$$

Illustration 33 : Find the derivative of the function $y = \tan^3 \frac{x}{2}$.

Solution : We have $y' = 3\tan^2 \frac{x}{2}.\left(\tan\frac{x}{2}\right)' = 3\tan^2 \frac{x}{2}\sec^2 \frac{x}{2}.\left(\frac{x}{2}\right)' = \frac{3}{2}\tan^2 \frac{x}{2}.\sec^2 \frac{x}{2}$

Illustration 34 Find the derivative of the function y ($y > 0$) defined by the equation $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$

Solution : Resolving this equation with respect to y and taking the plus sign, by hypothesis, we shall obtain our function in the explicit form $y = \frac{b}{a}\sqrt{a^2 - x^2}$.

Its differentiation is rather simple now.
But in some cases the given equation cannot be resolved with respect to y by elementary means and we have to regard y as an implicit function of x. Therefore we are going to indicate another method of finding the derivative of the implicit function y. Namely, assuming that instead of y its explicit expression is substituted in the given equation, we get an identity

$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$, where y is a function of x. Obviously, if two functions are identically equal to each other, then their derivatives are also equal to each other. Therefore, taking the derivatives of both sides of the foregoing identity and applying the rule for differentiation of a composite function, we have

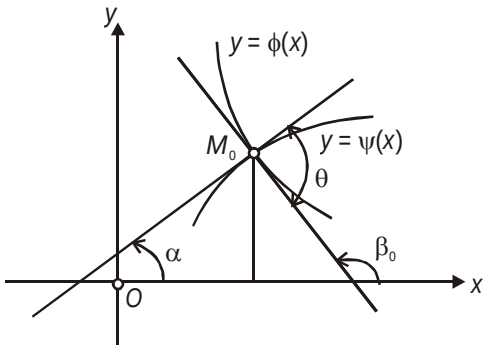
$\frac{2x}{a^2} + \frac{2y}{b^2}y' = 0$, hence $y' = -\frac{b^2 x}{a^2 y}$.

Illustration 35 : Let y be a positive or a negative implicit function of x defined by the equation $y^2 = 2px$.

Solution : Assuming that y is substitute by the corresponding explicit expression and differentiating with respect to x the left- and right-hand sides of the obtained identity, we have $2yy' = 2p$.

Hence, $y' = \frac{p}{y}$

Note :- If two functions $\phi(x)$ and $\psi(x)$ are equal to each



other not identically but for a certain value x_0 of the argument $\phi(x_0) = \psi(x_0)$, then from this, generally speaking, it does not follow that

$\phi'(x_0) = \psi'(x_0)$

This is clearly seen from figure, where $\phi'(x_0) = \tan\alpha_0$ and $\psi'(x_0) = \tan\beta_0$

Thus, in the general case one cannot differentiate an equality termwise.

Illustration 36 : If $x^2 + y^2 + xy = 2$, find dy/dx.

Solution : $x^2 + y^2 + xy = 2$, Differentiating both sides w.r.t. x we get,

$$\frac{d}{dx}(x^2) + \frac{d}{dx}(y^2) + \frac{d}{dx}(xy) = \frac{d}{dx}(2)$$

or $\quad 2x + 2y\frac{dy}{dx} + \left\{\frac{dx}{dx}\right\}y + x\left\{\frac{d}{dx}y\right\} = 0 \quad$ or $\quad 2x + 2y\frac{dy}{dx} + 1.y + x.\frac{dy}{dx} = 0$

$$(2y + x)\frac{dy}{dx} = -(2x + y) \quad \Rightarrow \quad \frac{dy}{dx} = -\frac{(2x + y)}{(2y + x)}$$

Illustration 37 : Differentiate $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ with respect to $\tan^{-1} x$.

Solution :
$$\frac{\frac{d}{dx}\sin^{-1}\left(\frac{2x}{1+x^2}\right)}{\frac{d}{dx}\tan^{-1}(x)} = \frac{\frac{d}{dx}\left(2\tan^{-1} x\right)}{\frac{d}{dx}\left(\tan^{-1} x\right)} = \frac{2\frac{d}{dx}(\tan^{-1} x)}{\frac{d}{dx}\left(\tan^{-1} x\right)} = 2$$

Illustration 38 : If $y = \tan^{-1}\left(\frac{\sin x}{1+\cos x}\right)$ find dy/dx.

Solution : $y = \tan^{-1}\left(\frac{2\sin\frac{x}{2}\cos\frac{x}{2}}{2\cos^2\frac{x}{2}}\right) = \tan^{-1}\left(\tan\frac{x}{2}\right) = \frac{x}{2} \qquad \therefore \frac{dy}{dx} = \frac{1}{2}$

Illustration 39 : Find the differential coefficient of $\tan^{-1}\left(\frac{\sin x + \cos x}{\cos x - \sin x}\right)$ w.r.t 'x'.

Solution : Let $y = \tan^{-1}\left(\frac{\sin x + \cos x}{\cos x - \sin x}\right) = \tan^{-1}\left(\frac{1+\tan x}{1-\tan x}\right) = \tan^{-1}\left(\tan\left(\frac{\pi}{4} + x\right)\right) = \frac{\pi}{4} + x$

$\therefore \frac{dy}{dx} = 1$

Illustration 40 : If $y = \sqrt{\cos x + \sqrt{\cos x + \sqrt{\cos x + ...}}}$ then find $\frac{dy}{dx}$

Solution : $y = \sqrt{\cos x + y} \Rightarrow y^2 = \cos x + y \Rightarrow 2y.\frac{dy}{dx} = -\sin x + \frac{dy}{dx}$

$\therefore \quad (2y - 1)\frac{dy}{dx} = -\sin x \quad \therefore \quad \frac{dy}{dx} = \frac{\sin x}{1-2y}$

EXERCISE

Q.1 If $y = \sin^{-1}\sqrt{\sin x}$, then $\frac{dy}{dx}$ equals-

(a) $\frac{2\sqrt{\sin x}}{\sqrt{1+\sin x}}$ (b) $\frac{\sqrt{\sin x}}{\sqrt{1-\sin x}}$ (c) $\frac{1}{2}\sqrt{1+\operatorname{cosec} x}$ (d) $\frac{1}{2}\sqrt{1-\operatorname{cosec} x}$

Q.2 $\frac{d}{dx}\left[\tan^{-1}\sqrt{\frac{1-\cos x}{1+\cos x}}\right]$ equals-

(a) 1/2 (b) 1 (c) sec x (d) cosec x

Q.3 If $x = at^2$ and $y = 2at$, then $xy\frac{d^2y}{dx^2}$ equals -

(a) a (b) $-a$ (c) 0 (d) $-a/xy$

Q.4 If $y = \tan^{-1}\left(\frac{4x}{1-4x^2}\right)$, then dy/dx equals -

(a) $\frac{4}{1+4x^2}$ (b) $\frac{1}{1+4x^2}$ (c) $\frac{16}{1+4x^2}$ (d) None of these

Q.5 The derivative of $\sin^{-1}x$ with respect to x^2 is -

(a) $\frac{1}{x\sqrt{1-x^2}}$ (b) $-\frac{1}{2\sqrt{1-x^2}}$ (c) $\frac{1}{2x\sqrt{1-x^2}}$ (d) $-\frac{1}{\sqrt{1-x^2}}$

Q.6 $\frac{d}{dx}(\log_{10}x)$ equals-

(a) 1/x (b) $(1/x)\log_{10}e$ (c) $\log \log_e x$ (d) None of these

Q.7 The differential coefficient of $\log \tan(x/2 + \pi/4)$ is -

(a) cosec x (b) cos x (c) sec x (d) sin x

Q.8 If $y = \log(x^x)$, then dy/dx equals -

(a) $\log(ex)$ (b) $\log(e/x)$ (c) $\log(x/e)$ (d) 1

Q.9 If $y = \log\sqrt{\left(\frac{1+\cos x}{1-\cos x}\right)}$, then the value of dy/dx is -

(a) sec x (b) cosec x (c) $-$ sec x (d) $-$ cosec x

Q.10 The derivative of $\log_a x + \log x^a$ is equal to -

(a) $\frac{1}{x}(a + \log_a e)$ (b) $\frac{1}{x}(a + \log_a a)$ (C) $\frac{a}{x} + \log_a e$ (d) None of these

Q.11 If $y = e^{x+e^x+e^x+.....\infty}$, then $\frac{dy}{dx}$ equals

(a) y (b) $\frac{1}{y}$ (c) $\frac{1}{1-y}$ (d) $\frac{y}{1-y}$

ANSWERS

1.	(c)	2.	(a)	3.	(b)	4.	(a)	5.	(c)	6. (b)
7.	(c)	8.	(a)	9.	(d)	10.	(a)	11.	(d)	

17. SUCCESSIVE DIFFERENTIATION

If y is a function of x and is differentiable with respect to x, then its derivative dy/dx can be found which is known as derivative of first order. If the first derivative dy/dx is also differentiable function, then it can be further differentiate with respect to x and this derivative is denoted by d^2y/dx^2 which is called the second derivative of y with respect to x further if d^2y/dx^2 is also differentiable then its derivative is called third derivative of y which is denoted by d^3y/dx^3. Similarly the nth derivative of y is denoted by d^ny/dx^n. All these derivative are called as successive derivative of y and this process is known as successive differentiation.

We also use the following symbols for the successive derivatives of $y = f(x)$:

y_1,	y_2, y_3,.....,	y_n,.....	
y',	y'', y''',.....,	y^n,.....	
Dy,	D^2y,	D^3y,.....,	D^ny,.....

where ($D = d/dx$)

dy/dx,	d^2y/dx^2, d^3y/dx^3.....	d^ny/dx^n.....	
$f'(x)$	$f''(x)$	$f'''(x)$,.....	$f^n(x)$,.....

If $y = f(x)$ then the value of the nth order derivative at $x = a$ is usually denoted by

$\left(\frac{d^ny}{dx^n}\right)_{x=a}$ or $(y^n)_{x=a}$ or $(y_n)_{x=a}$ or $f^n(a)$

Illustration 41 : Let $y = a \sin mx + b \cos mx$, then find $\frac{d^2y}{dx^2}$.

Solution : $\frac{dy}{dx} = am \cos mx - bm \sin mx$

$\therefore \quad \frac{d^2y}{dx^2} = -am^2 \sin mx - bm^2 \cos mx = -m^2 [a \sin mx + b \cos mx] = -m^2y$

Illustration 42 : If $x = t^2$, $y = t^3$, then find $\frac{d^2y}{dx^2}$

Solution : $\frac{dx}{dt} = 2t$, $\frac{dy}{dt} = 3t^2$

$\therefore \quad \frac{dy}{dx} = \frac{3t^2}{2t} = \frac{3}{2}t \quad \therefore \quad \frac{d^2y}{dx^2} = \frac{3}{2}.\frac{dt}{dx} = \frac{3}{2}\frac{1}{dx/dt} = \frac{3}{2.2t} = \frac{3}{4t}$

Illustration 43 : If $y = ax^{n+1} + bx^{-n}$, then $x^2 \frac{d^2y}{dx^2} =$

Solution : $y = ax^{n+1} + bx^{-n}$

$\frac{dy}{dx} = (n+1)ax^n - n\,bx^{-n-1}$

$$\frac{d^2y}{dx^2} = (n+1)n\,ax^{n-1} + n\,(n+1)\,bx^{-n-2}$$

$$\therefore\ x^2\,\frac{d^2y}{dx^2} = (n+1)\,na.\,x^{n+1} + n\,(n+1)\,b\,x^{-n} = n\,(n+1)\,[ax^{n+1} + bx^{-n}] = n\,(n+1)y$$

18. n^{th} DERIVATIVES OF SOME STANDARD FUNCTIONS

➤ $\frac{d^n}{dx^n}\sin(ax+b) = a^n \sin\left(\frac{n\pi}{2} + ax + b\right)$

Proof :

$y = \sin(ax+b)$

$\Rightarrow\ y_1 = a\cos(ax+b) = a\sin\left(\frac{\pi}{2} + ax + b\right)$

$\Rightarrow\ y_2 = -a^2\sin(ax+b) = a^2\sin\left(\frac{2\pi}{2} + ax + b\right)$

$\Rightarrow\ y_3 = -a^3\cos(ax+b) = a^3\sin\left(\frac{3\pi}{2} + ax + b\right)$

.....

.....

$\Rightarrow\ y_n = a^n\sin\left(\frac{n\pi}{2} + ax + b\right)$

Similarly $\frac{d^n}{dx^n}\cos(ax+b) = a^n\cos\left(\frac{n\pi}{2} + ax + b\right)$

➤ $\frac{d^n}{dx^n}(ax+b)^m = \frac{m!}{(m-n)!}\,a^n(ax+b)^{m-n}$, where $m < n$

Proof : $y = (ax+b)^m$

$\Rightarrow\ y_1 = a.\,m(ax+1)^{m-1}$

$\Rightarrow\ y_2 = a^2.\,m(m-1)\,(ax+b)^{m-2}$

$\Rightarrow\ y_3 = a^3 m\,(m-1)\,(m-2)\,(ax+b)^{m-3}$

.....

.....

$\Rightarrow\ y_n = a^n m(m-1)(m-2)\ldots(m-n+1)\,(ax+b)^{m-n}$ ..... (1)

Multiply and divide by (m – n)!

$$y_n = \frac{a^n m!(ax+b)^{m-n}}{(m-n)!}$$

Particular cases

(A) When $m = n$, $D^n\{(ax+b)^n\} = a^n.\,n!$

(B) When $a = 1$, $b = 0$ then $y = x^m$

$$\therefore\ D^n(x^m) = m(m-1)\ldots(m-n+1)x^{m-n} = \frac{m!}{(m-n)!}x^{m-n}$$

(C) When $a = 1$, $b = 0$ & $m = n$ then $y = x^n$

$\therefore\ D^n(x^n) = n!$

(D) When $m = -1$, $y = \frac{1}{(ax+b)}$ from equation (1)

$$y_n = a^n(-1)(-2)(-3)....(-n)(ax+b)^{-1-n} = a^n(-1)^n(1.2.3.....n)(ax+b)^{-1-n} = \frac{a^n(-1)^n n!}{(ax+b)^{n+1}}$$

Illustration 44 : If $y = \frac{1}{a-x}$ then find y_n.

Solution : $y_n = D^n\left(\frac{1}{a-x}\right) = D^n[-(x-a)^{-1}] = D^n(x-a)^{-1}$

$$= -\frac{(-1)^n n!}{(x-a)^{n+1}} = \frac{(-1)(-1)^n n!}{(-1)^{n+1}(a-x)^{n+1}} = \frac{n!}{(a-x)^{n+1}} \quad [\because D^n(ax+b)^{-1} = \frac{(-1)^n n! a^n}{(ax+b)^{n+1}}]$$

(3) $\frac{d^n}{dx^n}(\log(ax+b)) = \frac{(-1)^{n-1}(n-1)!a^n}{(ax+b)^n}$

Proof : $y_1 = \frac{a}{ax+b} = a.(ax+b)^{-1}$

$y_2 = -a^2(ax+b)^{-2}$

.....

.....

$y_n = \frac{(-1)^{n-1}(n-1)!a^n}{(ax+b)^n}$

Illustration 45 : n^{th} derivative of $\log\{(ax+b).(cx+d)\}$ is

(a) $(-1)^{n-1}(n-1)!\left[\frac{a^n}{(ax+b)^n} + \frac{c^n}{(cx+d)^n}\right]$

(b) $(-1)^n(n-1)!\left[\frac{a^n}{(ax+b)^n} + \frac{c^n}{(cx+d)^n}\right]$

(c) $(-1)^n n!\left[\frac{a^n}{(ax+b)^n} + \frac{c^n}{(cx+d)^n}\right]$

(d) None of these

Solution : (a) $D^n \log\{(ax+b)(cx+d)\} = D^n\{\log(ax+b)\} + D^n\{\log(cx+d)\}$

$$= \frac{(-1)^{n-1}(n-1)!a^n}{(ax+b)^n} + \frac{(-1)^{n-1}(n-1)!c^n}{(cx+d)^n} = (-1)^n(n-1)!\left[\frac{a^n}{(ax+b)^n} + \frac{c^n}{(cx+d)^n}\right]$$

(4) $\frac{d^n}{dx^n}(e^{ax}) = a^n e^{ax}$

Proof : $y = e^{ax}$

$y_1 = a.e^{ax}$

$y_2 = a^2.e^{ax}$

.....

.....

$y_n = a^n e^{ax}$

Illustration 46 : n^{th} derivative of e^{ax+b} is

(a) e^{ax+b} (b) $a^n e^{ax+b}$ (c) $b^n e^{ax+b}$ (d) $a^n b^n e^{ax+b}$

Solution : (b) $y = e^{ax+b}$

$y_1 = ae^{ax+b}$

$y_2 = a^2 e^{ax+b}$

.....

$y_n = a^n e^{ax+b}$

(5) $\dfrac{d^n(a^x)}{dx^n} = a^x (\log a)^n$

Proof $y_1 = a^x \log a$

$y_2 = a^x(\log a)^2$

$y_3 = a^x(\log a)^3$

.....

.....

$y_n = a^x (\log a)^n$

Illustration 47 : n^{th} derivative of 2^{3x+4} is

(a) $2^{3x+4} (\log_e 2)$ (b) $2^{3x+4} (\log_e 2)^n$

(c) $3^n (2^{3x+4}) (\log_e 2)^n$ (d) None of these

Solution (c) $y = 2^{3x+4}$

$y_1 = 2^{3x+4}. \log_e 2.3$

$y_2 = 2^{3x+4} (\log_e 2)^2. 3^2$

.....

$y_n = 2^{3x+4} (\log_e 2)^n. 3^n$

(6) $\dfrac{d^n}{dx^n} (e^{ax} \sin (bx + c) = r^n e^{ax} \sin (bx + c + n\phi)$

where $r = \sqrt{a^2 + b^2}$; $\phi = \tan^{-1} \dfrac{b}{a}$

Proof : $y = e^{ax} \sin (bx + c)$

$y_1 = e^{ax} b \cos (bx + c) + \sin (bx + c) a e^{ax}$; Put $a = r\cos\phi$, $b = r \sin \phi$

$y_1 = e^{ax} r (\sin (bx + c + \phi)$;

where $r = \sqrt{a^2 + b^2}$, $\phi = \tan^{-1} b/a$

$y_2 = r^2 e^{ax} \sin (bx + c + 2\phi)$

$y_3 = r^3 e^{ax} \sin (bx + c + 3\phi)$

.....

.....

$y_n = r^n e^{ax} \sin (bx + c + n\phi)$

Similarly $\dfrac{d^n}{dx^n} \{e^{ax} \cos (bx + c)\} = r^n e^{ax} \cos (bx + c + n\phi)$

Illustration 48 : $D^n (e^x \sin x)$ equals

(a) $(\sqrt{2})^n e^x \sin\left(x + \dfrac{n\pi}{4}\right)$ (b) $2^n e^x \cos\left(x + \dfrac{n\pi}{4}\right)$

(c) $(\sqrt{2})^n e^x \cos\left(x + \dfrac{n\pi}{4}\right)$ (d) $2^n e^x \sin\left(x + \dfrac{n\pi}{4}\right)$

Solution : (a) We Know that

$D^n [e^{ax} \sin(bx + c)] = r^n e^{ax} \sin (bx + c + n\phi)$

where $r = \sqrt{a^2 + b^2}$, $\phi = \tan^{-1}\left(\frac{b}{a}\right)$

Put $a = 1, b = 1, c = 0$; $D^n [e^x \sin x] = (\sqrt{2})^n e^x \sin\left(x + \frac{n\pi}{4}\right)$

n^{th} DERIVATIVE USING PARTIAL FRACTIONS :

For finding the nth derivative of fractional expressions whose numerator and denominator are rational algebraic expression, firstly we resolve them into partial fractions and then we find nth derivative by using the formula giving the nth derivative of $\frac{1}{ax+b}$.

Illustration 49 : n^{th} derivative of $\frac{1}{(1-5x+6x^2)}$ is

(a) $(-1)^n \left[\frac{2^{n+1}}{(2x-1)^{n+1}} - \frac{3^{n+1}}{(3x-1)^{n-1}}\right]$ (b) $(-1)^n n!\left[\frac{2^{n-1}}{(2x-1)^{n+1}} - \frac{3^{n-1}}{(3x-1)^{n+1}}\right]$

(c) $(-1)^n n !\left[\frac{2^{n+1}}{(2x-1)^{n+1}} - \frac{3^{n+1}}{(3x-1)^{n+1}}\right]$ (d) None of these

Solution : (c) $y = \frac{1}{1-5x+6x^2} = \frac{1}{(2x-1)(3x-1)} = \frac{2}{2x-1} - \frac{3}{3x-1}$

$\therefore \quad y_n = (-1)^n n ! \left[\frac{2^{n+1}}{(2x-1)^{n+1}} - \frac{3^{n+1}}{(3x-1)^{n+1}}\right]$

Illustration 50 : n^{th} derivative of $\frac{x^4}{(x-1)(x-2)}$ is (where $n > 2$)

(a) $(-1)^n n!\left[\frac{16}{(x-2)^{n+1}} - \frac{1}{(x-1)^{n+1}}\right]$ (b) $(-1)^n n!\left[\frac{1}{(x-2)^{n+1}} - \frac{1}{(x-1)^{n+1}}\right]$

(c) $(-1)^n n!\left[\frac{16}{(x-2)^{n+1}} + \frac{1}{(x-1)^{n+1}}\right]$ (d) None of these

Solution : (a) Let $y = \frac{x^4}{(x-1)(x-2)} = x^2 + 3x + 7 + \frac{16}{x-2} - \frac{1}{x-1}$

$\therefore \quad y_n = 0 + 0 + 0 + 16. \frac{(-1)^n n!}{(x-2)^{n+1}} - \frac{(-1)^n n!}{(x-1)^{n+1}} = (-1)^n n!\left[\frac{16}{(x-2)^{n+1}} - \frac{1}{(x-1)^{n+1}}\right]$

19. LEIBNITZ'S THEOREM :

G.W. Libnitz, a German mathematician gave a method for evaluating the nth differential coefficient of the product of two functions. This method is known as Leibnitz's Theorem.

Statement of the theorem:– If u and v are two functions of x such that their nth derivative exist, then

$D^n(uv) = {}^nC_0 (D^n u)v + {}^nC_1 D^{n-1}.u. Dv + {}^nC_2 D^{n-2}. u. D^2v + ...+ {}^nC_r D^{n-r}. u. D^r v + ... + u. (D^n v)$

Note :- The success in finding the nth derivative by this theorem lies in the proper selection of first and second function. Here first function should be selected whose nth derivative can be found by standard formulae. Second function should be such that on successive differentiation, at some stage, it becomes zero so that we need not to write further terms.

Illustration 51 : Find n^{th} derivative of $x^2 \sin 3x$

Solution : Let $u = \sin 3x$, $v = x^2$

$\therefore \quad D^n (\sin 3x.x^2) = D^n (\sin 3x)x^2 + {}^nC_1 D^{n-1} (\sin 3x).\ 2x + {}^nC_2 D^{n-2} (\sin 3x).\ 2$

$$= x^2 \sin\left(3x + \frac{n\pi}{2}\right).\ 3^n + 2nx.\ 3^{n-1} \sin\left[3x + \frac{1}{2}(n-1)\pi\right] + 3^{n-2}\ n\ (n-1)$$

$$\sin\left\{3x + \frac{1}{2}(n-2)\pi\right\}$$

Illustration 52 : Find n^{th} derivative of $x^3 e^x$

Solution : $y = x^3 e^x$

$D^n (e^x.\ x^3) = D^n (e^x).\ x^3 + {}^nC_1 D^{n-1} (e^x).\ 3x^2 + {}^nC_2 D^{n-2} (e^x)\ 6x + {}^nC_3 D^{n-3} (e^x).\ 6$

$= e^x x^3 + 3n\ e^x x^2 + n\ (n-1)\ e^x.\ 3x + n\ (n-1)\ (n-2)\ ,\ e^x$

PHYSICAL MEANING OF THE SECOND DERIVATIVE

We can find the velocity of motion with the aid of the first-order derivative. To compute the acceleration of motion, we have to take advantage of the second-order derivative.

Let the law of motion of a point M along the x-axis be expressed by the equation x = f (t). Let at the moment t the point M have the velocity v, and at the moment $t + \Delta t$ the velocity $v + \Delta v$.

Thus during the time interval Δt the velocity changed by the increment Δv. The ratio $\frac{\Delta v}{\Delta t}$ is called the average acceleration of rectilinear motion for time interval Dt. The limit of this ratio as $\Delta t \to 0$. i.e.,

$\lim\limits_{\Delta t \to 0} \frac{\Delta v}{\Delta t} = v'(t)$ is called the acceleration of the point M at a given moment t. Denoting the acceleration by j, we may write $j = v'(t)$

But $v = f'(t)$. Therefore $v'(t) = [f'(t)]' = f''(t)$.

Thus, we have $j = f''(t)$. i.e. the acceleration of rectilinear motion of a point is equal to the second derivative of the distance with respect to the time.

Student Task

Q.1 If $y = x^2 \log x$, then value of y_n is -

(a) $\frac{(-1)^{n-1}(n-3)!}{x^{n-2}}$ (b) $\frac{(-1)^{n-1}(n-3)!}{x^{n-2}}.2$ (c) $\frac{(-1)^{n-1}(n-2)!}{x^{n-2}}$ (d) None of these

Q.2 If $y = \tan^{-1}\left(\frac{\log(e/x^2)}{\log(ex^2)}\right) + \tan^{-1}\left(\frac{3+2\log x}{1-6\log x}\right)$ then $\frac{d^2y}{dx^2}$ is –

(a) 2 (b) 1 (c) 0 (d) – 1

Q.3 The nth derivative of $\frac{x}{1-x}$

(a) $\frac{(-1)^n n!}{(1-x)^{n+1}}$ (b) $\frac{n!}{(1-x)^{n+1}}$ (c) $\frac{(-1)^n}{(1-x)^{n+1}}$ (d) $\frac{1}{(1-x)^{n+1}}$

Q.4 The nth derivative of $e^{2\ -3x}$ is -

(a) $3^n\ e^{2-3x}$ (b) $(-3)^n\ e^{2-3x}$ (c) $\frac{1}{3^n}e^{2-3x}$ (d) $\frac{1}{(-3)^n}e^{2-3x}$

Q.5 If $y = \cos^2 x$, then y_n equals to-

(a) $2^{n-1}\cos\left(2x + \frac{1}{2}n\pi\right)$ (b) $-2^{n-1}\cos\left(2x + \frac{1}{2}n\pi\right)$

(c) $2^n \cos\left(2x + \frac{1}{2}n\pi\right)$ (d) $-2^n\cos\left(2x + \frac{1}{2}n\pi\right)$

Q.6 If $y^2 = ax^2 + bx + c$ where a, b, c are constants, then $y^3\frac{d^2y}{dx^2}$ is equal to

(a) a constant (b) a function of x

(c) a functions of y (d) a functions of x and y both

ANSWERS

1. (b) 2. (c) 3. (b) 4. (b) 5. (a) 6. (a)

20. L' HOPITAL'S RULE :

Consider the ratio $f(x)=\frac{\phi(x)}{\psi(x)}$, where the function $\phi(x)$ and $\psi(x)$ are defined and differentiable in a neighbourhood U_a of a point a excluding, perhaps, the point a itself. It may happen that as $x \to a$ both functions tend to 0 or to ∞, i.e. both these functions are either infinitely small or infinitely large as $x \to a$. Then we say that at the point a the function f (x) has an indeterminancy of the form

$$\frac{0}{0} \text{ or } \frac{\infty}{\infty}, \qquad \text{..... (1)}$$

respectively. In this case, using the derivative $\phi'(x)$ and $\psi'(x)$, we can formulate a simple rule for finding the limit of the function $f(x)$ as $x \to a$ i.e. give a method for evaluating certain indeterminancies of the form. This rule is usually associated with the name of the French mathematician L' Hopital's. (For simplicity some authors write as L' Hospital's rule)

Theorem : The limit of the ratio of two infinitely small or infinitely large functions is equal to the limit of the ratio of their derivatives (finite or infinite) if the latter exists (in the indicated sense).

Proof : We shall consider here only indeterminate forms $\frac{0}{0}$ and, for the sake of simplicity, assume that the function $\phi(x)$ and $\psi(x)$, as well as their derivatives $\phi'(x)$ and $\psi'(x)$, are continuous at the point a and $\psi'(a) \neq 0$. The proof of another case of L' Hopital's rule referring to indeterminate forms $\frac{\infty}{\infty}$ is more complicated.

And so, let $\lim_{x\to a}\phi(x)=\phi(a)=0$ ..... (2)

and $\lim_{x\to a}\psi(x)=\psi(a)=0$ ..... (2′)

The difference $\phi(x) - \phi(x)$ may be regarded as the increment of the function $\phi(x)$ at the point a corresponding to the increment of the argument $\Delta x = x - a$. Therefore,

$$\lim_{x\to a}\frac{\phi(x)-\phi(a)}{x-a}=\phi'(a) \qquad \text{..... (3)}$$

analogously, $\lim_{x\to a}\frac{\psi(x)-\psi(a)}{x-a}=\psi'(a)\neq 0$ ..... (3')

Taking into consideration formulas (2) and (2') for $x \neq a$, we get

$$\frac{\phi(x)}{\psi(x)}=\frac{\phi(x)-\phi(a)}{\psi(x)-\psi(a)}=\frac{\frac{\phi(x)-\phi(a)}{x-a}}{\frac{\psi(x)-\psi(a)}{x-a}}$$

Hence, passing to the limit $x \to a$ and applying formulas (3) and (3'), we shall have

$$\lim_{x\to a}\frac{\phi(x)}{\psi(x)}=\frac{\phi'(a)}{\psi'(a)} \qquad \text{..... (4)}$$

But we assumed that the derivative $\phi'(x)$ and $\psi'(x)$ are continuous as $x \to a$ and $\psi'(a) \neq 0$, therefore

$$\lim_{x\to a}\frac{\phi'(x)}{\psi'(x)}=\frac{\lim_{x\to a}\phi'(x)}{\lim_{x\to a}\psi'(x)}=\frac{\phi'(a)}{\psi'(a)} \qquad \text{..... (5)}$$

Comparing formulas (4) and (5), we obtain L'Hopital's rule

$$\lim_{x\to a}\frac{\phi(x)}{\psi(x)}=\lim_{x\to a}\frac{\phi'(x)}{\psi'(x)} \qquad \text{..... (6)}$$

Note :- The right-hand side of formula (6) is the limit of the ratio of the derivatives, but not of the derivative of the ratio of the functions.

From above discussion we can rewrite L' Hopital rules as

L' Hopital's rule (First Form) :

Suppose that $f(a) = g(a) = 0$, $f'(a)$ and $g'(a)$ exist, and that $g'(a) \neq 0$.

Then $\lim_{x\to a}\frac{f(x)}{g(x)}=\frac{f'(a)}{g'(a)}$.

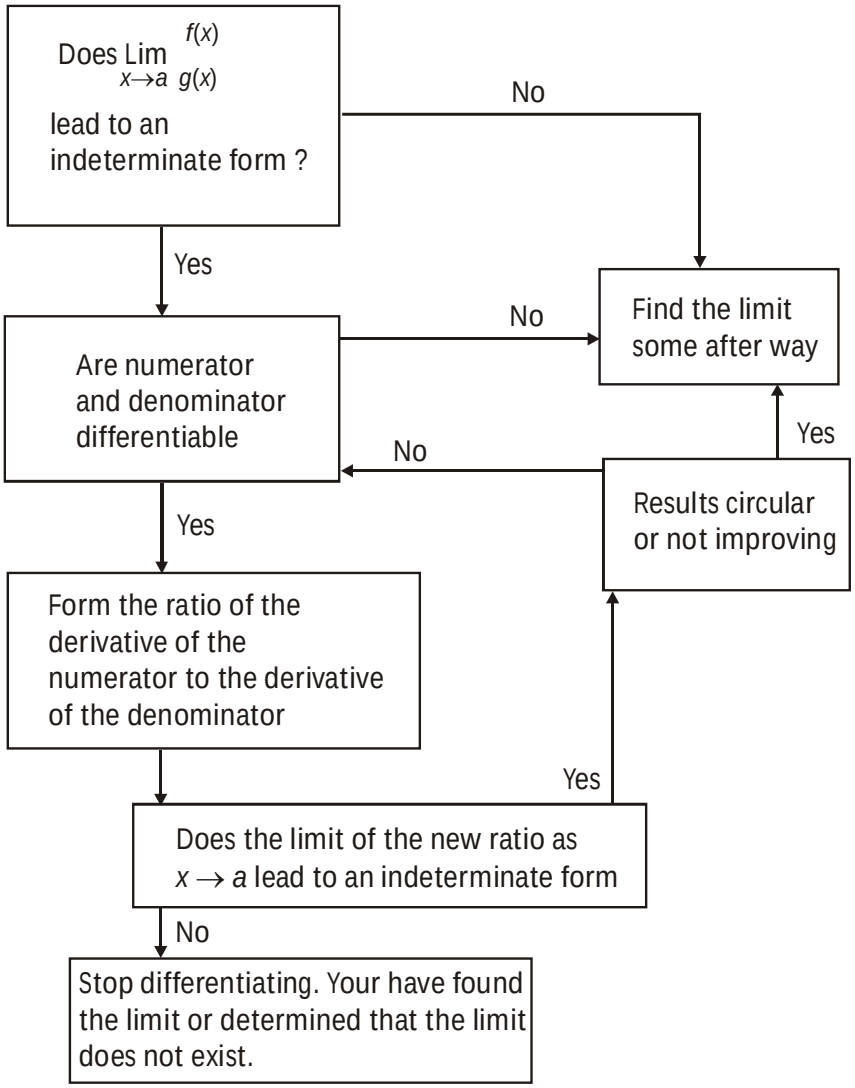
L' Hospital's rule (Stronger Form) :

Suppose that f(a) = g(a) = 0 and that f and g are differentiable on an open interval I containing a.

Suppose also that $g'(x) \neq 0$ on I if $x \neq a$. Then $\lim_{x\to a}\frac{f(x)}{g(x)}=\lim_{x\to a}\frac{f'(x)}{g'(x)}$.

If the limit on the right exists (or is ∞ or $-\infty$)

Working steps



We can sometimes use L' Hopital's rule of evaluate limits of indeterminate form $0.\infty$ by rewriting the expressions in the form of a quotient to which L' Hopital's rule applies. In particular, if the product uv has indeterminate form $0.\infty$, then by writing either

$$u(x)v(x) = \frac{u(x)}{1/v(x)} \quad \text{or} \quad u(x)v(x) = \frac{v(x)}{1/u(x)}$$

we can express uv in indeterminate form 0/0 to ∞/∞, respectively.

Illustration 53 : Evaluate $\lim_{x\to 0} x\cot x$

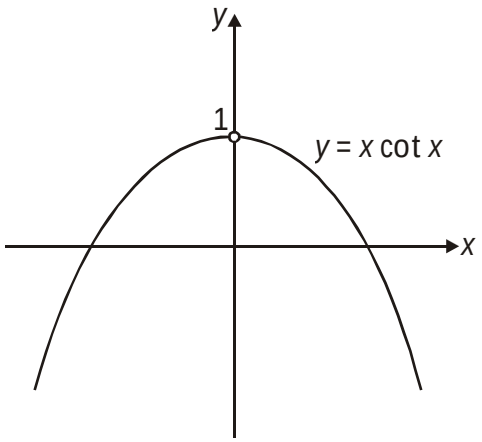
Solution : The product x cot x has indeterminate form $0.\infty$ at $x = 0$.
Rewriting the product as a quotient, we have

$$\lim_{x\to 0} x\cot x \underset{(\text{Rewriting})}{=} \lim_{x\to 0}\frac{\cot x}{1/x} \underset{(\infty/\infty\ L'H)}{=} \lim_{x\to 0}\frac{-\operatorname{cosec}^2 x}{-1/x^2}$$

At this point see that application of L' Hopital's rule to the quotient has given us an even more complicated quotient, so we abandon this effect and try rewriting the product as a different quotient, we then obtain

$$\lim_{x\to 0} x\cot x \underset{(\text{Rewriting})}{=} \lim_{x\to 0}\frac{x}{\tan x} \underset{(0/0,\ L'H)}{=} \lim_{x\to 0}\frac{1}{\sec^2 x} = 1$$

The graph of $x \cot x$ for x near zero is given in figure.



Note :-

- **(i)** Be sure to check that the expression has indeterminate form either 0/0 or ∞/∞ before applying L'Hopital's rule.
- **(ii)** Don't forget to try simplifying before L'Hopital's rule.
- **(iii)** We can simplify the differentiation in some problems by excluding factors that have finite, nonzero limits from the application of L'Hopital's rule.
- **(iv)** We may need to apply L'Hopital's rule more than once to evaluate a limit, but we should check that we are making progress.

Illustration 54 : Find $\lim_{x\to 0} \frac{2^x - 1}{\sin x}$

Solution : Here, for $x = 0$, both the numerator and the denominator of the fraction are equal to zero, i.e. as $x \to 0$ we have an indeterminate form $\frac{0}{0}$. Applying L'Hopital's rule (6), we get

$$\lim_{x\to 0} \frac{2^x - 1}{\sin x} = \lim_{x\to 0} \frac{2^x \ln 2}{\cos x} = \frac{\ln 2}{1} = \ln 2$$

Illustration 55 : Find $\lim_{x\to +\infty} \frac{x^2}{e^x}$

Solution : As $x \to +\infty$ we have indeterminate form of the type $\frac{\infty}{\infty}$. Applying L' Hospital rule twice, we obtain

$$\lim_{x\to +\infty} \frac{x^2}{e^x} = \lim_{x\to +\infty} \frac{2x}{e^x} = \lim_{x\to +\infty} \frac{2}{e^x} = 0.$$

Thus as $x \to +\infty$, the exponential function e^x is increasing more rapidly than the power function x^2.

Besides the above indicated types of indeterminates $\frac{0}{0}$ and $\frac{\infty}{\infty}$, there exist other indeterminate forms.

For instance, if $\phi(x) = \phi(x)\ \psi(x)$,

where $\phi(x) \to 0$ and $\psi(x) \to 0$ as $x \to a$ the function $f(x)$ represents an indeterminate form of the type $0.\infty$. Another example is the function $\phi(x) = \phi(x) - \psi(x)$ where $\phi(x) \to +\infty$ and $\psi(x) \to +\infty$ as $x \to a$. Here as $x \to a$ we obtain an indeterminancy of the form $\infty - \infty$. Other indeterminate forms, that is forms of the type ∞^0, 1^∞ and $0°$, are also possible. To evaluate these forms, we try to reduce them to the principal forms $\frac{0}{0}$ or $\frac{\infty}{\infty}$ which are then evaluate using L'Hopital rule.

Illustration 56 : Find $\lim_{x\to+0} x\ln x$

Solution : Here we have an indeterminate form of the type $0.\infty$. Rewriting the given expression in the form

$$\lim_{x\to+0} x\ln x = \lim_{x\to+0} \frac{\ln x}{1/x},$$

We get an indeterminate form the type $\frac{\infty}{\infty}$. Hence, applying L'Hopital rule, we find

$$\lim_{x\to+0} x\ln x = \lim_{x\to+0} \frac{1/x}{-1/x^2} = \lim_{x\to+0} (-x) = 0$$

Student Task

Evaluate the following limits :

Q.1 $\lim_{x\to1} \frac{x^2 - x.\ell nx + \ell nx - 1}{x-1}$

Q.2 $\lim_{x\to3} \frac{e^x - e^3}{x-3}$

Q.3 $\lim_{x\to0} \frac{e^{\sin x} - \sin x - 1}{x^2}$

Q.4 $\lim_{x\to0} \left(\frac{e^x - 1 - x}{x^2} \right)$

Q.5 $\lim_{x\to1} \frac{1 + \log x - x}{1 - 2x + x^2}$

Q.6 $\lim_{x\to\infty} \frac{x^n}{e^x}$ for all values of n (n integer)

ANSWERS

1. 2 2. e^3 3. 1/2 4. 1/2 5. –1/2 6. 0

USEFUL TIPS

1. $\lim_{n\to\infty} \left(\frac{\sqrt{n^2+n} - 1}{n} \right)^{2\sqrt{n^2+n}-1} = e^{-1}$

2. $\lim_{x\to0} \left(\frac{x - 1 + \cos x}{x} \right)^{1/x} = e^{-1/2}$

3. $\lim_{x\to0} \frac{\tan^2 x - x^2}{x^2 \tan^2 x} = \frac{2}{3}$

4. $\lim_{n\to\infty}\frac{[1.x]+[2.x]+[3.x]+.....+[n.x]}{n^2} = \frac{x}{2}$ where [.] denotes the greatest integer function.

5. $\lim_{x\to 1}\frac{1-x+\ln x}{1+\cos\pi x}=-\frac{1}{\pi^2}$

6. $\lim_{y\to 0}\left[\lim_{x\to\infty}\frac{\exp\left(x\ln\left(1+\frac{ay}{x}\right)\right)-\exp\left(x\ln\left(1+\frac{bx}{y}\right)\right)}{y}\right]=a-b$

7. If $f(n,\theta)\prod_{r=1}^{n}\left(1-\tan^2\frac{\theta}{2^r}\right)$, then $\lim_{n\to\infty} f(n,\theta)=\frac{\theta}{\tan\theta}$

8. Through a point *A* on a circle, a chord *AP* is drawn and on the tangent at *A* a point *T* is taken such that
AT = *AP*. If *TP* produced meet the diameter through *A* at *Q*, the limiting value of *AQ* when *P* moves upto A is double the diameter of the circle.

9. $\lim_{n\to\infty}\left(\frac{1}{\sqrt{n^2}}+\frac{1}{\sqrt{n^2+1}}+\frac{1}{\sqrt{n^2+2}}+....+\frac{1}{\sqrt{n^2+2n}}\right)=2$

10. $\lim_{n\to\infty}\frac{1}{1+n^2}+\frac{2}{2+n^2}+.....+\frac{n}{n+n^2}=\frac{1}{2}$

11. If $F(x)=\begin{vmatrix} f(x) & g(x) & h(x)\\ \ell(x) & m(x) & n(x)\\ u(x) & v(x) & w(x)\end{vmatrix}$, where *f*, *g*, *h*, ℓ, *m*, *n*, *u*, *v*, *w* are differentiable functions of *x* then

$$F'(x)=\begin{vmatrix} f'(x) & g'(x) & h'(x)\\ \ell(x) & m(x) & n(x)\\ u(x) & v(x) & w(x)\end{vmatrix}+\begin{vmatrix} f(x) & g(x) & h(x)\\ \ell'(x) & m'(x) & n'(x)\\ u(x) & v(x) & w(x)\end{vmatrix}+\begin{vmatrix} f(x) & g(x) & h(x)\\ \ell(x) & m(x) & n(x)\\ u'(x) & v'(x) & w'(x)\end{vmatrix}..$$

12. If $y = \sec 4x$ and $x = \tan^{-1}(t)$, then $\frac{dy}{dx}=\frac{16t(1-t^4)}{(1-6t^2+t^4)^2}$

13. If $x=\frac{1+\ln t}{t^2}$ and $x=\frac{3+2\ln t}{t}$ then $y\frac{dy}{dx}=2x\left(\frac{dy}{dx}\right)^2+1$.

14. If $\sqrt{1-x^6}+\sqrt{1-y^6}=a^3(x^3-y^3)$, then $\frac{dy}{dx}=\frac{x^2}{y^2}\sqrt{\frac{1-y^6}{1-x^6}}$

15. If $y=\tan^{-1}\frac{u}{\sqrt{1-u^2}}$ and $x=\sec^{-1}\frac{1}{2u^2-1}, u\in\left(0,\frac{1}{\sqrt{2}}\right)\cup\left(\frac{1}{\sqrt{2}},1\right)$, then $2\frac{dy}{dx}+1=0$.

16. If $y=\tan\frac{y}{2}-\ln\left[\frac{\left(1+\tan\frac{y}{2}\right)^2}{\tan\frac{y}{2}}\right]$, then $\frac{dy}{dx}=\frac{1}{2}\sin y\,(1+\sin y+\cos y)$.

17. If $\sin y = x \sin(a + y)$, then $\frac{dy}{dx} = \frac{\sin a}{1 - 2x\cos a + x^2}$.

18. If $y = \cos^{-1}\sqrt{\frac{\cos 3x}{\cos^3 x}}$, then $\frac{dy}{dx} = \sqrt{\frac{6}{\cos 2x + \cos 4x}}, \sin x > 0$

19. If $|a_1 \sin x + a_2 \sin 2x + ... + a_n \sin nx| \le |\sin x|$ for $x \in R$, then
$|a_1 + 2a_1 + 3a_3 + ... + na_n| \le 1$.

20. If $y = 1 + \frac{x_1}{x - x_1} + \frac{x_2.x}{(x - x_1)(x - x_2)} + \frac{x_3.x^2}{(x - x_1)(x - x_2)(x - x_3)} + ...\text{upto}\,(n+1)$ terms then

$$\frac{dy}{dx} = \frac{y}{x}\left[\frac{x_1}{x_1 - x} + \frac{x_2}{x_2 - x} + \frac{x_3}{x_3 - x} + ... + \frac{x_n}{x_n - x}\right]$$

21. $\cos x + \cos 3x + \cos 5x + ... + \cos(2n - 1)x = \frac{\sin 2nx}{2\sin x}$, $x \ne K\pi$, $K \in I$ and

$$\sin x + 3\sin 3x + 5\sin 5x ..... + (2n - 1)\sin(2n - 1)x = \frac{[(2n+1)\sin(2n-1)x - (2n-1)\sin(2n+1)x]}{4\sin^2 x}.$$

MISCELLANEOUS EXAMPLES

Example 1 :

Evaluate $\lim_{x\to\infty} \frac{\ell n\, x}{x}$.

Solution :

As $0 \le \ln x \le \sqrt{x}$ $(x > 1)$

$\Rightarrow$ $0 \le \frac{\ell n\, x}{x} \le \frac{1}{\sqrt{x}}$ $(x > 1)$

$$0 \le \lim_{x\to\infty} \frac{\ell n\, x}{x} \le \lim_{x\to\infty} \frac{1}{\sqrt{x}} = 0 \Rightarrow \lim_{x\to\infty} \frac{\ell n\, x}{x} = 0$$

Example 2 :

Evaluate $\lim_{x\to 0} \frac{\sin\left(\pi\cos^2 x\right)}{x^2}$.

Solution :

$$\lim_{x\to 0} \frac{\sin\left(\pi\cos^2 x\right)}{x^2} = \lim_{x\to 0} \frac{\sin\left\{\pi(1 - \sin^2 x)\right\}}{x^2} = \lim_{x\to 0} \frac{\sin\left(\pi - \pi\sin^2 x\right)}{x^2}$$

$$= \lim_{x\to 0} \frac{\sin\left(\pi\sin^2 x\right)}{x^2} = \lim_{x\to 0}\left\{\frac{\sin\left(\pi\sin^2 x\right)}{\pi\sin^2 x} \times \frac{\pi}{1} \times \frac{\sin^2 x}{x^2}\right\}$$

$$= \lim_{x\to 0}\frac{\sin\left(\pi\sin^2 x\right)}{\pi\sin^2 x}\times\pi\times\lim_{x\to 0}\frac{\sin^2 x}{x^2} = 1\times\pi\times 1 = \pi$$

Example 3 :

If $F(x) = \lim_{n\to\infty}\frac{x^{2n}f(x)+g(x)}{x^{2n}+1}$, find $F(x)$ in terms of $f(x)$ and $g(x)$

Solution :

$$\lim_{n\to\infty}x^{2n} = \begin{cases}0, & if \quad x^2<1\\ 1, & if \quad x^2=1\\ \infty, & if \quad x^2>1\end{cases} = \begin{cases}0, & if \quad -1<x<1\\ 1, & if \quad x=\pm 1\\ \infty, & if \quad x<-1\ or, x>1\end{cases} \quad \therefore \quad F(x)=\lim_{n\to\infty}\frac{x^{2n}.f(x)+g(x)}{x^{2n}+1}$$

$$= 0\begin{cases}\dfrac{0.f(x)+g(x)}{0+1}, & -1<x<1\\ \dfrac{1.f(x)+g(x)}{1+1}, & x=\pm 1\\ \dfrac{f(x)+\dfrac{g(x)}{x^{2n}}}{1+\dfrac{1}{x^{2n}}}, & x<-1 or x>1\end{cases} \quad \therefore \quad F(x)=\begin{cases}g(x), & if -1<x<1\\ \dfrac{f(x)+g(x)}{2}, & if\ x=\pm 1\\ f(x), & if\ x<-1\ or\ x>1\end{cases}$$

Example 4 :

$$\lim_{x\to 0}\frac{x(2^x-1)}{1-\cos x} = ?$$

Solution :

The given limit

$$= \lim_{x\to 0}\frac{2^x-1}{x}.\frac{x^2}{1-\cos x} = \lim_{x\to 0}\frac{2^x-1}{x}.\lim_{x\to 0}\frac{x^2}{2\sin^2\frac{x}{2}} = \log 2.\ 2\lim_{x\to 0}\left(\frac{x/2}{\sin(x/2)}\right)^2 = 2\log 2$$

Example 5 :

If $[x]$ denotes the integral part of x, then find $\lim_{n\to\infty}\frac{\left[1^2x\right]+\left[2^2x\right]+\ldots+\left[n^2x\right]}{n^3}$

Solution :

Let $S_n = [1^2x] + [2^2x] + \ldots + [n^2x]$

$x - 1 < [x] \le x$

$\therefore \quad 1^2x - 1 < [1^2x] \le 1^2x$

$2^2x - 1 < [2^2x] \le 2^2x$

$3^2x - 1 < [3^2x] \le 3^2x$

. . . . . .

. . . . . .

$n^2x - 1 < [n^2x] \le n^2x$

$\therefore \quad (1^2 + 2^2 + ... + n^2)\, x - n < S_n \le (1^2 + 2^2 + ... + n^2)\, x$

$\Rightarrow \quad \frac{n(n+1)(2n+1)}{6n^3}x - \frac{n}{n^3} < \frac{S_n}{n^3} \le \lim_{n\to\infty} \frac{n(n+1)(2n+1)}{6n^3}.x$

$\Rightarrow \quad \lim_{n\to\infty}\left\{\frac{n(n+1)(2n+1)}{6n^3}x - \frac{1}{n^2}\right\} < \lim_{n\to\infty}\frac{S_n}{n^3} \le \frac{n(n+1)(2n+1)}{6n^3}.x$

$\Rightarrow \quad \frac{x}{3} < \lim_{n\to\infty}\frac{S_n}{n^3} \le \frac{x}{3}$. Hence required limit i.e. $\lim_{n\to\infty}\frac{S_n}{n^3} = \frac{x}{3}$

Example 6 :

The integer n for which $\lim_{x\to 0}\frac{(\cos x - 1)(\cos x - e^x)}{x^n}$ is a finite nonzero number is –

(A) 1 (B) 2 (C) 3 (D) 4

Solution : (C)

n cannot be negative integer for then the limit = 0

$$Limit = \lim_{x\to 0}\frac{2\sin^2\frac{x}{2}}{2^2.\left(\frac{x}{2}\right)^2}.\frac{e^x - \cos x}{x^{n-2}} = \frac{1}{2}\lim_{x\to 0}\frac{e^x - \cos x}{x^{n-2}},\ (n \neq 1 \text{ for then the limit} = 0)$$

$$= \frac{1}{2}\lim_{x\to 0}\frac{e^x + \sin x}{(n-2)\,x^{n-3}}$$

So, if $n = 3$, the limit is $\frac{1}{2(n-2)}$ which is finite.

If $n = 4$, the limit is infinite.

Example 7 :

If $f(x) = \frac{x + |x|}{x}$, then $\lim_{x\to 0} f(x)$ = ?

Solution :

$$\text{LHL} = \lim_{h\to 0}\frac{-h + |h|}{-h} = \lim_{h\to 0}(0) = 0$$

$$\text{RHL} = \lim_{h\to 0}\frac{h + |h|}{h} = 2$$

LHL $\neq$ RHL $\Rightarrow$ does not exist

Example 8 :

Find $\lim_{x\to 0}\left(1^{\text{cosec}^2 x} + 2^{\text{cosec}^2 x} + ... + n^{\text{cosec}^2 x}\right)^{\sin^2 x}$

Solution :

Let $y = \text{cosec}^2 x$

Required limit = $\lim_{y\to\infty}(1^y + 2^y + ... n^y)^{1/y}$ [∞^0 form]

$$= \lim_{y\to\infty}\left(n^y\right)^{1/y}\left[\left(\frac{1}{n}\right)^y + \left(\frac{2}{n}\right)^y + ... + \left(\frac{n-1}{n}\right)^y + 1\right]^{1/y}$$

$$= \lim_{y\to\infty} n\left[\left(\frac{1}{n}\right)^y + \left(\frac{2}{n}\right)^y + ... + \left(\frac{n-1}{n}\right)^y + 1\right]^{1/y} = n.\,1^0 = n.\,1 = n$$

Example 9 :

$$\lim_{x\to1} \frac{\sqrt{1-\cos 2(x-1)}}{x-1}$$

(A) exists and it equals $\sqrt{2}$

(B) exists and it equals $-\sqrt{2}$

(C) does not exist because $(x-1) \to 0$

(D) does not exist because left hand limit is not equal to right hand limit

Solution : (D)

$$\lim_{x\to1} \frac{\sqrt{1-\cos 2(x-1)}}{x-1} = \lim_{x\to1} \frac{\sqrt{2\sin^2(x-1)}}{x-1} = \lim_{x\to1} \frac{\sqrt{2}\,|\sin(x-1)|}{x-1}$$

$$\text{LHL} = \lim_{x\to1^-} \frac{\sqrt{2}\,|\sin(x-1)}{x-1} = \lim_{h\to0} -\frac{\sqrt{2}\,|\sin(1-h-1)|}{(1-h-1)}$$

$$= \lim_{h\to0} \frac{\sqrt{2}\,|-\sinh|}{-h} = -\sqrt{2}\lim_{h\to0}\frac{\sinh}{h} = -\sqrt{2}.1 = -\sqrt{2}$$

$$\text{RHL} = \lim_{x\to1^+} \frac{\sqrt{2}\,|\sin(x-1)|}{(x-1)} = \lim_{h\to0} \frac{\sqrt{2}\,|\sin(1+h-1)|}{(1+h-1)}$$

$$= \lim_{h\to0} \frac{\sqrt{2}\,|\sinh|}{h} = \sqrt{2}\lim_{h\to0}\frac{\sinh}{h} = \sqrt{2}.1 = \sqrt{2}$$

Since LHL $\neq$ RHL,

$\therefore$ $\lim\limits_{x\to1} \dfrac{\sqrt{1-\cos 2(x-1)}}{x-1}$ does not exist.

Example 10 :

If α and β be the roots of equation $ax^2 + bx + c = 0$, find $\lim\limits_{x\to\frac{1}{\alpha}} \sqrt{\dfrac{1-\cos\left(cx^2+bx+a\right)}{2(1-\alpha x)^2}}$

Solution :

Since α and β are the roots of equation $ax^2 + bx + c = 0$, therefore roots of equation.

$cx^2 + bx + a = 0$ will be $\dfrac{1}{\alpha}$ and $\dfrac{1}{\beta}$ $\quad\therefore\quad cx^2 + bx + a = c\left(x-\dfrac{1}{\alpha}\right)\left(x-\dfrac{1}{\beta}\right)$

$$\text{Required limit} = \lim_{x\to\frac{1}{\alpha}} \sqrt{\frac{1-\cos\left\{c(x-1/\alpha)(x-1/\beta)\right\}}{2(1-\alpha x)^2}} = \lim_{x\to\frac{1}{\alpha}} \left|\frac{\sin\left\{\frac{c}{2}\left(x-\frac{1}{\alpha}\right)\left(x-\frac{1}{\beta}\right)\right\}}{1-\alpha x}\right|$$

$$= \lim_{x\to\frac{1}{\alpha}} \left| \frac{\sin\left\{\frac{c}{2}\left(x-\frac{1}{\alpha}\right)\left(x-\frac{1}{\beta}\right)\right\}}{\frac{c}{2}\left(x-\frac{1}{\alpha}\right)\left(x-\frac{1}{\beta}\right)} \cdot \frac{c(\alpha x-1)(\beta x-1)}{2\alpha\beta(1-\alpha x)} \right| = \left|\frac{c}{2\alpha\beta}\left(\frac{\beta}{\alpha}-1\right)\right| = \left|\frac{c}{2\alpha}\left(\frac{1}{\alpha}-\frac{1}{\beta}\right)\right|$$

Example 11 :

Show that $\lim_{x\to-2} \frac{\tan\pi x}{x+2} + \lim_{x\to\infty}\left(1+\frac{1}{x^2}\right)^x > 3.$

Solution :

$$\lim_{x\to-2} \frac{\tan\pi x}{x+2} = \lim_{y\to0} \frac{\tan\ \pi(y-2)}{y} = \pi$$

and $\lim_{x\to\infty}\left(1+\frac{1}{x^2}\right)^{x^2(1/x)} = e^{\lim_{x\to\infty}\left(1+\frac{1}{x^2}\right)^{x^2} \lim_{x\to\infty}\frac{1}{x}} = e^0 = 1$

$\therefore \quad \lim_{x\to\infty} \frac{\tan\pi x}{x+2} + \lim_{x\to\infty}\left(1+\frac{1}{x^2}\right)^x = \pi + 1 > 3.$

UNSOLVED EXERCISE

Section-A
(Straight objective type with single option correct)

Topic : 1 $\frac{0}{0}$ and $\frac{\infty}{\infty}$ $\infty-\infty$ Based Question

Q.1 Let a = min (x^2 + 2x + 3, $x \in R$) and $b = \lim_{\theta\to0}\frac{1-\cos\theta}{\theta^2}$. Then the value of $\sum_{r=0}^{n} a^r.b^{n-r}$ is

(a) $\frac{2^{n+1}-1}{3.2^n}$ (b) $\frac{2^{n+1}+1}{3.2^n}$

(c) $\frac{4^{n+1}-1}{3.2^n}$ (d) $\frac{4^{n+1}+1}{3.2^n}$

Q.2 If $S_n = \sum_{k=1}^{n} a_k$ and $\lim_{n\to\infty} a_n = a$, then $\lim_{n\to\infty} \frac{(S_{n+1}-S_n)}{\sqrt{\left(\sum_{k=1}^{n} k^3\right)}}$ is equal to

(a) 0 (b) a

(c) $a\sqrt{2}$ (d) $2a$

Q.3 If $A_i = \frac{x-a_i}{|x-a_i|}, i = 1,2,3.....,n$ and if a_1 < a_2 < a_3 ..... < a_n. Then $\lim_{x\to a_m} (A_1A_2.....A_n), 1 \le m \le n$

(a) Is equal to $(-1)^m$ (b) Is equal to $(-1)^{m+1}$
(c) Is equal to $(-1)^{m-1}$ (d) Does not exist

Q.4 $\lim_{x\to\infty} \frac{\sqrt{x^2+1}-\sqrt[3]{x^3+1}}{\sqrt[4]{x^4+1}-\sqrt[5]{x^4+1}}$ equals

(a) 1 (b) 0
(c) –1 (d) ∞

Q.5 The value of $\lim_{x\to1} \frac{x+x^2....+x^n-n}{x-1}$ is

(a) n (b) $\frac{n+1}{2}$

(c) $\frac{n(n+1)}{2}$ (d) $\frac{n(n-1)}{2}$

Q.6 $\lim_{x\to\infty}\frac{\sqrt{x^2+1}-\sqrt[3]{x^3+1}}{\sqrt[4]{x^4+1}-\sqrt[5]{x^4+1}}$ equals

(a) 1 (b) 0
(c) –1 (d) None of these

Topic : 2 1^∞ , LHL and RHL

Q.1 The value of $\lim_{x\to 0}$ $[x^2 + x + \sin x]$, where [.] denotes the greatest integer function

(a) is equal to –1 (b) Is equal to 0
(c) Is equal to 1 (d) Does not exist

Q.2 The function $f(x)=[x].\cos\frac{2x-1}{2}\pi$, where [.] denotes the greatest integer function, is discontinuous at

(a) All x
(b) All integer points
(c) no x
(d) x which is not an integer

Q.3 Among $\lim_{n\to\infty}\sec^{-1}\left(\frac{x}{\sin x}\right)$ and $\lim_{n\to\infty}\sec^{-1}\left(\frac{\sin x}{x}\right)$

(a) First exists second does not exist
(b) Second exists first does not exist
(c) Both first and second exist
(d) Neither first nor second exists

Q.4 $\lim_{x\to 0}\frac{\tan\left([-\pi^2]x^2\right)-\left(\tan[-\pi^2]\right)x^2}{\sin^2 x}$ equals

(a) 10 – tan 10 (b) tan 10 – 10
(c) –tan 10 + 10 (d) – 10 – tan 10

Q.5 $\lim_{x\to -1+}\frac{\sqrt{\pi}-\sqrt{\cos^{-1}x}}{\sqrt{x+1}}$ is equal to

(a) $1/\sqrt{2}$ (b) $1/\sqrt{2\pi}$
(c) $1/\sqrt{\pi}$ (d) 0

Q.6 If $[x]$ denotes the greatest integer less than or equal to x, then $\lim_{n\to\infty}\frac{1}{n^2}\sum_{k=1}^{n}[kx]$ equals

(a) $x/2$ (b) $x/3$
(c) x (d) 0

Topic : 3 Coninuity at a point Based Question

Q.1 $f(x)$ is continuous function on the real line. Given that $x^2+(f(x)-2)x-\sqrt{3}.f(x)+2\sqrt{3}-3=0$, then the value of $f(\sqrt{3})$ is

(a) cannot be determined
(b) $2(1-\sqrt{3})$
(c) zero
(d) $\frac{2(\sqrt{3}-2)}{\sqrt{3}}$

Q.2 If $f(x)=\begin{cases}x^k\cos(1/x), & x\neq 0\\ 0, & x=0\end{cases}$ is continuous at $x = 0$, then

(a) $k < 0$ (b) $k > 0$
(c) $k = 0$ (d) $k \geq 0$

Q.3 The function $f(x) = a[x + 1] + b[x – 1]$, where $[x]$ is the greatest integer function then the value of $a + b$ for which $f(x)$ is continuous at $x = 1$.

(a) 0 (b) 1
(c) 2 (d) –1

Q.4 If the function $f(x) = \begin{cases}1, & x\leq 2\\ ax+b, & 2<x<4\\ 7, & x\geq 4\end{cases}$ is continuous at x = 2 and 4, then the values of a and b are

(a) 3, 5 (b) 3, – 5
(c) 0, 3 (d) 0, 5

Q.5 Which of the following is true –

(a) If $\{f(x) + g(x)\}$ is continuous at $x = a$, then $f(x)$ and $g(x)$ are both separately continuous at $x = a$
(b) If $f(x)\,g(x)$ is continuous at $x = a$, then $f(x)$ and $g(x)$ are separately continuous at $x = a$
(c) If $f(x)\,g(x)$ and $f(x)$ are discontinuous at $x = a$, then $g(x)$ is continuous at $x = a$
(d) If both $f(x)$ and $g(x)$ are discontinuous at $x = a$ then $f(x) + g(x)$ may not be discontinuous at $x = a$

Q.6 If $f(x) = \begin{cases} x\sin 1/x & , \ x \neq 0 \\ k & , \ x = 0 \end{cases}$ is continuous at $x = 0$, then the value of k will be

(a) 1 (b) –1
(c) 0 (d) None of these

Q.7 Function $f(x) = |x - 1| + |x - 1| + \cos x$, where $x \in [0, 4]$ is not continuous at number of points–

(a) 1 (b) 2
(c) 3 (d) 0

Q.8 The value of $f(0)$ for which $f(x) = \frac{a - \sqrt{x + a^2}}{\sin bx}$ $(a > 0)$ is continuous at $x = 0$ is

(a) 1/2ab (b) $-1/ab$
(c) – a/2b (d) $-1/2ab$

Q.9 The set of points of discontinuity of the function $1/\log |x|$ is –

(a) {–1, 0, 1} (b) {0}
(c) {0, 1} (d) None of these

Q.10 The values of a, b and c which make the function $f(x) = \begin{cases} \frac{\sin(a+1)x + \sin x}{x}, & x < 0 \\ c, & x = 0 \\ \frac{\sqrt{x + bx^2} - \sqrt{x}}{bx^{3/2}}, & x > 0 \end{cases}$

continuous at $x = 0$ are –

(a) $a = \frac{-3}{2}, c = \frac{1}{2}, b = 0$

(b) $a = \frac{3}{2}, c = \frac{1}{2}, b \neq 0$

(c) $a = \frac{-3}{2}, c = \frac{1}{2}, b \neq 0$

(d) None of these

Topic : 4 Coninuity in Domine Based Question

Q.1 If $f(x) = \text{sgn}(\cos 2x - 2\sin x + 3)$, where sgn ( ) is the signum function, then $f(x)$ –

(a) is continuous over its domain
(b) has a missing point discontinuity
(c) has removable discontinuity
(d) has irremovable discontinuity

Q.2 $f(x) = [\tan^{-1} x]$, where [.] denotes the greatest integer function, is discontinuous at

(a) $x = \frac{\pi}{4}, -\frac{\pi}{4}$ and 0

(b) $x = \frac{\pi}{3}, -\frac{\pi}{3}$ and 0

(c) $x = \tan 1, -\tan 1$ and 0
(d) none of these

Q.3 Let $f(x) = \lim_{x \to \infty} \sin^{2n} x$, then number of point(s) where f(x) is discontinuous is –

(a) 0 (B) 1
(c) 2 (D) infinitely many

Q.4 Let $f(x) = \max \{|5 - x|, 7^{-|x|}\}$. Then the function is –

(a) discontinuous at 2 points
(b) discontinuous at ∞ points
(c) continuous everywhere (d) none of these

Q.5 Indicate all correct alternatives if, $f(x) = \frac{x}{2} - 1$, then on the interval $[0, \pi]$

(a) $\tan(f(x))$ and $\frac{1}{f(x)}$ are both continuous

(b) $\tan(f(x))$ and $\frac{1}{f(x)}$ are both discontinuous

(c) $\tan(f(x))$ and $f^{-1}(x)$ are both continuous

(d) $\frac{1}{f(x)}$ is continuous but $\tan(f(x))$ is not

Section-B
(Previous Years Questions)

Q.1 $\lim_{x\to\pi/2}\frac{\cot x-\cos x}{(\pi-2x)^3}$ equals

(a) $\frac{1}{24}$ (b) $\frac{1}{16}$

(c) $\frac{1}{8}$ (d) $\frac{1}{4}$

Q.2 $\lim_{x\to0}\frac{\sin(\pi\cos^2 x)}{x^2}$ is equal to

(a) $\frac{\pi}{2}$ (b) 1

(c) $-\pi$ (d) π

Q.3 $\lim_{x\to0}\frac{(1-\cos2x)(3+\cos x)}{x\tan4x}$ is equal to

(a) 4 (b) 3

(c) 2 (d) $\frac{1}{2}$

Q.4 If $\lim_{x\to\infty}\left(\frac{x^2+x+1}{x+1}-ax-b\right)=4$, then

(a) a = 1, b = 4 (b) a = 1, b = – 4

(c) a = 2, b = – 3 (d) a = 2, b = 3

Q.5 Let $\alpha(a)$ and $\beta(a)$ be the roots of the equation $(\sqrt[3]{1+a}-1)x^2-(\sqrt{1+a}-1)x+(\sqrt[6]{1+a}-1)=0$, where a > – 1. Then, $\lim_{a\to0^+}\alpha(a)$ and $\lim_{a\to0^+}\beta(a)$ are

(a) $-\frac{5}{2}$ and 1 (b) $-\frac{1}{2}$ and – 1

(c) $-\frac{7}{2}$ and 2 (d) $-\frac{9}{2}$ and 3

Q.6 $\lim_{h\to0}\frac{f(2h+2+h^2)-f(2)}{f(h-h^2+1)-f(1)}$, given that f'(2) = 6 and f'(1) = 4

(a) does not exist (b) is equal to – 3/2

(c) is equal to 3/2 (d) is equal to 3

Q.7 If $\lim_{x\to0}\frac{\{(a-n)nx-\tan x\}\sin nx}{x^2}$ = 0, where *n* is non-zero real number, then *a* is equal to

(a) 0 (b) $\frac{n+1}{n}$

(c) n (d) $n+\frac{1}{n}$

Q.8 The integer *n* for which $\lim_{x\to0}\frac{(\cos x-1)(\cos x-e^x)}{x^n}$ is a finite non-zero number, is

(a) 1 (b) 2

(c) 3 (d) 4

Q.9 $\lim_{x\to0}\frac{x\tan2x-2x\tan x}{(1-\cos2x)^2}$ is

(a) 2 (b) – 2

(c) $\frac{1}{2}$ (d) $-\frac{1}{2}$

Q.10 $\lim_{x\to1}\frac{\sqrt{1-\cos2(x-1)}}{(x-1)}$

(a) exists and it equals $\sqrt{2}$

(b) exists and it equals $-\sqrt{2}$

(c) does not exist because $x-1\to0$

(d) does not exist because left hand limit is not equal to right hand limit

Q.11 The value of $\lim_{x\to0}\frac{\sqrt{\frac{1}{2}(1-\cos^2 x)}}{x}$ is

(a) 1 (b) – 1

(c) 0 (d) None of the above

Q.12 If f (x) = $\begin{cases}\frac{\sin[x]}{[x]}, & [x]\neq0\\ 0, & [x]\neq0\end{cases}$ where, [x] denotes the greatest integer less than or equal to x, then $\lim_{x\to0}$ f(x) equals

(a) 1 (b) 0

(c) – 1 (d) None of these

Q.13 $\lim_{n\to\infty}\left(\frac{1}{1-n^2}+\frac{2}{1-n^2}+....+\frac{n}{1-n^2}\right)$ is equal to

(a) 0 (b) $-\frac{1}{2}$

(c) $\frac{1}{2}$ (d) None of these

Q.14 If f (a) = 2, f' (a) = 1, g(a) = – 1, g'(a) = 2, then the value of $\lim_{x\to a}\frac{g(x)f(a)-g(a)f(x)}{x-a}$ is

(a) -5 (b) $\frac{1}{5}$

(c) 5 (d) None of these

Q.15 If G(x) = $-\sqrt{25-x^2}$, then $\lim_{x\to1}\frac{G(x)-G(1)}{x-1}$ has the value

(a) $\frac{1}{\sqrt{24}}$ (b) $\frac{1}{5}$

(c) $-\sqrt{24}$ (d) None of these

Q.16 Let p = $\lim_{x\to0^+}(1+\tan^2\sqrt{x})^{1/2x}$, then log p is equal to

(a) 2 (b) 1

(c) $\frac{1}{2}$ (d) $\frac{1}{4}$

Q.17 If $\lim_{x\to0}[1+x\log(1+b^2)]^{1/x} = 2b\sin^2\theta$, b > 0 and $\theta \in (-\pi, \pi]$, then the value of θ is

(a) $\pm\frac{\pi}{4}$ (b) $\pm\frac{\pi}{3}$

(c) $\pm\frac{\pi}{6}$ (d) $\pm\frac{\pi}{8}$

Q.18 For x > 0, $\lim_{x\to0}\left[(\sin x)^{1/x}+\left(\frac{1}{x}\right)^{\sin x}\right]$ is

(a) 0 (b) – 1

(c) 1 (d) 2

Q.19 Let $f : R \to R$ be such that $f(1) = 3$ and $f'(1) = 6$. Then, $\lim_{x\to0}\left[\frac{f(1+x)}{f(1)}\right]^{1/x}$ equals:

(a) 1 (b) $e^{1/2}$

(c) e^2 (d) e^3

Q.20 For $x\in R$, $\lim_{x\to\infty}\left(\frac{x-3}{x+2}\right)^x$ is equal to

(a) e (b) e^{-1}

(c) e^{-5} (d) e^5

Q.21 $\lim_{x\to\frac{\pi}{4}}\frac{\int_2^{\sec^2 x} f(t)\,dt}{x^2-\frac{\pi^2}{16}}$ equals

(a) $\frac{8}{\pi}f(2)$ (b) $\frac{2}{\pi}f(2)$

(c) $\frac{2}{\pi}f\left(\frac{1}{2}\right)$ (d) 4f (2)

Q.22 $\lim_{n\to\infty}\frac{1}{n}\sum_{r=1}^{2n}\frac{r}{\sqrt{n^2+r^2}}$ equals

(a) $1+\sqrt{5}$ (b) $\sqrt{5}-1$

(c) $-1+\sqrt{2}$ (d) $1+\sqrt{2}$

Q.23 The function f (x) = $[x]^2 - [x^2]$ (where, [x] is the greatest integer less than or equal to x), is discontinuous at

(a) all integers
(b) all integers except 0 and 1
(c) all integers except 0
(d) all integers except 1

Q.24 Let [.] denotes the greatest integer function and f (x) = $[\tan^2 x]$, then

(a) $\lim_{x\to0}$ f (x) does not exist
(b) f (x) is continuous at x = 0
(c) f (x) is not differentiable at x = 0
(d) f' (0) = 1

Q.25 The function f (x) = [x] $\cos\left(\frac{2x-1}{2}\right)\pi$, [·] denotes the greatest integer function, is discontinuous at

(a) all *x*
(b) all integer points
(c) no *x*
(d) *x* which is not an integer

Q.26 If f (x) = $x(\sqrt{x}+\sqrt{x+1})$, then

(a) *f* (x) is continuous but not differentiable at x = 0
(b) *f* (x) is differentiable at x = 0
(c) *f* (x) is not differentiable at x = 0
(d) None of the above

Q.27 The function f (x) = $\frac{\log(1+ax)-\log(1-bx)}{x}$ is not defined at x = 0. The value which should be assigned to *f* at x = 0, so that it is continuous at x = 0, is

(a) a – b (b) a + b

(c) log a + log b (d) None of these

Q.28 If f (x) = $\frac{1}{2}x - 1$, then on the interval $[0, \pi]$

(a) tan [f (x)] and 1/f (x) are both continuous
(b) tan [f(x)] and 1/f (x) are both discontinuous
(c) tan [f (x)] and f^{-1} (x) are both continuous
(d) tan [f (x)] is continuous but 1/f (x) is not continuous

Q.29 For $x \in R$, $f(x) = |\log 2 - \sin x|$ and $g(x) = f(f(x))$, then
(a) g is not differentiable at x = 0
(b) g'(0) = cos (log 2)
(c) g'(0) = – cos (log 2)
(d) g is differentiable at x = 0 and g'(0) = – sin (log 2)

Q.30 If f and g are differentiable functions in (0, 1) satisfying f (0) = 2 = g(1), g(0) = 0 and f(1) = 6, then for some $c \in]0,1[$
(a) 2f'(c) = g'(c) (b) 2f'(c) = 3g'(c)
(c) f'(c) = g'(c) (d) f'(c) = 2g'(c)

Q.31 Let $f(x) = \begin{cases} x^2 \left|\cos\frac{\pi}{x}\right|, & x \neq 0 \\ 0, & x = 0 \end{cases}$, $x \in R$, then *f* is
(a) differentiableboth at x = 0 and at x = 2
(b) differentiable at x = 0 but not differentiable at x = 2
(c) not differentiable at x = 0 but differentiable at x = 2
(d) differentiable neither at x = 0 nor at x = 2

Q.32 Let $f(x) = \begin{cases} x^2 \left|\cos\frac{\pi}{x}\right|, & x \neq 0 \\ 0, & x = 0 \end{cases}$, $x \in R$, then *f* is
(a) differentiableboth at x = 0 and at x = 2
(b) differentiable at x = 0 but not differentiable at x = 2
(c) differentiable at x = 0 but differentiable at x = 2
(d) differentiable neither at x = 0 nor at x = 2

Q.33 Let $g(x) = \frac{(x-1)^n}{\log\cos^m(x-1)}$; $0 < x < 2$, m and n are integers, $m \neq 0$, $n > 0$ and let p be the left hand derivative of $|x - 1|$ at x = 1. If $\lim_{x \to 1^+} g(x) = p$, then
(a) n = 1, m = 1 (b) n = 1, m = – 1
(c) n = 2, m = 2 (d) n > 2, m = n

Q.34 If *f* is a differentiable function satisfying $f\left(\frac{1}{n}\right) = 0$, $\forall\ n \geq 1$, $n \in I$, then
(a) f (x) = 0, $x \in (0, 1]$
(b) f'(0) = 0 = f (0)
(c) f (0) = 0 but f' (0) not necessarily zero
(d) $|f(x)| \leq 1$, $x \in (0, 1]$

Q.35 Let f (x) = ||x| – 1|, then points where, f (x) is not differentiable is/are
(a) 0, ± 1 (b) ± 1
(c) 0 (d) 1

Q.36 The domain of the derivative of the functions
$f(x) = \begin{cases} \tan^{-1} x, & \text{if } |x| \leq 1 \\ \frac{1}{2}(|x| - 1), & \text{if } |x| > 1 \end{cases}$ is
(a) R – {0} (b) R – {1}
(c) R – {1} (d) R – {1, 1}

Q.37 Which of the following functions is differentiable at x = 0?
(a) cos (|x|) + |x| (b) cos (|x|) – |x|
(c) sin (|x|) + |x| (d) sin (|x|) – |x|

Q.38 The left hand derivative of f (x) = [x] sin (π x) at x = k, k is an integer, is
(a) $(-1)^k (k - 1)\pi$
(b) $(-1)^{k-1} (k - 1)\pi$
(c) $(-1)^k k\pi$
(d) $(-1)^{k-1} k\pi$

Q.39 Let $f : R \to R$ be any function. Define $g : R \to R$ by $g(x) = |f(x)|$, $\forall\ x$. Then, g is
(a) onto if *f* is onto
(b) one-one if *f* is one-one
(c) continuous if *f* is continuous
(d) differentiable if *f* is differentiable

Q.40 The function $f(x) = (x^2 - 1)|x^2 - 3x + 2| + \cos(|x|)$ is not differentiable at
(a) – 1 (b) 0
(c) 1 (d) 2

Q.41 The set of all points, where the function f (x) $= \frac{x}{1+|x|}$ is differentiable, is
(a) $(-\infty, \infty)$ (b) $[0, \infty)$
(c) $(-\infty, 0) \cup (0, \infty)$ (d) $(0, \infty)$

Section-C
(School / Board pattern)

Q.1 Let $f(x) = \begin{cases} \frac{1-\sin\pi x}{1+\cos 2\pi x} & , \ 0 < x < \frac{1}{2} \\ k & , \ x = \frac{1}{2} \\ \frac{\sqrt{2x-1}}{\sqrt{4+\sqrt{2x-1}}-2} & , \ x > \frac{1}{2} \end{cases}$.

Determine the value of k, if possible, so that the function is continuous at $x = 1/2$.

Q.2 Let $[x]$ denote the greatest integer function and $f(x)$ be defined in a neighborhood of 2 by

$$f(x) = \begin{cases} \frac{\exp\left((x+2)\frac{1}{4}[x+1]\ \ell n 4\right)-16}{4^x-16} & , \ x < 2 \\ A\,\frac{1-\cos(x-2)}{(x-2)\tan(x-2)} & , \ 2 < x < 2+\pi/2 \end{cases}.$$

Find the values of A and $f(2)$ in order that $f(x)$ may be continuous at $x = 2$.

Q.3 Determine the values of a, b & c for which the function $f(x) = \begin{cases} \frac{\sin(a+1)x+\sin x}{x} & \text{for} \quad x<0 \\ c & \text{for} \quad x=0 \\ \frac{\left(x+bx^2\right)^{1/2}-x^{1/2}}{bx^{3/2}} & \text{for} \quad x>0 \end{cases}$

is continuous at $x = 0$.

Q.4 Find the values of A and B for which function is continuous $f(x) = \begin{cases} (1+|\cos x|)^{\frac{AB}{|\cos x|}}, & \pi < x < \frac{3\pi}{2} \\ e^A e^B, & x = \frac{3\pi}{2} \\ e^{\frac{\cot 2x}{\cot 8x}}, & \frac{3\pi}{2} < x < \frac{25\pi}{16} \end{cases}$

at $x = 3\pi/2$

Q.5 Find the values of a and b. If $f(x)$ is continuous at $x = 0$, where $f(x) = \begin{cases} 3, & x = 0 \\ \left(1+\frac{ax+bx^3}{x^2}\right)^{\frac{1}{x}}, & x \neq 0 \end{cases}$.

Q.6 Let $f(x) = \begin{cases} \frac{\ell n\cos x}{\sqrt[4]{1+x^2}-1} & \text{if } x>0 \\ \frac{e^{\sin 4x}-1}{\ell n(1+\tan 2x)} & \text{if } x<0 \end{cases}$ is it possible to define f(0) to make the function continuous at $x = 0$. If yes what is the value of $f(0)$.

Q.7 The function $f(x) = \begin{cases} ax(x-1)+b & \text{when } x<1 \\ x-1 & \text{when } 1\le x\le 3 \\ px^2+qx+2 & \text{when } x>3 \end{cases}$.

Find the values of the constants a, b, p, q so that $f(x)$ is continuous for all x

Q.8 If $f(x)$ be continuous function in $[0, 2\pi]$ & $f(0) = f(2\pi)$ then prove that there exists point such that $f(c) = f(c+\pi)$.

Q.9 Suppose $f:[0,\ 1]\to R$ is continuous and

$f(0) = f(1)$. Prove that there exists $c\in\left[0, \frac{1}{2}\right]$ such that $f(c) = f\left(c+\frac{1}{2}\right)$.

Q.10 Let $f: [0,1] \to [0, 1]$ be a continuous function. Prove that $f(x) = x$ for some $x\in[0, 1]$.

Q.11 Prove that: $f(x)=\begin{cases} \frac{x^2}{2}, & \text{if } 0\le x\le 1 \\ 2x^2-3x+\frac{3}{2}, & \text{if } 1<x\le 2 \end{cases}$ is continuous at $x=1$

Q.12 $f(x)=\begin{cases} 3x-8, & \text{if } x\le 5 \\ 2k, & \text{if } x>5 \end{cases}$ is continuous at $x=5$. Then find the value of k.

Q.13 $f(x)=\begin{cases} \frac{2^{x+2}-16}{4^x-16}, & \text{if } x\ne 2 \\ k, & \text{if } x=2 \end{cases}$ is continuous at $x=2$. Then find the value of k.

Q.14 $f(x)=\begin{cases} \frac{\sqrt{1+kx}-\sqrt{1-kx}}{x}, & \text{if } -1\le x<0 \\ \frac{2x+1}{x-1} & \text{if } 0\le x\le 1 \end{cases}$ is continuous $x=0$. Then find the value of k.

Q.15 Prove that the function f defined by $f(x)=\begin{cases} \frac{x}{|x|+2x^2}, & \text{if } x\ne 0 \\ k, & \text{if } x=0 \end{cases}$ remains discontinuous at $x=0$, regardless the choice of K.

Q.16 Find the values of a and b such that the function f defined by $f(x)=\begin{cases} \frac{x-4}{|x-4|}+a, & \text{if } x<4 \\ a+b, & \text{if } x=4 \\ \frac{x-4}{|x-4|}+b, & \text{if } x>4 \end{cases}$

is a continuous function at x =4.

Q.17 Prove that: $f(x)=\begin{cases} x^2\sin\frac{1}{x}, & \text{if } x\ne 0 \\ 0, & \text{if } x=0 \end{cases}$ is differentiable at $x=0$.

Q.18 If $y=(\cos x)^{(\cos x)^{(\cos x)\ldots\ldots\infty}}$, then show that $\frac{dy}{dx}=\frac{y^2\tan x}{y\log\cos x-1}$.

Q.19 Find the values of p and q, so that $f(x)=\begin{cases} x^2+3x+p, & \text{if } x\le 1 \\ qx+2, & \text{if } x>1 \end{cases}$ is

Q.20 If $x=\sin t$ and $y=\sin pt$, then prove that $x=\sin t$ and $y=\sin pt$, then prove that

$(1-x^2)\frac{d^2y}{dx^2}-x\frac{dy}{dx}+p^2y=0.$

ANSWER KEY

CHAPTER-01 (RELATIONS & FUNCTIONS)

SECTION-A (Straight objective type questions)

Topic 1: Type of Functions Based Questions

1. (c) **2.** (a) **3.** (b) **4.** (d) **5.** (a) **6.** (d) **7.** (d)
8. (d) **9.** (c) **10.** (a) **11.** (a) **12.** (c) **13.** (d) **14.** (b)
15. (b) **16.** (d) **17.** (d) **18.** (b) **19.** (a)

Topic 2: Composition of Functions Based Questions

1. (d) **2.** (b) **3.** (d) **4.** (d) **5.** (b) **6.** (b) **7.** (c)
8. (a) **9.** (d)

Topic 3: Inverse Functions Based Questions

1. (b) **2.** (b) **3.** (b) **4.** (b) **5.** (a) **6.** (b) **7.** (a)
8. (b) **9.** (a) **10.** (d) **11.** (b) **12.** (a) **13.** (c) **14.** (b)
15. (a) **16.** (c) **17.** (b) **18.** (c) **19.** (d)

Topic 4: Perodic Even & Odd Function Based Question

1. (b) **2.** (c) **3.** (a) **4.** (a) **5.** (d) **6.** (c)

SECTION-B (Previous Years Questions)

1. (b) **2. (a)** **3. (c)** **4. (d)** **5. (d)** **6. (d)** **7. (c)**
8. (c) **9. (c)** **10. (a)** **11. (d)** **12. (b)** **13. (d)** **14. (b)**
15. (a) **16. (d)** **17. (d)** **18. (b)** **19. (d)** **20. (b)** **21. (a)**
22. (a) **23. (c)** **24. (b)** **25. (d)** **26. (a)** **27. (b)** **28. (b)**
29. (a)

SECTION-C (School / Board Pattern)

1. $[-5, 5]$ **2.** $f^{-1} = \{(b, a)(d, b)(a, c)(c, d)\}$ **3.** $\alpha = 2, \beta = -1$ **5.** $\left[\frac{1}{3}, 1\right]$

7. D = {1, 2,....., 20}, R = {1, 3,.....,35} R is nither reflecive nor Symmetric and nor transitive
9. (i) transitive, (ii) symmetric, (iii) equivalence, (iv) transitive
10. {(1, 4), (2, 5), (3, 6), (4, 7), (5, 8), (6, 9)}
11. (i) not commutative (ii) Commutative (iii) not commutative (iv) commutative **12.** (i)

CHAPTER-02 (INVERSE TRIGONOMETRIC FUNCTIONS)

SECTION-A (Straight objective type questions)

Topic: 1 Domain and Range Based Questions

1. (a) **2. (c)** **3. (a)** **4. (a)** **5. (d)**

Topic: 2 Property of inverse Based Questions

1. (b) **2. (c)** **3. (c)** **4. (c)** **5. (a)** **6. (a)** **7. (c)**
8. (a) **9. (c)** **10. (a)** **11. (c)** **12. (c)** **13. (a)** **14. (d)**
15. (b) **16. (b)** **17. (a)** **18. (c)** **19. (c)**

Topic: 3 Sum and Differenence Formula Based Questions

1. (a) **2. (d)** **3. (d)** **4. (c)** **5. (b)** **6. (a)** **7. (c)**
8. (c) **9. (d)**

SECTION-B (Previous Years Questions)

1. (c) **2. (b,c,d)** **3. (b)** **4. (c)** **5. (a)** **6. (b)** **7. (a)**
8. (c) **9. (d)** **10. (b)**

SECTION-C (School / Board Pattern)

1. 0 **2.** $\frac{\pi}{12}$ **4. 0, – 1** **8.** $\tan^{-1}\frac{4}{3} - x$ **10.** $\frac{\pi}{4}$ **12.** $\frac{a_n - a_1}{1 + a_n \cdot a_1}$

CHAPTER-03 (MATRICES AND DETERMINANTS)

SECTION-A (Straight objective type questions)

TOPIC: 1 Types of Matrices , Addition, Subtraction and Transpose of Matrix Based Questions

1. (b)	**2. (d)**	**3. (b)**	**4. (b)**	**5. (b)**	**6. (d)**	**7. (b)**
8. (c)	**9. (c)**	**10. (c)**	**11. (b)**	**12. (b)**	**13. (a)**	**14. (b)**
15. (d)	**16. (d)**	**17. (a)**	**18. (d)**	**19. (a)**	**20. (d)**	

TOPIC: 2 Properties of Determinants Based Questions

1. (c)	**2. (a)**	**3. (d)**	**4. (d)**	**5. (c)**	**6. (a)**	**7. (b)**
8. (c)	**9. (d)**	**10. (c)**	**11. (d)**	**12. (d)**	**13. (c)**	**14. (a)**
15. (c)	**16. (c)**	**17. (a)**				

TOPIC: 3 Adjoint and inverse of a Matrix Based Questions

1. (c)	**2. (b)**	**3. (b)**	**4. (d)**	**5. (a)**	**6. (b)**	**7. (b)**
8. (c)	**9. (b)**					

TOPIC: 4 Solvling system of Equation Based Questions

1. (c)	**2. (c)**	**3. (c)**	**4. (a)**	**5. (a)**	**6. (c)**	**7. (d)**
8. (d)	**9. (a)**	**10. (d)**				

SECTION-B (Previous Years Questions)

1. (a)	**2. (b)**	**3. (d)**	**4. (a)**	**5. (a)**	**6. (d)**	**7. (a)**
8. (c)	**9. (d)**	**10. (c)**	**11. (c)**	**12. (a)**	**13. (b)**	**14. (b)**
15. (b)	**16. (b)**	**17. (b)**	**18. (b)**	**19. (b)**	**20. (b)**	**21. (a)**
22. (c)	**23. (d)**	**24. (c)**	**25. (d)**	**26. (a)**	**27. (b)**	**28. (b)**
29. (a)	**30. (d)**					

SECTION-C (School / Board type questions)

1. $x=1, y=2$ **6.** $x=\pm\frac{1}{\sqrt{2}}, y=\pm\frac{1}{\sqrt{6}}, z=\pm\frac{1}{\sqrt{3}}$ **7.** $3(x+y+z)(yz+yx+xz)$. **9.** $x=y=z=1$

10. $x=2, y=-1, z=4$ **11.** $(a^2+b^2+c^2-3abc)[(a-b)^2+(b-c)^2+(c-a)^2]$

CHAPTER-04 (CONTINUITY & DIFFERENTIABILITY)

SECTION-A (Straight objective type questions)

Topic: 1 $\frac{0}{0}$ and $\frac{\infty}{\infty}\infty-\infty$ **type based questions**

1. (c)	**2. (a)**	**3. (d)**	**4. (b)**	**5. (c)**	**6. (b)**

Topic: 2 1^∞, LHL and RHL based questions

1. (d)	**2. (c)**	**3. (a)**	**4. (b)**	**5. (b)**	**6. (a)**

Topic: 3 Coninuity at a point based questions

1. (b)	**2. (b)**	**3. (a)**	**4. (b)**	**5. (b)**	**6. (c)**	**7. (d)**
8. (d)	**9. (c)**	**10. (a)**				

Topic: 4 Coninuity in domine based questions

1. (c)	**2. (c)**	**3. (d)**	**4. (c)**	**5. (c)**

Section-B (Previous Years Questions)

1. (b)	**2. (d)**	**3. (c)**	**4. (b)**	**5. (b)**	**6. (d)**	**7. (d)**
8. (c)	**9. (b)**	**10. (c)**	**11. (d)**	**12. (d)**	**13.(b)**	**14. (c)**
15. (a)	**16. (c)**	**17. (d)**	**18. (c)**	**19. (c)**	**20. (c)**	**21. (a)**
22. (b)	**23. (b)**	**24. (b)**	**25. (c)**	**26. (c)**	**27.(b)**	**28. (b)**
29. (b)	**30. (d)**	**31. (b)**	**32. (c)**	**33. (b)**	**34. (a)**	**35. (d)**
36. (d)	**37. (a)**	**38. (b)**	**39. (c)**	**40. (d)**	**41. (a)**	

SECTION-C (School / Board type questions)

1. $k = \phi$ **2.** $A = 1, f(2) = 1/2$ **3.** $a = -3/2,\ b \in R - \{0\}, c = 1/2$ **4.** a = 2, b = 2

5. a = 0, b = ln 3 **6.** No, non-removable discontinuity **7.** $a \neq 1$, b = 0, q = – 1, p = 1/3

12. k = 7/2 **13.** k = 1/2 **14.** k = – 1 **16.** b = – 1 and a = 1 **19.** p = 3, q = 5